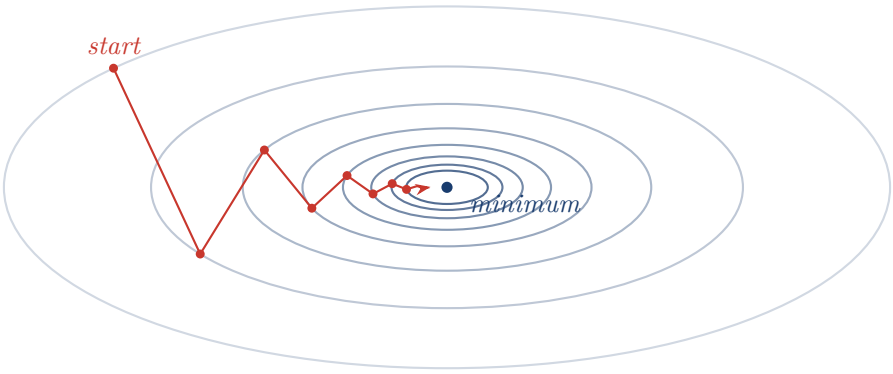
Mathematics for Machine Learning and Deep Learning

From Absolute Beginner to Prerequisites



Gradient descent finding the bottom of a valley: the central idea of Chapter 8, built from everything that comes before it.

Lecture Notes

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How to Use These Notes

Who these notes are for

These notes are written for someone who wants to understand machine learning (ML) and deep learning (DL) properly, but who currently knows little or no mathematics beyond everyday arithmetic: adding, subtracting, multiplying and dividing numbers. If you have ever opened a machine learning paper, a textbook or even a tutorial and been stopped by a wall of symbols such as $\nabla_{\boldsymbol{\theta}} L$, $\boldsymbol{W}^{\top}\boldsymbol{x}$, $\mathbb{E}[X]$ or $\sum_{i=1}^{n}$, these notes are meant to take you from that point to the point where those symbols feel like old friends.

Nothing is assumed. Every concept is introduced from first principles, with a reason for why it exists, a picture of what it means, concrete numerical examples, and only then the general rule. When we say "from absolute beginner", we mean it: Chapter 1 starts with what a number is.

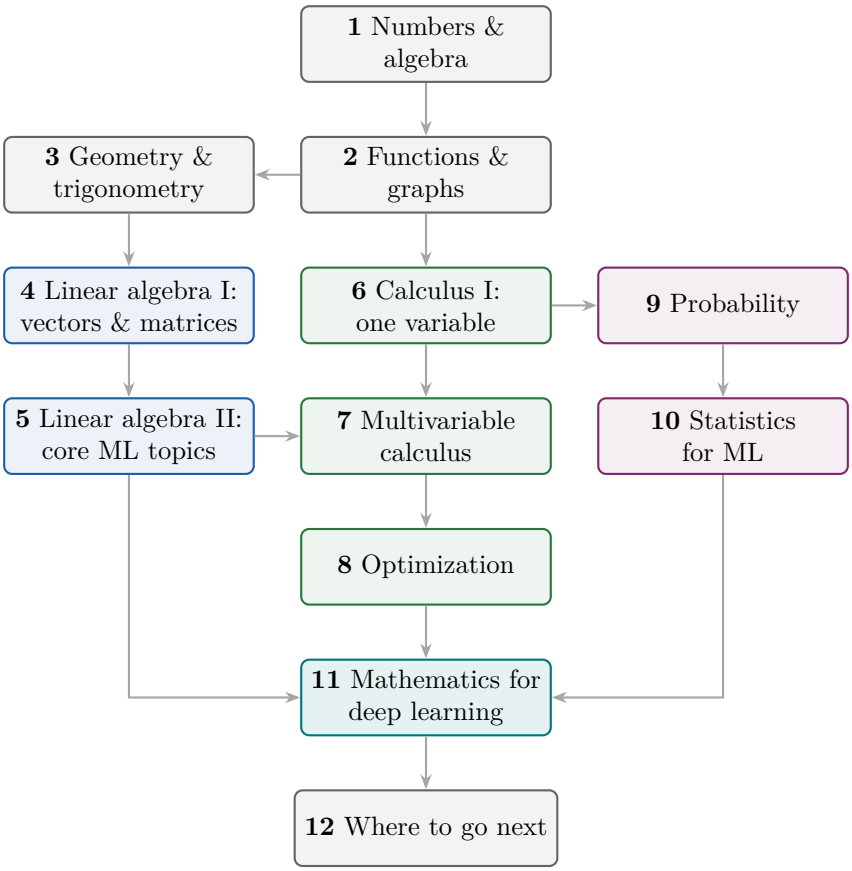
At the same time, the destination is ambitious. By the last chapter you will have met every piece of mathematics that a standard first course in machine learning or deep learning takes for granted: linear algebra (vectors, matrices, eigenvalues, the singular value decomposition), calculus (derivatives, gradients, the chain rule, Taylor approximations), optimization (gradient descent and its relatives), probability and statistics (distributions, expectation, Bayes' theorem, maximum likelihood, the bias–variance trade-off), and the specific tools of deep learning (softmax, cross-entropy, matrix calculus, backpropagation and attention).

What these notes are not

They are not a course on programming, and they are not a complete treatment of any branch of mathematics. Where a topic is deep (for example, the theory of limits or of measure-theoretic probability), we give the intuition that a practitioner needs and point you to where the rigorous story is told. The final chapter, *Where to Go Next*, suggests what to read after these notes.

How the notes are organised

The notes are divided into twelve chapters that build on one another. The diagram below shows which chapters depend on which. The three main "strands" are linear algebra (left), calculus and optimization (middle) and probability and statistics (right); they come together in Chapter 11, where we assemble the mathematics of a neural network.



If you are a complete beginner, read the chapters in order. If you already feel comfortable with some material (say, you remember high-school algebra), skim the corresponding chapter, try its exercises, and move on as soon as they feel easy. Each chapter opens with a list of what you will learn, which you can use as a checklist.

The boxes

The text is interspersed with coloured boxes. Each colour always means the same thing, so that you can find your way around quickly.

Motivation. Opens every chapter and many sections: *why* do we need this idea, and where does it show up in machine learning?

Definition. The precise meaning of a new word or symbol. Definitions are the vocabulary of mathematics; it pays to read them twice.

Theorem / Proposition / Corollary. A fact that follows from the definitions. Often we explain why it is true, sometimes with a short proof.

Example. A worked example with concrete numbers. Try to solve it yourself before reading the *Solution*.

Intuition / Picture this. An informal explanation, an analogy, or a mental picture. These are often the most important boxes of all.

Why this matters for ML. How the idea is used in machine learning or deep learning.

Common pitfall. A mistake that many people make, and how to avoid it.

Notation. How a new symbol is written and read aloud.

Remark. A side comment: a subtlety, a generalisation, or a historical note. Safe to skip on a first reading.

Algorithm. A recipe, written as a list of steps, that a computer (or you, with a pencil) can follow.

Exercise. A problem for you to solve. Exercises appear at the end of most sections. Every exercise has a solution in Appendix C; the small link in the bottom-right corner of each exercise box takes you straight to it, and the exercise number in the appendix takes you back.

New technical terms are printed in **bold colour** where they are first explained, and every such term is listed in the index at the end of the notes.

How to study mathematics

Mathematics is not a spectator sport. The following habits make an enormous difference.

- **Read slowly, with pencil and paper.** A page of mathematics can take much longer to read than a page of a novel. That is normal. When the text says "it follows that", check that it does.
- **Work through every example yourself.** Cover the solution, try the problem, then compare.
- **Do the exercises.** Understanding comes from doing. Start with the first few exercises of each section; they are the easiest.
- **Say the symbols out loud.** Every formula is a sentence. The *Notation* boxes tell you how to read new symbols, and Appendix A collects them all.
- **Expect confusion, and push through it.** Being confused is not a sign that you are bad at mathematics; it is what learning mathematics feels like. Come back to a difficult section after a day or two and it will often have become clear.
- **Connect to code if you can.** If you program, implement small examples (for instance in Python with NumPy). Seeing a formula produce the same number as your hand calculation is a powerful check.

Reference material at the end

Appendix A lists all notation and the Greek alphabet, Appendix B is a compact formula sheet for revision, and Appendix C contains solutions to all exercises.

Chapter 1

Numbers, Arithmetic, and Algebraic Thinking

Motivation. To a computer, everything in machine learning is a number. A photograph of a cat is a grid of numbers that record how bright each tiny dot (pixel) is. A sentence is turned into a list of numbers before a language model reads it. A house is described by numbers such as its floor area, its number of bedrooms and its age. A machine learning *model* is a recipe that turns input numbers into output numbers ("this picture is a cat with probability 0.97"), and the recipe contains adjustable numbers, called *parameters*, that are tuned during training.

So before anything else we need to be fluent with numbers: what kinds of numbers exist, how to calculate with them, and, crucially, how to reason about numbers we do not know yet. That last skill is called *algebra*. It is the language in which every formula of machine learning is written.

In this chapter you will learn to...

- distinguish natural numbers, integers, rational numbers and real numbers, and use set and interval notation;
- calculate confidently with negative numbers, fractions, decimals and percentages;
- use exponents, roots and scientific notation (such as 3×10^{-4});
- evaluate expressions using the correct order of operations;
- read and write summation notation $\sum$, the most common symbol in machine learning formulas;
- solve linear equations, simple systems of equations, and linear inequalities;
- work with absolute values and interpret them as distances.

1.1 Kinds of numbers

Numbers were invented to answer questions. "How many sheep do I have?" needs one kind of number; "how much do I owe?" needs another; "how long is the diagonal of this square?" needs yet another. Each new kind of question enlarged the family of numbers. We will meet the family members in the order in which they are usually discovered.

1.1.1 Natural numbers

The numbers we use for counting, $1, 2, 3, 4, \ldots$, are called the **natural numbers**. The three dots mean "and so on forever": there is no largest natural number, because you can always add one more.

Notation. A collection of objects is called a **set**, and the objects in it are its **elements**. We list the elements of a set between curly braces. The set of natural numbers has a

special symbol, a "blackboard bold" letter N:

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}.$$

We write $5 \in \mathbb{N}$, read "5 *is an element of* $\mathbb{N}$" or simply "5 is in $\mathbb{N}$". We write $-2 \notin \mathbb{N}$, read "-2 is not in $\mathbb{N}$".

Remark. Some books include 0 in the natural numbers and some do not. Nothing deep depends on this choice. In these notes, $\mathbb{N}$ starts at 1.

1.1.2 Integers

Counting numbers cannot describe a temperature of five degrees below zero, a debt of \$20, or a position 3 metres below sea level. For that we need zero and the negative numbers. The **integers** are the whole numbers, positive, negative and zero:

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

(The letter $\mathbb{Z}$ comes from the German word *Zahlen*, "numbers".)

The best way to picture the integers is on a **number line**: a straight line with 0 in the middle, the positive numbers spaced evenly to the right and the negative numbers to the left (Figure 1.1). Numbers further to the right are larger. So -1 is larger than -5, even though 5 is larger than 1: being 1 degree below zero is warmer than being 5 degrees below zero.

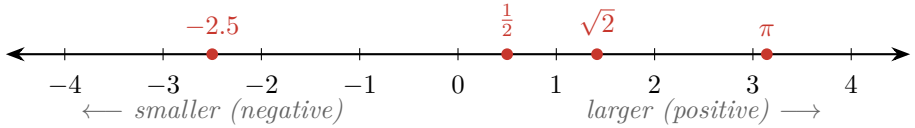


Figure 1.1: The number line. Every real number corresponds to exactly one point, and every point to exactly one real number. Integers are marked with ticks; a few non-integers are marked in red.

1.1.3 Rational numbers

Sharing 3 pizzas among 4 people gives each person $\frac{3}{4}$ of a pizza, which is not a whole number. A **rational number** is any number that can be written as a ratio (a fraction) of two integers,

$$\frac{p}{q}, \qquad \text{where } p \text{ and } q \text{ are integers and } q \neq 0.$$

Examples are $\frac{3}{4}$, $-\frac{5}{2}$, $\frac{7}{1} = 7$ and $0.25 = \frac{1}{4}$. The set of rational numbers is written $\mathbb{Q}$ (for *quotient*). Every integer is rational, because $n = \frac{n}{1}$.

When written as decimals, rational numbers either stop ($\frac{1}{4} = 0.25$) or repeat a pattern forever ($\frac{1}{3} = 0.333\ldots$, $\frac{1}{7} = 0.142857142857\ldots$).

1.1.4 Irrational and real numbers

The ancient Greeks discovered, to their dismay, that some lengths are *not* ratios of integers. The diagonal of a square with sides of length 1 has length $\sqrt{2} = 1.41421356\ldots$, and it can be proved that no fraction p/q equals $\sqrt{2}$ exactly. Numbers like this, whose decimal expansions never stop and never repeat, are called **irrational**. Another famous example is $\pi = 3.14159265\ldots$, the ratio of a circle's circumference to its diameter.

Putting the rational and irrational numbers together gives the **real numbers**, written $\mathbb{R}$. These are *all* the points on the number line, with no gaps. Almost all the numbers in machine learning are real numbers: weights, probabilities, distances, losses and so on.

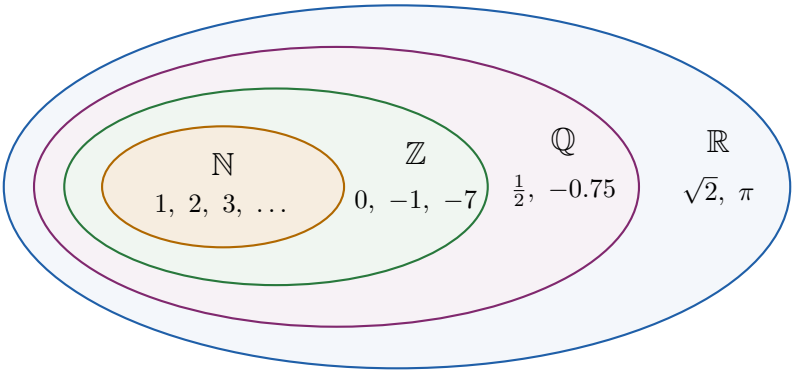


Figure 1.2: The number systems sit inside one another: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. Each ring shows a few numbers that belong to that set but not to the smaller ones inside it.

Notation. If every element of a set A is also an element of a set B, we say A is a **subset** of B and write $A \subset B$. Figure 1.2 shows that

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

We can also describe a set by a property its elements satisfy, using **set-builder notation**:

$$\{x \in \mathbb{R} : x > 0\}$$

is read "the set of all real numbers x *such that* x is greater than 0". The colon (sometimes written as a vertical bar |) means "such that".

1.1.5 Intervals

We often need to talk about all the real numbers between two values, for example "all probabilities", which lie between 0 and 1. Such a set is called an **interval**. A square bracket means the endpoint is included; a round bracket means it is excluded.

Notation	Meaning	Name
$[a, b]$	all x with $a \leq x \leq b$	closed interval
(a, b)	all x with $a < x < b$	open interval
$[a, b)$	all x with $a \leq x < b$	half-open interval
(a, ∞)	all x with $x > a$	unbounded interval
$(-\infty, b]$	all x with $x \leq b$	unbounded interval

The symbol ∞ ("infinity") is not a number; (a, ∞) simply means that the interval goes on forever to the right, so an infinity symbol always gets a round bracket. On a number line we draw a filled dot for an included endpoint and an open dot for an excluded one (Figure 1.3).

Why this matters for ML (What numbers look like inside a computer)**.** A grayscale image is stored as a grid of integers from 0 (black) to 255 (white). Before feeding it to

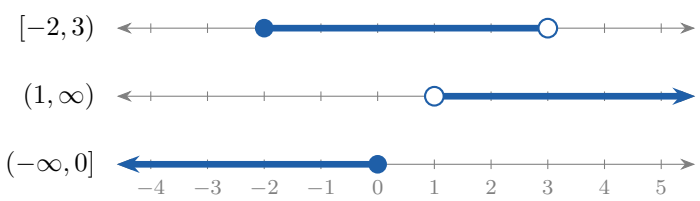


Figure 1.3: Three intervals drawn on the number line. Filled dots are included endpoints, open dots are excluded endpoints.

a neural network we usually divide every value by 255, so that all inputs lie in the interval $[0, 1]$. Class labels such as "cat", "dog", "bird" are stored as integers $0, 1, 2$. Everything else (weights, predictions, probabilities, losses) is a real number.

A computer cannot store infinitely many decimal digits, so it stores real numbers *approximately*, using a format called *floating point*. The common 32-bit format keeps about 7 significant decimal digits. This is usually plenty, but it means that extremely small numbers can be rounded to 0. We will see in Chapter 2 that this is one reason why machine learning works with *logarithms* of probabilities.

Exercises for Section 1.1

Exercise 1.1. For each number, list every set among $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ that contains it: 7, -3, 0, 0.25, $-\frac{2}{3}$, $\sqrt{9}$, $\sqrt{5}$, π.

Solution on p. 239 →

Exercise 1.2. Write each set in interval notation and sketch it on a number line: all x with $-2 \le x < 5$; all x with $x > 3$; all x with $x \le 0$; all x with $0 \le x \le 1$.

Solution on p. 239 →

Exercise 1.3. Show that 0.75 and $0.272727\ldots$ (with "27" repeating forever) are rational numbers by writing each as a fraction of integers. *Hint for the second: call the number x, compute $100x$, and subtract x.*

Solution on p. 239 →

Exercise 1.4. A grayscale pixel is stored as an integer from 0 to 255. How many different values can a pixel take? If we rescale by dividing by 255, in which interval do the rescaled values lie, and what is the rescaled value of a pixel equal to 51?

Solution on p. 239 →

1.2 Arithmetic and its laws

The four basic operations are addition ($+$), subtraction ($-$), multiplication ($\times$) and division ($\div$ or $/$). The result of an addition is called a **sum**, of a subtraction a **difference**, of a multiplication a **product** and of a division a **quotient**. In algebra the multiplication sign is often written as a dot, $3 \cdot 4$, or left out entirely when letters are involved: $3x$ means $3 \times x$, and ab means $a \times b$.

1.2.1 The laws of arithmetic

A handful of rules govern all calculations. They look obvious with small numbers, but they are exactly what lets us rearrange complicated formulas later on. In what follows, a, b and

c stand for any real numbers.

Theorem 1.1 (Laws of arithmetic). For all real numbers a, b, c:

(a) **Commutative laws** (order does not matter): $a + b = b + a$ and $ab = ba$.
(b) **Associative laws** (grouping does not matter): $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
(c) **Distributive law** (multiplication spreads over addition): $a(b + c) = ab + ac$.
(d) **Identities**: $a + 0 = a$ and $a \cdot 1 = a$.
(e) **Inverses**: $a + (-a) = 0$, and if $a \neq 0$, $a \cdot \frac{1}{a} = 1$.

The number $-a$ is the **additive inverse** (or negative) of a, and $\frac{1}{a}$ is the **reciprocal** of a.

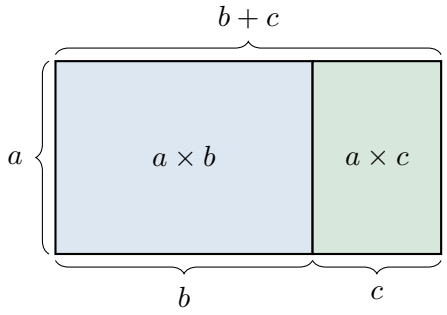


Figure 1.4: The distributive law as areas. The whole rectangle has area $a \times (b + c)$; its two pieces have areas $a \times b$ and $a \times c$.

Intuition (The distributive law is about areas). Think of $a \times (b + c)$ as the area of a rectangle of height a and width $b + c$ (Figure 1.4). Cutting it into two rectangles of widths b and c does not change the total area, which is therefore $ab + ac$. For instance, $3 \times (4 + 2) = 3 \times 6 = 18$ and $3 \times 4 + 3 \times 2 = 12 + 6 = 18$. The distributive law is also the trick behind mental arithmetic: $7 \times 102 = 7 \times 100 + 7 \times 2 = 714$.

1.2.2 Negative numbers

On the number line, adding a positive number moves you to the right and subtracting a positive number moves you to the left. So $-3 + 5 = 2$ (start at -3, move 5 steps right) and $2 - 6 = -4$ (start at 2, move 6 steps left).

Subtracting a negative number is the same as adding a positive one:

$$5 - (-3) = 5 + 3 = 8.$$

One way to see this: subtraction asks "how far is it from -3 up to 5?", and on the number line that distance is 8. Another way: taking away a debt of \$3 leaves you \$3 richer.

For multiplication and division the rules of signs are:

positive × positive = positive	$3 \times 4 = 12$
positive × negative = negative	$3 \times (-4) = -12$
negative × negative = positive	$(-3) \times (-4) = 12$

and exactly the same rules hold for division, e.g. $(-12) \div (-4) = 3$.

Intuition (Why is negative times negative positive?)**.** Look at the pattern as the first number decreases by 1 each time:

$$3\times(-2)=-6,\quad 2\times(-2)=-4,\quad 1\times(-2)=-2,\quad 0\times(-2)=0,\quad (-1)\times(-2)=\,?$$

Each answer is 2 larger than the one before, so the pattern forces $(-1)\times(-2)=2$. The rule "negative times negative is positive" is the only rule that keeps the distributive law true; any other choice would break arithmetic.

1.2.3 Division by zero

Division undoes multiplication: $12\div 3=4$ because $3\times 4=12$. So $6\div 0$ would have to be a number that gives 6 when multiplied by 0. But anything times 0 is 0, so no such number exists.

Common pitfall (Never divide by zero)**.** Division by zero is *undefined*: expressions such as $\frac{5}{0}$ have no meaning. (On the other hand, $\frac{0}{5}=0$ is perfectly fine.) In code, dividing by zero produces an error or a special "not a number" value, and it is a common source of bugs in machine learning, for example when normalising data whose spread happens to be zero. Practitioners therefore often add a tiny number such as 10^{-8} to a denominator to keep it away from zero.

Common pitfall (Subtraction and division have no commutative or associative law)**.** $5-3\neq 3-5$ and $8\div 2\neq 2\div 8$. Grouping matters too: $(10-5)-2=3$, but $10-(5-2)=7$. If you want to rearrange subtractions freely, rewrite them as additions of negatives: $10-5-2=10+(-5)+(-2)$, and now any order gives 3.

Why this matters for ML (Linearity)**.** The distributive law $a(b+c)=ab+ac$ is the simplest example of a property called *linearity*: scaling a sum is the same as summing the scaled parts. Linearity is the central idea of linear algebra (Chapter 4), and linear algebra is the engine that runs every neural network.

Exercises for Section 1.2

Exercise 1.5. Compute: (a) $-7+12$; (b) $4-9$; (c) $-3-(-8)$; (d) $(-6)\times 4$; (e) $(-5)\times(-7)$; (f) $(-36)\div(-4)$; (g) $0\div 5$.

Solution on p. 239 →

Exercise 1.6. Use the distributive law to compute 7×98 and 25×44 in your head.

Solution on p. 239 →

Exercise 1.7. Give a numerical example showing that division is not associative, that is, that $(a\div b)\div c$ can differ from $a\div(b\div c)$.

Solution on p. 239 →

Exercise 1.8. A friend claims that $\frac{0}{0}=1$ "because anything divided by itself is 1". Explain what is wrong.

Solution on p. 239 →

1.3 Fractions, decimals and percentages

1.3.1 What a fraction means

A **fraction** $\frac{a}{b}$ has a **numerator** a (on top) and a **denominator** b (on the bottom, never zero). The denominator says *into how many equal parts* the whole is cut; the numerator says *how many* of those parts we take. So $\frac{3}{4}$ of a pizza means: cut the pizza into 4 equal slices and take 3 of them. A fraction is also simply a division: $\frac{3}{4} = 3 \div 4 = 0.75$.

1.3.2 Equivalent fractions and simplifying

Cutting each slice in half gives 8 slices, of which we now have 6: $\frac{3}{4} = \frac{6}{8}$. In general, multiplying or dividing the numerator *and* the denominator by the same non-zero number does not change the value of a fraction:

$$\frac{a}{b} = \frac{a \times k}{b \times k} \qquad (k \neq 0).$$

To *simplify* a fraction, divide top and bottom by any common factor until none is left: $\frac{18}{24} = \frac{18 \div 6}{24 \div 6} = \frac{3}{4}$.

1.3.3 Multiplying and dividing fractions

To multiply fractions, multiply the numerators and multiply the denominators:

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}.$$

The word "of" usually signals multiplication: half of a third is $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$.

To divide by a fraction, multiply by its reciprocal (flip it upside down):

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}.$$

Why? Ask how many quarters fit into 3: each whole contains 4 quarters, so $3 \div \frac{1}{4} = 3 \times 4 = 12$.

1.3.4 Adding and subtracting fractions

You can only add fractions directly if they count slices of the same size, that is, if they have the same denominator: $\frac{2}{7} + \frac{3}{7} = \frac{5}{7}$. Otherwise, first rewrite them with a *common denominator*.

Example 1.2 (Adding fractions)**.** Compute $\frac{3}{4} + \frac{5}{6}$.

Solution. A common denominator is 12, since both 4 and 6 divide 12. Then $\frac{3}{4} = \frac{9}{12}$ and $\frac{5}{6} = \frac{10}{12}$, so

$$\frac{3}{4} + \frac{5}{6} = \frac{9}{12} + \frac{10}{12} = \frac{19}{12}.$$

As a check, $0.75 + 0.8333\ldots = 1.5833\ldots$, and $19 \div 12 = 1.5833\ldots$.

Common pitfall (Adding tops and bottoms)**.** It is tempting to write $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$. This is wrong: the answer must be bigger than $\frac{1}{2}$, but $\frac{2}{5}$ is smaller. The correct calculation is $\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.

Example 1.3 (Dividing fractions)**.** Compute $\frac{2}{3} \div \frac{4}{9}$.

Solution. Multiply by the reciprocal: $\frac{2}{3} \times \frac{9}{4} = \frac{18}{12} = \frac{3}{2}$.

1.3.5 Decimals

Decimals are fractions whose denominators are powers of ten: $0.7 = \frac{7}{10}$, $0.25 = \frac{25}{100}$, $0.125 = \frac{125}{1000}$. To turn a fraction into a decimal, divide: $\frac{3}{8} = 3 \div 8 = 0.375$.

1.3.6 Percentages

"Per cent" means "per hundred", so a **percentage** is a fraction with denominator 100: $35\% = \frac{35}{100} = 0.35$. To find a percentage of a quantity, multiply: 18% of 250 is $0.18 \times 250 = 45$.

The **percentage change** from an old value to a new value is

$$\text{percentage change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100\%.$$

Example 1.4 (Percentage changes are not symmetric)**.** A price rises from 80 to 100 and then falls back from 100 to 80. What are the two percentage changes?

Solution. Rise: $\frac{100-80}{80} \times 100\% = \frac{20}{80} \times 100\% = 25\%$. Fall: $\frac{80-100}{100} \times 100\% = -20\%$. The same absolute change of 20 is a 25% rise but only a 20% fall, because the percentages are computed relative to different starting values.

Common pitfall (Percent versus percentage points)**.** Suppose a classifier's accuracy improves from 90% to 95%. The accuracy has risen by 5 **percentage points**, which is a *relative* increase of $\frac{5}{90} \approx 5.6\%$. Meanwhile the *error rate* has fallen from 10% to 5%, a relative reduction of 50%: the model makes half as many mistakes. All three statements describe the same improvement. When you read "a 50% improvement" in a paper, always ask: 50% of *what*?

1.3.7 Ratios

A **ratio** compares two quantities. If a data set contains 750 normal emails and 250 spam emails, the ratio of normal to spam is $750 : 250$, which simplifies to $3 : 1$. Out of every 4 emails, 3 are normal and 1 is spam, so the *fraction* of spam is $\frac{1}{4} = 25\%$.

Why this matters for ML. Fractions and percentages are everywhere in machine learning: accuracy ($\frac{\text{correct predictions}}{\text{all predictions}}$), train/test splits ("use 80% of the data for training"), class imbalance ("spam is 1 in 4 emails") and dropout rates ("switch off 10% of the neurons"). Being precise about *what* a percentage is a percentage *of* avoids many misunderstandings.

Exercises for Section 1.3

Exercise 1.9. Simplify: (a) $\frac{18}{24}$; (b) $\frac{35}{49}$; (c) $\frac{120}{36}$.

Solution on p. 239 →

Exercise 1.10. Compute and simplify: (a) $\frac{2}{5}+\frac{1}{3}$; (b) $\frac{7}{8}-\frac{1}{6}$; (c) $\frac{3}{4}\times\frac{8}{9}$; (d) $\frac{5}{6}\div\frac{10}{3}$.

Solution on p. 239 →

Exercise 1.11. (a) What is 15% of 80? (b) 45 is what percentage of 180? (c) A price rises from 40 to 50; what is the percentage change? (d) It then falls back to 40; what is the percentage change now?

Solution on p. 239 →

Exercise 1.12. Model A has an accuracy of 80% and model B an accuracy of 84%. (a) By how many percentage points is B better? (b) What is the relative increase in accuracy? (c) What is the relative reduction in the error rate?

Solution on p. 239 →

Exercise 1.13. A data set of 1000 emails has a $3:1$ ratio of normal emails to spam. How many spam emails are there? If we keep 80% of the data for training and the split preserves the ratio, how many spam emails are in the training set?

Solution on p. 240 →

1.4 Exponents, roots and scientific notation

1.4.1 Powers

Repeated addition is multiplication: $3+3+3+3=4\times 3$. Repeated multiplication is *exponentiation*:

$$3^4 = 3\times 3\times 3\times 3 = 81.$$

In a^n, the number a is the **base** and n is the **exponent** (or **power**); we read a^n as "a to the power n". Two powers have their own names: a^2 is "a *squared*" (the area of a square with side a) and a^3 is "a *cubed*" (the volume of a cube with side a).

1.4.2 The laws of exponents

Counting factors explains every rule. For example, $2^3\times 2^2=(2\times 2\times 2)\times(2\times 2)=2^5$: three factors plus two factors make five factors.

Theorem 1.5 (Laws of exponents)**.** For real numbers $a, b\neq 0$ and exponents m, n:

$$a^m\cdot a^n = a^{m+n}, \qquad \frac{a^m}{a^n}=a^{m-n}, \qquad (a^m)^n = a^{mn},$$
$$(ab)^n = a^n b^n, \qquad \left(\frac{a}{b}\right)^n=\frac{a^n}{b^n}, \qquad a^0=1, \qquad a^{-n}=\frac{1}{a^n}.$$

The last two rules may look strange: what does it mean to multiply 2 by itself zero times, or minus three times? They are *definitions*, chosen so that the other rules keep working. Watch the pattern as we repeatedly divide by 2:

$$2^3=8,\quad 2^2=4,\quad 2^1=2,\quad 2^0=1,\quad 2^{-1}=\tfrac{1}{2},\quad 2^{-2}=\tfrac{1}{4},\quad 2^{-3}=\tfrac{1}{8}.$$

Each step down in the exponent divides by 2. Continuing the pattern forces $2^0=1$ and $2^{-n}=\frac{1}{2^n}$.

1.4.3 Roots and fractional exponents

The **square root** of a number $a \geq 0$, written $\sqrt{a}$, is the non-negative number whose square is a: $\sqrt{9} = 3$ because $3^2 = 9$. (Note that $(-3)^2 = 9$ as well, but the symbol $\sqrt{\ }$ always means the non-negative root.) Similarly the *cube root* $\sqrt[3]{a}$ is the number whose cube is a: $\sqrt[3]{8} = 2$.

Roots are powers with fractional exponents. Since $(a^{1/2})^2 = a^{(1/2)\cdot 2} = a^1 = a$ by the power rule, the number $a^{1/2}$ must be $\sqrt{a}$. In general,

$$a^{1/n} = \sqrt[n]{a}, \qquad a^{m/n} = \left(\sqrt[n]{a}\right)^m.$$

For example, $8^{2/3} = (\sqrt[3]{8})^2 = 2^2 = 4$, and $27^{-1/3} = \frac{1}{\sqrt[3]{27}} = \frac{1}{3}$.

Common pitfall (Powers do not distribute over sums)**.** $(a+b)^2$ is **not** a^2+b^2. With numbers: $(3+4)^2 = 7^2 = 49$, but $3^2+4^2 = 9+16 = 25$. The correct expansion, $(a+b)^2 = a^2+2ab+b^2$, is pictured in Figure 1.5: the missing $24 = 2\times 3\times 4$ is the area of the two rectangles. Likewise $\sqrt{a+b} \neq \sqrt{a}+\sqrt{b}$: $\sqrt{9+16} = 5$, but $\sqrt{9}+\sqrt{16} = 7$. Powers and roots *do* distribute over products: $(ab)^2 = a^2b^2$ and $\sqrt{ab} = \sqrt{a}\sqrt{b}$ for $a, b \geq 0$.

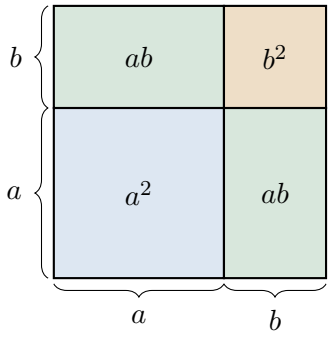


Figure 1.5: A square with side $a+b$ has area $(a+b)^2 = a^2+2ab+b^2$: besides the squares a^2 and b^2 there are two rectangles of area ab.

Common pitfall (Negative signs and powers)**.** -3^2 means $-(3^2) = -9$, because the power applies only to the 3. If you want to square -3, write $(-3)^2 = 9$. Also, the square root of a negative number, such as $\sqrt{-4}$, is not a real number (no real number squared is negative). Finally, $\sqrt{x^2} = |x|$, not x: for example $\sqrt{(-5)^2} = \sqrt{25} = 5$. (The absolute value $|x|$ is explained in Section 1.10.)

1.4.4 Scientific notation

Machine learning involves both very large and very small numbers. Writing them out in full is error-prone, so we use **scientific notation**: a number between 1 and 10 multiplied by a power of 10.

$$0.0003 = 3\times 10^{-4}, \qquad 175\,000\,000\,000 = 1.75\times 10^{11}.$$

The exponent counts how many places the decimal point moves: right for positive exponents, left for negative ones. To multiply numbers in scientific notation, multiply the leading parts and add the exponents:

$$(3\times 10^{-4})\times(2\times 10^{6}) = 6\times 10^{2} = 600.$$

In computer code, 3×10^{-4} is usually written `3e-4`.

Why this matters for ML. Typical numbers in deep learning: a learning rate of `3e-4`, a weight decay of `1e-2`, a small "safety" constant ϵ of `1e-8` added to denominators, and models with 1.75×10^{11} parameters (the size of the GPT-3 language model). Being able to read and compare such numbers at a glance is a basic practical skill.

Exercises for Section 1.4

Exercise 1.14. Evaluate: (a) $2^3 \times 2^4$; (b) $\frac{5^6}{5^4}$; (c) $(3^2)^3$; (d) 10^0; (e) 4^{-2}; (f) $9^{1/2}$; (g) $8^{2/3}$; (h) $16^{-3/4}$.

Solution on p. 240 →

Exercise 1.15. Simplify, assuming $x \neq 0$: (a) $\frac{x^3 \cdot x^5}{x^2}$; (b) $(2x^2)^3$; (c) $\sqrt{x^4}$; (d) $\frac{(x^{-1})^2}{x^{-3}}$.

Solution on p. 240 →

Exercise 1.16. True or false? Give a counterexample for each false statement. (a) $(a+b)^2 = a^2+b^2$; (b) $\sqrt{ab} = \sqrt{a}\sqrt{b}$ for $a, b \geq 0$; (c) $\sqrt{a+b} = \sqrt{a}+\sqrt{b}$; (d) $(ab)^3 = a^3b^3$; (e) $2^{a+b} = 2^a + 2^b$.

Solution on p. 240 →

Exercise 1.17. Write 0.00005 and 3 400 000 in scientific notation, and compute $(4 \times 10^3) \times (2.5 \times 10^{-5})$.

Solution on p. 240 →

Exercise 1.18. A language model has 7×10^9 parameters, each stored using 2 bytes. How many bytes are needed to store the model? Express the answer in gigabytes, where 1 gigabyte $= 10^9$ bytes.

Solution on p. 240 →

1.5 Order of operations

What is $2 + 3 \times 4$? If you add first you get 20; if you multiply first you get 14. Mathematics needs a single agreed answer, and the convention is that multiplication comes first, so $2 + 3 \times 4 = 14$. The full convention, the **order of operations**, is:

1. **Parentheses** (brackets): work out anything inside brackets first, innermost brackets first.
2. **Exponents** (powers and roots).
3. **Multiplication and division**, from left to right.
4. **Addition and subtraction**, from left to right.

Many people remember this as PEMDAS or BODMAS. Note that multiplication and division have *equal* rank and are done left to right, and likewise for addition and subtraction.

Example 1.6 (Order of operations)**.** Evaluate (a) $8 - 2 \times 3 + 4$; (b) $20 \div 4 \times 5$; (c) 2×3^2; (d) $(2+3)^2 - 4 \times 5$; (e) $3 + 12 \div (5 - 2) \times 2$.

Solution. (a) Multiply first: $8 - 6 + 4$; then left to right: $2 + 4 = 6$. (b) Left to right: $20 \div 4 = 5$, then $5 \times 5 = 25$ (not $20 \div 20 = 1$). (c) Power first: $2 \times 9 = 18$ (not $6^2 = 36$). (d) Brackets, then power: $5^2 - 4 \times 5 = 25 - 20 = 5$. (e) Brackets: $3 + 12 \div 3 \times 2$; then left to right: $12 \div 3 = 4$, $4 \times 2 = 8$; finally $3 + 8 = 11$.

A fraction bar groups like brackets: $\dfrac{6+4}{2}$ means $(6+4) \div 2 = 5$, not $6 + 4 \div 2 = 8$.

Common pitfall (Ambiguous expressions)**.** Expressions such as $1/2x$ are ambiguous: is it $\frac{1}{2x}$ or $\frac{1}{2}x$? Different people (and different calculators) disagree. When writing, always add brackets to remove any doubt: $1/(2x)$ or $(1/2)x$.

Remark (Order of operations in code)**.** Programming languages follow the same conventions. In Python, powers are written `**`, and `-2**2` evaluates to `-4`, exactly like -2^2 in mathematics, because the power is applied before the minus sign.

Exercises for Section 1.5

Exercise 1.19. Evaluate: (a) $6 + 4 \times 3 - 2$; (b) $(6 + 4) \times (3 - 2)$; (c) $48 \div 6 \div 2$; (d) $2 \times 5^2 - 3^2$; (e) $-3^2 + (-3)^2$; (f) $(8 - 2 \times 3)^3$.

Solution on p. 240 →

Exercise 1.20. Insert brackets into $2 + 3 \times 4 - 1$ so that its value is (a) 19; (b) 11; (c) 15.

Solution on p. 240 →

Exercise 1.21. Evaluate $\dfrac{2^3 + 4}{6 - 2 \times 3 + 3}$ and $\sqrt{3^2 + 4^2}$.

Solution on p. 240 →

1.6 Variables, expressions and algebraic thinking

1.6.1 Variables

Algebra begins with a simple but powerful idea: use a letter to stand for a number. The letter is called a **variable**. It may stand for a number we do not know yet ("find x"), or for a number that can change ("the price depends on the size x of the house"), or for *any* number at all ("for every x, $x + 0 = x$").

An **expression** is a combination of numbers, variables and operations, such as $3x + 2$ or $x^2 - 5x + 6$. *Evaluating* an expression means substituting a value for each variable and calculating. For $x = 4$, $3x + 2 = 3 \times 4 + 2 = 14$.

Intuition (A variable is a labelled box)**.** Think of a variable as a box with a name written on it. The expression $3x + 2$ is a recipe: "take whatever is in box x, multiply it by 3, then add 2". The recipe makes sense before we know what is in the box. This is exactly how a machine learning model is written down: as a recipe in terms of inputs (boxes filled by the data) and parameters (boxes filled during training).

1.6.2 Terms, coefficients and like terms

In $5x^2 - 3x + 7$, the pieces separated by $+$ or $-$ signs are called **terms**: $5x^2$, $-3x$ and 7. The number in front of a variable is its **coefficient** (5 and -3 here), and a term without a variable is a **constant**. **Like terms** contain the same variables raised to the same powers, and they can be combined by adding their coefficients:

$$3x + 5 + 2x - 1 = (3 + 2)x + (5 - 1) = 5x + 4.$$

This is just the distributive law in reverse: $3x + 2x = (3 + 2)x$. Unlike terms, such as x and x^2, cannot be combined.

1.6.3 Expanding brackets

Expanding means multiplying out brackets using the distributive law: $3(x + 2) = 3x + 6$ and $-2(x - 5) = -2x + 10$ (careful with the signs). For a product of two brackets, multiply every term in the first bracket by every term in the second:

$$(x + 2)(x + 3) = x \cdot x + x \cdot 3 + 2 \cdot x + 2 \cdot 3 = x^2 + 5x + 6.$$

Figure 1.6 shows this as the area of a rectangle with sides $x + 2$ and $x + 3$.

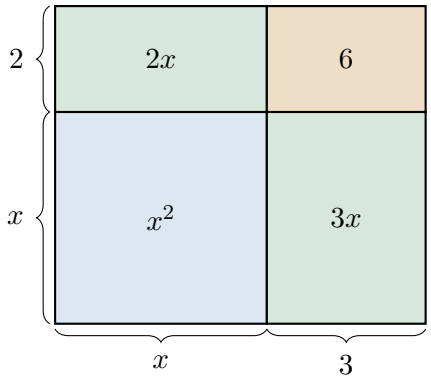


Figure 1.6: The area model for $(x + 2)(x + 3)$. The four pieces have areas x^2, $3x$, $2x$ and 6, giving $x^2 + 5x + 6$.

Three products appear so often that they are worth memorising:

$$\begin{aligned}(a + b)^2 &= a^2 + 2ab + b^2,\\ (a - b)^2 &= a^2 - 2ab + b^2,\\ (a + b)(a - b) &= a^2 - b^2.\end{aligned}$$

The second identity will reappear many times, because machine learning is full of squared differences such as $(y - \hat{y})^2$.

1.6.4 Factoring

Factoring is expanding in reverse: writing an expression as a product. Pull out a common factor, $6x + 9 = 3(2x + 3)$, or find two brackets that multiply to the expression, $x^2 + 5x + 6 = (x + 2)(x + 3)$. To factor $x^2 + bx + c$, look for two numbers that *add* to b and *multiply* to c; here $2 + 3 = 5$ and $2 \times 3 = 6$. Factoring will help us solve equations involving x^2 in Chapter 2.

1.6.5 Formulas

A **formula** is an equation that tells you how to compute one quantity from others. Examples are the area of a rectangle, $A = \ell w$; the conversion from degrees Celsius to degrees Fahrenheit, $F = \frac{9}{5}C + 32$; and the prediction of a simple model of house prices,

$$\text{predicted price} = w \times \text{size} + b,$$

where w and b are numbers that the model learns from data.

Why this matters for ML (Models are expressions with parameters)**.** In the house-price formula, "size" is an *input* (it changes from house to house) while w and b are *parameters* (they are fixed once the model has been trained). In machine learning, w is called a **weight** and b a **bias**. A large neural network is the same idea on a huge scale: a very long expression with millions or billions of weights. Training means finding values of the weights that make the expression's outputs match the data.

Exercises for Section 1.6

Exercise 1.22. Evaluate $2x^2 - 3x + 1$ at $x = 3$, at $x = -1$ and at $x = \frac{1}{2}$.

Solution on p. 240 →

Exercise 1.23. Simplify: (a) $4a + 3b - a + 5b$; (b) $2(x - 3) - 3(x + 1)$; (c) $x(x + 2) - (x^2 - 4x)$.

Solution on p. 240 →

Exercise 1.24. Expand: (a) $(x + 4)(x - 2)$; (b) $(2x - 3)^2$; (c) $(x + 5)(x - 5)$; (d) $(a - b)(a^2 + ab + b^2)$.

Solution on p. 240 →

Exercise 1.25. Factor: (a) $12x + 18$; (b) $x^2 + 7x + 12$; (c) $x^2 - 9$; (d) $x^2 - x - 6$.

Solution on p. 240 →

Exercise 1.26. A model predicts house prices with the formula price $= 3000 \times$ size $+$ 50 000, with size in square metres. What does it predict for a house of 80 square metres? If the true price is 300 000, what is the error true $-$ predicted?

Solution on p. 240 →

1.7 Summation and product notation

Motivation. Machine learning learns from data sets with thousands or millions of examples, and its formulas constantly say "add this up over all the examples". Writing out $x_1 + x_2 + x_3 + \cdots + x_{1000000}$ is impractical, so mathematicians use a compact shorthand: the capital Greek letter sigma, $\sum$. You will see it in almost every machine learning formula, so it is worth getting completely comfortable with it now.

1.7.1 Subscripts

When we have many related numbers, we give them the same letter and tell them apart with a small number written below, called a **subscript** or **index**. A data set of four

measurements might be

$$x_1 = 2, \quad x_2 = 5, \quad x_3 = 3, \quad x_4 = 7.$$

We read x_3 as "x sub three" or "x three". A general element is written x_i, where i can be any of $1, 2, 3, 4$.

1.7.2 Sigma notation

Definition 1.7 (Summation notation)**.** For numbers $x_1, x_2, \ldots, x_n$,

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \cdots + x_n.$$

This is read "the sum of x_i for i from 1 to n". The variable i is the **index of summation**; it starts at the value below the $\sum$ and increases in steps of 1 up to the value on top.

Example 1.8 (Reading sums)**.** Compute (a) $\sum_{i=1}^{4} i$; (b) $\sum_{i=1}^{3} i^2$; (c) $\sum_{k=0}^{3} 2^k$; (d) $\sum_{i=1}^{4} x_i$ for the data $x_1 = 2, x_2 = 5, x_3 = 3, x_4 = 7$.

Solution. (a) $1+2+3+4 = 10$. (b) $1^2+2^2+3^2 = 1+4+9 = 14$. (c) $2^0+2^1+2^2+2^3 = 1+2+4+8 = 15$ (the index may start at 0). (d) $2+5+3+7 = 17$.

The name of the index does not matter: $\sum_{i=1}^{4} x_i$, $\sum_{j=1}^{4} x_j$ and $\sum_{k=1}^{4} x_k$ all mean the same thing. For this reason the index is called a *dummy variable*.

1.7.3 Rules for sums

Because a sum is just repeated addition, the laws of arithmetic give us some useful rules.

Proposition 1.9 (Rules for sums)**.** For numbers $x_1, \ldots, x_n$, $y_1, \ldots, y_n$ and a constant c:

$$\sum_{i=1}^{n} c\,x_i = c\sum_{i=1}^{n} x_i \qquad \text{(constants can be pulled out)},$$

$$\sum_{i=1}^{n} (x_i + y_i) = \sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i \qquad \text{(sums can be split)},$$

$$\sum_{i=1}^{n} c = n\,c \qquad \text{(adding } c \text{ to itself } n \text{ times)}.$$

Proof. The first rule is the distributive law: $cx_1 + cx_2 + \cdots + cx_n = c(x_1 + \cdots + x_n)$. The second rule just reorders the terms, which the commutative and associative laws allow. The third rule is the definition of multiplication. ∎

Common pitfall (What you cannot do with sums)**.** In general,

$$\sum_i x_i y_i \neq \Big(\sum_i x_i\Big)\Big(\sum_i y_i\Big) \qquad \text{and} \qquad \sum_i x_i^2 \neq \Big(\sum_i x_i\Big)^2.$$

For the data $2, 5, 3, 7$: $\sum_i x_i^2 = 4 + 25 + 9 + 49 = 87$, whereas $\big(\sum_i x_i\big)^2 = 17^2 = 289$.

1.7.4 The mean

The **mean** (average) of n numbers is their sum divided by n. It is written with a bar:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

For the data $2, 5, 3, 7$, $\bar{x} = \frac{17}{4} = 4.25$.

1.7.5 Product notation

The capital Greek letter pi, $\prod$, does for multiplication what $\sum$ does for addition:

$$\prod_{i=1}^{n} x_i = x_1 \times x_2 \times \cdots \times x_n.$$

For example $\prod_{i=1}^{4} i = 1 \times 2 \times 3 \times 4 = 24$. This particular product is so common that it has its own name and symbol: $n! = 1 \times 2 \times \cdots \times n$ is called "n **factorial**", so $4! = 24$. By convention, $0! = 1$.

Why this matters for ML (The mean squared error)**.** Suppose a model makes predictions $\hat{y}_1, \ldots, \hat{y}_n$ (read "y hat") for n data points whose true values are $y_1, \ldots, y_n$. The most common way to measure how wrong the model is, is the **mean squared error**:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

In words: for each data point, compute the error $y_i - \hat{y}_i$, square it (so that positive and negative errors both count, and big errors count a lot), and average over all data points. For true values $3, 5, 4, 6$ and predictions $2, 5, 5, 8$, the errors are $1, 0, -1, -2$, their squares are $1, 0, 1, 4$, and $\text{MSE} = \frac{6}{4} = 1.5$. Training a model usually means making a quantity like this as small as possible.

Products appear when we combine probabilities: the probability that a model assigns to an entire data set is often a product $\prod_{i=1}^{n} p_i$ of the probabilities of the individual examples (Chapter 10).

Exercises for Section 1.7

Exercise 1.27. Compute: (a) $\sum_{i=1}^{5}(2i-1)$; (b) $\sum_{k=1}^{4} k^2$; (c) $\sum_{j=0}^{3} 3$; (d) $\prod_{i=1}^{5} i$.

Solution on p. 240 →

Exercise 1.28. Write in sigma notation: (a) $1 + 2 + \cdots + 100$; (b) $x_1^2 + x_2^2 + \cdots + x_n^2$; (c) $w_1x_1 + w_2x_2 + w_3x_3$.

Solution on p. 240 →

Exercise 1.29. Let $x = (1, 4, 2, 5)$ and $y = (3, 0, 1, 2)$, meaning $x_1 = 1$, $y_1 = 3$, and so on. Compute $\sum_i x_i$, $\sum_i y_i$, $\sum_i x_i y_i$ and $\left(\sum_i x_i\right)\left(\sum_i y_i\right)$.

Solution on p. 240 →

Exercise 1.30. Use the rules for sums to show that the deviations from the mean always add up to zero: $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$.

Solution on p. 240 →

Exercise 1.31. True values are $y = (2, 4, 6)$ and a model predicts $\hat{y} = (3, 4, 4)$. Compute the mean squared error and the mean absolute error $\frac{1}{n}\sum_i |y_i - \hat{y}_i|$. (The absolute value $|\cdot|$ simply removes a minus sign; see Section 1.10.)

Solution on p. 241 →

1.8 Solving equations

An **equation** states that two expressions are equal, for example $3x + 5 = 20$. A **solution** is a value of the variable that makes the statement true. Here $x = 5$ is a solution, because $3 \times 5 + 5 = 20$.

1.8.1 The balance principle

Picture this. Imagine an old-fashioned balance scale with $3x + 5$ in the left pan and 20 in the right pan, perfectly level. If you remove 5 from *both* pans, it stays level: $3x = 15$. If you then divide the contents of *both* pans into three equal parts and keep one part of each, it still stays level: $x = 5$. Solving an equation means doing the same thing to both sides until the variable stands alone.

The allowed moves are: add or subtract the same number on both sides; multiply or divide both sides by the same *non-zero* number; and simplify either side. Each move undoes one operation that was applied to x, in reverse order.

Example 1.10 (Linear equations)**.** Solve (a) $3x + 5 = 20$; (b) $5x - 3 = 2x + 9$; (c) $\frac{x}{4} + 1 = 3$; (d) $\frac{x+1}{3} = \frac{x-1}{2}$.

Solution. (a) Subtract 5: $3x = 15$. Divide by 3: $x = 5$. *Check:* $3 \times 5 + 5 = 20$. ✓

(b) Gather the x terms on one side by subtracting $2x$: $3x - 3 = 9$. Add 3: $3x = 12$. Divide by 3: $x = 4$. *Check:* left side $5 \times 4 - 3 = 17$, right side $2 \times 4 + 9 = 17$. ✓

(c) Subtract 1: $\frac{x}{4} = 2$. Multiply by 4: $x = 8$.

(d) Clear the fractions by multiplying both sides by 6, a common multiple of the denominators: $2(x + 1) = 3(x - 1)$. Expand: $2x + 2 = 3x - 3$. Subtract $2x$ and add 3: $5 = x$. *Check:* $\frac{6}{3} = 2 = \frac{4}{2}$. ✓

Always check your answer by substituting it back into the *original* equation. It takes seconds and catches most mistakes.

Equations of the form $ax + b = c$ are called **linear equations**, because (as we will see in Chapter 2) their graphs are straight lines. Every linear equation with $a \neq 0$ has exactly one solution, $x = \frac{c-b}{a}$. Some equations that look linear have no solution at all ($x + 1 = x + 2$ simplifies to the false statement $1 = 2$) or are true for every x ($2(x + 1) = 2x + 2$ simplifies to $2 = 2$; such an equation is called an *identity*).

1.8.2 Rearranging formulas

The same moves let us solve a formula for any one of its variables. To convert Fahrenheit to Celsius, solve $F = \frac{9}{5}C + 32$ for C: subtract 32, then multiply by $\frac{5}{9}$:

$$C = \tfrac{5}{9}(F - 32).$$

A body temperature of 98.6 °F is therefore $\frac{5}{9} \times 66.6 = 37$ °C.

1.8.3 Two equations in two unknowns

Sometimes we have two unknowns and two facts about them. For example: two numbers add up to 10 and differ by 2. What are they? Writing x and y for the numbers,

$$\begin{cases} x + y = 10, \\ x - y = 2. \end{cases}$$

This is a **system of linear equations**. Two standard methods solve it.

Elimination. Add the two equations: the y's cancel and we get $2x = 12$, so $x = 6$. Substituting into the first equation gives $y = 4$.

Substitution. Solve one equation for one variable and substitute into the other. From the second equation, $x = y + 2$; substituting into the first gives $(y + 2) + y = 10$, so $2y = 8$, $y = 4$, and $x = 6$.

Example 1.11 (Fitting a line through two points)**.** Find numbers w and b so that the formula $y = wx + b$ gives $y = 3$ when $x = 1$ and gives $y = 7$ when $x = 3$.

Solution. Substituting the two data points gives two equations:

$$\begin{cases} w + b = 3, \\ 3w + b = 7. \end{cases}$$

Subtracting the first equation from the second eliminates b: $2w = 4$, so $w = 2$. Then $b = 3 - w = 1$. The formula is $y = 2x + 1$. *Check:* $2 \times 1 + 1 = 3$ and $2 \times 3 + 1 = 7$. ✓

Why this matters for ML (From two equations to millions)**.** Example 1.11 is a miniature version of training a model: we chose the parameters w and b so that the model reproduces the data. With two data points and two parameters we can fit the data exactly. Real data sets have many more data points than parameters, so there are more equations than unknowns and usually no exact solution. We will then look for the parameters that make the errors as small as possible, a problem called *least squares* (Chapter 5). Large systems of equations are written and solved with matrices (Chapter 4).

Exercises for Section 1.8

Exercise 1.32. Solve and check: (a) $4x - 7 = 13$; (b) $2(x + 3) = 5x - 3$; (c) $\frac{x}{3} - 2 = 4$; (d) $\frac{2x-1}{5} = \frac{x+2}{3}$.

Solution on p. 241 →

Exercise 1.33. Rearrange each formula for the indicated variable: (a) $v = u + at$ for t; (b) $A = \frac{1}{2}bh$ for h; (c) $z = \frac{x-\mu}{\sigma}$ for x. (The letters μ and σ are the Greek letters "mu" and "sigma". The quantity z is the *z-score* of statistics, which you will meet in Chapter 9.)

Solution on p. 241 →

Exercise 1.34. Solve the systems: (a) $x + y = 7$, $x - y = 3$; (b) $3x + 2y = 16$, $x - 2y = 0$; (c) $2x + y = 5$, $4x + 2y = 10$; (d) $x + y = 1$, $x + y = 3$.

Solution on p. 241 →

Exercise 1.35. Find the line $y = wx + b$ that passes through the points $(x, y) = (2, 5)$ and $(4, 9)$.

Solution on p. 241 →

Exercise 1.36. A theatre sells adult tickets at 12 and child tickets at 7. A group buys 20 tickets for a total of 190. How many adult tickets did they buy?

Solution on p. 241 →

1.9 Inequalities

1.9.1 Inequality symbols

An **inequality** compares two quantities that need not be equal:

Symbol	Read as	Example
$a < b$	a is less than b	$-3 < 2$
$a > b$	a is greater than b	$7 > 4$
$a \leq b$	a is less than or equal to b	$5 \leq 5$
$a \geq b$	a is greater than or equal to b	$0.3 \geq 0$
$a \neq b$	a is not equal to b	$1 \neq 2$

On the number line, $a < b$ means that a lies to the left of b. (A memory aid: the symbol always opens towards the larger number.)

1.9.2 Solving inequalities

Inequalities are solved almost exactly like equations: you may add or subtract the same number on both sides, and multiply or divide both sides by the same *positive* number. There is one crucial exception.

Theorem 1.12 (Multiplying an inequality by a negative number)**.** If $a < b$ and $c < 0$, then $ca > cb$. In words: multiplying or dividing both sides of an inequality by a negative number *reverses* the direction of the inequality.

Intuition. Multiplying by -1 reflects the number line in the point 0, like a mirror. Since $2 < 5$, the point 2 is to the left of 5; after reflection, -2 is to the *right* of -5, so $-2 > -5$. Order is reversed by the mirror.

Example 1.13 (Solving an inequality)**.** Solve $3 - 2x \geq 7$.

Solution. Subtract 3 from both sides: $-2x \geq 4$. Divide by -2, which is negative, so the inequality flips: $x \leq -2$. The solution set is the interval $(-\infty, -2]$:



Check a value inside the set, say $x = -3$: $3 + 6 = 9 \geq 7$ ✓; and one outside, say $x = 0$: $3 \geq 7$ is false. ✓

A *compound inequality* such as $-1 < 2x + 3 \leq 9$ is solved by operating on all three parts at once: subtract 3 to get $-4 < 2x \leq 6$, then divide by 2 to get $-2 < x \leq 3$, i.e. the interval $(-2, 3]$.

1.9.3 Maximum and minimum

The **maximum** of two numbers is the larger of them, and the **minimum** is the smaller:

$$\max(3, 7) = 7, \qquad \min(3, 7) = 3, \qquad \max(-2, -5) = -2.$$

More generally, max and min of a list of numbers pick out its largest and smallest elements.

Why this matters for ML (Inequalities in machine learning)**.** Inequalities express constraints: a probability p always satisfies $0 \leq p \leq 1$, and a learning rate must be positive. The maximum function also defines the most popular building block of modern neural networks, the **rectified linear unit**, or ReLU:

$$\mathrm{ReLU}(x) = \max(0, x).$$

It passes positive numbers through unchanged and replaces negative numbers by 0. So $\mathrm{ReLU}(2.5) = 2.5$ and $\mathrm{ReLU}(-1.3) = 0$. Another example is *clipping*, which forces a number into an interval $[-c, c]$ using $\min(\max(x, -c), c)$; "gradient clipping" uses this idea to stop training from blowing up.

Exercises for Section 1.9

Exercise 1.37. Solve, and write the answer in interval notation: (a) $2x + 5 < 11$; (b) $4 - 3x \leq 10$; (c) $-3 \leq 2x - 1 < 7$.

Solution on p. 241 →

Exercise 1.38. Evaluate: (a) $\max(-2, 5)$; (b) $\min(-2, 5)$; (c) $\mathrm{ReLU}(-3.7)$; (d) $\mathrm{ReLU}(2.1)$; (e) $\max(1, 4, -6, 2)$.

Solution on p. 241 →

Exercise 1.39. For which values of x is $\mathrm{ReLU}(x) = x$? For which is $\mathrm{ReLU}(x) = 0$?

Solution on p. 241 →

Exercise 1.40. The clipping function with $c = 1$ is $\mathrm{clip}(x) = \min(\max(x, -1), 1)$. Compute $\mathrm{clip}(-3)$, $\mathrm{clip}(0.5)$ and $\mathrm{clip}(2)$, and describe in words what the function does.

Solution on p. 241 →

Exercise 1.41. Starting from the true statement $3 < 8$, multiply both sides by -2. Is $-6 < -16$ true? Explain what goes wrong and how to fix it.

Solution on p. 241 →

1.10 Absolute value

Definition 1.14 (Absolute value)**.** The **absolute value** of a real number x is

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

In practice: the absolute value removes a minus sign, if there is one. $|5| = 5$, $|-5| = 5$, $|0| = 0$. (In the second case the definition says $|-5| = -(-5) = 5$.)

Intuition (Absolute value is distance)**.** $|x|$ is the distance from x to 0 on the number line, and distances are never negative. More usefully, $|a - b|$ is the distance between the numbers a and b (Figure 1.7). For example, the distance between -2 and 3 is $|-2-3| = |-5| = 5$, and it does not matter in which order we subtract: $|3-(-2)| = 5$ too.

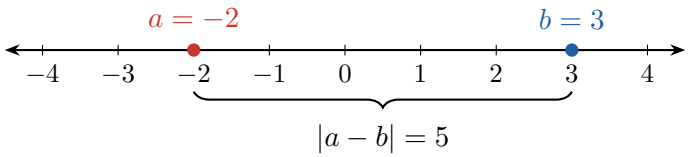


Figure 1.7: The distance between two numbers a and b on the number line is $|a - b|$.

Proposition 1.15 (Properties of absolute value)**.** For all real numbers a and b:

(a) $|a| \geq 0$, and $|a| = 0$ only when $a = 0$;
(b) $|ab| = |a|\,|b|$ and, if $b \neq 0$, $|a/b| = |a|/|b|$;
(c) $|a + b| \leq |a| + |b|$ (the **triangle inequality**);
(d) $\sqrt{a^2} = |a|$.

The triangle inequality says that going from 0 to $a + b$ can never be longer than going first a distance $|a|$ and then a distance $|b|$. Equality holds when a and b have the same sign; when the signs differ, the two steps partly cancel: $|3 + (-5)| = 2$, while $|3| + |-5| = 8$. The name will make more sense in Chapter 4, where the same inequality holds for the sides of a triangle.

1.10.1 Equations and inequalities with absolute values

Reading absolute values as distances makes these problems easy.

Example 1.16 (Absolute value equations and inequalities)**.** Solve (a) $|x - 3| = 2$; (b) $|x - 3| < 2$; (c) $|x - 3| \geq 2$; (d) $|2x + 1| \leq 5$.

Solution. (a) "The distance from x to 3 is 2": $x = 1$ or $x = 5$.

(b) "The distance from x to 3 is less than 2": $1 < x < 5$, the interval $(1, 5)$.

(c) "At least 2 away from 3": $x \leq 1$ or $x \geq 5$.

(d) $|u| \leq 5$ means $-5 \leq u \leq 5$. With $u = 2x + 1$: $-5 \leq 2x + 1 \leq 5$, so $-6 \leq 2x \leq 4$, i.e. $-3 \leq x \leq 2$.

Common pitfall (Absolute value does not split over sums)**.** $|a + b|$ is not in general equal to $|a| + |b|$ (see the example above). Also, $-|x|$ is always ≤ 0, while $|-x| = |x|$ is always ≥ 0; these are different things.

Why this matters for ML (Two ways of measuring errors)**.** If a model predicts $\hat{y}$ and the true value is y, the size of the error is $|y - \hat{y}|$, regardless of whether the prediction was too high or too low. Averaging over a data set gives the **mean absolute error**,

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|,$$

an alternative to the mean squared error of Section 1.7. Squaring punishes large errors much more than small ones (an error of 10 contributes 100 to the MSE but only 10 to the MAE), so the two measures lead to models with different behaviour. In Chapter 5 absolute values reappear in the "L^1 norm", which is used to encourage models to set many weights to exactly zero.

Exercises for Section 1.10

Exercise 1.42. Evaluate: $|-8|$, $|3 - 10|$, $|-2| \cdot |5|$, $|-4 + 9|$, $-|-6|$.

Solution on p. 241 →

Exercise 1.43. Solve: (a) $|x + 2| = 5$; (b) $|x - 1| \leq 3$; (c) $|2x - 4| > 6$.

Solution on p. 241 →

Exercise 1.44. What is the distance between -3.5 and 2 on the number line?

Solution on p. 241 →

Exercise 1.45. Check the triangle inequality for $a = 4$, $b = -7$. For which pairs of numbers does $|a + b| = |a| + |b|$ hold exactly?

Solution on p. 241 →

Exercise 1.46. True values are $y = (10, 12, 8)$ and predictions are $\hat{y} = (11, 9, 8)$. Compute the MAE and the MSE. Which data point dominates the MSE, and why?

Solution on p. 241 →

1.11 Chapter summary

Key takeaways

- The number systems are nested: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. Machine learning works almost entirely with real numbers, which are stored approximately by computers.
- The commutative, associative and distributive laws let us rearrange calculations. Never divide by zero.
- Fractions: find a common denominator to add; multiply straight across; divide by multiplying by the reciprocal. Always ask what a percentage is a percentage *of*.
- Exponent laws: $a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, $a^0 = 1$, $a^{-n} = 1/a^n$, $a^{1/n} = \sqrt[n]{a}$. But $(a+b)^2 \neq a^2 + b^2$.
- Order of operations: brackets, exponents, multiplication and division (left to right), addition and subtraction (left to right).
- Variables are named boxes for numbers; a model is an expression with inputs and parameters (weights and biases).
- $\sum_{i=1}^{n} x_i$ means $x_1 + \cdots + x_n$. Constants can be pulled out of sums and sums can be split, but a sum of products is not a product of sums. The mean is $\bar{x} = \frac{1}{n}\sum_i x_i$.
- Solve equations by doing the same thing to both sides. Two unknowns need two equations (elimination or substitution).
- Inequalities work like equations, except that multiplying by a negative number reverses them. $\mathrm{ReLU}(x) = \max(0, x)$.
- $|a - b|$ is the distance between a and b; mean absolute error and mean squared error are two ways to measure a model's mistakes.

In the next chapter we turn expressions into *functions*: machines that take an input and produce an output. Functions are the mathematical name for what a machine learning model is.

Chapter 2

Functions and Graphs

Motivation. What *is* a machine learning model? Strip away the jargon and the answer is short: a model is a *function*. It takes an input (an image, a sentence, the features of a house) and produces an output (a label, a translation, a price). Learning from data means searching for a function that produces the right outputs for the inputs we have seen, and, hopefully, for new inputs too.

Functions are therefore the central objects of this whole subject. In this chapter we learn what functions are, how to picture them as graphs, and meet the handful of function families that appear again and again in machine learning: linear functions, polynomials, exponentials and logarithms, and the "activation functions" used inside neural networks. We also learn two ways of building new functions from old ones, *composition* and *inversion*. Composition is exactly how deep neural networks are built.

In this chapter you will learn to...

- explain what a function is, using the words domain, codomain and range, and the notation $f(x)$;
- plot and read graphs of functions;
- work with linear functions, slopes and intercepts, and interpret the slope as a rate of change;
- solve quadratic equations and find the minimum of a quadratic function;
- describe exponential growth and decay and the special number e;
- use logarithms and their rules, and explain why machine learning prefers log-probabilities;
- compose functions, find inverse functions, and predict how parameters shift and stretch a graph;
- recognise the common activation functions: step, sigmoid, tanh, ReLU and softplus.

2.1 What is a function?

Picture this. A function is a machine. You drop an input into the top, the machine applies a fixed rule, and an output comes out of the bottom. Drop in the same input twice, and you get the same output twice. A "squaring machine" turns 3 into 9 and -4 into 16. A "cat detector" turns a photograph into one of the words *cat* or *not cat*.

Definition 2.1 (Function). A **function** f from a set A to a set B is a rule that assigns to *each* element x of A *exactly one* element of B, called $f(x)$. The set A is the **domain** of f (the allowed inputs), and B is the **codomain** (where the outputs live). The set of outputs that actually occur, $\{f(x) : x \in A\}$, is the **range** of f.

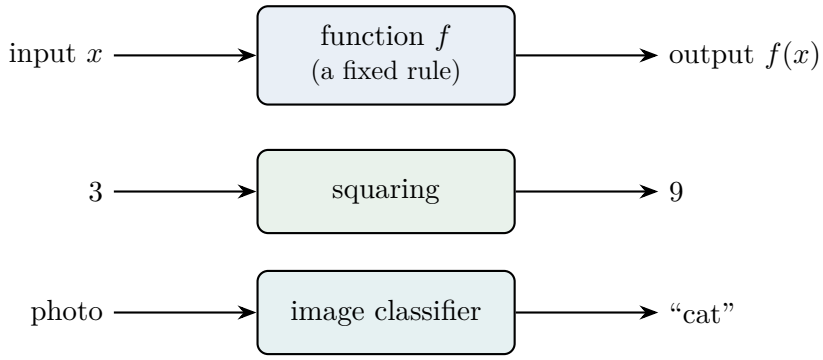


Figure 2.1: A function is a machine that turns each input into exactly one output.

Notation. We write $f\colon A \to B$, read "f is a function from A to B". The output for input x is written $f(x)$ and read "f of x". A rule is often given by a formula, such as $f(x) = x^2$; equivalently $x \mapsto x^2$, read "x maps to x^2". In graphs and applications we often write $y = f(x)$ and call x the **input** (or independent variable) and y the **output** (or dependent variable).

Common pitfall ($f(x)$ is not f times x). The brackets in $f(x)$ do *not* mean multiplication. $f(3)$ means "the output of f when the input is 3". Likewise $f(a+1)$ is *not* $f(a)+1$: if $f(x) = x^2$, then $f(a+1) = (a+1)^2 = a^2 + 2a + 1$, which is quite different from $f(a)+1 = a^2+1$. To evaluate f at any input, replace *every* x in the formula by that input, in brackets.

Two words in the definition deserve emphasis. *Each* input must get an output, so the formula $1/x$ does not define a function on all of $\mathbb{R}$ (there is no output for $x = 0$); it defines a function on the domain $\{x \in \mathbb{R} : x \neq 0\}$. And *exactly one* output: the rule "y is a number whose square is x" is not a function, because the input 4 would have two outputs, 2 and -2.

Example 2.2 (Domains and ranges). Find a suitable domain and the range of (a) $f(x) = x^2$; (b) $g(x) = \sqrt{x}$; (c) $h(x) = \frac{1}{x-2}$.

Solution. (a) Any real number can be squared, so the domain is $\mathbb{R}$. Squares are never negative, and every non-negative number is a square, so the range is $[0, \infty)$.

(b) Square roots of negative numbers are not real, so the domain is $[0, \infty)$; the range is also $[0, \infty)$.

(c) We cannot divide by zero, so $x = 2$ must be excluded: the domain is $\{x \in \mathbb{R} : x \neq 2\}$. The output can be any non-zero number (to get output y, take $x = 2 + \frac{1}{y}$), so the range is $\{y \in \mathbb{R} : y \neq 0\}$.

2.1.1 Ways to describe a function

A function can be described by a formula ($f(x) = 2x+1$), by a table of inputs and outputs, by a graph (Section 2.2), in words ("double it and add one"), or by a computer program. A trained neural network is a function described by a program together with a long list of numbers (its weights). Often no simple formula exists at all.

2.1.2 Functions of several inputs

Many functions take more than one input. The area of a rectangle, $A(\ell, w) = \ell w$, takes two. A house-price model might take three:

$$\text{price}(\text{size}, \text{bedrooms}, \text{age}) = 3000 \cdot \text{size} + 10\,000 \cdot \text{bedrooms} - 500 \cdot \text{age} + 20\,000.$$

In Chapter 4 we will bundle the inputs into a single object, a *vector*, and in Chapter 7 we will learn calculus for such functions. Until then we mostly study functions of one input.

Why this matters for ML (Learning is searching for a function)**.** Supervised machine learning starts from examples of input–output pairs $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$: photos with their labels, houses with their prices. We imagine that there is some unknown function that produced the outputs from the inputs, and we try to find a function f with $f(x_i) \approx y_i$ for all i ("$\approx$" means "approximately equal"). The set of candidate functions we search through, for example "all straight lines" or "all neural networks of a certain size", is called the **hypothesis space**. Note that a model *must* be a function: the same input always gives the same output.

Exercises for Section 2.1

Exercise 2.1. Which of the following define a function from $\mathbb{R}$ to $\mathbb{R}$? Explain. (a) $f(x) = 3x - 1$; (b) "y is a number with $y^2 = x$"; (c) $f(x) = \frac{1}{x}$; (d) $f(x) = |x|$.

Solution on p. 242 →

Exercise 2.2. Let $f(x) = x^2 - 3x$. Compute $f(0)$, $f(2)$, $f(-1)$, $f(a)$ and $f(a+1)$.

Solution on p. 242 →

Exercise 2.3. Find the largest possible domain of (a) $\sqrt{x-2}$; (b) $\frac{1}{x+3}$; (c) $\sqrt{4-x^2}$.

Solution on p. 242 →

Exercise 2.4. Find the range of $f(x) = x^2 + 1$ and of $g(x) = -|x|$, both with domain $\mathbb{R}$.

Solution on p. 242 →

Exercise 2.5. A spam filter reads an email and outputs either "spam" or "not spam". What are its domain and codomain? Why would it be a problem if the filter were not a function?

Solution on p. 242 →

2.2 Graphs of functions

To draw a function we use two perpendicular number lines: a horizontal *x-axis* for inputs and a vertical *y-axis* for outputs. Every pair of numbers (x, y) is a point in this plane: go x units to the right (left if x is negative) and y units up (down if y is negative). Chapter 3 studies this *Cartesian plane* in detail.

Definition 2.3 (Graph)**.** The **graph** of a function f is the set of all points $(x, f(x))$ with x in the domain of f.

The simplest way to sketch a graph is to make a table of values, plot the points and join them with a smooth curve.

Example 2.4 (Plotting a graph)**.** Sketch the graph of $f(x) = x^2 - 2x - 3$ for $-2 \le x \le 4$.

Solution. A table of values:

x	-2	-1	0	1	2	3	4
$f(x)$	5	0	-3	-4	-3	0	5

For instance $f(-2) = 4 + 4 - 3 = 5$. Plotting the points and joining them gives the U-shaped curve in Figure 2.2.

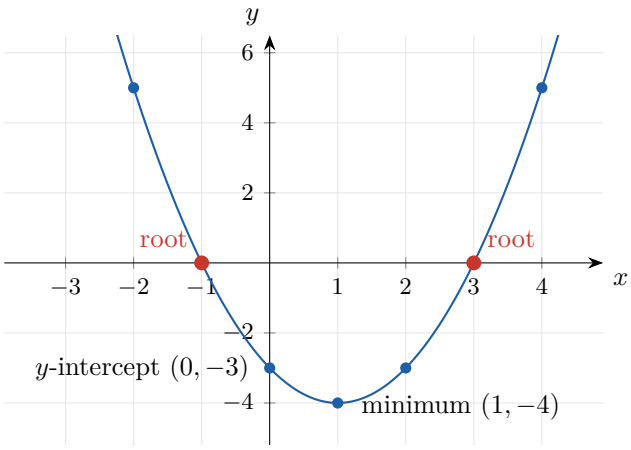


Figure 2.2: The graph of $f(x) = x^2 - 2x - 3$. It crosses the x-axis at the roots $x = -1$ and $x = 3$, crosses the y-axis at -3, and has its lowest point at $(1, -4)$.

2.2.1 Reading a graph

A graph shows at a glance much of what we want to know about a function:

- the value $f(a)$ is the height of the graph above (or below) the point a on the x-axis;
- the **roots** (or *zeros*) are the inputs where $f(x) = 0$, i.e. where the graph meets the x-axis;
- the **y-intercept** $f(0)$ is where the graph meets the y-axis;
- f is **increasing** where the graph goes up as we move right, and **decreasing** where it goes down;
- the lowest and highest points show the *minimum* and *maximum* values.

In Figure 2.2, f decreases until $x = 1$, then increases; its minimum value is -4, reached at $x = 1$; and $f(x) < 0$ exactly when $-1 < x < 3$.

Remark (The vertical line test)**.** Because a function gives exactly one output for each input, every vertical line meets the graph of a function at most once. A curve that some vertical line crosses twice, such as a circle, is not the graph of a function $y = f(x)$.

Why this matters for ML (Reading training curves)**.** The most-viewed graph in machine learning practice is the *training curve*: the loss (the model's error) plotted against the number of training steps (Figure 2.3). A healthy curve goes down quickly and then flattens out. A curve that goes up, oscillates wildly or flattens out immediately tells an experienced practitioner that something is wrong. Reading such graphs is a daily skill.

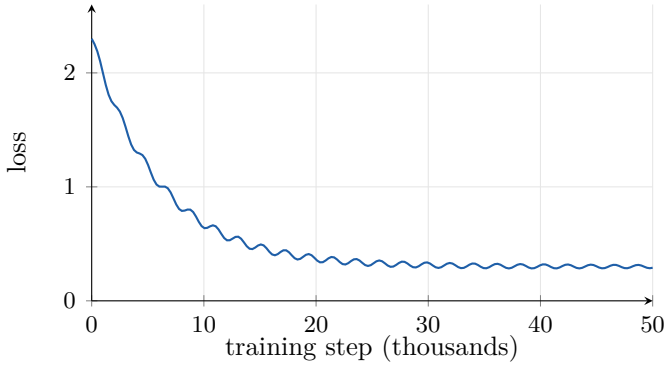


Figure 2.3: A typical training curve: the loss falls quickly at first and then levels off. The small wiggles come from the randomness of training.

Exercises for Section 2.2

Exercise 2.6. Make a table of values of $f(x) = 4 - x^2$ for $x = -3, -2, \ldots, 3$ and sketch the graph. Where are the roots, the y-intercept, and the maximum?

Solution on p. 242 →

Exercise 2.7. Use Figure 2.2 to answer: for which x is $x^2 - 2x - 3 > 0$? On which interval is the function increasing?

Solution on p. 242 →

Exercise 2.8. Is the circle consisting of all points with $x^2 + y^2 = 1$ the graph of a function $y = f(x)$? Can you split it into two pieces that are graphs of functions?

Solution on p. 242 →

2.3 Linear functions

Definition 2.5 (Linear function)**.** A function of the form

$$f(x) = mx + b,$$

where m and b are fixed numbers, is called a **linear function**. Its graph is a straight line. The number m is the **slope** and b is the **intercept** (the y-intercept, since $f(0) = b$).

Remark. Strictly speaking, mathematicians call $mx + b$ an *affine* function and reserve "linear" for the case $b = 0$; we will meet this distinction in Chapter 4. In everyday usage, and in most of machine learning, both are called linear.

2.3.1 Slope

The slope measures how steep the line is. Pick any two points (x_1, y_1) and (x_2, y_2) on the line. The *run* is the horizontal change $x_2 - x_1$, the *rise* is the vertical change $y_2 - y_1$, and

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}.$$

For a straight line this ratio is the same whichever two points we pick.

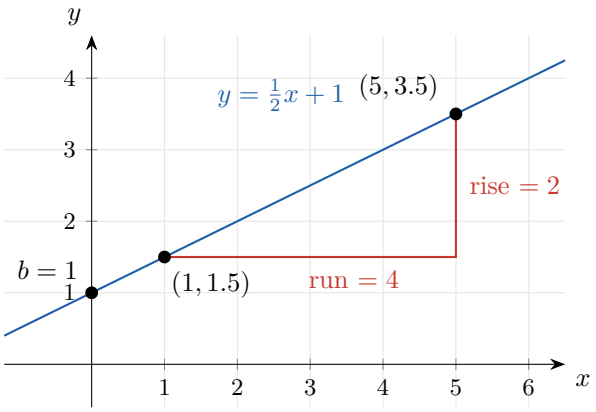


Figure 2.4: The slope is rise over run: here $m = \frac{2}{4} = \frac{1}{2}$. The line crosses the y-axis at the intercept $b = 1$.

Intuition (Slope is a rate of change)**.** The slope tells you *how much the output changes when the input increases by one unit.* If $y = \frac{1}{2}x + 1$, then every step of 1 to the right raises y by $\frac{1}{2}$. A positive slope means the line goes up from left to right, a negative slope means it goes down, a slope of 0 gives a horizontal line, and a larger $|m|$ means a steeper line. Slopes carry units: if x is a house size in square metres and y a price in euros, then m is in euros per square metre. In Chapter 6, the idea of slope will be extended from straight lines to curves; the result is called the *derivative*.

Example 2.6 (The line through two points)**.** Find the linear function whose graph passes through $(1, 2)$ and $(4, 11)$.

Solution. The slope is $m = \frac{11-2}{4-1} = \frac{9}{3} = 3$. So $f(x) = 3x + b$, and since $f(1) = 2$ we need $3 + b = 2$, so $b = -1$. The function is $f(x) = 3x - 1$. *Check:* $f(4) = 12 - 1 = 11$. ✓

A convenient alternative is the *point–slope form*: the line with slope m through the point (x_1, y_1) is

$$y - y_1 = m(x - x_1).$$

Example 2.7 (Celsius and Fahrenheit)**.** The formula $F = 1.8C + 32$ is a linear function of C. Its slope 1.8 says that each degree Celsius is worth 1.8 degrees Fahrenheit; its intercept 32 is the Fahrenheit temperature at $0\,^\circ\text{C}$, the freezing point of water.

2.3.2 Special lines

A horizontal line $y = c$ has slope 0: the output does not depend on the input at all (a *constant function*). A vertical line $x = c$ is not the graph of a function (it fails the vertical

line test), and its slope is undefined because the run is 0. Two lines are *parallel* exactly when they have the same slope. When $b = 0$, the output is proportional to the input: doubling x doubles y.

Why this matters for ML (Linear regression). The simplest machine learning model is **linear regression** with one input:

$$\hat{y} = wx + b.$$

In machine learning the slope is called the *weight* w and the intercept the *bias* b. The weight measures how sensitive the prediction is to the input: $w = 3000$ euros per square metre means that each extra square metre adds 3000 euros to the predicted price. Training finds the line that best fits a cloud of data points (Chapters 5 and 8).

Exercises for Section 2.3

Exercise 2.9. Find the slope and intercept of the line through $(1, 2)$ and $(3, -4)$.

Solution on p. 242 →

Exercise 2.10. Write the equation of the line with slope -2 through the point $(3, 1)$ in the form $y = mx + b$.

Solution on p. 242 →

Exercise 2.11. Using $F = 1.8C + 32$: (a) by how much does F change when C increases by 10? (b) At what temperature do the Celsius and Fahrenheit scales show the same number?

Solution on p. 242 →

Exercise 2.12. A model predicts price $= 2500 \times \text{size} + 40\,000$. Interpret the numbers 2500 and 40 000. By how much does the predicted price rise if the size increases by 12 square metres? Is the intercept meaningful?

Solution on p. 242 →

Exercise 2.13. Are the lines $y = 2x + 1$ and $y = 2x - 5$ parallel? Where do the lines $y = 2x + 1$ and $y = -x + 4$ meet?

Solution on p. 242 →

2.4 Polynomials and quadratic functions

Definition 2.8 (Polynomial). A **polynomial** is a function of the form

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where n is a whole number ≥ 0 and the *coefficients* $a_0, \ldots, a_n$ are real numbers with $a_n \neq 0$. The highest power n is the **degree**.

Constant functions have degree 0, linear functions have degree 1, *quadratic* functions have degree 2 (for example $x^2 - 2x - 3$), and *cubic* functions have degree 3 (for example $x^3 - 3x$).

2.4.1 Quadratic functions

A **quadratic function** has the form $f(x) = ax^2 + bx + c$ with $a \neq 0$. Its graph is a **parabola**: a U shape that opens upwards if $a > 0$ and downwards if $a < 0$. An upward parabola has a lowest point, and a downward one a highest point; this turning point is called the **vertex**. The larger $|a|$, the narrower the U.

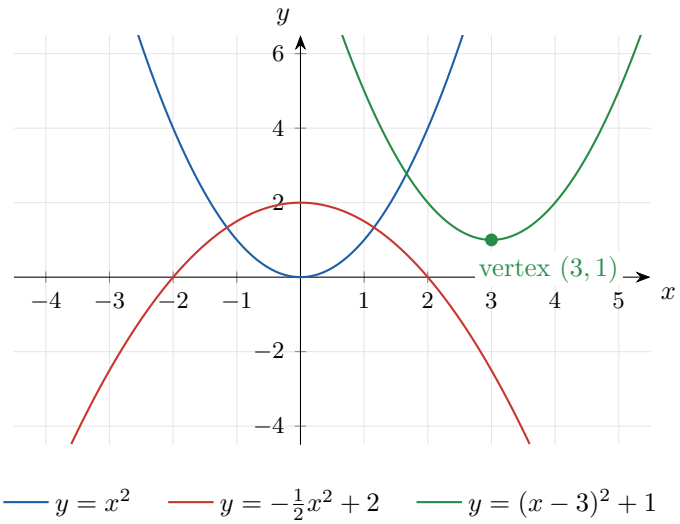


Figure 2.5: Three parabolas. $y = x^2$ opens upwards with vertex at the origin; $y = -\frac{1}{2}x^2 + 2$ opens downwards and is wider; $y = (x-3)^2 + 1$ is the graph of x^2 moved 3 units right and 1 unit up.

2.4.2 Solving quadratic equations

The roots of a quadratic function are the solutions of $ax^2 + bx + c = 0$. Sometimes we can find them by factoring (Section 1.6): since $x^2 - 2x - 3 = (x-3)(x+1)$, the equation $x^2 - 2x - 3 = 0$ becomes $(x-3)(x+1) = 0$. A product is zero only if one of its factors is zero, so $x = 3$ or $x = -1$, matching Figure 2.2.

When factoring is not obvious, there is a formula that always works.

Theorem 2.9 (The quadratic formula)**.** The solutions of $ax^2 + bx + c = 0$ (with $a \neq 0$) are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The symbol $\pm$ ("plus or minus") means that there is one solution with $+$ and one with $-$. The number $D = b^2 - 4ac$ is the **discriminant**: if $D > 0$ there are two real solutions, if $D = 0$ there is exactly one, and if $D < 0$ there are no real solutions (the parabola does not reach the x-axis).

Example 2.10 (Using the quadratic formula)**.** Solve $2x^2 - 4x - 1 = 0$.

Solution. Here $a = 2$, $b = -4$, $c = -1$, so $D = (-4)^2 - 4 \cdot 2 \cdot (-1) = 16 + 8 = 24 > 0$: two solutions,

$$x = \frac{4 \pm \sqrt{24}}{4} = 1 \pm \frac{\sqrt{6}}{2}, \qquad \text{i.e. } x \approx 2.225 \text{ or } x \approx -0.225.$$

2.4.3 Completing the square: where is the minimum?

For machine learning, the most important fact about a quadratic is the location of its minimum. A technique called **completing the square** rewrites $ax^2 + bx + c$ in *vertex form* $a(x - h)^2 + k$, from which the vertex (h, k) can be read off directly.

Example 2.11 (Finding the minimum of a quadratic)**.** Find the minimum value of $f(x) = x^2 - 6x + 10$.

Solution. The identity $(x - 3)^2 = x^2 - 6x + 9$ contains the first two terms of f. So

$$f(x) = (x^2 - 6x + 9) + 1 = (x - 3)^2 + 1.$$

A square is never negative, so $(x - 3)^2 \geq 0$, with equality only at $x = 3$. Hence $f(x) \geq 1$ for every x, and the minimum value 1 is reached at $x = 3$. The vertex is $(3, 1)$ (the green parabola in Figure 2.5).

In general, the halfway point between the two terms x^2 and bx gives the recipe: $x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \frac{b^2}{4}$. Applying this to $ax^2 + bx + c$ shows that

Proposition 2.12 (Vertex of a parabola)**.** The vertex of $f(x) = ax^2 + bx + c$ is at $x = -\frac{b}{2a}$. If $a > 0$, this is where f attains its minimum; if $a < 0$, its maximum.

Remark. Completing the square on $ax^2 + bx + c = 0$ in general is exactly how the quadratic formula is derived. You can try it: divide by a, complete the square, and take square roots of both sides.

2.4.4 Higher-degree polynomials

Polynomials of higher degree can wiggle more: a polynomial of degree n has at most n roots and at most $n - 1$ turning points (Figure 2.6). For large $|x|$, the highest-power term dominates, so the graph eventually shoots off to $+\infty$ or $-\infty$.

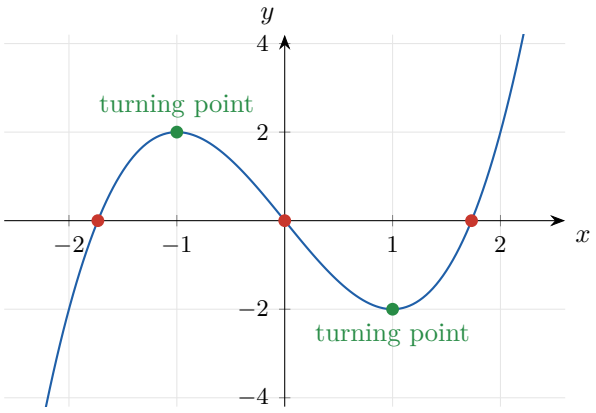


Figure 2.6: The cubic $y = x^3 - 3x$ has three roots (0 and $\pm\sqrt{3}$, in red) and two turning points (in green).

Why this matters for ML (Squared error is a quadratic)**.** Suppose we want to summarise the three numbers $y_1 = 2$, $y_2 = 4$, $y_3 = 9$ by a single number c, and we measure the quality of c by the total squared error

$$L(c) = (2 - c)^2 + (4 - c)^2 + (9 - c)^2.$$

Expanding, $L(c) = 3c^2 - 30c + 101$: a quadratic in c with $a = 3 > 0$. By Proposition 2.12 its minimum is at $c = \frac{30}{2 \cdot 3} = 5$, which is exactly the mean $\frac{2+4+9}{3}$. This is no coincidence: the mean is always the number that minimises the squared error, a fact we will prove with calculus in Chapter 6. More generally, the loss of a linear regression model is a quadratic function of its parameters, which is why it is so easy to minimise.

Exercises for Section 2.4

Exercise 2.14. Give the degree and leading coefficient (the coefficient of the highest power) of (a) $4x^3 - x + 7$; (b) $(x + 1)(x - 2)$; (c) 5.

Solution on p. 242 →

Exercise 2.15. Solve by factoring: (a) $x^2 - 5x + 6 = 0$; (b) $x^2 - 16 = 0$; (c) $x^2 + 3x = 0$.

Solution on p. 242 →

Exercise 2.16. Use the discriminant to decide how many real solutions each equation has, and find them: (a) $x^2 + x + 1 = 0$; (b) $x^2 - 6x + 9 = 0$; (c) $2x^2 - 3x - 2 = 0$; (d) $x^2 + 4x + 1 = 0$.

Solution on p. 242 →

Exercise 2.17. Complete the square to find the vertex and the minimum or maximum value of (a) $f(x) = x^2 + 4x + 7$; (b) $g(x) = -x^2 + 6x - 5$.

Solution on p. 243 →

Exercise 2.18. For the data $y = (1, 3, 8)$, write the total squared error $L(c) = \sum_{i=1}^{3} (y_i - c)^2$ as a quadratic in c, find the value of c that minimises it, and compare with the mean.

Solution on p. 243 →

2.5 Exponential functions

Motivation. Some quantities grow by *adding* a fixed amount each step (a salary that rises by 1000 per year). Others grow by *multiplying* by a fixed factor each step (bacteria that double every hour, money earning interest, a rumour spreading on social media). Multiplicative growth and decay are described by exponential functions. In machine learning, exponentials turn arbitrary scores into positive numbers (and so into probabilities), describe how learning rates are decayed, and explain why signals can explode or vanish in deep networks.

Definition 2.13 (Exponential function)**.** For a fixed number $a > 0$ with $a \neq 1$, the

function

$$f(x) = a^x$$

is called an **exponential function** with base a. Thanks to the rules of Section 1.4, a^x makes sense for every real number x: for example $2^{-1} = \frac{1}{2}$ and $2^{1/2} = \sqrt{2}$.

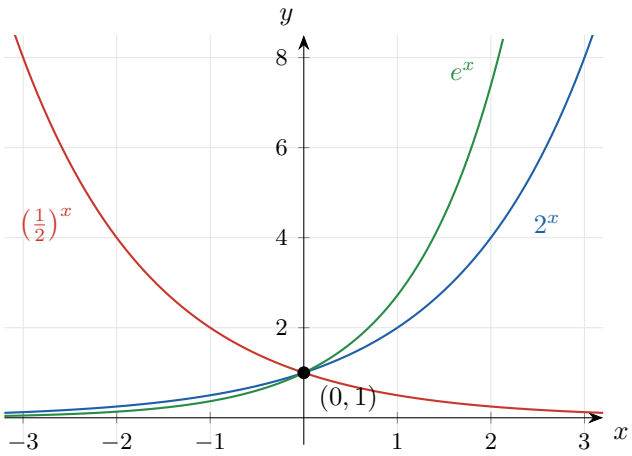


Figure 2.7: Exponential functions. Every graph passes through $(0, 1)$ and stays above the x-axis. 2^x and e^x grow, $(\frac{1}{2})^x$ decays.

The graphs in Figure 2.7 show the key properties:

- $a^x > 0$ for every x: an exponential is never zero or negative;
- $a^0 = 1$, so every graph passes through $(0, 1)$;
- if $a > 1$ the function increases (*exponential growth*); if $0 < a < 1$ it decreases (*exponential decay*);
- $a^{x+1} = a \cdot a^x$: increasing the input by 1 *multiplies* the output by a. (For a linear function, increasing the input by 1 *adds* the slope to the output.)

Intuition (Exponential growth eventually beats everything)**.** Compare x^2 with 2^x:

x	1	2	3	4	5	6	7	8	9	10
x^2	1	4	9	16	25	36	49	64	81	100
2^x	2	4	8	16	32	64	128	256	512	1024

At first they are similar, but after $x = 4$ the exponential races ahead, and the gap keeps widening: 2^{30} is over a billion while $30^2 = 900$. Any exponential with base greater than 1 eventually overtakes any polynomial.

2.5.1 Growth and decay models

A quantity that starts at N_0 and is multiplied by a in every unit of time is described by $N(t) = N_0 a^t$.

Example 2.14 (Doubling and halving)**.** (a) A colony of 100 bacteria doubles every hour. How many are there after 5 hours? (b) A substance has a *half-life* of 10 years: every 10 years, half of it decays. What fraction remains after 30 years?

Solution. (a) $N(t) = 100 \cdot 2^t$, so $N(5) = 100 \cdot 32 = 3200$. (b) 30 years is three half-lives,

so the remaining fraction is $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$. As a formula, the amount after t years is $A_0\left(\frac{1}{2}\right)^{t/10}$.

2.5.2 The number e

One base is so useful that it has its own name. Suppose a bank pays 100% interest per year on 1 euro. Paid once at the end of the year, you get 2 euros. Paid in two halves of 50%, you get $(1+\frac{1}{2})^2 = 2.25$. Paid in n instalments of $\frac{100}{n}$%, you get $\left(1+\frac{1}{n}\right)^n$:

n	1	2	4	12	365	10 000	1 000 000
$\left(1+\frac{1}{n}\right)^n$	2	2.25	2.4414	2.6130	2.7146	2.71815	2.718280

The more often interest is paid, the more you get, but the amounts approach a limit: the number

$$e = 2.718281828\ldots,$$

called **Euler's number**. Like π, it is irrational. The function e^x is called *the* exponential function, also written $\exp(x)$. Its special property, which we will see in Chapter 6, is that at every point its slope equals its value. This makes e the most natural base for calculus, and it is the base used throughout machine learning.

Why this matters for ML (Exponentials in machine learning)**.** Exponentials appear in machine learning in at least three ways.

- **Making things positive.** A neural network produces arbitrary real-valued scores, but probabilities must be positive. Since $e^x > 0$ for every x, applying e^x to the scores makes them positive; this is the first step of the *softmax* function (Chapter 11).
- **Learning-rate decay.** A common schedule multiplies the learning rate by a fixed factor after every epoch: $\eta_t = \eta_0 \gamma^t$ with, say, $\eta_0 = 0.1$ and $\gamma = 0.95$. After 20 epochs the rate is $0.1 \times 0.95^{20} \approx 0.036$.
- **Exploding and vanishing.** Multiplying many numbers that are slightly larger or smaller than 1 gives exponential behaviour: $1.1^{100} \approx 13\,781$ but $0.9^{100} \approx 0.000\,027$. In a deep network, signals pass through many layers and get multiplied many times, which is why they can *explode* or *vanish* (Chapter 11).

Exercises for Section 2.5

Exercise 2.19. Evaluate 2^5, 2^{-3}, 10^0, $\left(\frac{1}{3}\right)^{-2}$, $8^{1/3}$ and e^0.

Solution on p. 243 →

Exercise 2.20. A colony of 500 cells triples every hour. Write a formula for the number of cells after t hours and compute it for $t = 4$.

Solution on p. 243 →

Exercise 2.21. A substance with a half-life of 8 days starts at 160 grams. Write a formula for the amount after t days, and find the amount after 24 days.

Solution on p. 243 →

Exercise 2.22. Compare x^3 and 2^x at $x = 10$ and $x = 20$.

Solution on p. 243 →

Exercise 2.23. A signal is multiplied by 0.5 in each of 50 layers of a network. What fraction of it is left? What if it is multiplied by 1.5 in each layer?

Solution on p. 243 →

2.6 Logarithms

An exponential answers the question "what do I get if I multiply a by itself x times?". A logarithm answers the reverse question: "how many times must I multiply a by itself to get y?".

Definition 2.15 (Logarithm). Let $a > 0$ with $a \neq 1$, and let $y > 0$. The **logarithm** of y to base a, written $\log_a y$, is the number x such that $a^x = y$:

$$\log_a y = x \quad \text{means exactly} \quad a^x = y.$$

So $\log_2 8 = 3$ because $2^3 = 8$; $\log_{10} 1000 = 3$ because $10^3 = 1000$; $\log_{10} 0.01 = -2$ because $10^{-2} = 0.01$; and $\log_a 1 = 0$ for every base, because $a^0 = 1$.

Notation. The logarithm with base e is the **natural logarithm**, written ln: $\ln y = \log_e y$. Base 10 is common in science and base 2 in computer science and information theory. **In machine learning papers and code, "log" with no base almost always means the natural logarithm** ln; for example, `np.log` in Python computes ln.

2.6.1 The rules of logarithms

Because logarithms are exponents, the laws of exponents (Theorem 1.5) turn into laws of logarithms.

Theorem 2.16 (Laws of logarithms). For $x, y > 0$, any real number p, and any valid base:

$$\begin{aligned}
\log(xy) &= \log x + \log y && \text{(products become sums)},\\
\log\left(\tfrac{x}{y}\right) &= \log x - \log y && \text{(quotients become differences)},\\
\log\left(x^p\right) &= p \log x && \text{(powers become multiples)},\\
\log_a x &= \frac{\ln x}{\ln a} && \text{(change of base)}.
\end{aligned}$$

Why the first rule holds. Let $u = \log_a x$ and $v = \log_a y$, so $x = a^u$ and $y = a^v$. Then $xy = a^u a^v = a^{u+v}$, which says exactly that $\log_a(xy) = u + v$. The other rules are proved in the same way from the other exponent laws. ∎

The first rule is the reason logarithms were invented (in the 1600s): they turn hard multiplications into easy additions.

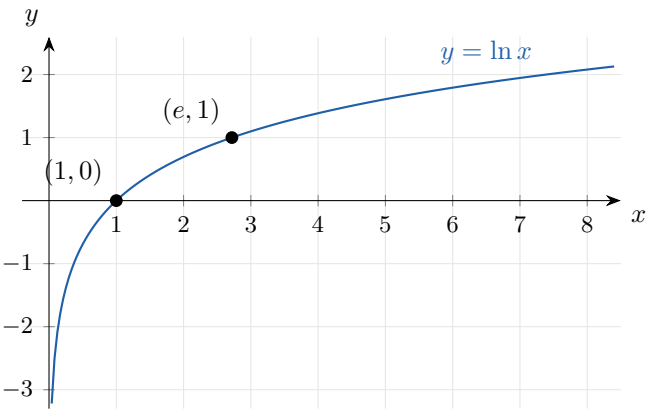


Figure 2.8: The natural logarithm. It is defined only for $x > 0$, it is negative for $0 < x < 1$, zero at $x = 1$, and it grows very slowly: $\ln x$ reaches 2 only at $x = e^2 \approx 7.39$. As x approaches 0, $\ln x$ plunges towards $-\infty$.

Common pitfall (What logarithms do not do)**.** $\log(x + y)$ is *not* $\log x + \log y$; logarithms turn *products* into sums, not sums into sums. For example, $\ln(1 + 1) = \ln 2 \approx 0.693$, but $\ln 1 + \ln 1 = 0$. Also, $\log 0$ and the logarithm of a negative number are undefined: in code, `log(0)` returns $-\infty$ and `log(-1)` returns "not a number".

Example 2.17 (Solving exponential equations)**.** Solve $2^x = 20$ and $e^{2x} = 7$.

Solution. Take logarithms of both sides. From $2^x = 20$: $\ln(2^x) = \ln 20$, so $x \ln 2 = \ln 20$ and $x = \frac{\ln 20}{\ln 2} \approx \frac{2.996}{0.693} \approx 4.32$. (Check: $2^4 = 16$ and $2^5 = 32$, so the answer must lie between 4 and 5.) From $e^{2x} = 7$: $2x = \ln 7$, so $x = \frac{1}{2} \ln 7 \approx 0.973$.

2.6.2 Logarithmic scales

Logarithms measure *orders of magnitude*: $\log_{10}$ of a number is roughly its number of digits. When values range over many orders of magnitude we often plot them on a **logarithmic scale**, where equal distances represent equal *ratios*: the distance from 10^{-4} to 10^{-3} equals the distance from 10^{-3} to 10^{-2}.

Why this matters for ML (Why machine learning loves logarithms)**.** Logarithms are everywhere in machine learning, for three main reasons.

- **Products of probabilities underflow.** The probability that a model assigns to a whole data set is a product of many small numbers. With 1000 examples, each given probability 0.01, the product is 10^{-2000}, far below the smallest number a computer can represent, so it becomes 0. Its logarithm, however, is a perfectly ordinary number: $\ln\left(0.01^{1000}\right) = 1000 \ln 0.01 \approx -4605$. So we work with sums of log-probabilities (*log-likelihoods*) instead of products of probabilities (Chapter 10).
- **Loss functions.** The most widely used loss for classification, the cross-entropy, is $-\ln p$, where p is the probability the model gave to the correct answer (Chapter 11). It is 0 when $p = 1$ and grows without limit as $p \to 0$.
- **Hyperparameter search.** Learning rates are usually tried on a logarithmic scale, e.g. $10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}$, because what matters is their order of

magnitude.

Exercises for Section 2.6

Exercise 2.24. Evaluate: $\log_2 32$, $\log_{10} 0.001$, $\log_3 81$, $\ln e^5$, $\log_5 1$, $\log_2 \frac{1}{8}$.

Solution on p. 243 →

Exercise 2.25. (a) Write $\ln\left(\frac{x^2 y}{z}\right)$ in terms of $\ln x$, $\ln y$ and $\ln z$. (b) Simplify $\ln 8 - \ln 2$ and express it as a multiple of $\ln 2$.

Solution on p. 243 →

Exercise 2.26. Solve $3^x = 50$ and $5e^x = 40$. Give answers to three decimal places.

Solution on p. 243 →

Exercise 2.27. A learning rate starts at 0.1 and is multiplied by 0.9 after every epoch, so $\eta_t = 0.1 \times 0.9^t$. After how many epochs does it first drop below 0.01?

Solution on p. 243 →

Exercise 2.28. A model gives probability 0.05 to each of 1000 independent examples. Compute the logarithm of the product of these probabilities, and use it to find the power of 10 that the product is roughly equal to.

Solution on p. 243 →

2.7 Composition of functions

Functions can be chained: the output of one becomes the input of the next.

Definition 2.18 (Composition)**.** The **composition** of f and g, written $f \circ g$ (read "f after g" or "f composed with g"), is the function

$$(f \circ g)(x) = f(g(x)).$$

First apply g, then apply f to the result.

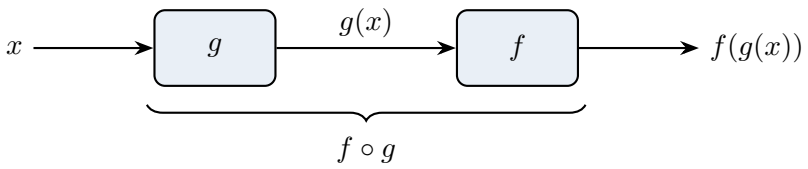


Figure 2.9: Composition is an assembly line: x goes through g first, and the result goes through f.

Example 2.19 (Order matters)**.** Let $g(x) = 2x + 1$ and $f(x) = x^2$. Find $f(g(x))$ and $g(f(x))$, and evaluate both at $x = 3$.

Solution. $f(g(x)) = (2x+1)^2$, while $g(f(x)) = 2x^2 + 1$. At $x = 3$: $f(g(3)) = f(7) = 49$, but $g(f(3)) = g(9) = 19$. In general $f \circ g \neq g \circ f$.

It is just as important to be able to *decompose* a complicated function into simple pieces. For example, $h(x) = e^{-x^2}$ is $f(g(x))$ with inner function $g(x) = -x^2$ and outer function $f(u) = e^u$. In Chapter 6 the *chain rule* will compute slopes of compositions piece by piece, and that rule is the mathematical heart of how neural networks are trained.

Proposition 2.20 (Composing linear functions gives a linear function)**.** If $f(x) = ax + b$ and $g(x) = cx + d$, then $f(g(x)) = a(cx + d) + b = (ac)x + (ad + b)$, which is again linear.

Why this matters for ML (Neural networks are compositions)**.** A neural network with three layers computes

$$\text{output} = f_3(f_2(f_1(\text{input}))) = (f_3 \circ f_2 \circ f_1)(\text{input}),$$

where each *layer* f_k is a fairly simple function. "Deep" learning simply means composing many layers. Each layer consists of a linear part followed by a simple non-linear function called an *activation function* (Section 2.10). The non-linear part is essential: Proposition 2.20 shows that a composition of linear functions is just one linear function, so without activation functions a deep network would be no more powerful than a single straight line. We will see the many-input version of this fact in Chapter 4.

Exercises for Section 2.7

Exercise 2.29. Let $f(x) = x + 3$ and $g(x) = x^2$. Compute $f(g(2))$, $g(f(2))$, and formulas for $f(g(x))$ and $g(f(x))$.

Solution on p. 243 →

Exercise 2.30. Write each function as a composition of simpler functions: (a) $h(x) = \sqrt{3x + 1}$; (b) $k(x) = \dfrac{1}{1 + e^{-x}}$.

Solution on p. 243 →

Exercise 2.31. A "network" has one hidden value $h = 2x - 1$ and output $y = -3h + 4$. (a) Write y as a function of x. (b) Now insert a ReLU: $h = \text{ReLU}(2x - 1)$, $y = -3h + 4$. Compute y for $x = 0, 1, 2$, and decide whether y is still a linear function of x.

Solution on p. 243 →

2.8 Inverse functions

An inverse function undoes what a function does. If $f(x) = 2x + 3$ ("double, then add 3"), the inverse is "subtract 3, then halve": $f^{-1}(y) = \frac{y-3}{2}$.

Definition 2.21 (Inverse function)**.** A function $f\colon A \to B$ has an **inverse function** $f^{-1}\colon B \to A$ if

$$f^{-1}(f(x)) = x \text{ for all } x \in A \qquad \text{and} \qquad f(f^{-1}(y)) = y \text{ for all } y \in B.$$

How to find an inverse. Write $y = f(x)$ and solve for x in terms of y. For $y = 2x + 3$: $y - 3 = 2x$, so $x = \frac{y-3}{2}$, which is $f^{-1}(y)$.

Common pitfall (f^{-1} is not $1/f$). The -1 in f^{-1} is *not* an exponent. For $f(x) = 2x + 3$, $f^{-1}(y) = \frac{y-3}{2}$, whereas $\frac{1}{f(x)} = \frac{1}{2x+3}$; these are completely different functions.

2.8.1 When does an inverse exist?

To undo f we must be able to tell, from the output, which input produced it. So f must never send two different inputs to the same output; such a function is called **one-to-one**. The function $f(x) = x^2$ on $\mathbb{R}$ is not one-to-one, because $f(2) = f(-2) = 4$: from the output 4 we cannot tell whether the input was 2 or -2. But if we *restrict the domain* to $[0, \infty)$, the function becomes one-to-one and its inverse is the square root, $f^{-1}(y) = \sqrt{y}$. (Graphically: a function is one-to-one when every horizontal line meets its graph at most once.)

2.8.2 Graphs of inverse functions

If (a, b) is on the graph of f, then $f(a) = b$, so $f^{-1}(b) = a$ and (b, a) is on the graph of f^{-1}. Swapping the coordinates of every point reflects the graph in the diagonal line $y = x$ (Figure 2.10).

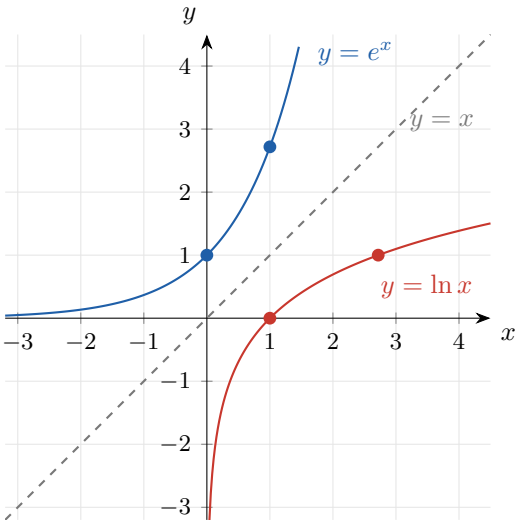


Figure 2.10: e^x and $\ln x$ are inverse functions, so their graphs are mirror images in the line $y = x$. The point $(0, 1)$ on e^x corresponds to $(1, 0)$ on $\ln x$, and $(1, e)$ to $(e, 1)$.

Theorem 2.22 (Exponential and logarithm are inverses)**.** For every real x, $\ln(e^x) = x$, and for every $x > 0$, $e^{\ln x} = x$. The same holds for any base: $\log_a(a^x) = x$ and $a^{\log_a x} = x$.

This is just the definition of the logarithm restated, but it is extremely useful for solving equations, as we saw in Section 2.6.

Why this matters for ML (Sigmoid and logit)**.** The **sigmoid function**

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

(the Greek letter σ is "sigma") squashes any real number z into the interval $(0, 1)$, so its output can be read as a probability: $\sigma(0) = 0.5$, $\sigma(2) \approx 0.881$, $\sigma(-2) \approx 0.119$. Its

inverse is the **logit** function,

$$\text{logit}(p) = \ln \frac{p}{1-p}, \qquad 0 < p < 1.$$

To check, solve $p = \frac{1}{1+e^{-z}}$ for z: $1 + e^{-z} = \frac{1}{p}$, so $e^{-z} = \frac{1-p}{p}$, so $-z = \ln \frac{1-p}{p}$, i.e. $z = \ln \frac{p}{1-p}$. The quantity $\frac{p}{1-p}$ is the *odds* (a probability of 0.8 means odds of 4 to 1), so the logit is the "log-odds". In logistic regression and neural network classifiers, the raw scores z before the sigmoid are called *logits* for this reason.

Exercises for Section 2.8

Exercise 2.32. Find the inverse of each function and check your answer: (a) $f(x) = 5x - 2$; (b) $g(x) = \frac{x-3}{4}$; (c) $h(x) = x^3 + 1$.

Solution on p. 243 →

Exercise 2.33. Is $f(x) = x^2 + 1$ one-to-one on $\mathbb{R}$? On $[0, \infty)$? Find the inverse in the second case, including its domain.

Solution on p. 243 →

Exercise 2.34. Compute $\sigma(0)$, $\text{logit}(0.5)$ and $\text{logit}(0.9)$. Then check that the sigmoid undoes the logit, by computing $\sigma(\text{logit}(0.9))$.

Solution on p. 244 →

Exercise 2.35. Data are often standardised with $z = \frac{x-\mu}{s}$, where $\mu = 50$ and $s = 10$. Find the inverse transformation. A model trained on standardised data predicts $z = 1.3$; what is the prediction on the original scale?

Solution on p. 244 →

2.9 How parameters move graphs

Machine learning models contain parameters, and changing a parameter changes the graph of the model. It is very useful to be able to predict how. Starting from any function f and numbers $c > 0$:

New function	Effect on the graph of f
$f(x) + c$	shift *up* by c
$f(x) - c$	shift *down* by c
$f(x - c)$	shift *right* by c
$f(x + c)$	shift *left* by c
$c\,f(x)$	stretch vertically by the factor c
$f(cx)$	squeeze horizontally by the factor c (if $c > 1$)
$-f(x)$	reflect in the x-axis (upside down)
$f(-x)$	reflect in the y-axis (left–right)

Common pitfall (Horizontal shifts go the "wrong" way)**.** $f(x-3)$ moves the graph 3 units to the *right*, not the left. Reason: the new function takes at $x=3$ the value that f took at 0, so every feature of the graph now happens 3 units later. We saw this with $(x-3)^2+1$, whose vertex is at $x=+3$ (Figure 2.5).

Why this matters for ML (What the weight and bias of a neuron do)**.** A single artificial neuron with one input computes $\sigma(wx+b)$. Writing $wx+b = w\left(x+\frac{b}{w}\right)$ shows that:

- the weight w squeezes the sigmoid horizontally: larger $|w|$ gives a steeper, more switch-like transition, and a negative w flips it;
- the bias b shifts the graph horizontally: the transition (where the output is 0.5) happens where $wx+b=0$, i.e. at $x=-\frac{b}{w}$.

Figure 2.11 shows this. By choosing w and b, the neuron can place a soft "switch" wherever it is needed; combining many such switches lets a network build very complicated functions.

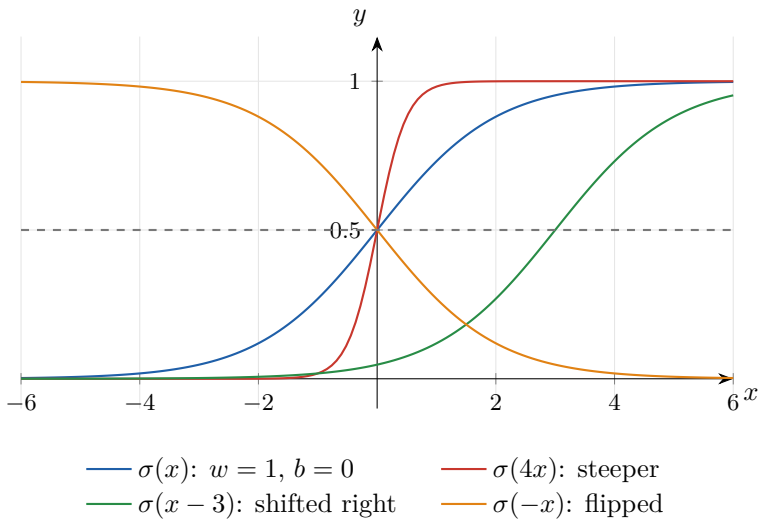


Figure 2.11: The sigmoid $\sigma(wx+b)$ for different weights and biases. The weight controls steepness and direction; the bias controls where the transition (output 0.5, dashed line) happens.

Exercises for Section 2.9

Exercise 2.36. Describe how to obtain the graph of $g(x)=(x+1)^2-4$ from the graph of x^2. Where is its vertex? Where are its roots?

Solution on p. 244 →

Exercise 2.37. Starting from $f(x)=|x|$, write down the function whose graph is shifted 3 units to the right and stretched vertically by a factor of 2.

Solution on p. 244 →

Exercise 2.38. A neuron computes $\sigma(4x - 2)$. At which input is its output 0.5? Is it steeper or less steep than $\sigma(x)$? What changes if the weight becomes -4 (with the same bias)?

Solution on p. 244 →

Exercise 2.39. Describe the graphs of e^{x-2}, e^{-x}, $-e^{x}$ and $e^{x} - 1$ in terms of the graph of e^{x}.

Solution on p. 244 →

2.10 A gallery of functions used in machine learning

2.10.1 Piecewise functions

Some functions are defined by different formulas on different parts of the domain; these are called **piecewise functions**. We have already met two: the absolute value $|x|$ (Definition 1.14) and $\mathrm{ReLU}(x) = \max(0, x)$, which equals 0 for $x < 0$ and x for $x \geq 0$. Another is the **step function**,

$$\mathrm{step}(x) = \begin{cases} 1 & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

2.10.2 Activation functions

Neural networks use a small number of non-linear functions, called **activation functions**, after each linear step. Figure 2.12 shows the most important ones.

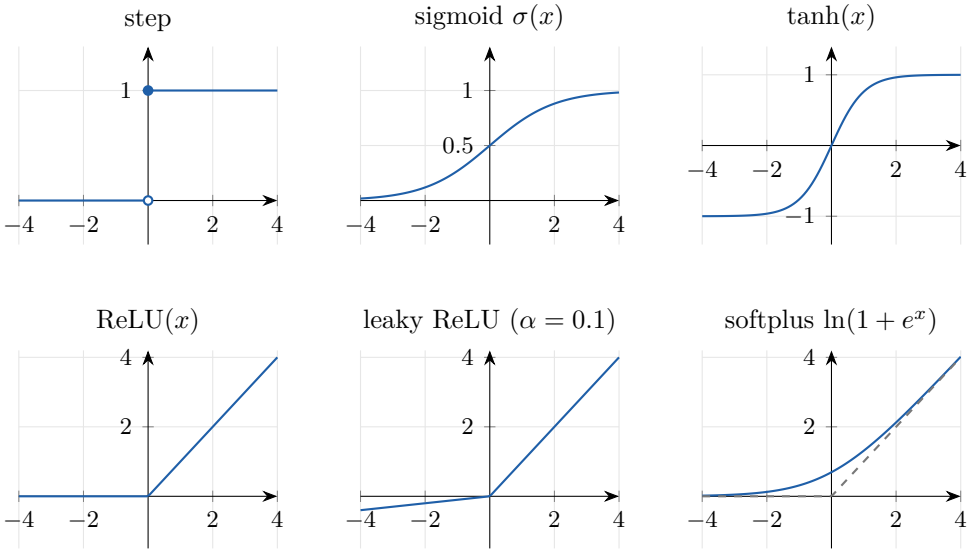


Figure 2.12: Common activation functions. The dashed line in the last panel is the ReLU, which softplus smooths out.

- **Step.** Used in the earliest artificial neuron, the perceptron (1958): the neuron either fires (1) or not (0). Its graph has a jump, which (as we will see in Chapter 6) makes it useless for gradient-based training.
- **Sigmoid**, $\sigma(x) = \frac{1}{1+e^{-x}}$. A smooth version of the step, with outputs in $(0, 1)$. Still used to output probabilities for yes/no questions.
- **Hyperbolic tangent**, $\tanh(x) = \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$. Shaped like the sigmoid but with outputs in $(-1, 1)$ and centred at 0. In fact $\tanh(x) = 2\sigma(2x) - 1$.

- **ReLU**, $\max(0, x)$. Simple and fast, and the default choice in modern networks since about 2012.
- **Leaky ReLU**, $\max(\alpha x, x)$ for a small α such as 0.01 or 0.1: like ReLU but with a small slope for negative inputs, so that the neuron never becomes completely "dead".
- **Softplus**, $\ln(1 + e^x)$: a smooth version of ReLU. Modern networks also use other smooth variants of ReLU with names like GELU and SiLU, which look similar.

Why this matters for ML (Why so many activation functions?)**.** All activation functions do the same basic job: they bend the otherwise straight-line behaviour of a layer, so that compositions of layers can represent curved, complicated functions. They differ in their ranges (probabilities need $(0, 1)$), in whether they are smooth, and above all in how easily the network can be trained through them, a question about their slopes that we will understand in Chapters 6 and 11.

Exercises for Section 2.10

Exercise 2.40. Compute $\sigma(0)$, $\tanh(0)$, $\mathrm{ReLU}(0)$ and $\mathrm{softplus}(0)$.

Solution on p. 244 →

Exercise 2.41. Show that $\sigma(-x) = 1 - \sigma(x)$ for every x. Interpret this for a classifier whose output $\sigma(z)$ is the probability of the class "yes".

Solution on p. 244 →

Exercise 2.42. Show that $\tanh(x) = 2\sigma(2x) - 1$.

Solution on p. 244 →

Exercise 2.43. Compute the leaky ReLU with $\alpha = 0.01$ at $x = -5$, 0 and 3. Then compute $\mathrm{softplus}(5)$ and $\mathrm{softplus}(-5)$ and compare with $\mathrm{ReLU}(5)$ and $\mathrm{ReLU}(-5)$.

Solution on p. 244 →

2.11 Chapter summary

Key takeaways

- A function assigns to each input exactly one output; it has a domain, a codomain and a range. A machine learning model is a function, and learning is a search for a good function.
- The graph of f is the set of points $(x, f(x))$; roots, intercepts, increasing/decreasing behaviour and extreme values can be read from it.
- Linear functions $mx + b$ have constant slope m (rise over run, a rate of change) and intercept b. In ML they are written $wx + b$.
- Quadratics ax^2+bx+c have parabolic graphs with vertex at $x = -\frac{b}{2a}$; roots come from factoring or the quadratic formula. Squared-error losses are quadratics, minimised by the mean.
- Exponentials a^x model multiplicative growth and decay; the natural base is $e \approx 2.718$. Logarithms undo exponentials and turn products into sums, which is why ML works with log-probabilities.
- Composition $f(g(x))$ chains functions; neural networks are long compositions.

Linear functions composed stay linear, so non-linear activations are essential.

- Inverse functions undo functions; e^x and $\ln x$ are inverses, and so are the sigmoid and the logit.
- $f(x-c)$ shifts right, $f(x)+c$ shifts up, $f(cx)$ squeezes. In a neuron $\sigma(wx+b)$, w sets the steepness and b the location.
- Key activation functions: step, sigmoid, tanh, ReLU, leaky ReLU, softplus.

So far all our functions had one input. Real data has many features, and a data point is best thought of as a point in a space of many dimensions. The next chapter builds the geometric intuition we need for this: coordinates, distances, angles and trigonometry.

Chapter 3

Geometry, Coordinates and Trigonometry

Motivation. Machine learning is surprisingly geometric. A data point with d features is a point in a d-dimensional space, and a data set is a cloud of such points. Two data points are "similar" when they are close together, so we need a way to measure *distance*. Two word embeddings mean similar things when they point in similar directions, so we need *angles*. A simple classifier draws a *line* (or a flat "hyperplane") through the space with one class on each side. And modern language models encode the position of each word with *sines and cosines* and with *rotations*.

This chapter builds the geometric toolkit: coordinates, distances, lines and circles, angles and trigonometry, and rotations. It is also the bridge to linear algebra: almost every idea here reappears in Chapter 4 in the language of vectors.

In this chapter you will learn to...

- locate points in the plane, in space and, by analogy, in n dimensions;
- compute distances with the Pythagorean theorem in any number of dimensions, and use them for nearest-neighbour classification;
- work with lines in the form $w_1x_1 + w_2x_2 + b = 0$ and decide on which side of a line a point lies;
- work with circles, and recognise data that no line can separate;
- measure angles in degrees and radians;
- define and use sine, cosine and tangent, their key values and graphs, and the law of cosines;
- convert between Cartesian and polar coordinates, and rotate points.

3.1 Coordinates

3.1.1 The Cartesian plane

In Section 2.2 we drew graphs on two perpendicular number lines. This setting is called the **Cartesian plane**, after the French philosopher and mathematician René Descartes. The horizontal line is the **x-axis**, the vertical line the **y-axis**, and they cross at the **origin** O. Each point P is described by an ordered pair of **coordinates** (x, y): start at the origin, move x units horizontally (right if $x > 0$, left if $x < 0$) and y units vertically (up if $y > 0$, down if $y < 0$). The axes divide the plane into four **quadrants**, numbered anticlockwise (Figure 3.1).

Common pitfall (Order matters)**.** The pair (x, y) is *ordered*: $(3, 2)$ and $(2, 3)$ are different points. By convention the horizontal coordinate always comes first.

3.1.2 Three dimensions

To describe a position in space we need a third axis, the *z-axis*, perpendicular to the other two. A point is then a triple (x, y, z) (Figure 3.2). You can think of a room: x is the

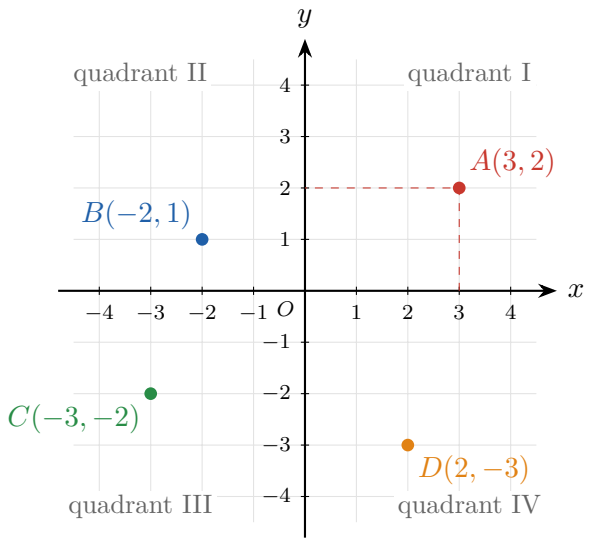


Figure 3.1: The Cartesian plane. The point $A(3, 2)$ is reached by moving 3 units right and 2 units up from the origin (dashed lines).

distance along one wall, y along another wall and z the height above the floor.

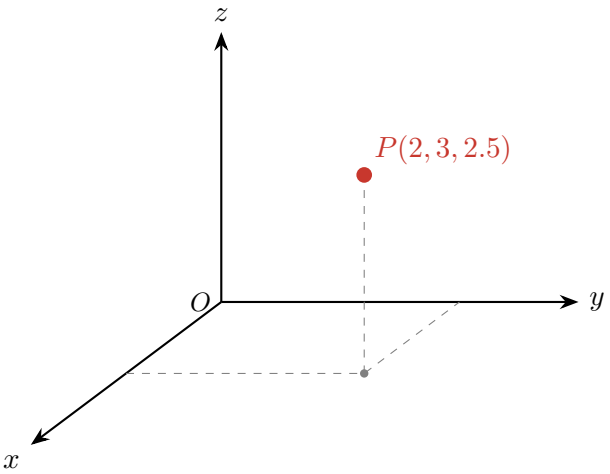


Figure 3.2: A point in three-dimensional space: go 2 units along the x-axis, 3 along the y-axis, then 2.5 up.

3.1.3 Many dimensions

Nothing stops us from writing down lists of four, ten or a million numbers. Although we cannot *draw* a four-dimensional space, we can work with it exactly as with two or three dimensions, using the same formulas. A point in **n-dimensional space** is an ordered list of n real numbers, $(x_1, x_2, \ldots, x_n)$.

> **Notation.** The set of all ordered pairs of real numbers is written $\mathbb{R}^2$ (read "R two"), the set of all triples $\mathbb{R}^3$, and the set of all ordered lists of n real numbers $\mathbb{R}^n$. So $(3, 2) \in \mathbb{R}^2$ and $(1, -4, 0.5, 7) \in \mathbb{R}^4$. When there are many coordinates we number them with subscripts, $x_1, x_2, \ldots, x_n$, rather than using different letters.

Intuition (How to think about high dimensions)**.** Nobody can picture 784 dimensions. Mathematicians and machine learning researchers think in two or three dimensions, and then check that the *formulas* carry over. Almost all of the geometry in this chapter does carry over, which is exactly why it is so useful. (A few things do behave strangely in high dimensions; for example, in a very high-dimensional space almost all pairs of random points are roughly the same distance apart. Such surprises are called the *curse of dimensionality*.)

Why this matters for ML (Feature space)**.** If each data point is described by d numbers (its *features*), then each data point is a point in $\mathbb{R}^d$, called the **feature space**, and a data set is a cloud of points in it. A house described by (size, bedrooms, age, distance to centre) is a point in $\mathbb{R}^4$. A 28×28 grayscale image of a handwritten digit, read pixel by pixel, is a list of 784 numbers and hence a point in $\mathbb{R}^{784}$. When there are only two features we can draw the data as a *scatter plot* (Figure 3.5 below), and much of our intuition comes from such pictures.

Exercises for Section 3.1

Exercise 3.1. In which quadrant, or on which axis, does each point lie? $(4, -1)$, $(-3, -5)$, $(0, 2)$, $(-1, 6)$.

Solution on p. 244 →

Exercise 3.2. A data set contains 1000 grayscale images of 32×32 pixels. Viewed as points, in which space $\mathbb{R}^n$ do they live? What if the images are in colour, with three numbers (red, green, blue) per pixel?

Solution on p. 244 →

Exercise 3.3. Describe in words: (a) the set of points (x, y) in the plane with $x = 2$; (b) the set with $y > 0$; (c) the set of points (x, y, z) in space with $z = 0$.

Solution on p. 244 →

3.2 Distance and the Pythagorean theorem

3.2.1 The Pythagorean theorem

A triangle with a right angle ($90°$) is called a **right triangle**. The side opposite the right angle is the longest side, called the **hypotenuse**; the other two sides are the *legs*.

Theorem 3.1 (Pythagorean theorem)**.** In a right triangle with legs of lengths a and b and hypotenuse of length c,

$$a^2 + b^2 = c^2$$

Proof. Arrange four copies of the triangle inside a square of side $a + b$ as in Figure 3.3. The area of the big square can be computed in two ways. Directly, it is $(a + b)^2 = a^2 + 2ab + b^2$. As a sum of pieces, it is four triangles of area $\frac{1}{2}ab$ each, plus the tilted square of area c^2: in total $2ab + c^2$. Setting the two expressions equal gives $a^2 + 2ab + b^2 = 2ab + c^2$, so $a^2 + b^2 = c^2$. ∎

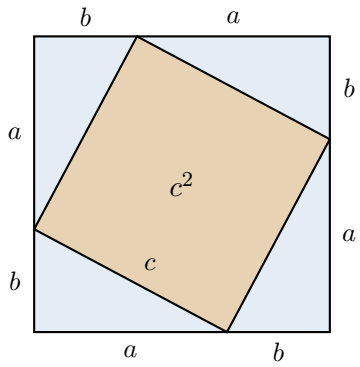


Figure 3.3: Why the Pythagorean theorem is true. The big square has side $a+b$ and contains four copies of the right triangle (blue) around a tilted square of side c (orange).

The theorem also works backwards: if three lengths satisfy $a^2+b^2=c^2$, the triangle has a right angle. The classic example is the 3–4–5 triangle: $9+16=25$.

3.2.2 Distance between two points

To find the distance between $P(x_1,y_1)$ and $Q(x_2,y_2)$, draw the right triangle whose legs run horizontally and vertically (Figure 3.4). The legs have lengths $|x_2-x_1|$ and $|y_2-y_1|$, so by Pythagoras:

Definition 3.2 (Distance in the plane)**.** The **distance** between the points (x_1,y_1) and (x_2,y_2) is

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}.$$

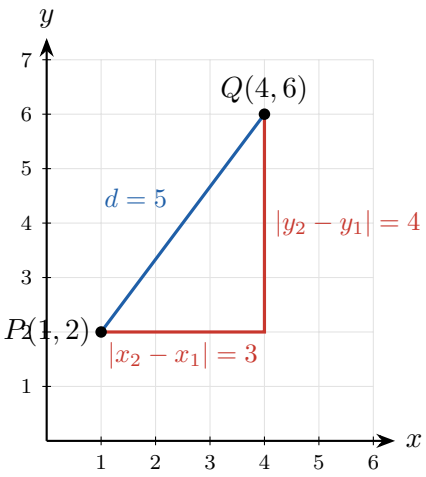


Figure 3.4: The distance between $P(1,2)$ and $Q(4,6)$ is the hypotenuse of a right triangle with legs 3 and 4, so $d=\sqrt{9+16}=5$. The red path along the grid has length $3+4=7$: this is the Manhattan distance.

Example 3.3 (Distances in two and three dimensions)**.** Find the distance (a) between $(1,2)$ and $(4,6)$; (b) between $(1,2,3)$ and $(4,6,15)$.

Solution. (a) $\sqrt{(4-1)^2+(6-2)^2}=\sqrt{9+16}=5$. (b) In three dimensions we apply

Pythagoras twice: first within the floor to get the horizontal distance $\sqrt{3^2+4^2}=5$, then with the height difference 12: $\sqrt{5^2+12^2}=13$. Equivalently, $\sqrt{3^2+4^2+12^2}=\sqrt{169}=13$.

The pattern continues to any number of dimensions: square the differences in each coordinate, add them up and take the square root.

Definition 3.4 (Euclidean distance in $\mathbb{R}^n$)**.** The **Euclidean distance** between $\boldsymbol{x}=(x_1,\dots,x_n)$ and $\boldsymbol{y}=(y_1,\dots,y_n)$ is

$$d(\boldsymbol{x},\boldsymbol{y})=\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2}$$

(Bold letters such as $\boldsymbol{x}$ denote whole lists of numbers; this notation is explained properly in Chapter 4.) The *midpoint* of the segment from $\boldsymbol{x}$ to $\boldsymbol{y}$ is found by averaging each coordinate: in the plane, the midpoint of (x_1,y_1) and (x_2,y_2) is $\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$.

Remark (Other ways to measure distance)**.** In a city laid out on a grid you cannot walk diagonally through buildings. The walking distance between two street corners is the **Manhattan distance** $\sum_i|x_i-y_i|$, the length of the red path in Figure 3.4. Different notions of distance lead to different algorithms; we will study them systematically as *norms* in Chapter 5.

Why this matters for ML (Nearest neighbours)**.** The **k-nearest neighbours** (k-NN) classifier is perhaps the simplest learning algorithm of all: to classify a new point, find the k training points closest to it and take a majority vote of their labels. Everything rests on the distance formula.

Suppose the training data are

point	P_1	P_2	P_3	P_4	P_5
coordinates	$(1,1)$	$(2,1)$	$(4,3)$	$(5,4)$	$(3,4.5)$
label	A	A	B	B	B

and we want to classify the new point $\boldsymbol{q}=(2,2)$. The distances from $\boldsymbol{q}$ are $d(\boldsymbol{q},P_1)=\sqrt{1+1}\approx1.41$, $d(\boldsymbol{q},P_2)=1$, $d(\boldsymbol{q},P_3)=\sqrt{4+1}\approx2.24$, $d(\boldsymbol{q},P_4)=\sqrt{9+4}\approx3.61$ and $d(\boldsymbol{q},P_5)=\sqrt{1+6.25}\approx2.69$. The three nearest neighbours are P_2 (A), P_1 (A) and P_3 (B), so with $k=3$ the vote is 2 to 1 and $\boldsymbol{q}$ is classified as A.

Common pitfall (Features on different scales)**.** Suppose houses are described by (size in m^2, number of bedrooms). For $A=(100,4)$, $B=(110,1)$ and $C=(130,4)$, the distance formula gives $d(A,B)=\sqrt{100+9}\approx10.4$ and $d(A,C)=\sqrt{900+0}=30$, so B counts as much more similar to A than C is, even though B has three fewer bedrooms. The size feature dominates simply because its numbers are bigger. Before using distances, features are usually *standardised* to comparable scales (Chapter 9).

Exercises for Section 3.2

Exercise 3.4. Find the distance between $(-2, 3)$ and $(4, -5)$, and the midpoint of the segment joining them.

Solution on p. 244 →

Exercise 3.5. Find the distance between $(1, 0, 2)$ and $(3, 4, -2)$.

Solution on p. 244 →

Exercise 3.6. Compute the Euclidean and the Manhattan distance between $\boldsymbol{u} = (1, 2, 3, 4)$ and $\boldsymbol{v} = (2, 0, 3, 7)$.

Solution on p. 244 →

Exercise 3.7. Is the triangle with vertices $(1, 1)$, $(4, 2)$ and $(3, 5)$ a right triangle? Use the converse of the Pythagorean theorem.

Solution on p. 245 →

Exercise 3.8. Using the training data in the nearest-neighbour example above, classify the point $(3.5, 3)$ with $k = 1$ and with $k = 3$.

Solution on p. 245 →

3.3 Lines and circles

3.3.1 The general equation of a line

In Section 2.3 we wrote lines as $y = mx + b$. That form cannot describe vertical lines, and it treats x and y unequally. A more symmetric form, which works for every line in the plane, is

$$w_1 x_1 + w_2 x_2 + b = 0,$$

where we have renamed the coordinates x_1 and x_2 (as is common in machine learning) and where w_1 and w_2 are not both zero. For instance, the line $x_2 = 2x_1 + 1$ can be rewritten as $2x_1 - x_2 + 1 = 0$, and the vertical line $x_1 = 3$ is $x_1 - 3 = 0$ (with $w_2 = 0$). If $w_2 \neq 0$ we can always solve for x_2 to get back the slope–intercept form: $x_2 = -\frac{w_1}{w_2} x_1 - \frac{b}{w_2}$.

3.3.2 Which side of a line?

The general form has a wonderful extra feature. Every point (x_1, x_2) of the plane gives a number $w_1 x_1 + w_2 x_2 + b$. On the line this number is zero, on one side of the line it is positive, and on the other side it is negative.

Example 3.5 (Testing the side of a line)**.** For the line $x_1 + x_2 - 7 = 0$, on which side do the points $(2, 3)$ and $(5, 4)$ lie?

Solution. For $(2, 3)$: $2 + 3 - 7 = -2 < 0$. For $(5, 4)$: $5 + 4 - 7 = 2 > 0$. They lie on opposite sides. The negative side is the one containing the origin (since $0 + 0 - 7 < 0$), i.e. the side "below" the line.

Why this matters for ML (Linear classifiers)**.** Figure 3.5 shows the idea behind a

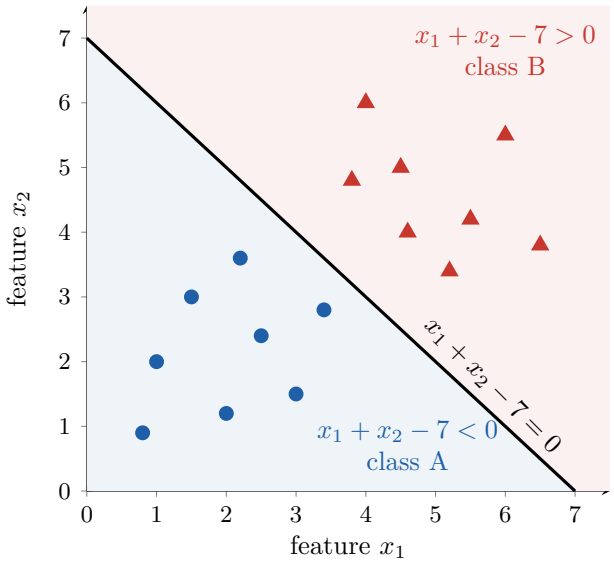


Figure 3.5: A data set with two features and two classes (blue circles and red triangles). The line $x_1 + x_2 - 7 = 0$ separates them: the sign of $x_1 + x_2 - 7$ tells us the class.

linear classifier: choose numbers w_1, w_2, b and predict

$$\text{class} \begin{cases} \text{B} & \text{if } w_1x_1 + w_2x_2 + b > 0, \\ \text{A} & \text{if } w_1x_1 + w_2x_2 + b < 0. \end{cases}$$

The line $w_1x_1+w_2x_2+b = 0$ is the **decision boundary**. This is exactly the *perceptron*, the ancestor of all neural networks, and it is also how logistic regression and support vector machines make decisions. With d features the boundary $w_1x_1+\cdots+w_dx_d+b = 0$ is a flat $(d-1)$-dimensional surface called a **hyperplane**. Learning means finding weights w_i and a bias b that put the boundary in a good place. In Chapter 4 we will see that (w_1, w_2) is a vector perpendicular to the boundary.

3.3.3 Circles

A **circle** is the set of all points at a fixed distance r, the *radius*, from a fixed point (a, b), the *centre*. By the distance formula, a point (x, y) is on the circle exactly when $\sqrt{(x-a)^2 + (y-b)^2} = r$, or, squaring both sides:

$$(x-a)^2 + (y-b)^2 = r^2.$$

The circle of radius 1 centred at the origin, $x^2 + y^2 = 1$, is the **unit circle**; it will be the stage for trigonometry in Section 3.5. Points *inside* the circle satisfy $(x-a)^2+(y-b)^2 < r^2$ and points outside satisfy $> r^2$.

Example 3.6 (Centre and radius)**.** Find the centre and radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$.

Solution. Complete the square in x and in y (Example 2.11): $x^2 - 4x = (x-2)^2 - 4$ and $y^2+6y = (y+3)^2-9$. So the equation becomes $(x-2)^2-4+(y+3)^2-9-12 = 0$, i.e.

$$(x-2)^2 + (y+3)^2 = 25.$$

The centre is $(2, -3)$ and the radius is $\sqrt{25} = 5$.

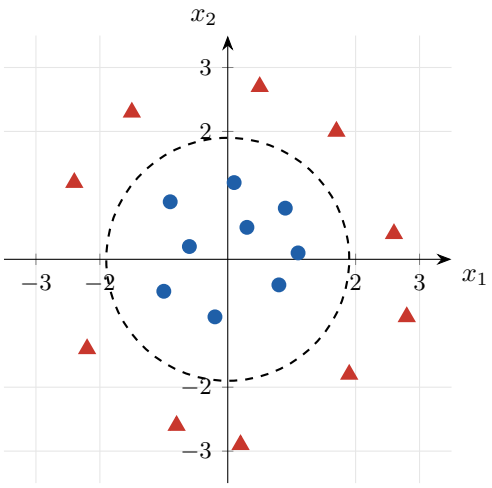


Figure 3.6: Two classes that no straight line can separate: one class inside a circle, the other outside. The dashed circle $x_1^2 + x_2^2 = 1.9^2$ separates them perfectly.

Why this matters for ML (When a line is not enough)**.** In Figure 3.6 one class sits inside a circle and the other outside. Any straight line leaves points of both classes on at least one of its sides, so no linear classifier can succeed: the data are not **linearly separable**. Two remedies exist. We can invent a new feature, such as $r^2 = x_1^2 + x_2^2$ (the squared distance from the origin), after which a single threshold $r^2 < 1.9^2$ separates the classes; this is the idea behind *feature engineering* and *kernel methods*. Or we can use a model, such as a neural network, that learns curved boundaries by itself.

Exercises for Section 3.3

Exercise 3.9. Write the line $2x + 3y = 6$ in slope–intercept form and find where it crosses the axes.

Solution on p. 245 →

Exercise 3.10. For the line $2x_1 - x_2 + 1 = 0$, decide on which side (positive, negative, or on the line) each point lies: $(1, 1)$, $(0, 3)$, $(1, 3)$.

Solution on p. 245 →

Exercise 3.11. A perceptron with weights $w_1 = 2$, $w_2 = -1$ and bias $b = -1$ predicts class 1 when $w_1 x_1 + w_2 x_2 + b > 0$ and class 0 when it is negative. Classify $(1, 0)$, $(0, 1)$ and $(2, 3)$, and write the decision boundary in the form $x_2 = m x_1 + c$.

Solution on p. 245 →

Exercise 3.12. Find the centre and radius of the circle $x^2 + y^2 + 2x - 8y + 8 = 0$. Is the point $(1, 2)$ inside, on, or outside the circle?

Solution on p. 245 →

Exercise 3.13. In Figure 3.6, replace each point (x_1, x_2) by the single number $s = x_1^2 + x_2^2$. Explain why, after this transformation, the two classes can be separated by a single threshold. What threshold works?

Solution on p. 245 →

3.4 Angles: degrees and radians

An **angle** measures an amount of turning. Picture a ray (a half-line) starting at the origin and pointing along the positive x-axis, and rotate it about the origin. By convention, *anticlockwise* rotation counts as positive and clockwise rotation as negative.

3.4.1 Degrees

The familiar unit divides a full turn into 360 **degrees**: a quarter turn (a right angle) is 90°, a half turn (a straight line) is 180°. Angles larger than 360° are allowed (more than one full turn), and so are negative angles.

3.4.2 Radians

Mathematicians prefer a different unit, which turns out to make calculus much simpler. Put the angle at the centre of the unit circle. The **radian** measure of the angle is the *length of the arc* of the unit circle that it cuts out (Figure 3.7).

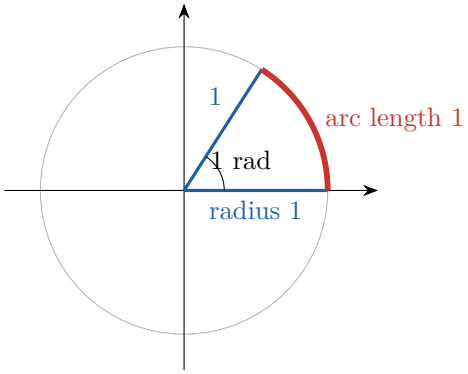


Figure 3.7: One radian is the angle that cuts off an arc of length 1 on the unit circle. It is about 57.3°.

The whole unit circle has circumference 2π, so a full turn is 2π radians. Hence

$$360^\circ = 2\pi \text{ rad}, \qquad 180^\circ = \pi \text{ rad},$$

and to convert:

$$\text{radians} = \text{degrees} \times \frac{\pi}{180}, \qquad \text{degrees} = \text{radians} \times \frac{180}{\pi}.$$

One radian is $\frac{180^\circ}{\pi} \approx 57.3^\circ$. The angles that appear most often are worth memorising:

degrees	0°	30°	45°	60°	90°	120°	180°	270°	360°
radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{3\pi}{2}$	2π

On a circle of radius r, an angle of θ radians cuts off an arc of length $s = r\theta$ (the unit-circle arc, scaled by r). This simple formula is one reason radians are natural. (The Greek letter θ, "theta", is the traditional name for an angle.)

> **Remark** (Radians in practice)**.** In calculus, formulas such as "the slope of $\sin x$ is $\cos x$" (Chapter 6) are true only when x is measured in radians. For the same reason, the functions `sin`, `cos` and so on in every programming language expect radians: `math.sin(30)` computes the sine of 30 radians, not of 30°. From now on, angles are in radians unless a degree sign is shown.

Exercises for Section 3.4

Exercise 3.14. Convert to radians: 45°, 120°, 270°, 1°.

Solution on p. 245 →

Exercise 3.15. Convert to degrees: $\frac{\pi}{6}$, $\frac{5\pi}{4}$, 2 radians.

Solution on p. 245 →

Exercise 3.16. Through what angle, in radians, does the minute hand of a clock turn in 20 minutes? Should the angle be counted as positive or negative?

Solution on p. 245 →

Exercise 3.17. How long is the arc cut off by an angle of 0.8 radians on a circle of radius 5? What angle cuts off an arc of length π on a circle of radius 2?

Solution on p. 245 →

3.5 Trigonometry

Motivation. Trigonometry relates angles to lengths. In machine learning its most important consequence is a way to measure the angle between two directions (the *cosine similarity* of Chapter 4), but sines and cosines also describe anything periodic, such as waves, seasons and positions in a sequence, and they describe rotations.

3.5.1 Sine, cosine and tangent in a right triangle

In a right triangle, fix one of the two acute angles and call it θ. Name the sides relative to it: the *hypotenuse* (opposite the right angle), the *opposite* side (across from θ) and the *adjacent* side (next to θ, not the hypotenuse) (Figure 3.8).

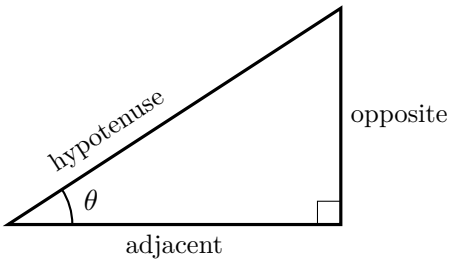


Figure 3.8: The sides of a right triangle, named relative to the angle θ.

Definition 3.7 (Sine, cosine, tangent)**.** For an acute angle θ in a right triangle,

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

These ratios depend only on the angle, not on the size of the triangle: a bigger triangle with the same angles is just a scaled copy, and scaling does not change ratios. Many people remember the definitions with the mnemonic *SOH–CAH–TOA*: Sine is Opposite over Hypotenuse, Cosine is Adjacent over Hypotenuse, Tangent is Opposite over Adjacent. Note that $\tan\theta = \frac{\sin\theta}{\cos\theta}$.

Example 3.8 (Using trigonometry to find lengths)**.** A 5 m ladder leans against a wall, making an angle of 70° with the ground. How high up the wall does it reach, and how far is its foot from the wall?

Solution. The ladder is the hypotenuse. The height is the side opposite the 70° angle: $5\sin 70° \approx 5 \times 0.940 = 4.70\,\text{m}$. The distance from the wall is the adjacent side: $5\cos 70° \approx 5 \times 0.342 = 1.71\,\text{m}$. (Check with Pythagoras: $4.70^2 + 1.71^2 \approx 25.0 = 5^2$.)

3.5.2 Special angles

Two triangles give exact values for the most common angles. Cutting a square of side 1 along its diagonal gives a right triangle with two 45° angles and hypotenuse $\sqrt{2}$, so $\sin 45° = \cos 45° = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$. Cutting an equilateral triangle of side 2 in half gives a right triangle with angles 30° and 60°, sides 1 and $\sqrt{3}$ and hypotenuse 2. From these:

θ	0	$\frac{\pi}{6}$ (30°)	$\frac{\pi}{4}$ (45°)	$\frac{\pi}{3}$ (60°)	$\frac{\pi}{2}$ (90°)
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

3.5.3 The unit circle: trigonometry for every angle

Right triangles only give us angles between 0 and 90°. To define sine and cosine for *every* angle, we use the unit circle. Rotate a radius of the unit circle anticlockwise from the positive x-axis by the angle θ. It ends at a point P on the circle (Figure 3.9).

Definition 3.9 (Sine and cosine on the unit circle)**.** If the point P on the unit circle is reached by rotating through the angle θ, then

$$P = (\cos\theta, \sin\theta).$$

In words: $\cos\theta$ is the x-coordinate and $\sin\theta$ the y-coordinate of P.

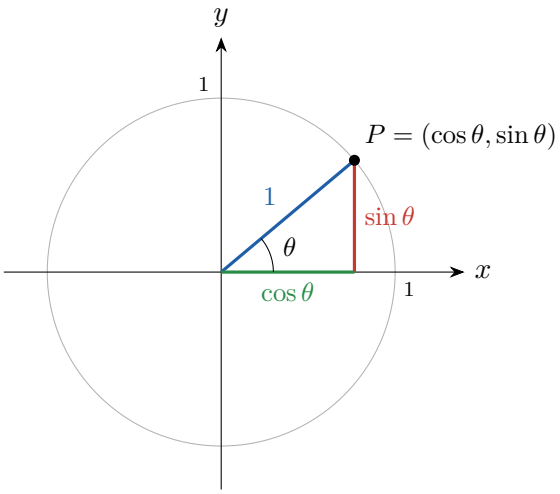


Figure 3.9: The unit-circle definition of sine and cosine. For an acute angle this agrees with the triangle definition, because the hypotenuse has length 1.

For an angle between 0 and $\frac{\pi}{2}$, the green and red segments in Figure 3.9 form a right triangle with hypotenuse 1, so this new definition agrees with the old one. But it works for every angle. For example, $\theta = \pi$ (half a turn) lands on $(-1, 0)$, so $\cos\pi = -1$ and $\sin\pi = 0$. In the second quadrant cosine is negative and sine positive; in the third both are negative; in the fourth cosine is positive and sine negative. By symmetry, for instance, $\sin 150^\circ = \sin 30^\circ = \frac{1}{2}$ while $\cos 150^\circ = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$.

Because P lies on the unit circle, $x^2 + y^2 = 1$, which gives the most important trigonometric identity.

Theorem 3.10 (Pythagorean identity)**.** For every angle θ,

$$\sin^2\theta + \cos^2\theta = 1.$$

(Here $\sin^2\theta$ means $(\sin\theta)^2$.)

3.5.4 Graphs of sine and cosine

As θ increases, the point P goes round and round the circle, so the values of $\sin\theta$ and $\cos\theta$ repeat every 2π: the functions are **periodic** with *period* 2π. Their graphs are waves that oscillate between -1 and 1 (Figure 3.10). The cosine wave is the sine wave shifted left by $\frac{\pi}{2}$: $\cos\theta = \sin\left(\theta + \frac{\pi}{2}\right)$.

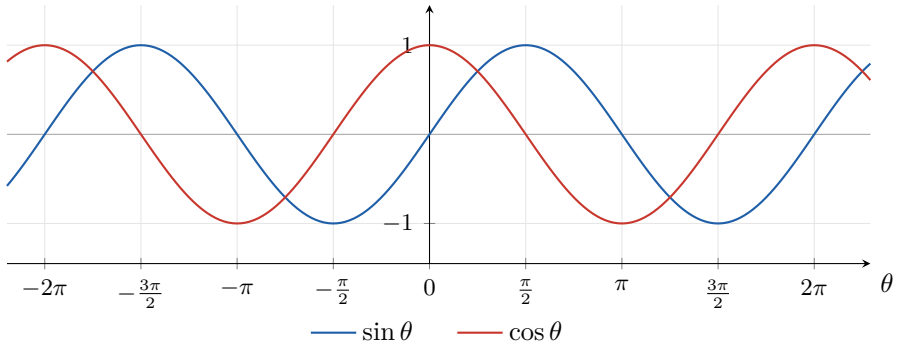


Figure 3.10: The sine and cosine functions. Both repeat every 2π and take values between -1 and 1.

More general waves have the form $y = A\sin(\omega x + \varphi)$. The *amplitude* A stretches the wave vertically, the *angular frequency* ω ("omega") squeezes it horizontally, so that the period becomes $\frac{2\pi}{\omega}$, and the *phase* φ ("phi") shifts it, just as in Section 2.9.

3.5.5 Inverse trigonometric functions

Given $\cos\theta$, can we recover θ? Cosine is not one-to-one (many angles share the same cosine), so as in Section 2.8 we restrict the domain. The **arccosine** $\arccos y$ (written `acos` in code, or $\cos^{-1} y$) is the angle in $[0, \pi]$ whose cosine is y; for example $\arccos\left(\frac{1}{2}\right) = \frac{\pi}{3}$ and $\arccos(0) = \frac{\pi}{2}$. Similarly arcsin returns angles in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and arctan angles in $(-\frac{\pi}{2}, \frac{\pi}{2})$. We will use arccos to compute angles between vectors in Chapter 4.

3.5.6 The law of cosines

The Pythagorean theorem only applies to right triangles. Its generalisation to *any* triangle is the following.

Theorem 3.11 (Law of cosines)**.** In a triangle with sides a, b, c, where γ ("gamma") is the angle between the sides a and b (that is, opposite to c),

$$c^2 = a^2 + b^2 - 2ab\cos\gamma.$$

Proof. Put the vertex with angle γ at the origin and the side b along the positive x-axis, so that its other end is $(b, 0)$. By the unit-circle definition (scaled by a), the end of side a is $(a\cos\gamma, a\sin\gamma)$. The side c joins these two points, so by the distance formula

$$\begin{aligned} c^2 &= (a\cos\gamma - b)^2 + (a\sin\gamma)^2 \\ &= a^2\cos^2\gamma - 2ab\cos\gamma + b^2 + a^2\sin^2\gamma \\ &= a^2(\cos^2\gamma + \sin^2\gamma) + b^2 - 2ab\cos\gamma = a^2 + b^2 - 2ab\cos\gamma, \end{aligned}$$

using the Pythagorean identity in the last step. ∎

When $\gamma = 90°$, $\cos\gamma = 0$ and we recover Pythagoras. When γ is larger than $90°$, $\cos\gamma < 0$ and c is longer than Pythagoras would predict; when γ is smaller, c is shorter.

Example 3.12 (Finding an angle from three sides)**.** A triangle has sides 4, 5 and 6. Find its largest angle.

Solution. The largest angle is opposite the longest side, $c = 6$. Solving the law of cosines for $\cos\gamma$:

$$\cos\gamma = \frac{a^2 + b^2 - c^2}{2ab} = \frac{16 + 25 - 36}{2 \cdot 4 \cdot 5} = \frac{5}{40} = 0.125,$$

so $\gamma = \arccos(0.125) \approx 82.8°$.

3.5.7 Sums of angles

We will need two more formulas, which tell us the sine and cosine of a sum of angles. We state them without proof (a short proof using matrices appears in the exercises of Chapter 4).

Theorem 3.13 (Angle addition formulas)**.** For all angles α and β,

$$\begin{aligned} \cos(\alpha + \beta) &= \cos\alpha\cos\beta - \sin\alpha\sin\beta, \\ \sin(\alpha + \beta) &= \sin\alpha\cos\beta + \cos\alpha\sin\beta. \end{aligned}$$

Why this matters for ML (Sines and cosines in modern models)**.** Here are two places where trigonometry turns up in machine learning.

- **Similarity as an angle.** Two lists of numbers can be viewed as arrows from the origin, and the cosine of the angle between them measures how much they point the same way: 1 for the same direction, 0 for perpendicular, -1 for opposite. This *cosine similarity*, computed with the dot product of Chapter 4, is the standard way to compare word and sentence embeddings, and it is used by search engines and recommendation systems.
- **Encoding positions.** A transformer (the architecture behind large language models) reads all the words of a sentence at once, so it must be told where

each word is. The original transformer added to the word at position p a list of numbers of the form $\sin(p\,\omega_k)$ and $\cos(p\,\omega_k)$ for many different frequencies ω_k: fast waves distinguish neighbouring positions, slow waves distinguish distant ones.

Exercises for Section 3.5

Exercise 3.18. A right triangle has hypotenuse 10 and an angle of $30°$. Find the lengths of the side opposite and the side adjacent to this angle.

Solution on p. 245 →

Exercise 3.19. The sun is $35°$ above the horizon and a tree casts a shadow 12 m long. How tall is the tree?

Solution on p. 245 →

Exercise 3.20. Using the unit circle, give exact values of $\sin 150°$, $\cos 150°$, $\sin 225°$, $\cos 300°$, $\tan 45°$ and $\sin(-30°)$.

Solution on p. 245 →

Exercise 3.21. If $\sin\theta = 0.6$ and θ is between 0 and $\frac{\pi}{2}$, find $\cos\theta$ and $\tan\theta$ without a calculator.

Solution on p. 245 →

Exercise 3.22. A triangle has two sides of lengths 5 and 7 with an angle of $60°$ between them. How long is the third side?

Solution on p. 245 →

Exercise 3.23. Give the amplitude and the period of $y = 3\sin(2x)$ and of $y = \cos\left(\frac{x}{2}\right)$.

Solution on p. 245 →

3.6 Polar coordinates and rotations

3.6.1 Polar coordinates

A point in the plane can also be described by *how far* it is from the origin and *in which direction*. These are its **polar coordinates** (r, θ): $r \geq 0$ is the distance from the origin and θ is the angle measured anticlockwise from the positive x-axis. Scaling Definition 3.9 by r gives the conversion:

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

and in the other direction

$$r = \sqrt{x^2 + y^2}, \qquad \tan\theta = \frac{y}{x}.$$

Care is needed when finding θ: the equation $\tan\theta = \frac{y}{x}$ does not distinguish between opposite directions, such as $(1, 1)$ and $(-1, -1)$, so you must check the quadrant. Programming languages provide a function `atan2(y, x)` that does this automatically.

Example 3.14 (Converting coordinates)**.** (a) Find the polar coordinates of $(-1, \sqrt{3})$. (b) Find the Cartesian coordinates of the point with $r = 2$, $\theta = \frac{\pi}{6}$.

Solution. (a) $r = \sqrt{1+3} = 2$. The point is in the second quadrant, with $\cos\theta = \frac{x}{r} = -\frac{1}{2}$ and $\sin\theta = \frac{y}{r} = \frac{\sqrt{3}}{2}$, so $\theta = 120^\circ = \frac{2\pi}{3}$. (b) $x = 2\cos\frac{\pi}{6} = \sqrt{3}$ and $y = 2\sin\frac{\pi}{6} = 1$.

3.6.2 Rotating a point

How do we rotate a point (x, y) about the origin by an angle α? In polar coordinates this is trivial: the distance r stays the same and the angle θ becomes $\theta + \alpha$. So the rotated point is $(r\cos(\theta+\alpha), r\sin(\theta+\alpha))$. Expanding with the angle addition formulas (Theorem 3.13),

$$r\cos(\theta+\alpha) = r\cos\theta\cos\alpha - r\sin\theta\sin\alpha = x\cos\alpha - y\sin\alpha,$$
$$r\sin(\theta+\alpha) = r\sin\theta\cos\alpha + r\cos\theta\sin\alpha = x\sin\alpha + y\cos\alpha.$$

Theorem 3.15 (Rotation formula)**.** Rotating the point (x, y) anticlockwise about the origin by the angle α gives the point (x', y') with

$$x' = x\cos\alpha - y\sin\alpha, \qquad y' = x\sin\alpha + y\cos\alpha$$

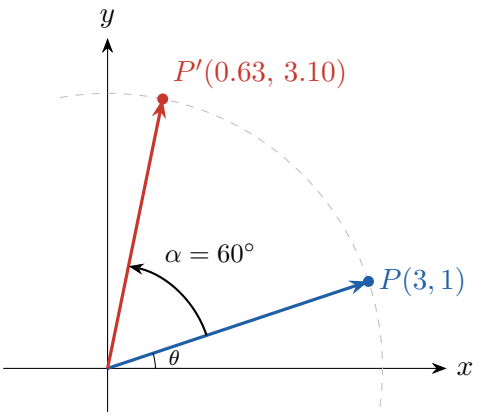


Figure 3.11: Rotating $P(3,1)$ anticlockwise by 60° about the origin gives P'. The distance from the origin, $\sqrt{10}$, does not change (dashed circle).

Example 3.16 (Rotations)**.** (a) Rotate $(2,1)$ by 90°. (b) Rotate $(3,1)$ by 60° (Figure 3.11).

Solution. (a) $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$, so $x' = 2\cdot 0 - 1\cdot 1 = -1$ and $y' = 2\cdot 1 + 1\cdot 0 = 2$: the result is $(-1, 2)$. (b) $\cos 60^\circ = 0.5$ and $\sin 60^\circ \approx 0.866$, so $x' \approx 3(0.5) - 1(0.866) = 0.634$ and $y' \approx 3(0.866) + 1(0.5) = 3.098$. Check: $0.634^2 + 3.098^2 \approx 10.0 = 3^2 + 1^2$, so the distance from the origin has not changed.

Notice that the new coordinates are obtained from the old ones by multiplying by fixed numbers and adding. Transformations of this kind are called *linear transformations*, and in Chapter 4 we will write the rotation compactly as a *matrix*.

Why this matters for ML (Rotations in machine learning)**.** Rotations appear in *data augmentation*, where training images are randomly rotated so that a model learns that a rotated cat is still a cat. More surprisingly, many current large language models use *rotary position embeddings*: the numbers that represent the word at position p

are grouped in pairs, and each pair is rotated by an angle proportional to p. Because rotating by $p\alpha$ and by $q\alpha$ leaves an angle of $(p-q)\alpha$ between the results, comparisons between two words then depend only on how far apart they are, which is exactly the information the model needs.

Exercises for Section 3.6

Exercise 3.24. Convert to polar coordinates (give θ in radians): $(1, 1)$, $(0, -3)$, $(-2, 0)$.

Solution on p. 246 →

Exercise 3.25. Convert to Cartesian coordinates: $r = 4$, $\theta = \frac{3\pi}{4}$; and $r = 1$, $\theta = \pi$.

Solution on p. 246 →

Exercise 3.26. Rotate the point $(3, 4)$ anticlockwise by $30°$, and check that its distance from the origin is unchanged.

Solution on p. 246 →

Exercise 3.27. Explain, using polar coordinates, why rotating by α and then by β is the same as rotating once by $\alpha + \beta$. What does rotating by α and then by $-\alpha$ do?

Solution on p. 246 →

Exercise 3.28. A model represents position p by rotating the point $(1, 0)$ by the angle $p \times 10°$. Find the points for positions $p = 1$ and $p = 3$, and the angle between them. Would the angle change for positions $p = 11$ and $p = 13$?

Solution on p. 246 →

3.7 Chapter summary

Key takeaways

- Points in the plane are pairs (x, y); in space, triples; in general, lists of n numbers in $\mathbb{R}^n$. A data point with d features is a point in the feature space $\mathbb{R}^d$.
- Pythagoras: $a^2 + b^2 = c^2$. The Euclidean distance between $\boldsymbol{x}$ and $\boldsymbol{y}$ is $\sqrt{\sum_i (x_i - y_i)^2}$ in any dimension; it is the basis of nearest-neighbour methods, and features should be on comparable scales before it is used.
- A line can be written $w_1 x_1 + w_2 x_2 + b = 0$; the sign of $w_1 x_1 + w_2 x_2 + b$ tells which side a point is on. This is how linear classifiers decide. Circles are $(x-a)^2 + (y-b)^2 = r^2$; some data sets need curved boundaries.
- Angles are measured in degrees or radians; $180° = \pi$. Code and calculus use radians.
- $\sin\theta$ and $\cos\theta$ are the coordinates of the point at angle θ on the unit circle; $\sin^2\theta + \cos^2\theta = 1$; both are periodic with period 2π.
- Law of cosines: $c^2 = a^2 + b^2 - 2ab\cos\gamma$.
- Polar coordinates (r, θ) with $x = r\cos\theta$, $y = r\sin\theta$. Rotation by α: $(x\cos\alpha - y\sin\alpha,\ x\sin\alpha + y\cos\alpha)$.

We now have numbers, functions and geometry. The next chapter combines them into

the single most important mathematical tool in machine learning: linear algebra, the mathematics of vectors and matrices.

Chapter 4

Linear Algebra I: Vectors and Matrices

Motivation. Open the source code of any machine learning system and you will find the same two kinds of object everywhere: *vectors* (lists of numbers) and *matrices* (tables of numbers). A data point is a vector of features; a data set is a matrix with one row per data point; an image is a matrix of pixels; the weights of a neural network layer form a matrix; and the computation done by that layer is "multiply the input vector by the weight matrix, add a bias vector, apply an activation function". Graphics processing units (GPUs), the hardware on which modern AI runs, are essentially machines for multiplying matrices very fast.

Linear algebra, the mathematics of vectors and matrices, therefore gives us two things at once: a compact *notation* in which a single short formula describes millions of arithmetic operations, and a geometric *picture* in which a matrix is a transformation that stretches, rotates and squashes space. This chapter covers the basics; Chapter 5 covers the deeper ideas that machine learning relies on.

In this chapter you will learn to...

- represent data as vectors, and picture vectors as arrows;
- add and scale vectors and form linear combinations;
- compute lengths, distances, dot products and angles between vectors;
- store data sets and images as matrices;
- multiply a matrix by a vector and by another matrix, and explain both operations in two complementary ways;
- interpret a matrix as a linear transformation of space;
- use the transpose, the identity matrix, inverses and 2×2 and 3×3 determinants;
- write a system of linear equations as $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and solve it by Gaussian elimination.

4.1 Vectors

4.1.1 Two ways to see a vector

The word *vector* is used for two ideas that turn out to be the same thing.

- **A vector is an arrow.** In physics a vector is a quantity with a size and a direction, such as a displacement ("3 m east and 2 m north") or a velocity. We draw it as an arrow. Only its length and direction matter, not where it is drawn: the three arrows in Figure 4.1 all represent the same vector.
- **A vector is a list of numbers.** In computer science and data science a vector is an ordered list of numbers, such as the features of a house: $(120, 3, 15)$ for $120\,\mathrm{m}^2$, 3 bedrooms and 15 years of age.

The bridge between the two views is the coordinate system of Chapter 3: the arrow that starts at the origin and ends at the point $(3, 2)$ corresponds to the list $(3, 2)$. The numbers say how far the arrow goes in each coordinate direction.

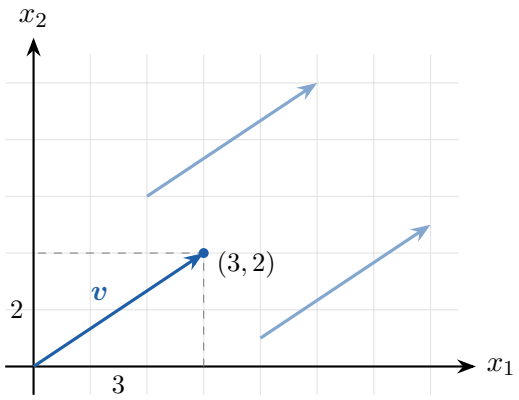


Figure 4.1: The vector $\boldsymbol{v} = (3, 2)$ drawn as an arrow from the origin. The two paler arrows have the same length and direction, so they represent the same vector: "3 to the right and 2 up".

Definition 4.1 (Vector)**.** A **vector** in $\mathbb{R}^n$ is an ordered list of n real numbers, called its **components** (or entries). We write vectors as columns:

$$\boldsymbol{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}.$$

The number n is the *dimension* of the space $\mathbb{R}^n$. Two vectors are equal when all their components are equal.

Notation. Vectors are written in **bold** lower-case letters: $\boldsymbol{v}$, $\boldsymbol{x}$, $\boldsymbol{w}$. Their components are written in ordinary italic with a subscript: v_1, x_3, w_i. Single numbers are called **scalars**, to distinguish them from vectors, and are written in ordinary italic: a, c, λ. (On paper, where bold is hard to write, people often write $\vec{v}$ or underline the letter.) To save space we sometimes write a column vector on one line as $\boldsymbol{v} = (v_1, \ldots, v_n)$; it still means a column. The **zero vector** $\mathbf{0} = (0, \ldots, 0)$ has every component equal to 0.

Why this matters for ML (Everything is a vector)**.** In machine learning, almost every object is stored as a vector.

- **Tabular data:** a house is $\boldsymbol{x} = (120, 3, 15, 2.5)$: size, bedrooms, age and distance to the centre.
- **Images:** a 28×28 grayscale image becomes a vector in $\mathbb{R}^{784}$ by reading its pixels row by row ("flattening").
- **Words:** language models represent each word (or word piece) by a learned vector called an **embedding**, typically with hundreds or thousands of components.
- **Categories:** a category out of k possibilities is encoded as a **one-hot vector** with a single 1: with the categories cat, dog, bird, "dog" is $(0, 1, 0)$.
- **Model parameters:** all the weights of a model can be listed in one long vector, usually called $\boldsymbol{\theta}$ ("theta").

Exercises for Section 4.1

Exercise 4.1. Write a feature vector for a house of $120\,\mathrm{m}^2$ with 3 bedrooms, aged 15 years and 2.5 km from the city centre. In which space $\mathbb{R}^n$ does it live?

Solution on p. 246 →

Exercise 4.2. Write the one-hot vectors for the categories red, green and blue (in that order). What is the sum of the entries of any one-hot vector?

Solution on p. 246 →

Exercise 4.3. Which vector is represented by the arrow from $P(1,4)$ to $Q(5,1)$? How long is the arrow?

Solution on p. 246 →

4.2 Adding and scaling vectors

4.2.1 Vector addition

Vectors of the same dimension are added component by component:

$$\boldsymbol{u}+\boldsymbol{v}=\begin{pmatrix}u_1\\ \vdots\\ u_n\end{pmatrix}+\begin{pmatrix}v_1\\ \vdots\\ v_n\end{pmatrix}=\begin{pmatrix}u_1+v_1\\ \vdots\\ u_n+v_n\end{pmatrix}.$$

For example, $(3,1)+(1,2)=(4,3)$. Geometrically, addition means "follow one arrow, then the other": place the tail of $\boldsymbol{v}$ at the tip of $\boldsymbol{u}$; the sum goes from the tail of $\boldsymbol{u}$ to the tip of $\boldsymbol{v}$. Doing it in the other order gives the same result, and the two routes form a parallelogram (Figure 4.2, left). Vectors of different dimensions cannot be added.

4.2.2 Scalar multiplication

Multiplying a vector by a scalar multiplies every component:

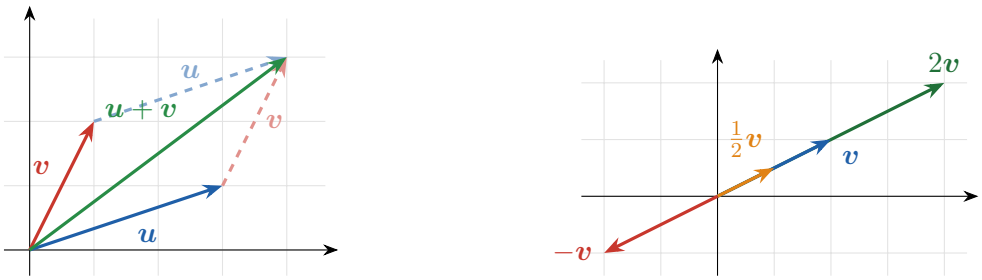
$$c\,\boldsymbol{v}=\begin{pmatrix}c\,v_1\\ \vdots\\ c\,v_n\end{pmatrix}.$$

Geometrically, $c\boldsymbol{v}$ points along the same line as $\boldsymbol{v}$ and is $|c|$ times as long; if $c<0$ it points the opposite way (Figure 4.2, right). The vector $(-1)\boldsymbol{v}$ is written $-\boldsymbol{v}$, and subtraction is defined by $\boldsymbol{u}-\boldsymbol{v}=\boldsymbol{u}+(-\boldsymbol{v})$, i.e. component by component. Geometrically, $\boldsymbol{u}-\boldsymbol{v}$ is the arrow from the tip of $\boldsymbol{v}$ to the tip of $\boldsymbol{u}$.

These operations obey the same laws as ordinary arithmetic (Theorem 1.1): $\boldsymbol{u}+\boldsymbol{v}=\boldsymbol{v}+\boldsymbol{u}$, $(\boldsymbol{u}+\boldsymbol{v})+\boldsymbol{w}=\boldsymbol{u}+(\boldsymbol{v}+\boldsymbol{w})$, $\boldsymbol{v}+\boldsymbol{0}=\boldsymbol{v}$, $\boldsymbol{v}+(-\boldsymbol{v})=\boldsymbol{0}$, $c(\boldsymbol{u}+\boldsymbol{v})=c\boldsymbol{u}+c\boldsymbol{v}$, $(c+d)\boldsymbol{v}=c\boldsymbol{v}+d\boldsymbol{v}$ and $c(d\boldsymbol{v})=(cd)\boldsymbol{v}$. Any collection of objects with an addition and a scalar multiplication obeying these laws is called a **vector space**; $\mathbb{R}^n$ is the main example.

4.2.3 Linear combinations

Combining the two operations gives the single most important construction in linear algebra.



(a) The parallelogram rule: $(3,1)+(1,2)=(4,3)$. **(b)** Scalar multiples of $\boldsymbol{v}=(2,1)$.

Figure 4.2: The two basic vector operations.

Definition 4.2 (Linear combination)**.** A **linear combination** of vectors $\boldsymbol{v}_1,\dots,\boldsymbol{v}_k$ is any vector of the form

$$c_1\boldsymbol{v}_1+c_2\boldsymbol{v}_2+\cdots+c_k\boldsymbol{v}_k,$$

where $c_1,\dots,c_k$ are scalars, called the *coefficients* or *weights* of the combination.

Example 4.3 (A linear combination)**.** Compute $2\boldsymbol{u}-3\boldsymbol{v}$ for $\boldsymbol{u}=(1,0,2)$ and $\boldsymbol{v}=(2,-1,1)$.

Solution. $2\boldsymbol{u}=(2,0,4)$ and $3\boldsymbol{v}=(6,-3,3)$, so $2\boldsymbol{u}-3\boldsymbol{v}=(2-6,0+3,4-3)=(-4,3,1)$.

The vectors $\boldsymbol{e}_1=(1,0)$ and $\boldsymbol{e}_2=(0,1)$ are special: every vector in $\mathbb{R}^2$ is a linear combination of them, in exactly one way:

$$\begin{pmatrix}3\\2\end{pmatrix}=3\begin{pmatrix}1\\0\end{pmatrix}+2\begin{pmatrix}0\\1\end{pmatrix}=3\boldsymbol{e}_1+2\boldsymbol{e}_2.$$

In $\mathbb{R}^n$, the **standard basis vectors** $\boldsymbol{e}_1,\dots,\boldsymbol{e}_n$ have a 1 in one position and 0 elsewhere, and every vector is $\boldsymbol{v}=v_1\boldsymbol{e}_1+\cdots+v_n\boldsymbol{e}_n$. The components of a vector are just the coefficients needed to build it from the standard basis.

Intuition (Linear combinations are recipes)**.** Think of $\boldsymbol{v}_1,\dots,\boldsymbol{v}_k$ as ingredients and of the coefficients as amounts. If one portion of food A provides $(10,5,30)$ grams of (protein, fat, carbohydrate) and one portion of food B provides $(20,10,5)$, then two portions of A and one of B provide $2(10,5,30)+(20,10,5)=(40,20,65)$. Which nutrient totals can you reach by mixing A and B? That question, "which vectors are linear combinations of these?", is the subject of Chapter 5.

Why this matters for ML (Vector arithmetic in machine learning)**.** Adding and scaling vectors is the bread and butter of machine learning code.

- The *mean* of data vectors $\boldsymbol{x}_1,\dots,\boldsymbol{x}_n$ is the linear combination $\frac{1}{n}(\boldsymbol{x}_1+\cdots+\boldsymbol{x}_n)$, computed component by component. It is the "centre of mass" of the data cloud.
- Training updates all weights at once with a vector operation: gradient descent replaces $\boldsymbol{\theta}$ by $\boldsymbol{\theta}-\eta\,\boldsymbol{g}$, where $\boldsymbol{g}$ is a vector (the gradient, Chapter 7) and η a small scalar (Chapter 8).
- Word embeddings famously support analogies by vector arithmetic: the vector king − man + woman lies close to queen.

Exercises for Section 4.2

Exercise 4.4. Let $\boldsymbol{u} = (2, -1, 3)$ and $\boldsymbol{v} = (-1, 4, 0)$. Compute $\boldsymbol{u} + \boldsymbol{v}$, $\boldsymbol{u} - \boldsymbol{v}$, $3\boldsymbol{u}$ and $2\boldsymbol{u} - 3\boldsymbol{v}$.

Solution on p. 246 →

Exercise 4.5. Find scalars a and b such that $a(1, 1) + b(1, -1) = (5, 1)$.

Solution on p. 246 →

Exercise 4.6. Compute the mean of the data vectors $(1, 2)$, $(3, 6)$ and $(5, 1)$.

Solution on p. 246 →

Exercise 4.7. A model has weights $\boldsymbol{\theta} = (0.5, -1.2, 2.0)$. One step of gradient descent with learning rate $\eta = 0.1$ and gradient $\boldsymbol{g} = (0.4, -0.6, 1.0)$ replaces $\boldsymbol{\theta}$ by $\boldsymbol{\theta} - \eta\boldsymbol{g}$. Compute the new weights.

Solution on p. 246 →

4.3 Length, distance and unit vectors

The length of the arrow $\boldsymbol{v} = (3, 4)$ is the hypotenuse of a right triangle with legs 3 and 4, namely 5. In general, by the Pythagorean theorem in n dimensions (Definition 3.4):

Definition 4.4 (Length (norm) of a vector)**.** The **length**, or **Euclidean norm**, of $\boldsymbol{v} \in \mathbb{R}^n$ is

$$\|\boldsymbol{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

The *distance* between $\boldsymbol{u}$ and $\boldsymbol{v}$ is the length of their difference, $\|\boldsymbol{u} - \boldsymbol{v}\|$.

For example, $\|(1, -2, 2)\| = \sqrt{1 + 4 + 4} = 3$. Lengths behave as expected under scaling: $\|c\boldsymbol{v}\| = |c|\,\|\boldsymbol{v}\|$. (Doubling a vector doubles its length; reversing it does not change its length.)

A vector of length 1 is a **unit vector**. Any non-zero vector can be turned into a unit vector pointing in the same direction by dividing by its length, which is called **normalising** it:

$$\hat{\boldsymbol{v}} = \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|}.$$

For $\boldsymbol{v} = (3, 4)$, $\hat{\boldsymbol{v}} = (0.6, 0.8)$. A unit vector records a pure direction.

Theorem 4.5 (Triangle inequality)**.** For all vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$,

$$\|\boldsymbol{u} + \boldsymbol{v}\| \le \|\boldsymbol{u}\| + \|\boldsymbol{v}\|.$$

In the parallelogram picture (Figure 4.2), $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{u} + \boldsymbol{v}$ form a triangle, and the theorem says that one side of a triangle is never longer than the other two together. This is the inequality we met for numbers in Proposition 1.15, now in its original geometric setting. Equality holds only when $\boldsymbol{u}$ and $\boldsymbol{v}$ point in the same direction.

Why this matters for ML. Many systems normalise vectors before comparing them: search engines and retrieval systems store unit-length embeddings, so that only their directions matter. Lengths also measure the *size* of things: the length of the

weight vector is used to penalise overly large weights (Chapter 5), and the length of the gradient decides whether "gradient clipping" is needed (Chapter 8).

Exercises for Section 4.3

Exercise 4.8. Compute $\|(3,4)\|$, $\|(1,-2,2)\|$ and $\|(1,1,1,1)\|$.

Solution on p. 246 →

Exercise 4.9. Find the unit vector pointing in the same direction as $(6,-8)$.

Solution on p. 246 →

Exercise 4.10. Find the distance between $(1,2,3)$ and $(4,6,3)$.

Solution on p. 246 →

Exercise 4.11. Check the triangle inequality for $\boldsymbol{u} = (1,0)$ and $\boldsymbol{v} = (0,1)$. Then find two non-zero vectors for which it holds with equality.

Solution on p. 246 →

4.4 The dot product

Motivation. The dot product is the most important operation in this chapter. It multiplies two vectors and returns a single number. Algebraically it is a *weighted sum*, which is exactly what an artificial neuron computes. Geometrically it measures how much two vectors *point the same way*, which is how machine learning measures similarity.

Definition 4.6 (Dot product)**.** The **dot product** (or **inner product**) of $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ is the number

$$\boxed{\boldsymbol{u} \cdot \boldsymbol{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n = \sum_{i=1}^{n} u_i v_i}$$

Example 4.7 (Computing a dot product)**.** $(1,2,3) \cdot (4,-5,6) = 1 \cdot 4 + 2 \cdot (-5) + 3 \cdot 6 = 4 - 10 + 18 = 12$.

The dot product is commutative ($\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u}$), distributes over addition ($\boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \cdot \boldsymbol{v} + \boldsymbol{u} \cdot \boldsymbol{w}$), and lets scalars move freely ($(c\boldsymbol{u}) \cdot \boldsymbol{v} = c(\boldsymbol{u} \cdot \boldsymbol{v})$). Most importantly, the dot product of a vector with itself is its squared length:

$$\boldsymbol{v} \cdot \boldsymbol{v} = v_1^2 + \cdots + v_n^2 = \|\boldsymbol{v}\|^2.$$

4.4.1 The geometric meaning

Theorem 4.8 (Geometric formula for the dot product)**.** If θ is the angle between the non-zero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$, then

$$\boxed{\boldsymbol{u} \cdot \boldsymbol{v} = \|\boldsymbol{u}\| \, \|\boldsymbol{v}\| \cos\theta}$$

Proof. The vectors $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{u}-\boldsymbol{v}$ form a triangle in which θ is the angle between the sides $\boldsymbol{u}$ and $\boldsymbol{v}$. By the law of cosines (Theorem 3.11),

$$\|\boldsymbol{u}-\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2 - 2\|\boldsymbol{u}\|\,\|\boldsymbol{v}\|\cos\theta.$$

On the other hand, expanding with the rules of the dot product,

$$\|\boldsymbol{u}-\boldsymbol{v}\|^2 = (\boldsymbol{u}-\boldsymbol{v})\cdot(\boldsymbol{u}-\boldsymbol{v}) = \boldsymbol{u}\cdot\boldsymbol{u} - 2\,\boldsymbol{u}\cdot\boldsymbol{v} + \boldsymbol{v}\cdot\boldsymbol{v} = \|\boldsymbol{u}\|^2 - 2\,\boldsymbol{u}\cdot\boldsymbol{v} + \|\boldsymbol{v}\|^2.$$

Comparing the two expressions gives $\boldsymbol{u}\cdot\boldsymbol{v} = \|\boldsymbol{u}\|\|\boldsymbol{v}\|\cos\theta$. ∎

Solving for the angle:

$$\cos\theta = \frac{\boldsymbol{u}\cdot\boldsymbol{v}}{\|\boldsymbol{u}\|\,\|\boldsymbol{v}\|}.$$

Example 4.9 (The angle between two vectors)**.** Find the angle between $\boldsymbol{u} = (2, 1, -1)$ and $\boldsymbol{v} = (1, 3, 2)$.

Solution. $\boldsymbol{u}\cdot\boldsymbol{v} = 2 + 3 - 2 = 3$, $\|\boldsymbol{u}\| = \sqrt{6}$ and $\|\boldsymbol{v}\| = \sqrt{14}$. So $\cos\theta = \frac{3}{\sqrt{84}} \approx 0.327$ and $\theta = \arccos(0.327) \approx 70.9°$.

Intuition (What the sign of the dot product tells you)**.** Since lengths are positive, the sign of $\boldsymbol{u}\cdot\boldsymbol{v}$ is the sign of $\cos\theta$:

- $\boldsymbol{u}\cdot\boldsymbol{v} > 0$: the angle is less than 90°; the vectors point roughly the same way;
- $\boldsymbol{u}\cdot\boldsymbol{v} = 0$: the angle is exactly 90°; the vectors are perpendicular;
- $\boldsymbol{u}\cdot\boldsymbol{v} < 0$: the angle is more than 90°; the vectors point roughly in opposite directions.

Definition 4.10 (Orthogonal vectors)**.** Two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are **orthogonal** (perpendicular), written $\boldsymbol{u} \perp \boldsymbol{v}$, if $\boldsymbol{u}\cdot\boldsymbol{v} = 0$.

For example, $(1, 2)\cdot(-2, 1) = -2 + 2 = 0$, so these vectors are orthogonal. This is a remarkably cheap test for a right angle, and it works in any number of dimensions, where we cannot draw pictures.

4.4.2 Projections: the dot product as a shadow

If $\hat{\boldsymbol{v}}$ is a unit vector, then $\boldsymbol{u}\cdot\hat{\boldsymbol{v}} = \|\boldsymbol{u}\|\cos\theta$. By the right-triangle definition of cosine, this is the length of the "shadow" that $\boldsymbol{u}$ casts on the line through $\hat{\boldsymbol{v}}$ when light shines perpendicular to that line (Figure 4.3). It is called the **scalar projection** of $\boldsymbol{u}$ onto the direction $\hat{\boldsymbol{v}}$; it is negative if the shadow falls backwards. So the dot product with a unit vector answers the question "how far does $\boldsymbol{u}$ go in the direction $\hat{\boldsymbol{v}}$?".

Theorem 4.11 (Cauchy–Schwarz inequality)**.** For all $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$, $|\boldsymbol{u}\cdot\boldsymbol{v}| \le \|\boldsymbol{u}\|\,\|\boldsymbol{v}\|$.

This follows at once from Theorem 4.8, because $|\cos\theta| \le 1$. It guarantees that the "cosine" computed from the dot product always lies between -1 and 1, even in a million dimensions.

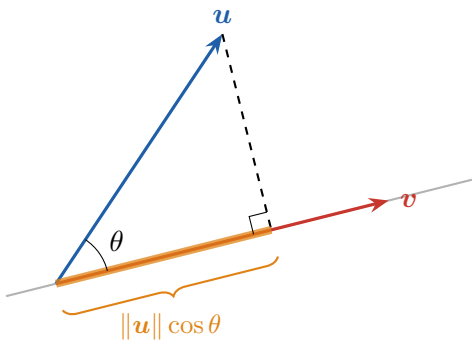


Figure 4.3: The scalar projection of $\boldsymbol{u}$ onto the direction of $\boldsymbol{v}$: the length of the orange "shadow" is $\|\boldsymbol{u}\| \cos\theta = \boldsymbol{u} \cdot \hat{\boldsymbol{v}}$.

4.4.3 The dot product in machine learning

Why this matters for ML (Neurons, similarity and attention)**.** Dot products are the workhorse of machine learning.

- **Weighted sums.** A linear model, and a single artificial neuron before its activation, compute

$$z = w_1 x_1 + \cdots + w_d x_d + b = \boldsymbol{w} \cdot \boldsymbol{x} + b.$$

  Each weight says how much its feature counts.
- **Cosine similarity.** The **cosine similarity** $\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{\|\boldsymbol{u}\|\|\boldsymbol{v}\|}$ compares directions while ignoring lengths. Take the made-up three-dimensional "embeddings"

$$\text{cat} = (0.9, 0.8, 0.1), \quad \text{dog} = (0.8, 0.9, 0.2), \quad \text{car} = (0.1, 0.2, 0.9).$$

  The similarity of cat and dog is $\frac{1.46}{1.208\times 1.221} \approx 0.99$, while that of cat and car is $\frac{0.34}{1.208\times 0.927} \approx 0.30$.
- **Attention.** In a transformer, each word produces a "query" vector $\boldsymbol{q}$ and a "key" vector $\boldsymbol{k}$, and the relevance of word j to word i is measured by the dot product $\boldsymbol{q}_i \cdot \boldsymbol{k}_j$. Attention is, at heart, a large table of dot products.

Proposition 4.12 (The weight vector is perpendicular to the decision boundary)**.** Let $\boldsymbol{w} \neq \boldsymbol{0}$. The vector $\boldsymbol{w}$ is perpendicular to the decision boundary $\{\boldsymbol{x} : \boldsymbol{w}\cdot\boldsymbol{x} + b = 0\}$ of a linear classifier.

Proof. Take any two points $\boldsymbol{x}$ and $\boldsymbol{x}'$ on the boundary. Then $\boldsymbol{w}\cdot\boldsymbol{x} + b = 0$ and $\boldsymbol{w}\cdot\boldsymbol{x}' + b = 0$; subtracting, $\boldsymbol{w}\cdot(\boldsymbol{x} - \boldsymbol{x}') = 0$. So $\boldsymbol{w}$ is orthogonal to every direction $\boldsymbol{x} - \boldsymbol{x}'$ that lies within the boundary. ∎

This explains the picture of Figure 3.5: there $\boldsymbol{w} = (1, 1)$, which points from the region of class A straight across the boundary into the region of class B. The quantity $\boldsymbol{w}\cdot\boldsymbol{x} + b$ increases fastest in the direction of $\boldsymbol{w}$.

Exercises for Section 4.4

Exercise 4.12. Compute $(2,-1,3)\cdot(4,0,-2)$ and $(1,1,1,1)\cdot(1,-1,1,-1)$. Are either of these pairs orthogonal?

Solution on p. 246 →

Exercise 4.13. Find the angle between (a) $(1,1)$ and $(-1,0)$; (b) $(1,2,2)$ and $(2,-2,1)$.

Solution on p. 247 →

Exercise 4.14. For which value of k is $(k,2)$ orthogonal to $(3,-6)$?

Solution on p. 247 →

Exercise 4.15. Using the made-up embeddings in the box above, compute the cosine similarity of "dog" and "car". Which pair of the three words is least similar?

Solution on p. 247 →

Exercise 4.16. For the line $3x_1 + 4x_2 - 12 = 0$, give a vector perpendicular to it. Then find the point of the line closest to the origin, and its distance from the origin. *Hint:* the closest point lies in the direction of the perpendicular vector from the origin, so it has the form $t(3,4)$.

Solution on p. 247 →

Exercise 4.17. A neuron has weights $\boldsymbol{w} = (0.5,-1,2)$, bias $b = 0.1$ and a sigmoid activation. Compute its output for the input $\boldsymbol{x} = (2,1,0.5)$.

Solution on p. 247 →

4.5 Matrices

Definition 4.13 (Matrix). A **matrix** is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ *matrix* (read "m by n"), and the set of all such matrices is written $\mathbb{R}^{m\times n}$. The entry in row i and column j of a matrix $\boldsymbol{A}$ is written a_{ij} (or A_{ij}):

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

Notation. Matrices are written in bold upper-case letters: $\boldsymbol{A}$, $\boldsymbol{W}$, $\boldsymbol{X}$. The size is always "rows × columns", and the first index of an entry is always the row. A vector in $\mathbb{R}^n$ is an $n\times 1$ matrix (a single column). A matrix can be viewed as a list of columns or as a stack of rows; both views will be useful.

For example,

$$\boldsymbol{A} = \begin{pmatrix} 3 & 0 & -1 \\ 2 & 5 & 4 \end{pmatrix}$$

is a 2×3 matrix with $a_{13} = -1$ and $a_{22} = 5$. Its second row is $(2, 5, 4)$ and its third column is $(-1, 4)$.

Why this matters for ML (Data sets and images are matrices)**.** A data set of n examples with d features each is stored as an $n \times d$ **data matrix** $\boldsymbol{X}$: one row per example, one column per feature. For three houses with features (size, bedrooms),

$$\boldsymbol{X} = \begin{pmatrix} 100 & 3 \\ 80 & 2 \\ 120 & 4 \end{pmatrix}.$$

A grayscale image is a matrix of pixel intensities. A colour image is three such matrices (red, green, blue) stacked together; arrays with more than two indices like this are called **tensors**, which is where the software library TensorFlow gets its name.

4.5.1 Special matrices

- A *square* matrix has as many rows as columns.
- The **zero matrix** $\mathbf{0}$ has all entries equal to 0.
- A **diagonal matrix** is square with zeros everywhere off the main diagonal (the entries $a_{11}, a_{22}, \dots$), for example $\left(\begin{smallmatrix} 2 & 0 \\ 0 & -1 \end{smallmatrix}\right)$.
- The **identity matrix** $\boldsymbol{I}$ (or $\boldsymbol{I}_n$) is the diagonal matrix with all diagonal entries equal to 1.
- A square matrix is **symmetric** if $a_{ij} = a_{ji}$ for all i, j, i.e. it is a mirror image of itself across the main diagonal, like $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$.
- A square matrix is *upper triangular* if all entries below the main diagonal are zero.

4.5.2 Adding and scaling matrices

Matrices of the same size are added, and multiplied by scalars, entry by entry, exactly like vectors:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 5 & 4 \end{pmatrix}.$$

Exercises for Section 4.5

Exercise 4.18. For $\boldsymbol{A} = \left(\begin{smallmatrix} 3 & 0 & -1 \\ 2 & 5 & 4 \end{smallmatrix}\right)$, give the size of $\boldsymbol{A}$, the entries a_{21} and a_{13}, the first row and the second column.

Solution on p. 247 →

Exercise 4.19. Which of these matrices are symmetric? $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 4 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 6 \end{smallmatrix}\right)$.

Solution on p. 247 →

Exercise 4.20. Compute $2\boldsymbol{A} - \boldsymbol{B}$ for $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $\boldsymbol{B} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

Solution on p. 247 →

Exercise 4.21. A data set consists of 500 grayscale images of 28×28 pixels. Each image is flattened into a row of the data matrix. What is the size of the data matrix?

Solution on p. 247 →

4.6 Multiplying a matrix by a vector

Definition 4.14 (Matrix–vector product)**.** Let $\boldsymbol{A}$ be an $m \times n$ matrix and $\boldsymbol{x} \in \mathbb{R}^n$. The product $\boldsymbol{Ax}$ is the vector in $\mathbb{R}^m$ whose i-th component is the dot product of the i-th row of $\boldsymbol{A}$ with $\boldsymbol{x}$:

$$(\boldsymbol{Ax})_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = \sum_{j=1}^{n} a_{ij}x_j.$$

The number of columns of $\boldsymbol{A}$ must equal the number of components of $\boldsymbol{x}$; the result has as many components as $\boldsymbol{A}$ has rows:

$$\underbrace{\boldsymbol{A}}_{m\times n}\ \underbrace{\boldsymbol{x}}_{n\times 1} = \underbrace{\boldsymbol{Ax}}_{m\times 1}.$$

Example 4.15 (Two ways to compute $\boldsymbol{Ax}$)**.** Compute $\boldsymbol{Ax}$ for $\boldsymbol{A} = \left(\begin{smallmatrix} 2 & 1 \\ -1 & 3 \\ 0 & 4 \end{smallmatrix}\right)$ and $\boldsymbol{x} = (3, 2)$.

Solution. Row view (dot products of the rows with $\boldsymbol{x}$):

$$\begin{pmatrix} 2 & 1 \\ -1 & 3 \\ 0 & 4 \end{pmatrix}\begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 2\cdot 3 + 1\cdot 2 \\ -1\cdot 3 + 3\cdot 2 \\ 0\cdot 3 + 4\cdot 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 3 \\ 8 \end{pmatrix}.$$

Column view (a linear combination of the columns, with the entries of $\boldsymbol{x}$ as coefficients):

$$3\begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 6 \\ 8 \end{pmatrix} = \begin{pmatrix} 8 \\ 3 \\ 8 \end{pmatrix}.$$

Intuition (Rows and columns)**.** Both views give the same answer, but they tell different stories.

- **Row view:** $\boldsymbol{Ax}$ is a list of m dot products, one for each row. If each row is the weight vector of a neuron, $\boldsymbol{Ax}$ computes all the neurons of a layer at once.
- **Column view:** $\boldsymbol{Ax}$ is a linear combination of the columns of $\boldsymbol{A}$, mixed in the amounts given by $\boldsymbol{x}$. So the possible outputs $\boldsymbol{Ax}$ are exactly the linear combinations of the columns, a fact that will be central in Chapter 5.

Matrix–vector multiplication is *linear*: for all vectors $\boldsymbol{u}, \boldsymbol{v}$ and scalars c,

$$\boldsymbol{A}(\boldsymbol{u} + \boldsymbol{v}) = \boldsymbol{Au} + \boldsymbol{Av}, \qquad \boldsymbol{A}(c\boldsymbol{u}) = c\,\boldsymbol{Au},$$

which follows directly from the distributive law.

Why this matters for ML (Predictions for a whole data set, and a neural network layer)**.** For a linear model $\hat{y} = \boldsymbol{w}\cdot\boldsymbol{x} + b$, the predictions for all n examples at once are $\hat{\boldsymbol{y}} = \boldsymbol{Xw} + b$, where $\boldsymbol{X}$ is the data matrix and b is added to every component. With

the house data above, $\boldsymbol{w} = (2000, 10\,000)$ and $b = 20\,000$:

$$\hat{\boldsymbol{y}} = \begin{pmatrix} 100 & 3 \\ 80 & 2 \\ 120 & 4 \end{pmatrix} \begin{pmatrix} 2000 \\ 10\,000 \end{pmatrix} + 20\,000 = \begin{pmatrix} 250\,000 \\ 200\,000 \\ 300\,000 \end{pmatrix}.$$

A **layer** of a neural network with weight matrix $\boldsymbol{W}$ (one row per neuron), bias vector $\boldsymbol{b}$ and activation f computes

$$\boxed{\boldsymbol{h} = f(\boldsymbol{W}\boldsymbol{x} + \boldsymbol{b})}$$

where f is applied to each component separately. For example, with $\boldsymbol{W} = \left(\begin{smallmatrix} 1 & -1 & 2 \\ 0 & -2 & 1 \end{smallmatrix}\right)$, $\boldsymbol{b} = (0.5, -1)$, $\boldsymbol{x} = (1, 2, 3)$ and $f = \mathrm{ReLU}$: $\boldsymbol{W}\boldsymbol{x} + \boldsymbol{b} = (5 + 0.5, -1 - 1) = (5.5, -2)$, and $\boldsymbol{h} = (5.5, 0)$.

Exercises for Section 4.6

Exercise 4.22. Compute $\left(\begin{smallmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \end{smallmatrix}\right) \left(\begin{smallmatrix} 1 \\ 2 \\ 3 \end{smallmatrix}\right)$.

Solution on p. 247 →

Exercise 4.23. Compute $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right) \left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$ using the row view and again using the column view.

Solution on p. 247 →

Exercise 4.24. Let $\boldsymbol{A}$ be a 3×2 matrix, $\boldsymbol{x} \in \mathbb{R}^3$ and $\boldsymbol{y} \in \mathbb{R}^2$. Which of $\boldsymbol{A}\boldsymbol{x}$ and $\boldsymbol{A}\boldsymbol{y}$ is defined, and in which space does it live?

Solution on p. 247 →

Exercise 4.25. A linear model has weights $\boldsymbol{w} = (2, -1)$ and bias $b = 0.5$. Compute its predictions for the data matrix $\boldsymbol{X} = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 0 \\ -1 & 1 \end{smallmatrix}\right)$.

Solution on p. 247 →

4.7 Matrices as linear transformations

A matrix $\boldsymbol{A}$ of size $m \times n$ turns each vector $\boldsymbol{x} \in \mathbb{R}^n$ into a vector $\boldsymbol{A}\boldsymbol{x} \in \mathbb{R}^m$, so it defines a function $\boldsymbol{x} \mapsto \boldsymbol{A}\boldsymbol{x}$. Thinking of matrices as functions that move points around, rather than as tables of numbers, is the key to understanding them.

Definition 4.16 (Linear transformation)**.** A function $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is a **linear transformation** (or *linear map*) if for all vectors $\boldsymbol{u}, \boldsymbol{v}$ and scalars c,

$$T(\boldsymbol{u} + \boldsymbol{v}) = T(\boldsymbol{u}) + T(\boldsymbol{v}) \qquad \text{and} \qquad T(c\boldsymbol{u}) = cT(\boldsymbol{u}).$$

Every matrix defines a linear transformation, as we just saw. Remarkably, the converse is also true, and the proof tells us how to find the matrix.

Theorem 4.17 (The columns of a matrix are the images of the basis vectors)**.** For any matrix $\boldsymbol{A}$, the j-th column of $\boldsymbol{A}$ is $\boldsymbol{A}\boldsymbol{e}_j$. Consequently, every linear transformation $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is given by a matrix, namely the matrix whose columns are $T(\boldsymbol{e}_1), \dots, T(\boldsymbol{e}_n)$.

Proof. By the column view, $\boldsymbol{A}\boldsymbol{e}_j$ is the combination of columns with coefficient 1 for column j and 0 for all others, which is column j itself. Now let T be linear and let $\boldsymbol{A}$ have columns $T(\boldsymbol{e}_1), \dots, T(\boldsymbol{e}_n)$. Every $\boldsymbol{x}$ is $x_1\boldsymbol{e}_1 + \cdots + x_n\boldsymbol{e}_n$, so by linearity $T(\boldsymbol{x}) = x_1T(\boldsymbol{e}_1) + \cdots + x_nT(\boldsymbol{e}_n)$, which by the column view is $\boldsymbol{A}\boldsymbol{x}$. ∎

Intuition (Watch where the basis vectors go)**.** To understand what a 2×2 matrix does, look at its two columns: they show where $\boldsymbol{e}_1 = (1, 0)$ and $\boldsymbol{e}_2 = (0, 1)$ end up. Everything else follows, because every vector is a combination of $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ and linearity preserves combinations. The unit square spanned by $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ becomes the parallelogram spanned by the two columns. Grid lines stay straight, parallel and evenly spaced, and the origin stays put. Figure 4.4 shows a gallery.

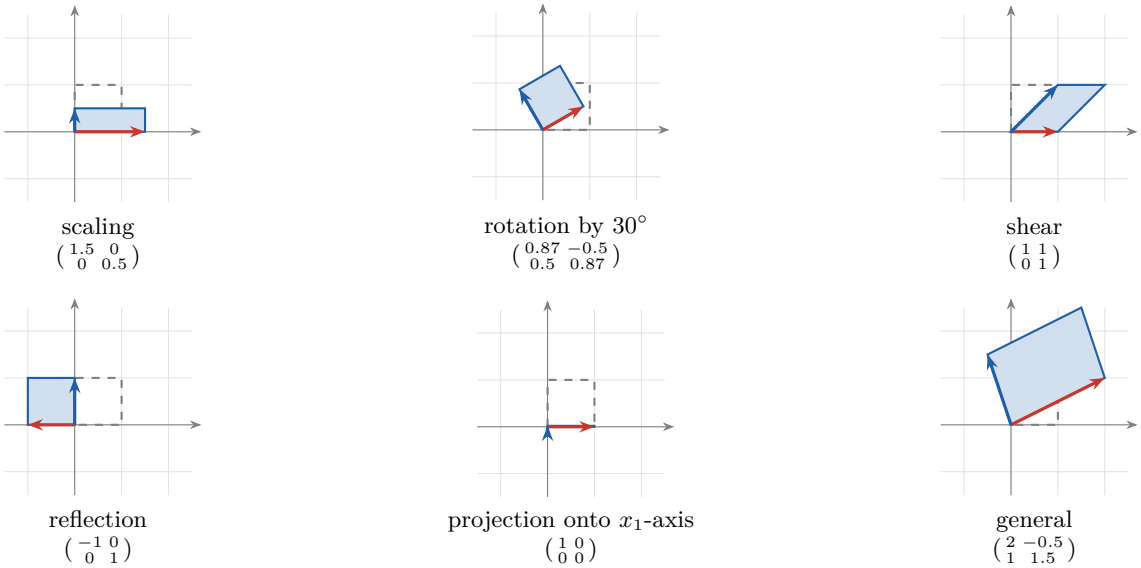


Figure 4.4: What 2×2 matrices do to the unit square (dashed). The red arrow is the image of $\boldsymbol{e}_1$ (the first column), the blue arrow the image of $\boldsymbol{e}_2$ (the second column), and the shaded parallelogram is the image of the unit square. The projection squashes the square flat onto the x_1-axis.

Example 4.18 (Finding the matrix of a transformation)**.** Find the matrix that rotates the plane anticlockwise by 90°, and apply it to $(2, 1)$.

Solution. Rotating by 90° sends $\boldsymbol{e}_1 = (1, 0)$ to $(0, 1)$ and $\boldsymbol{e}_2 = (0, 1)$ to $(-1, 0)$. These are the columns:

$$\boldsymbol{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \boldsymbol{R}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix},$$

matching the rotation formula of Theorem 3.15. In general, rotation by α is given by

the **rotation matrix**

$$\boldsymbol{R}_\alpha = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}.$$

Remark (Linear versus affine)**.** A linear transformation always sends $\mathbf{0}$ to $\mathbf{0}$, because $T(\mathbf{0}) = T(0 \cdot \mathbf{0}) = 0 \cdot T(\mathbf{0}) = \mathbf{0}$. So a *translation* $\boldsymbol{x} \mapsto \boldsymbol{x} + \boldsymbol{b}$ with $\boldsymbol{b} \neq \mathbf{0}$ is not linear. Maps of the form $\boldsymbol{x} \mapsto \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}$ are called **affine**: a linear transformation followed by a shift. A neural network layer before its activation function is affine; in machine learning this distinction is usually blurred and both are called "linear".

Why this matters for ML (A neural network transforms space)**.** Each layer of a neural network applies an affine map (which can rotate, stretch, shear and squash the data cloud) followed by a non-linear activation (which bends it). Layer after layer, a network can move tangled-up classes of data into positions where a simple linear boundary separates them. This geometric picture of "deforming space until the classes separate" is one of the best ways to think about what deep networks do.

Exercises for Section 4.7

Exercise 4.26. Find the 2×2 matrix that (a) reflects every point in the line $x_2 = x_1$; (b) doubles every vector; (c) rotates by $180°$.

Solution on p. 247 →

Exercise 4.27. Is $T(x_1, x_2) = (x_1 + 1, x_2)$ a linear transformation? Is $S(x_1, x_2) = (x_2, 3x_1)$? If so, give its matrix.

Solution on p. 247 →

Exercise 4.28. A linear transformation sends $\boldsymbol{e}_1$ to $(2, 1)$ and $\boldsymbol{e}_2$ to $(-1, 3)$. Write down its matrix and find the image of $(4, 2)$.

Solution on p. 247 →

4.8 Matrix multiplication

4.8.1 Composing transformations

What happens if we apply one linear transformation and then another? First $\boldsymbol{x} \mapsto \boldsymbol{B}\boldsymbol{x}$, then $\boldsymbol{A}$ applied to the result: $\boldsymbol{x} \mapsto \boldsymbol{A}(\boldsymbol{B}\boldsymbol{x})$. A composition of linear maps is linear, so by Theorem 4.17 it is given by a single matrix. We define the *product* $\boldsymbol{A}\boldsymbol{B}$ to be exactly that matrix, so that

$$(\boldsymbol{A}\boldsymbol{B})\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{B}\boldsymbol{x}) \quad \text{for every } \boldsymbol{x}.$$

Its j-th column is the image of $\boldsymbol{e}_j$, namely $\boldsymbol{A}(\boldsymbol{B}\boldsymbol{e}_j) = \boldsymbol{A}\boldsymbol{b}_j$, where $\boldsymbol{b}_j$ is the j-th column of $\boldsymbol{B}$.

Definition 4.19 (Matrix product)**.** Let $\boldsymbol{A}$ be $m \times n$ and $\boldsymbol{B}$ be $n \times p$, with columns $\boldsymbol{b}_1, \dots, \boldsymbol{b}_p$. Their **product** is the $m \times p$ matrix

$$\boldsymbol{A}\boldsymbol{B} = \begin{pmatrix} \boldsymbol{A}\boldsymbol{b}_1 & \boldsymbol{A}\boldsymbol{b}_2 & \cdots & \boldsymbol{A}\boldsymbol{b}_p \end{pmatrix},$$

whose (i, j) entry is the dot product of row i of $\boldsymbol{A}$ with column j of $\boldsymbol{B}$:

$$(\boldsymbol{AB})_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

Common pitfall (Shapes must match). $\boldsymbol{AB}$ is defined only when the number of *columns* of $\boldsymbol{A}$ equals the number of *rows* of $\boldsymbol{B}$:

$$\underbrace{\boldsymbol{A}}_{m \times \boxed{n}} \; \underbrace{\boldsymbol{B}}_{\boxed{n} \times p} = \underbrace{\boldsymbol{AB}}_{m \times p}.$$

The inner sizes must agree and the outer sizes give the size of the result. Checking shapes like this catches a large fraction of bugs in machine learning code, where "shape mismatch" is the most common error message of all.

Example 4.20 (Matrix multiplication is not commutative). Let $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $\boldsymbol{B} = \left(\begin{smallmatrix} 2 & 0 \\ 1 & 3 \end{smallmatrix}\right)$. Compute $\boldsymbol{AB}$ and $\boldsymbol{BA}$.

Solution. Each entry is "row of the left matrix · column of the right matrix":

$$\boldsymbol{AB} = \begin{pmatrix} 1\cdot 2 + 2\cdot 1 & 1\cdot 0 + 2\cdot 3 \\ 3\cdot 2 + 4\cdot 1 & 3\cdot 0 + 4\cdot 3 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 10 & 12 \end{pmatrix}, \qquad \boldsymbol{BA} = \begin{pmatrix} 2 & 4 \\ 10 & 14 \end{pmatrix}.$$

So $\boldsymbol{AB} \neq \boldsymbol{BA}$.

Intuition (Why order matters). $\boldsymbol{AB}$ means "first $\boldsymbol{B}$, then $\boldsymbol{A}$" (read right to left, like $f(g(x))$). Doing things in a different order usually gives a different result: rotating a picture and then stretching it horizontally is not the same as stretching it horizontally and then rotating it. Putting on socks and then shoes is not the same as shoes and then socks.

Theorem 4.21 (Rules of matrix multiplication). Whenever the products are defined:

(a) $(\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{BC})$ (associative);
(b) $\boldsymbol{A}(\boldsymbol{B} + \boldsymbol{C}) = \boldsymbol{AB} + \boldsymbol{AC}$ and $(\boldsymbol{A} + \boldsymbol{B})\boldsymbol{C} = \boldsymbol{AC} + \boldsymbol{BC}$ (distributive);
(c) $(c\boldsymbol{A})\boldsymbol{B} = c(\boldsymbol{AB}) = \boldsymbol{A}(c\boldsymbol{B})$;
(d) $\boldsymbol{AI} = \boldsymbol{IA} = \boldsymbol{A}$ (the identity does nothing);
(e) but in general $\boldsymbol{AB} \neq \boldsymbol{BA}$.

Associativity is clear from the composition picture: applying $\boldsymbol{C}$, then $\boldsymbol{B}$, then $\boldsymbol{A}$ is the same process however we group it. We write $\boldsymbol{A}^2$ for $\boldsymbol{AA}$, $\boldsymbol{A}^3$ for $\boldsymbol{AAA}$, and so on.

Common pitfall (Familiar rules that fail for matrices). Besides commutativity, another rule of ordinary numbers fails: $\boldsymbol{AB} = \boldsymbol{0}$ does *not* imply that $\boldsymbol{A} = \boldsymbol{0}$ or $\boldsymbol{B} = \boldsymbol{0}$. For example,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Also, $(\boldsymbol{A}+\boldsymbol{B})^2 = \boldsymbol{A}^2 + \boldsymbol{AB} + \boldsymbol{BA} + \boldsymbol{B}^2$, which equals $\boldsymbol{A}^2 + 2\boldsymbol{AB} + \boldsymbol{B}^2$ only if $\boldsymbol{AB} = \boldsymbol{BA}$.

Why this matters for ML (Why neural networks need non-linear activations)**.** Two layers without an activation function in between compute $\boldsymbol{W}_2(\boldsymbol{W}_1\boldsymbol{x}) = (\boldsymbol{W}_2\boldsymbol{W}_1)\boldsymbol{x}$: a single linear map with matrix $\boldsymbol{W}_2\boldsymbol{W}_1$. However many linear layers we stack, the result collapses into one linear layer, so depth would add nothing. This is the many-input version of Proposition 2.20, and it is why every layer of a neural network ends with a non-linear activation function.

Matrix products also make computation efficient. To push a whole batch of inputs (the rows of $\boldsymbol{X}$) through a layer at once, compute $\boldsymbol{X}\boldsymbol{W}^\top$ (the transpose $\top$ is explained in the next section): one big matrix product instead of many small ones, which is exactly what GPUs are good at. Multiplying an $m \times n$ by an $n \times p$ matrix takes mnp multiplications, which is why the size of these matrices dominates the cost of training large models.

Exercises for Section 4.8

Exercise 4.29. Compute $\boldsymbol{AB}$ and $\boldsymbol{BA}$ for $\boldsymbol{A} = \left(\begin{smallmatrix}1 & 0\\ 2 & 1\end{smallmatrix}\right)$ and $\boldsymbol{B} = \left(\begin{smallmatrix}3 & 1\\ 0 & 2\end{smallmatrix}\right)$.

Solution on p. 247 →

Exercise 4.30. Let $\boldsymbol{A}$ be 2×3, $\boldsymbol{B}$ be 3×4 and $\boldsymbol{C}$ be 4×2. Which of the products $\boldsymbol{AB}$, $\boldsymbol{BC}$, $\boldsymbol{CA}$, $\boldsymbol{AC}$, $\boldsymbol{ABC}$ are defined, and what are their sizes?

Solution on p. 247 →

Exercise 4.31. Two linear layers have weight matrices $\boldsymbol{W}_1 = \left(\begin{smallmatrix}1 & 2\\ 0 & 1\end{smallmatrix}\right)$ and $\boldsymbol{W}_2 = \begin{pmatrix}1 & -1\end{pmatrix}$ (a 1×2 matrix). Find the single matrix $\boldsymbol{W}_2\boldsymbol{W}_1$ that does the same job, and check that both give the same output for $\boldsymbol{x} = (1, 1)$.

Solution on p. 247 →

Exercise 4.32. (Angle addition formulas.) Rotating by β and then by α is the same as rotating by $\alpha + \beta$, so $\boldsymbol{R}_\alpha\boldsymbol{R}_\beta = \boldsymbol{R}_{\alpha+\beta}$. Multiply out $\boldsymbol{R}_\alpha\boldsymbol{R}_\beta$ and compare entries to derive the formulas of Theorem 3.13.

Solution on p. 248 →

Exercise 4.33. Show that $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)^2 = \left(\begin{smallmatrix}1 & 2\\ 0 & 1\end{smallmatrix}\right)$, and explain the result geometrically.

Solution on p. 248 →

4.9 The transpose

Definition 4.22 (Transpose)**.** The **transpose** of an $m \times n$ matrix $\boldsymbol{A}$ is the $n \times m$ matrix $\boldsymbol{A}^\top$ obtained by turning rows into columns: $(\boldsymbol{A}^\top)_{ij} = a_{ji}$.

For example,

$$\begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{pmatrix}^\top = \begin{pmatrix}1 & 4\\ 2 & 5\\ 3 & 6\end{pmatrix}.$$

A matrix is symmetric exactly when $\boldsymbol{A}^\top = \boldsymbol{A}$. The transpose of a column vector is a *row vector*, and this gives a very common way of writing dot products as matrix products:

$$\boldsymbol{u}^\top \boldsymbol{v} = \begin{pmatrix} u_1 & \cdots & u_n \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \boldsymbol{u} \cdot \boldsymbol{v}.$$

(Strictly, $\boldsymbol{u}^\top \boldsymbol{v}$ is a 1×1 matrix, which we identify with a number.) Machine learning papers write $\boldsymbol{w}^\top \boldsymbol{x}$ far more often than $\boldsymbol{w} \cdot \boldsymbol{x}$.

Theorem 4.23 (Rules for transposes). $(\boldsymbol{A}^\top)^\top = \boldsymbol{A}$, $(\boldsymbol{A} + \boldsymbol{B})^\top = \boldsymbol{A}^\top + \boldsymbol{B}^\top$, $(c\boldsymbol{A})^\top = c\boldsymbol{A}^\top$, and

$$(\boldsymbol{A}\boldsymbol{B})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top.$$

The order reverses in the last rule; you can check it by comparing shapes: if $\boldsymbol{A}$ is $m \times n$ and $\boldsymbol{B}$ is $n \times p$, then $(\boldsymbol{A}\boldsymbol{B})^\top$ is $p \times m$, and of the two candidates only $\boldsymbol{B}^\top \boldsymbol{A}^\top$ ($p \times n$ times $n \times m$) has the right shape.

Proposition 4.24. For any matrix $\boldsymbol{A}$, the matrix $\boldsymbol{A}^\top \boldsymbol{A}$ is square and symmetric. Its (i, j) entry is the dot product of columns i and j of $\boldsymbol{A}$.

Proof. $(\boldsymbol{A}^\top \boldsymbol{A})^\top = \boldsymbol{A}^\top (\boldsymbol{A}^\top)^\top = \boldsymbol{A}^\top \boldsymbol{A}$. The entry formula follows from "row i of $\boldsymbol{A}^\top$ (which is column i of $\boldsymbol{A}$) times column j of $\boldsymbol{A}$". ∎

4.9.1 The outer product

Multiplying a column by a row gives a whole matrix. For $\boldsymbol{u} \in \mathbb{R}^m$ and $\boldsymbol{v} \in \mathbb{R}^n$, the **outer product** $\boldsymbol{u}\boldsymbol{v}^\top$ is the $m \times n$ matrix with entries $u_i v_j$:

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 3 & -1 & 0 \\ 6 & -2 & 0 \end{pmatrix}.$$

Every row is a multiple of $\boldsymbol{v}^\top$ and every column is a multiple of $\boldsymbol{u}$. Compare the *inner* product $\boldsymbol{u}^\top \boldsymbol{v}$ (a number) with the *outer* product $\boldsymbol{u}\boldsymbol{v}^\top$ (a matrix).

Why this matters for ML. The matrices $\boldsymbol{X}^\top \boldsymbol{X}$ (a $d \times d$ matrix of dot products between feature columns) and $\boldsymbol{X}\boldsymbol{X}^\top$ (an $n \times n$ matrix of dot products between examples) appear everywhere: in the formula for linear regression (Chapter 5), in covariance matrices (Chapter 9) and in kernel methods. Outer products appear in backpropagation: the gradient of a loss with respect to a weight matrix is an outer product of two vectors (Chapter 11).

Exercises for Section 4.9

Exercise 4.34. Write down $\boldsymbol{A}^\top$ for $\boldsymbol{A} = \left(\begin{smallmatrix} 2 & 0 & -1 \\ 1 & 3 & 5 \end{smallmatrix}\right)$, and compute $\boldsymbol{A}\boldsymbol{A}^\top$.

Solution on p. 248 →

Exercise 4.35. Using Example 4.20, verify that $(\boldsymbol{A}\boldsymbol{B})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$.

Solution on p. 248 →

Exercise 4.36. For $\boldsymbol{u} = (1, 2)$ and $\boldsymbol{v} = (3, -1)$, compute $\boldsymbol{u}^\top \boldsymbol{v}$, $\boldsymbol{u}\boldsymbol{v}^\top$ and $\boldsymbol{v}\boldsymbol{u}^\top$.

Solution on p. 248 →

Exercise 4.37. $\boldsymbol{X}$ is an $n \times d$ data matrix. What are the sizes of $\boldsymbol{X}^\top \boldsymbol{X}$ and $\boldsymbol{X}\boldsymbol{X}^\top$? For a data set of a million examples with 100 features, which of the two is cheaper to store?

Solution on p. 248 →

4.10 Inverses and determinants

4.10.1 The inverse of a matrix

The identity matrix $\boldsymbol{I}$ is the transformation that does nothing: $\boldsymbol{I}\boldsymbol{x} = \boldsymbol{x}$. A transformation can be undone if there is another transformation that brings every point back to where it started.

Definition 4.25 (Inverse matrix)**.** A square matrix $\boldsymbol{A}$ is **invertible** if there is a matrix $\boldsymbol{A}^{-1}$, its **inverse**, such that

$$\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}.$$

A square matrix that has no inverse is called **singular**.

Some inverses are obvious from the geometry: the inverse of a rotation by α is the rotation by $-\alpha$, and the inverse of the scaling $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 0.5 \end{smallmatrix}\right)$ is $\left(\begin{smallmatrix} 0.5 & 0 \\ 0 & 2 \end{smallmatrix}\right)$. For 2×2 matrices there is a formula.

Theorem 4.26 (Inverse of a 2×2 matrix)**.** The matrix $\boldsymbol{A} = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is invertible if and only if $ad - bc \neq 0$, and then

$$\boldsymbol{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

(Swap the diagonal entries, change the signs of the off-diagonal entries, divide by $ad - bc$.) You can check the formula by multiplying: the product with $\boldsymbol{A}$ is $\frac{1}{ad-bc}\left(\begin{smallmatrix} ad-bc & 0 \\ 0 & ad-bc \end{smallmatrix}\right) = \boldsymbol{I}$.

Example 4.27 (Inverting a matrix)**.** Find the inverse of $\boldsymbol{A} = \left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right)$.

Solution. $ad - bc = 6 - 5 = 1$, so $\boldsymbol{A}^{-1} = \left(\begin{smallmatrix} 3 & -1 \\ -5 & 2 \end{smallmatrix}\right)$. *Check:* $\boldsymbol{A}\boldsymbol{A}^{-1} = \left(\begin{smallmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{smallmatrix}\right) = \boldsymbol{I}$. ✓

4.10.2 The determinant

The number $ad - bc$ deserves a name and an interpretation.

Definition 4.28 (Determinant of a 2×2 matrix)**.** The **determinant** of $\boldsymbol{A} = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is $\det \boldsymbol{A} = ad - bc$.

Intuition (The determinant measures change of area)**.** The matrix $\boldsymbol{A}$ turns the unit square (area 1) into the parallelogram spanned by its columns (a, c) and (b, d), and

the area of that parallelogram is $|ad - bc|$ (Figure 4.5). Because the transformation is linear, *every* region gets its area multiplied by the same factor $|\det \boldsymbol{A}|$. The sign records orientation: a negative determinant means the transformation flips the plane over, like a mirror.

- Scaling $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$: $\det = 6$, areas grow sixfold.
- Any rotation: $\det = \cos^2\alpha + \sin^2\alpha = 1$; areas are unchanged. A shear also has $\det = 1$.
- Reflection $\left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$: $\det = -1$; areas unchanged, orientation flipped.
- Projection $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$: $\det = 0$; the plane is squashed onto a line, and areas become 0.

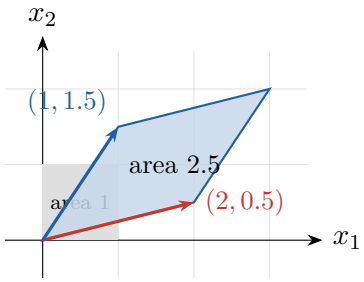


Figure 4.5: The matrix $\left(\begin{smallmatrix} 2 & 1 \\ 0.5 & 1.5 \end{smallmatrix}\right)$ maps the unit square to the parallelogram spanned by its columns. Its determinant $2 \times 1.5 - 1 \times 0.5 = 2.5$ is the area of the parallelogram.

This picture explains when a matrix is invertible. If $\det \boldsymbol{A} = 0$, the transformation squashes the plane onto a line (or a point). Many different points land on the same image, so there is no way to undo it: $\boldsymbol{A}$ is singular. For example, the columns of $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right)$ are parallel ($(2, 4) = 2(1, 2)$), its determinant is $4 - 4 = 0$, and it squashes the whole plane onto the line through $(1, 2)$.

4.10.3 Determinants of larger matrices

Determinants exist for square matrices of every size, with the same meaning: for a 3×3 matrix, $|\det \boldsymbol{A}|$ is the factor by which volumes are multiplied. A 3×3 determinant can be computed by *expansion along the first row*:

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \det \begin{pmatrix} d & f \\ g & i \end{pmatrix} + c \det \begin{pmatrix} d & e \\ g & h \end{pmatrix}.$$

Each entry of the first row multiplies the determinant of the 2×2 matrix left after deleting that entry's row and column, with alternating signs $+, -, +$. For example,

$$\det \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 1 & 1 & 2 \end{pmatrix} = 2(6 - 2) - 0(2 - 2) + 1(1 - 3) = 8 - 0 - 2 = 6.$$

For triangular matrices the determinant is simply the product of the diagonal entries. In practice, computers calculate determinants by elimination (Section 4.11), never by expansion, which becomes impossibly slow for large matrices.

Theorem 4.29 (Properties of determinants and inverses)**.** For square matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ of the same size:

(a) $\boldsymbol{A}$ is invertible if and only if $\det \boldsymbol{A} \neq 0$;
(b) $\det(\boldsymbol{AB}) = \det(\boldsymbol{A})\det(\boldsymbol{B})$ and $\det(\boldsymbol{A}^\top) = \det \boldsymbol{A}$;
(c) if $\boldsymbol{A}$ is invertible, $\det(\boldsymbol{A}^{-1}) = 1/\det \boldsymbol{A}$;
(d) if $\boldsymbol{A}$ and $\boldsymbol{B}$ are invertible, $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$ and $(\boldsymbol{A}^\top)^{-1} = (\boldsymbol{A}^{-1})^\top$.

The rule $\det(\boldsymbol{AB}) = \det \boldsymbol{A} \det \boldsymbol{B}$ is the area picture again: scaling areas by $\det \boldsymbol{B}$ and then by $\det \boldsymbol{A}$ scales them by the product. And to undo "first $\boldsymbol{B}$, then $\boldsymbol{A}$" you must first undo $\boldsymbol{A}$ and then undo $\boldsymbol{B}$, hence the reversed order in $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.

Common pitfall (There is no matrix division)**.** We never write $\frac{\boldsymbol{A}}{\boldsymbol{B}}$, because $\boldsymbol{AB}^{-1}$ and $\boldsymbol{B}^{-1}\boldsymbol{A}$ are generally different. Only square matrices can have inverses, and many square matrices (those with determinant 0) do not. Finally, even when an inverse exists, computing it explicitly is slow and can be numerically inaccurate; to solve $\boldsymbol{Ax} = \boldsymbol{b}$, software uses elimination instead (Section 4.11).

Why this matters for ML. Determinants appear in the formula for the multivariate Gaussian distribution (Chapter 9), where det of a covariance matrix measures the volume of a data cloud. They are also central to *normalising flows*, a family of generative models that transform simple random noise into complex data with invertible maps and must keep track of how volumes change.

Exercises for Section 4.10

Exercise 4.38. Find the determinant and the inverse of $\left(\begin{smallmatrix} 4 & 7 \\ 2 & 6 \end{smallmatrix}\right)$, and check your answer.

Solution on p. 248 →

Exercise 4.39. For which value of k is $\left(\begin{smallmatrix} k & 2 \\ 3 & 6 \end{smallmatrix}\right)$ singular? What does it do to the plane for that value?

Solution on p. 248 →

Exercise 4.40. Compute $\det \left(\begin{smallmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{smallmatrix}\right)$ by expansion along the first row, and compare with the product of the diagonal entries.

Solution on p. 248 →

Exercise 4.41. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be 2×2 matrices with $\det \boldsymbol{A} = 3$ and $\det \boldsymbol{B} = -2$. Find $\det(\boldsymbol{AB})$, $\det(\boldsymbol{A}^{-1})$ and $\det(2\boldsymbol{A})$.

Solution on p. 248 →

4.11 Systems of linear equations

4.11.1 From equations to matrices

In Section 1.8 we solved two equations in two unknowns. With matrices, any number of linear equations in any number of unknowns can be written as a single equation. The

system

$$\begin{cases} x_1 + 2x_2 + x_3 = 8 \\ 2x_1 - x_2 + 3x_3 = 9 \\ 3x_1 + x_2 - x_3 = 2 \end{cases} \quad \text{is} \quad \underbrace{\begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 3 \\ 3 & 1 & -1 \end{pmatrix}}_{\boldsymbol{A}} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_{\boldsymbol{x}} = \underbrace{\begin{pmatrix} 8 \\ 9 \\ 2 \end{pmatrix}}_{\boldsymbol{b}},$$

because each row of $\boldsymbol{Ax}$ is the left side of one equation. The matrix $\boldsymbol{A}$ holds the coefficients, $\boldsymbol{x}$ the unknowns and $\boldsymbol{b}$ the right-hand sides.

4.11.2 Two pictures of a linear system

Row picture. With two unknowns, each equation describes a line in the plane, and a solution is a point on all the lines at once. Two lines can meet in exactly one point, be parallel and never meet, or coincide (Figure 4.6). So a linear system has either exactly one solution, no solution, or infinitely many solutions; it can never have exactly two. The same holds in any dimension, where each equation describes a plane or a hyperplane.

Column picture. By the column view, $\boldsymbol{Ax} = \boldsymbol{b}$ asks: *which linear combination of the columns of* $\boldsymbol{A}$ *produces* $\boldsymbol{b}$*?* The unknowns are the amounts of each column in the recipe.

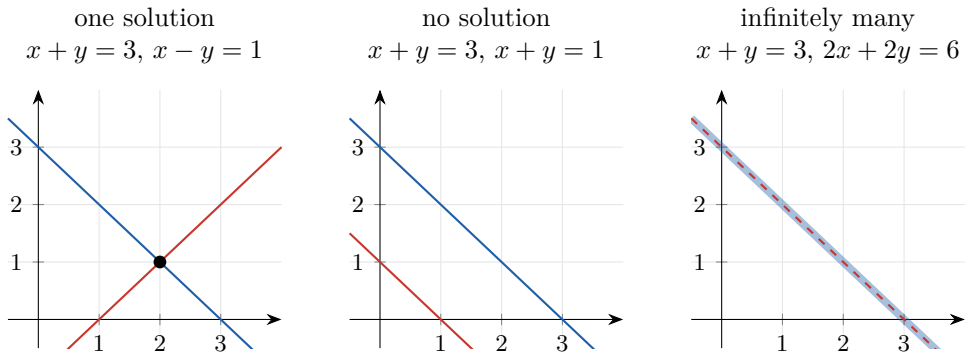


Figure 4.6: The row picture of three systems of two equations in two unknowns: lines that cross once, parallel lines, and two equations describing the same line.

4.11.3 Gaussian elimination

The standard method for solving linear systems, by hand or by computer, is **Gaussian elimination**. It is the elimination method of Section 1.8, organised systematically. We work with the **augmented matrix** $[\boldsymbol{A} \mid \boldsymbol{b}]$, which records the system without the variable names, and apply **elementary row operations**, none of which changes the solutions:

(i) swap two rows;
(ii) multiply a row by a non-zero number;
(iii) add a multiple of one row to another row.

The goal is a *triangular* ("row echelon") form in which each equation has fewer unknowns than the one above it; the last equation is then solved first and the others by *back substitution*.

Example 4.30 (Gaussian elimination)**.** Solve the 3×3 system above.

Solution. Start with the augmented matrix and eliminate x_1 from rows 2 and 3:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 2 & -1 & 3 & 9 \\ 3 & 1 & -1 & 2 \end{array}\right) \xrightarrow[R_3-3R_1]{R_2-2R_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -5 & 1 & -7 \\ 0 & -5 & -4 & -22 \end{array}\right).$$

Now eliminate x_2 from row 3:

$$\xrightarrow{R_3-R_2} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -5 & 1 & -7 \\ 0 & 0 & -5 & -15 \end{array}\right).$$

Back substitution: the last row says $-5x_3 = -15$, so $x_3 = 3$. The second row says $-5x_2 + x_3 = -7$, so $-5x_2 = -10$ and $x_2 = 2$. The first row says $x_1 + 2x_2 + x_3 = 8$, so $x_1 = 8 - 4 - 3 = 1$. The solution is $\boldsymbol{x} = (1, 2, 3)$.

Check in the original equations: $1 + 4 + 3 = 8$, $2 - 2 + 9 = 9$, $3 + 2 - 3 = 2$. ✓

If elimination produces a row $(0\ 0\ \cdots\ 0 \mid c)$ with $c \neq 0$, that row says $0 = c$, and the system has no solution. If it produces a row of zeros $(0\ \cdots\ 0 \mid 0)$, one equation was redundant, and there may be infinitely many solutions.

4.11.4 Solutions, inverses and determinants

For a square system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, the three cases are decided by the determinant:

Theorem 4.31 (Square systems)**.** If $\boldsymbol{A}$ is square and $\det \boldsymbol{A} \neq 0$, then $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ has exactly one solution for every $\boldsymbol{b}$, namely $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$. If $\det \boldsymbol{A} = 0$, the system has either no solution or infinitely many, depending on $\boldsymbol{b}$.

Indeed, multiplying $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ on the left by $\boldsymbol{A}^{-1}$ gives $\boldsymbol{A}^{-1}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$, i.e. $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$. This is a useful formula for thinking, even though (as noted above) software finds $\boldsymbol{x}$ by elimination rather than by computing $\boldsymbol{A}^{-1}$.

Why this matters for ML (More equations than unknowns)**.** In machine learning, systems are rarely square. Fitting a linear model $\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{w}$ to n data points with d features gives n equations $\boldsymbol{X}\boldsymbol{w} = \boldsymbol{y}$ in d unknowns, and usually n is much larger than d. Such an *overdetermined* system typically has no exact solution, because no line passes exactly through a noisy cloud of points. Instead we look for the $\boldsymbol{w}$ that makes $\boldsymbol{X}\boldsymbol{w}$ as close as possible to $\boldsymbol{y}$. This *least-squares* solution satisfies the square system

$$\boldsymbol{X}^\top \boldsymbol{X}\, \boldsymbol{w} = \boldsymbol{X}^\top \boldsymbol{y},$$

called the *normal equations*, which we derive geometrically in Chapter 5 and with calculus in Chapter 8.

Remark (Linear algebra in code)**.** In Python, the NumPy library provides all the operations of this chapter. Note that `@` is matrix multiplication, while `*` multiplies entry by entry.

```python
import numpy as np

A = np.array([[2, 1],
              [-1, 3],
              [0, 4]])          # a 3 x 2 matrix
x = np.array([3, 2])            # a vector in R^2
print(A @ x)                    # [8 3 8]
print(A.T)                      # the 2 x 3 transpose

B = np.array([[1, 2],
              [3, 4]])
print(np.linalg.det(B))         # -2.0 (up to rounding)
print(np.linalg.solve(B, np.array([5, 6])))   # [-4.   4.5]
```

Exercises for Section 4.11

Exercise 4.42. Write the system $2x - y = 1$, $x + 3y = 11$ in the form $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and solve it using the inverse of $\boldsymbol{A}$.

Solution on p. 248 →

Exercise 4.43. Solve by Gaussian elimination: $x_1 + x_2 + x_3 = 6$, $2x_1 + x_2 - x_3 = 1$, $x_1 - x_2 + 2x_3 = 5$.

Solution on p. 248 →

Exercise 4.44. How many solutions does each system have? (a) $x + 2y = 4$, $2x + 4y = 8$; (b) $x + 2y = 4$, $2x + 4y = 5$; (c) $x + 2y = 4$, $x - y = 1$.

Solution on p. 248 →

Exercise 4.45. We try to fit a line $y = w_1 x + w_0$ exactly through the three points $(1, 2)$, $(2, 3)$ and $(3, 5)$. Write the three equations in matrix form and show that they have no solution.

Solution on p. 248 →

4.12 Chapter summary

Key takeaways

- A vector is both an arrow and a list of numbers. Vectors are added and scaled component by component; linear combinations $c_1\boldsymbol{v}_1 + \cdots + c_k\boldsymbol{v}_k$ are the central construction.
- $\|\boldsymbol{v}\| = \sqrt{\boldsymbol{v} \cdot \boldsymbol{v}}$ is the length; $\boldsymbol{v}/\|\boldsymbol{v}\|$ is a unit vector; $\|\boldsymbol{u} - \boldsymbol{v}\|$ is a distance.
- The dot product $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{u}^\top \boldsymbol{v} = \sum_i u_i v_i = \|\boldsymbol{u}\|\|\boldsymbol{v}\| \cos\theta$ is a weighted sum and a measure of alignment; $\boldsymbol{u} \cdot \boldsymbol{v} = 0$ means orthogonal. Cosine similarity and neurons are dot products.
- An $m \times n$ matrix maps $\mathbb{R}^n$ to $\mathbb{R}^m$. $\boldsymbol{A}\boldsymbol{x}$ is a list of dot products (rows) and a combination of columns. The columns are the images of the standard basis vectors.
- $\boldsymbol{A}\boldsymbol{B}$ means "first $\boldsymbol{B}$, then $\boldsymbol{A}$"; entries are row · column; shapes must match; $\boldsymbol{A}\boldsymbol{B} \neq \boldsymbol{B}\boldsymbol{A}$ in general. Stacked linear layers collapse into one.

- $(\boldsymbol{AB})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$; $\boldsymbol{A}^\top \boldsymbol{A}$ is symmetric.
- $\det \boldsymbol{A}$ is the factor by which areas (volumes) change; $\boldsymbol{A}^{-1}$ exists exactly when $\det \boldsymbol{A} \neq 0$.
- $\boldsymbol{Ax} = \boldsymbol{b}$ has one, none or infinitely many solutions, found by Gaussian elimination. Machine learning systems are usually overdetermined and are solved by least squares.

Chapter 5

Linear Algebra II: Core Ideas for Machine Learning

Motivation. Chapter 4 gave us the vocabulary of linear algebra: vectors, matrices and the ways to combine them. This chapter is about the ideas that machine learning actually runs on. Which outputs can a linear model produce, and what happens when features are redundant (*span*, *independence*, *rank*)? How do we measure the size of a weight vector in order to penalise it (*norms*)? Why is fitting a line by least squares the same as dropping a perpendicular (*orthogonality* and *projections*)? Along which directions does a matrix simply stretch, and what does that say about data, training dynamics and curvature (*eigenvalues*)? How can a huge matrix be approximated by a small one (*singular value decomposition*)? And when does a quadratic function have a bowl shape with a unique minimum (*positive definite matrices*)? We finish by combining everything into *principal component analysis*, one of the most widely used tools in data science.

In this chapter you will learn to...

- describe the span of a set of vectors and the column space of a matrix;
- decide whether vectors are linearly independent, and recognise redundant features;
- find coordinates in a basis, and compute the rank of a matrix;
- use the L^1, L^2 and L^∞ norms and connect them to regularisation;
- work with orthonormal bases and orthogonal matrices, and apply Gram–Schmidt;
- project vectors onto lines and subspaces, and solve least-squares problems;
- compute eigenvalues and eigenvectors of 2×2 matrices and explain what they mean;
- describe the singular value decomposition and low-rank approximation;
- recognise positive definite matrices and the shapes of quadratic forms;
- carry out principal component analysis on a small data set.

5.1 Span and column space

Definition 5.1 (Span). The **span** of vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ is the set of all their linear combinations:

$$\operatorname{span}\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\} = \{c_1\boldsymbol{v}_1 + \cdots + c_k\boldsymbol{v}_k : c_1, \dots, c_k \in \mathbb{R}\}.$$

The span is everything you can reach using only the given directions, going forwards or backwards along each by any amount.

- The span of a single non-zero vector is the line through the origin in its direction: all multiples $c\boldsymbol{v}$.

- The span of two vectors in $\mathbb{R}^2$ that do not lie on a common line is the whole plane $\mathbb{R}^2$. (We saw in Section 4.2 that $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ reach every vector.)
- The span of two such vectors in $\mathbb{R}^3$ is a plane through the origin (Figure 5.1); a third vector outside that plane would extend the span to all of $\mathbb{R}^3$.
- The span of any vectors always contains the zero vector (take all coefficients equal to 0).

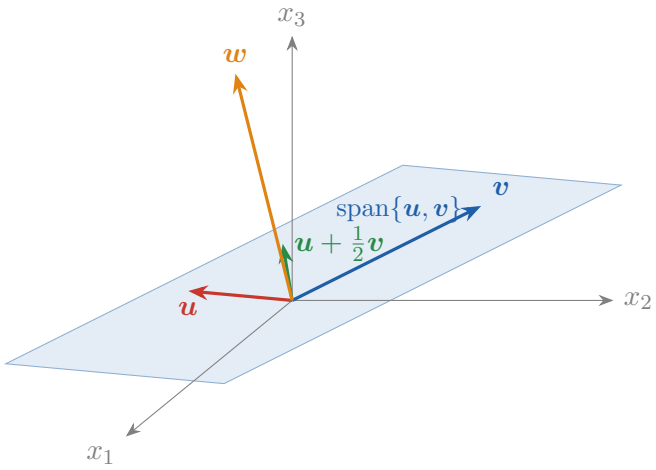


Figure 5.1: Two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\mathbb{R}^3$ span a plane through the origin; every linear combination such as $\boldsymbol{u} + \frac{1}{2}\boldsymbol{v}$ lies in it. The vector $\boldsymbol{w}$ sticks out of the plane, so it is not in the span.

Example 5.2 (Is a vector in the span?)**.** Is $\boldsymbol{b} = (5, 1, 4)$ in the span of $\boldsymbol{v}_1 = (1, 0, 1)$ and $\boldsymbol{v}_2 = (2, 1, 1)$? What about $\boldsymbol{b}' = (5, 1, 3)$?

Solution. We need a and b with $a(1, 0, 1) + b(2, 1, 1) = (5, 1, 4)$, i.e. $a + 2b = 5$, $b = 1$, $a + b = 4$. The second equation gives $b = 1$, the first then gives $a = 3$, and the third is satisfied: $3 + 1 = 4$. So $\boldsymbol{b} = 3\boldsymbol{v}_1 + \boldsymbol{v}_2$ is in the span. For $\boldsymbol{b}'$ the same reasoning gives $a = 3$, $b = 1$, but the third equation would need $a + b = 3$, which fails. So $\boldsymbol{b}'$ is not in the span.

Definition 5.3 (Column space)**.** The **column space** $C(\boldsymbol{A})$ of a matrix $\boldsymbol{A}$ is the span of its columns. By the column view of matrix–vector multiplication, it is exactly the set of all possible outputs $\boldsymbol{Ax}$. Consequently,

$$\boldsymbol{Ax} = \boldsymbol{b} \text{ has a solution } \iff \boldsymbol{b} \in C(\boldsymbol{A}).$$

A set of vectors that contains $\mathbf{0}$ and is closed under addition and scalar multiplication (adding or scaling vectors in the set never leads outside it) is called a **subspace**. Every span is a subspace: lines and planes through the origin are the typical examples.

Why this matters for ML (What a linear model can produce)**.** A linear model without bias predicts $\hat{\boldsymbol{y}} = \boldsymbol{Xw}$ for the whole training set. Whatever weights we choose, the vector of predictions lies in the column space of the data matrix $\boldsymbol{X}$. With n data points and d features, this is (at most) a d-dimensional subspace of $\mathbb{R}^n$. When $n > d$, the vector of true targets $\boldsymbol{y}$ almost never lies in it, so the model cannot fit the data

exactly. The best it can do is to find the point of $C(\boldsymbol{X})$ closest to $\boldsymbol{y}$: that is least squares (Section 5.6).

Exercises for Section 5.1

Exercise 5.1. Describe geometrically: (a) $\text{span}\{(1,2)\}$; (b) $\text{span}\{(1,2),(2,4)\}$; (c) $\text{span}\{(1,2),(0,1)\}$.

Solution on p. 249 →

Exercise 5.2. Is $(3,5,1)$ in $\text{span}\{(1,1,0),(1,2,1)\}$? Is $(3,4,1)$?

Solution on p. 249 →

Exercise 5.3. A linear model without bias, $\hat{y} = w_1 x_1 + w_2 x_2$, is fitted to three data points. In which set must the vector of the three predictions lie? Can the model fit every possible vector of three targets exactly?

Solution on p. 249 →

5.2 Linear independence

In Figure 5.1, adding $\boldsymbol{u} + \frac{1}{2}\boldsymbol{v}$ to the set $\{\boldsymbol{u}, \boldsymbol{v}\}$ would not enlarge the span: the new vector is redundant. Linear independence formalises "no redundancy".

Definition 5.4 (Linear independence)**.** Vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ are **linearly independent** if the only way to make

$$c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_k\boldsymbol{v}_k = \boldsymbol{0}$$

is with $c_1 = c_2 = \cdots = c_k = 0$. Otherwise they are **linearly dependent**.

If the vectors are dependent, some combination with a non-zero coefficient, say $c_1 \neq 0$, gives $\boldsymbol{0}$; then $\boldsymbol{v}_1 = -\frac{1}{c_1}(c_2\boldsymbol{v}_2 + \cdots + c_k\boldsymbol{v}_k)$ is a combination of the others. So: *vectors are dependent exactly when at least one of them is a linear combination of the rest*, i.e. redundant.

Example 5.5 (Testing independence)**.** (a) $(1,2)$ and $(2,4)$ are dependent, because $2(1,2) - (2,4) = \boldsymbol{0}$.

(b) $(1,0)$ and $(1,1)$ are independent: $c_1(1,0) + c_2(1,1) = (c_1 + c_2, c_2) = (0,0)$ forces $c_2 = 0$ and then $c_1 = 0$.

(c) $(1,0,0)$, $(0,1,0)$ and $(1,1,0)$ are dependent, because the third is the sum of the first two. All three lie in the plane $x_3 = 0$.

Theorem 5.6 (Practical tests for independence)**.** Let $\boldsymbol{A}$ be a matrix and let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ be vectors in $\mathbb{R}^n$.

(a) The columns of a matrix $\boldsymbol{A}$ are independent exactly when $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ has only the solution $\boldsymbol{x} = \boldsymbol{0}$.

(b) n vectors in $\mathbb{R}^n$ are independent exactly when the $n \times n$ matrix having them as columns has non-zero determinant.

(c) More than n vectors in $\mathbb{R}^n$ are always dependent. A set containing the zero vector is always dependent.

The first test is just the definition written with matrices; it can be checked by Gaussian elimination. The second combines it with Theorem 4.31. The third says that $\mathbb{R}^n$ has room for at most n independent directions.

Why this matters for ML (Redundant features). When some feature columns of a data matrix are linear combinations of others, the columns are dependent and the features are called *collinear*. Examples: the price in euros and the same price in dollars; or the total floor area together with the areas of each room, which add up to it. Then $\boldsymbol{X}\boldsymbol{w} = \boldsymbol{X}\boldsymbol{w}'$ for different weight vectors (move weight from one redundant feature to another), so the "best" weights are not unique, the matrix $\boldsymbol{X}^\top\boldsymbol{X}$ in the normal equations is singular, and numerical solvers become unstable. A classic trap is *one-hot encoding* combined with a constant column of ones for the bias: the one-hot columns always add up to the column of ones. This is known as the *dummy-variable trap*, and the usual fix is to drop one of the one-hot columns.

Exercises for Section 5.2

Exercise 5.4. Decide whether each set is linearly independent: (a) $(1,3),(2,6)$; (b) $(1,0,1),(0,1,1),(1,1,0)$; (c) $(1,2),(3,4),(5,6)$.

Solution on p. 249 →

Exercise 5.5. For which values of k are $(1,k)$ and $(k,4)$ linearly dependent?

Solution on p. 249 →

Exercise 5.6. A data set has a column of ones (for the bias) and three one-hot columns for the categories red, green, blue. Show that the four columns are linearly dependent.

Solution on p. 249 →

5.3 Basis, dimension and rank

5.3.1 Bases and coordinates

Definition 5.7 (Basis). A **basis** of a subspace S is a set of linearly independent vectors that spans S.

A basis is a minimal set of directions that reaches everything: spanning means "enough vectors", independence means "no redundant vectors". The standard basis $\boldsymbol{e}_1,\dots,\boldsymbol{e}_n$ is a basis of $\mathbb{R}^n$, but there are infinitely many others.

Theorem 5.8 (Unique coordinates). If $\boldsymbol{b}_1,\dots,\boldsymbol{b}_k$ is a basis of S, then every vector $\boldsymbol{v} \in S$ can be written as $\boldsymbol{v} = c_1\boldsymbol{b}_1 + \cdots + c_k\boldsymbol{b}_k$ in *exactly one* way. The numbers $c_1,\dots,c_k$ are the **coordinates** of $\boldsymbol{v}$ in this basis.

Proof. Spanning guarantees at least one way. If there were two, $\sum c_i\boldsymbol{b}_i = \sum c'_i\boldsymbol{b}_i$, then $\sum(c_i - c'_i)\boldsymbol{b}_i = \boldsymbol{0}$, and independence forces $c_i = c'_i$ for every i. ∎

Example 5.9 (Coordinates in a non-standard basis)**.** Find the coordinates of $\boldsymbol{v} = (4, 5)$ in the basis $\boldsymbol{b}_1 = (2, 1)$, $\boldsymbol{b}_2 = (1, 2)$.

Solution. Solve $a(2, 1) + b(1, 2) = (4, 5)$: $2a + b = 4$ and $a + 2b = 5$. Then $a = 1$ and $b = 2$, so $\boldsymbol{v} = \boldsymbol{b}_1 + 2\boldsymbol{b}_2$ and its coordinates in this basis are $(1, 2)$ (Figure 5.2).

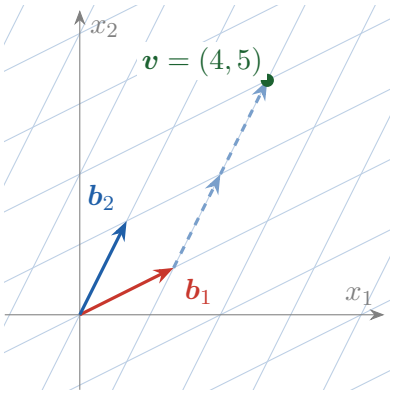


Figure 5.2: The grid of the basis $\boldsymbol{b}_1 = (2, 1)$, $\boldsymbol{b}_2 = (1, 2)$. To reach $\boldsymbol{v} = (4, 5)$, take one step of $\boldsymbol{b}_1$ and two steps of $\boldsymbol{b}_2$: in this basis $\boldsymbol{v}$ has coordinates $(1, 2)$.

Intuition (A basis is a coordinate system)**.** A vector is a geometric object; its coordinates are a *description* of it, and the description depends on the basis, just as the same distance is described by different numbers in metres and in feet. Changing the basis does not move the vector; it changes the grid we measure it against. Much of linear algebra, and much of machine learning, is about finding a basis in which data or a matrix looks simple. Eigenvectors (Section 5.7) and principal components (Section 5.10) are such bases.

5.3.2 Dimension

Every basis of a given subspace has the same number of vectors; this number is the **dimension** of the subspace. So $\mathbb{R}^n$ has dimension n, a line through the origin has dimension 1, and a plane through the origin has dimension 2, whatever space they sit in.

5.3.3 Rank

Definition 5.10 (Rank)**.** The **rank** of a matrix $\boldsymbol{A}$, written $\operatorname{rank} \boldsymbol{A}$, is the dimension of its column space, i.e. the maximum number of linearly independent columns.

A remarkable fact is that the maximum number of independent *rows* is always the same as the maximum number of independent columns. So the rank of an $m \times n$ matrix is at most $\min(m, n)$; a matrix reaching this bound is said to have *full rank*. Gaussian elimination computes the rank: it is the number of non-zero rows left in the triangular form.

Example 5.11 (Computing a rank)**.** Find the rank of $\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$.

Solution. The second row is twice the first, so elimination $(R_2 - 2R_1)$ turns it into a row of zeros, while the first and third rows are not multiples of each other. Two independent rows remain, so $\operatorname{rank} \boldsymbol{A} = 2$. Correspondingly, the three columns span

only a plane in $\mathbb{R}^3$: indeed the third column is the sum of the first two.

5.3.4 The null space and the rank–nullity theorem

The **null space** of $\boldsymbol{A}$ is the set of all $\boldsymbol{x}$ with $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$: the inputs that the transformation squashes to zero. For $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right)$, the null space is the line spanned by $(-2, 1)$.

Theorem 5.12 (Rank–nullity theorem)**.** For any $m \times n$ matrix $\boldsymbol{A}$,

$$\operatorname{rank} \boldsymbol{A} + \dim(\text{null space of } \boldsymbol{A}) = n.$$

In words: of the n input dimensions, some survive (they make up the column space) and the rest are squashed to zero (the null space). For the example just given, $1 + 1 = 2$.

5.3.5 Low-rank matrices

A matrix of rank 1 has all its columns on one line, so it is an outer product $\boldsymbol{u}\boldsymbol{v}^\top$ (Section 4.9). More generally, a matrix of rank r can be written as a sum of r outer products. Storing an $m \times n$ matrix needs mn numbers, but storing r outer products needs only $r(m + n)$, which is much less when r is small. This is the idea behind the low-rank approximations of Section 5.8.

Why this matters for ML (Low rank in modern deep learning)**.** Fine-tuning a large language model updates enormous weight matrices. The popular method *LoRA* (low-rank adaptation) keeps the original weights $\boldsymbol{W}$ frozen and learns a correction of low rank, $\boldsymbol{W} + \boldsymbol{B}\boldsymbol{A}$, where $\boldsymbol{B}$ is $m \times r$ and $\boldsymbol{A}$ is $r \times n$ with a small r such as 8. For a 4096×4096 matrix this means training $2 \times 4096 \times 8 = 65\,536$ numbers instead of about 16.8 million. More broadly, real data often has much lower "intrinsic dimension" than the number of features suggests: the set of natural images is a tiny, highly structured part of $\mathbb{R}^{784}$.

Exercises for Section 5.3

Exercise 5.7. Find the coordinates of $(1, 5)$ in the basis $(1, 1)$, $(-1, 1)$.

Solution on p. 249 →

Exercise 5.8. Find the rank of (a) $\left(\begin{smallmatrix} 1 & 2 \\ 3 & 6 \end{smallmatrix}\right)$; (b) $\left(\begin{smallmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{smallmatrix}\right)$; (c) $\left(\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{smallmatrix}\right)$.

Solution on p. 249 →

Exercise 5.9. Find a vector spanning the null space of $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right)$ and check the rank–nullity theorem.

Solution on p. 249 →

Exercise 5.10. A 1000×1000 matrix has rank 10. How many numbers are needed to store it as a sum of outer products, compared with storing all its entries?

Solution on p. 249 →

5.4 Norms: measuring size

The length $\|\boldsymbol{v}\|$ of Definition 4.4 is not the only sensible way to measure the size of a vector. Any function that behaves like a length is called a norm.

Definition 5.13 (Norm). A **norm** on $\mathbb{R}^n$ is a function $\boldsymbol{v} \mapsto \|\boldsymbol{v}\|$ such that for all vectors $\boldsymbol{u}, \boldsymbol{v}$ and scalars c:

(a) $\|\boldsymbol{v}\| \geq 0$, and $\|\boldsymbol{v}\| = 0$ only for $\boldsymbol{v} = \boldsymbol{0}$;
(b) $\|c\boldsymbol{v}\| = |c|\,\|\boldsymbol{v}\|$;
(c) $\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$ (triangle inequality).

Definition 5.14 (The L^p norms). For $\boldsymbol{v} \in \mathbb{R}^n$:

$$\begin{aligned} \|\boldsymbol{v}\|_1 &= |v_1| + \cdots + |v_n| && (L^1 \text{ or Manhattan norm}),\\ \|\boldsymbol{v}\|_2 &= \sqrt{v_1^2 + \cdots + v_n^2} && (L^2 \text{ or Euclidean norm}),\\ \|\boldsymbol{v}\|_\infty &= \max(|v_1|, \ldots, |v_n|) && (L^\infty \text{ or maximum norm}), \end{aligned}$$

and in general $\|\boldsymbol{v}\|_p = \left(\sum_i |v_i|^p\right)^{1/p}$ for $p \geq 1$.

For $\boldsymbol{v} = (3, -4, 0)$: $\|\boldsymbol{v}\|_1 = 7$, $\|\boldsymbol{v}\|_2 = 5$ and $\|\boldsymbol{v}\|_\infty = 4$. When no subscript is written, $\|\boldsymbol{v}\|$ means the Euclidean norm $\|\boldsymbol{v}\|_2$.

The shape of the *unit ball* $\{\boldsymbol{v} : \|\boldsymbol{v}\| \leq 1\}$ shows the character of each norm (Figure 5.3): a diamond for L^1, a disc for L^2 and a square for L^∞.

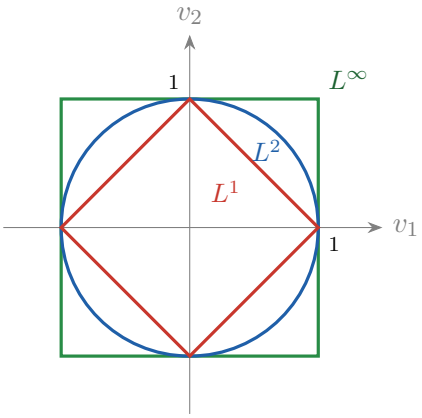


Figure 5.3: Unit balls in $\mathbb{R}^2$: the points with $\|\boldsymbol{v}\|_1 \leq 1$ (red diamond), $\|\boldsymbol{v}\|_2 \leq 1$ (blue disc) and $\|\boldsymbol{v}\|_\infty \leq 1$ (green square).

Remark (Counting non-zeros). The number of non-zero entries of $\boldsymbol{v}$ is sometimes called the "L^0 norm" $\|\boldsymbol{v}\|_0$, although it is not a norm (it fails rule 2). It measures *sparsity*: a vector with few non-zero entries is called **sparse**.

5.4.1 Matrix norms

The simplest way to measure the size of a matrix is to treat its entries as one long vector. This gives the **Frobenius norm**

$$\|\boldsymbol{A}\|_F = \sqrt{\sum_{i,j} a_{ij}^2}.$$

Another important measure is the largest factor by which $\boldsymbol{A}$ can stretch a vector,

$$\max_{\boldsymbol{x}\neq\boldsymbol{0}} \frac{\|\boldsymbol{A}\boldsymbol{x}\|}{\|\boldsymbol{x}\|},$$

called the *spectral norm*; we will see in Section 5.8 that it equals the largest singular value.

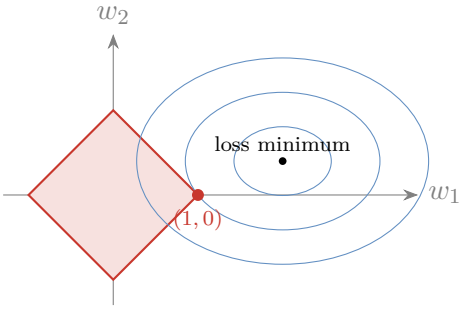
5.4.2 Norms in machine learning: regularisation

Why this matters for ML (Regularisation and other uses of norms)**.** Models with very large weights tend to fit noise in the training data (*overfitting*, Chapter 10). A standard remedy is **regularisation**: add a penalty on the size of the weights to the loss,

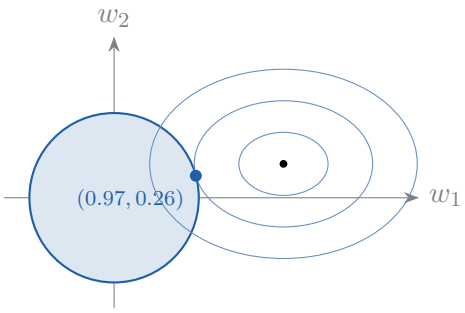
$$\text{loss}(\boldsymbol{w}) + \lambda\|\boldsymbol{w}\|_2^2 \quad (\textit{ridge or weight decay}), \qquad \text{loss}(\boldsymbol{w}) + \lambda\|\boldsymbol{w}\|_1 \quad (\textit{lasso}),$$

where $\lambda > 0$ controls the strength of the penalty. The two penalties behave very differently: the L^1 penalty tends to make many weights *exactly zero* (a sparse model that ignores irrelevant features), while the L^2 penalty shrinks all weights a little but rarely to zero. Figure 5.4 explains why: the corners of the L^1 diamond stick out, and they lie on the axes.

Norms appear in many other places: *gradient clipping* rescales the gradient $\boldsymbol{g}$ to $c\,\boldsymbol{g}/\|\boldsymbol{g}\|$ whenever $\|\boldsymbol{g}\| > c$, and *normalisation layers* rescale activation vectors to a standard size.



(a) L^1 constraint: the best point is a corner, $w_2 = 0$.



(b) L^2 constraint: the best point has both $w_1, w_2 \neq 0$.

Figure 5.4: Why L^1 regularisation produces sparse weights. The ellipses are contour lines of a loss with its minimum at $(2, 0.4)$. Restricting the weights to the unit ball of a norm, the best allowed point is where the smallest contour touches the ball. The diamond is usually touched first at a corner, where one weight is exactly zero; the disc is touched at a generic point.

Exercises for Section 5.4

Exercise 5.11. Compute the L^1, L^2 and L^∞ norms of $(1, -2, 2)$ and of $(-3, 0, 4, 0)$.

Solution on p. 249 →

Exercise 5.12. Which of the points $(0.6, 0.6)$, $(0.9, 0.2)$ and $(0.5, -0.5)$ lie in the unit ball of each of the three norms?

Solution on p. 249 →

Exercise 5.13. Compute the Frobenius norm of $\left(\begin{smallmatrix} 1 & 2 \\ -2 & 4 \end{smallmatrix}\right)$.

Solution on p. 249 →

Exercise 5.14. Gradient clipping with threshold $c = 1$ replaces $\boldsymbol{g}$ by $\boldsymbol{g}/\|\boldsymbol{g}\|$ whenever $\|\boldsymbol{g}\| > 1$. What happens to $\boldsymbol{g} = (3, 4)$ and to $\boldsymbol{g} = (0.3, 0.4)$?

Solution on p. 249 →

Exercise 5.15. For $\boldsymbol{w} = (2, -1, 0.5)$ and $\lambda = 0.1$, compute the ridge penalty $\lambda\|\boldsymbol{w}\|_2^2$ and the lasso penalty $\lambda\|\boldsymbol{w}\|_1$.

Solution on p. 249 →

5.5 Orthogonality

Definition 5.15 (Orthonormal vectors)**.** Vectors $\boldsymbol{q}_1, \dots, \boldsymbol{q}_k$ are **orthonormal** if they are pairwise orthogonal and all have length 1:

$$\boldsymbol{q}_i \cdot \boldsymbol{q}_j = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

The standard basis is orthonormal, and so are, for example, $\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$ and $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$, which are the standard basis rotated by 45°.

Proposition 5.16 (Orthogonal vectors are independent)**.** Non-zero, pairwise orthogonal vectors are linearly independent.

Proof. Suppose $c_1\boldsymbol{q}_1 + \cdots + c_k\boldsymbol{q}_k = \boldsymbol{0}$. Take the dot product of both sides with $\boldsymbol{q}_j$: all terms with $i \neq j$ vanish, leaving $c_j\, \boldsymbol{q}_j \cdot \boldsymbol{q}_j = 0$. Since $\boldsymbol{q}_j \cdot \boldsymbol{q}_j = \|\boldsymbol{q}_j\|^2 > 0$, we get $c_j = 0$, for every j. ∎

5.5.1 Coordinates in an orthonormal basis are easy

In a general basis, finding coordinates means solving a linear system (Example 5.9). In an orthonormal basis $\boldsymbol{q}_1, \dots, \boldsymbol{q}_n$ it is just a dot product: if $\boldsymbol{v} = c_1\boldsymbol{q}_1 + \cdots + c_n\boldsymbol{q}_n$, taking the dot product with $\boldsymbol{q}_j$ gives $c_j = \boldsymbol{v} \cdot \boldsymbol{q}_j$. So

$$\boxed{\boldsymbol{v} = (\boldsymbol{v} \cdot \boldsymbol{q}_1)\, \boldsymbol{q}_1 + (\boldsymbol{v} \cdot \boldsymbol{q}_2)\, \boldsymbol{q}_2 + \cdots + (\boldsymbol{v} \cdot \boldsymbol{q}_n)\, \boldsymbol{q}_n}$$

Each coordinate is the scalar projection of $\boldsymbol{v}$ onto one of the basis directions (Figure 4.3). For $\boldsymbol{v} = (5, 1)$ and the rotated basis above: $\boldsymbol{v} \cdot \boldsymbol{q}_1 = \frac{6}{\sqrt{2}} = 3\sqrt{2}$ and $\boldsymbol{v} \cdot \boldsymbol{q}_2 = \frac{4}{\sqrt{2}} = 2\sqrt{2}$, and indeed $3\sqrt{2}\,\boldsymbol{q}_1 + 2\sqrt{2}\,\boldsymbol{q}_2 = (3, 3) + (2, -2) = (5, 1)$.

5.5.2 Orthogonal matrices

Definition 5.17 (Orthogonal matrix)**.** A square matrix $\boldsymbol{Q}$ whose columns are orthonormal is called an **orthogonal matrix**. Equivalently, $\boldsymbol{Q}^{\top}\boldsymbol{Q} = \boldsymbol{I}$, which means

$$\boldsymbol{Q}^{-1} = \boldsymbol{Q}^{\top}.$$

The equivalence is just the definition of matrix multiplication: the (i, j) entry of $\boldsymbol{Q}^{\top}\boldsymbol{Q}$ is $\boldsymbol{q}_i \cdot \boldsymbol{q}_j$. Orthogonal matrices are wonderful to work with, because inverting them costs nothing.

Theorem 5.18 (Orthogonal matrices preserve lengths and angles)**.** If $\boldsymbol{Q}$ is orthogonal, then $\|\boldsymbol{Q}\boldsymbol{x}\| = \|\boldsymbol{x}\|$ and $(\boldsymbol{Q}\boldsymbol{x}) \cdot (\boldsymbol{Q}\boldsymbol{y}) = \boldsymbol{x} \cdot \boldsymbol{y}$ for all $\boldsymbol{x}, \boldsymbol{y}$. Also $\det \boldsymbol{Q} = \pm 1$.

Proof. $(\boldsymbol{Q}\boldsymbol{x}) \cdot (\boldsymbol{Q}\boldsymbol{y}) = (\boldsymbol{Q}\boldsymbol{x})^{\top}(\boldsymbol{Q}\boldsymbol{y}) = \boldsymbol{x}^{\top}\boldsymbol{Q}^{\top}\boldsymbol{Q}\boldsymbol{y} = \boldsymbol{x}^{\top}\boldsymbol{y}$; taking $\boldsymbol{y} = \boldsymbol{x}$ gives the statement about lengths. Finally, $1 = \det(\boldsymbol{Q}^{\top}\boldsymbol{Q}) = \det(\boldsymbol{Q})^2$. ∎

Geometrically, orthogonal matrices are the *rigid motions* that keep the origin fixed: rotations ($\det = 1$) and reflections ($\det = -1$). The rotation matrices $\boldsymbol{R}_{\alpha}$ of Chapter 4 are the main example.

5.5.3 The Gram–Schmidt process

Any set of independent vectors can be turned into an orthonormal set with the same span. The idea: take the vectors one at a time, subtract from each its components along the directions already chosen, and normalise what is left.

Algorithm 5.19 (Gram–Schmidt)**.** Given independent vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$:

Step 1. Set $\boldsymbol{q}_1 = \boldsymbol{v}_1 / \|\boldsymbol{v}_1\|$.

Step 2. For $j = 2, \dots, k$: remove the components along the previous directions, $\boldsymbol{w}_j = \boldsymbol{v}_j - \sum_{i<j} (\boldsymbol{v}_j \cdot \boldsymbol{q}_i)\, \boldsymbol{q}_i$, and normalise, $\boldsymbol{q}_j = \boldsymbol{w}_j / \|\boldsymbol{w}_j\|$.

Example 5.20 (Gram–Schmidt in the plane)**.** Orthonormalise $\boldsymbol{v}_1 = (3, 4)$, $\boldsymbol{v}_2 = (1, 0)$.

Solution. $\boldsymbol{q}_1 = \frac{1}{5}(3, 4) = (0.6, 0.8)$. Next, $\boldsymbol{v}_2 \cdot \boldsymbol{q}_1 = 0.6$, so $\boldsymbol{w}_2 = (1, 0) - 0.6\,(0.6, 0.8) = (0.64, -0.48)$, whose length is 0.8; hence $\boldsymbol{q}_2 = (0.8, -0.6)$. *Check:* $\boldsymbol{q}_1 \cdot \boldsymbol{q}_2 = 0.48 - 0.48 = 0$. ✓

Applied to the columns of a matrix $\boldsymbol{A}$, Gram–Schmidt produces the factorisation $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$, with $\boldsymbol{Q}$ having orthonormal columns and $\boldsymbol{R}$ upper triangular. Numerical software uses this *QR decomposition* to solve least-squares problems accurately.

Why this matters for ML. Orthogonal matrices neither stretch nor shrink vectors, so a signal multiplied by many of them neither explodes nor vanishes. This motivates *orthogonal initialisation* of the weights of deep and recurrent networks. Orthonormal bases are also what make principal component analysis (Section 5.10) convenient: the new coordinates of a data point are just dot products.

Exercises for Section 5.5

Exercise 5.16. Show that $\boldsymbol{q}_1 = \frac{1}{3}(1,2,2)$, $\boldsymbol{q}_2 = \frac{1}{3}(2,1,-2)$ and $\boldsymbol{q}_3 = \frac{1}{3}(2,-2,1)$ form an orthonormal basis of $\mathbb{R}^3$, and find the coordinates of $\boldsymbol{v} = (3,0,3)$ in this basis.

Solution on p. 250 →

Exercise 5.17. Apply Gram–Schmidt to $\boldsymbol{v}_1 = (1,1,0)$ and $\boldsymbol{v}_2 = (1,0,1)$.

Solution on p. 250 →

Exercise 5.18. Verify that the rotation matrix $\boldsymbol{R}_\alpha$ satisfies $\boldsymbol{R}_\alpha^\top \boldsymbol{R}_\alpha = \boldsymbol{I}$. What is $\boldsymbol{R}_\alpha^{-1}$?

Solution on p. 250 →

5.6 Projections and least squares

Motivation. Linear regression, the most basic model of machine learning, asks for the weights that make $\boldsymbol{X}\boldsymbol{w}$ as close as possible to the targets $\boldsymbol{y}$. This section shows that the answer is a *projection*: drop a perpendicular from $\boldsymbol{y}$ onto the column space of $\boldsymbol{X}$. The geometry gives the famous normal equations almost for free.

5.6.1 Projecting onto a line

Given a vector $\boldsymbol{b}$ and a line through the origin in the direction $\boldsymbol{a}$, which point $\boldsymbol{p}$ of the line is closest to $\boldsymbol{b}$? The picture (Figure 5.5) suggests the answer: the point where the error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ is perpendicular to the line.

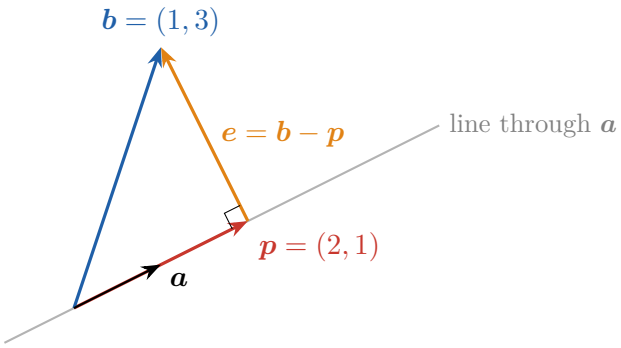


Figure 5.5: Projecting $\boldsymbol{b} = (1,3)$ onto the line spanned by $\boldsymbol{a}$. The projection $\boldsymbol{p}$ is the closest point of the line to $\boldsymbol{b}$; the error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ is perpendicular to the line.

Theorem 5.21 (Projection onto a line)**.** The projection of $\boldsymbol{b}$ onto the line spanned by $\boldsymbol{a} \neq \boldsymbol{0}$ is

$$\operatorname{proj}_{\boldsymbol{a}}(\boldsymbol{b}) = \frac{\boldsymbol{a}\cdot\boldsymbol{b}}{\boldsymbol{a}\cdot\boldsymbol{a}}\,\boldsymbol{a}$$

It is the point of the line closest to $\boldsymbol{b}$, and the error $\boldsymbol{b} - \operatorname{proj}_{\boldsymbol{a}}(\boldsymbol{b})$ is orthogonal to $\boldsymbol{a}$.

Proof. Points of the line are $t\boldsymbol{a}$. Requiring the error to be perpendicular to $\boldsymbol{a}$: $\boldsymbol{a}\cdot(\boldsymbol{b}-t\boldsymbol{a}) = 0$, i.e. $\boldsymbol{a}\cdot\boldsymbol{b} - t\,\boldsymbol{a}\cdot\boldsymbol{a} = 0$, so $t = \frac{\boldsymbol{a}\cdot\boldsymbol{b}}{\boldsymbol{a}\cdot\boldsymbol{a}}$. To see that this point is the closest, take any other point

$\boldsymbol{q}$ on the line. The vectors $\boldsymbol{b}-\boldsymbol{p}$ and $\boldsymbol{p}-\boldsymbol{q}$ are perpendicular (the second lies along the line), so by Pythagoras $\|\boldsymbol{b}-\boldsymbol{q}\|^2 = \|\boldsymbol{b}-\boldsymbol{p}\|^2 + \|\boldsymbol{p}-\boldsymbol{q}\|^2 \geq \|\boldsymbol{b}-\boldsymbol{p}\|^2$. ■

Example 5.22 (Projecting onto a line). Project $\boldsymbol{b} = (1,3)$ onto the line spanned by $\boldsymbol{a} = (2,1)$.

Solution. $\boldsymbol{a}\cdot\boldsymbol{b} = 2+3 = 5$ and $\boldsymbol{a}\cdot\boldsymbol{a} = 5$, so $\boldsymbol{p} = 1\cdot\boldsymbol{a} = (2,1)$. The error is $\boldsymbol{e} = (1,3)-(2,1) = (-1,2)$, and indeed $\boldsymbol{e}\cdot\boldsymbol{a} = -2+2 = 0$. The distance from $\boldsymbol{b}$ to the line is $\|\boldsymbol{e}\| = \sqrt{5}$.

In matrix form, $\operatorname{proj}_{\boldsymbol{a}}(\boldsymbol{b}) = \boldsymbol{P}\boldsymbol{b}$ with the **projection matrix** $\boldsymbol{P} = \frac{\boldsymbol{a}\boldsymbol{a}^\top}{\boldsymbol{a}^\top\boldsymbol{a}}$. Projecting twice changes nothing, so $\boldsymbol{P}^2 = \boldsymbol{P}$.

5.6.2 Projecting onto a subspace

Now project $\boldsymbol{b} \in \mathbb{R}^n$ onto the column space of an $n \times d$ matrix $\boldsymbol{A}$ with independent columns. The projection has the form $\boldsymbol{p} = \boldsymbol{A}\hat{\boldsymbol{x}}$ for some coefficient vector $\hat{\boldsymbol{x}}$. As before, the error $\boldsymbol{b}-\boldsymbol{A}\hat{\boldsymbol{x}}$ must be perpendicular to the whole subspace, i.e. to every column of $\boldsymbol{A}$. Collecting these d conditions "column $\cdot$ error $= 0$" into one matrix equation:

$$\boldsymbol{A}^\top(\boldsymbol{b}-\boldsymbol{A}\hat{\boldsymbol{x}}) = \boldsymbol{0}.$$

Theorem 5.23 (Normal equations). The vector $\hat{\boldsymbol{x}}$ that minimises $\|\boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\|$ satisfies the **normal equations**

$$\boxed{\boldsymbol{A}^\top\boldsymbol{A}\,\hat{\boldsymbol{x}} = \boldsymbol{A}^\top\boldsymbol{b}}$$

If the columns of $\boldsymbol{A}$ are independent, $\boldsymbol{A}^\top\boldsymbol{A}$ is invertible and $\hat{\boldsymbol{x}} = (\boldsymbol{A}^\top\boldsymbol{A})^{-1}\boldsymbol{A}^\top\boldsymbol{b}$.

(The name comes from "normal" in the sense of perpendicular.) The minimiser is called the **least-squares** solution, because $\|\boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\|^2$ is a sum of squared errors.

5.6.3 Linear regression as a projection

Example 5.24 (Fitting a line by least squares). In the last exercise of Chapter 4 we found that no line passes through $(1,2)$, $(2,3)$ and $(3,5)$. Find the line $y = w_1x + w_0$ that fits them best in the least-squares sense.

Solution. The equations $w_1x_i + w_0 = y_i$ are $\boldsymbol{X}\boldsymbol{w} = \boldsymbol{y}$ with

$$\boldsymbol{X} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{pmatrix}, \qquad \boldsymbol{w} = \begin{pmatrix} w_1 \\ w_0 \end{pmatrix}, \qquad \boldsymbol{y} = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}.$$

(The column of ones multiplies the intercept w_0.) Then

$$\boldsymbol{X}^\top\boldsymbol{X} = \begin{pmatrix} 14 & 6 \\ 6 & 3 \end{pmatrix}, \qquad \boldsymbol{X}^\top\boldsymbol{y} = \begin{pmatrix} 23 \\ 10 \end{pmatrix}.$$

Using the 2×2 inverse formula ($\det = 42-36 = 6$):

$$\hat{\boldsymbol{w}} = \frac{1}{6}\begin{pmatrix} 3 & -6 \\ -6 & 14 \end{pmatrix}\begin{pmatrix} 23 \\ 10 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 9 \\ 2 \end{pmatrix} = \begin{pmatrix} 1.5 \\ 0.333\ldots \end{pmatrix}.$$

The best line is $y = 1.5x + \frac{1}{3}$ (Figure 5.6). Its predictions are $(1.83, 3.33, 4.83)$ and the errors (*residuals*) are $(0.17, -0.33, 0.17)$, with sum of squares $\frac{1}{6}$. As the theory predicts, the residual vector is orthogonal to both columns of $\boldsymbol{X}$: $1(0.17)+2(-0.33)+3(0.17) = 0$ and $0.17 - 0.33 + 0.17 = 0$.

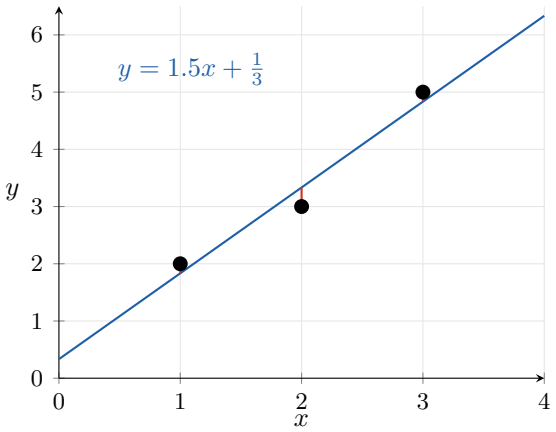


Figure 5.6: The least-squares line through three points. The red segments are the residuals; the line minimises the sum of their squares.

Why this matters for ML (Linear regression in one sentence)**.** *Least-squares linear regression projects the target vector $\boldsymbol{y}$ onto the column space of the data matrix $\boldsymbol{X}$.* The fitted values $\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{w}}$ are the projection, the residuals $\boldsymbol{y} - \hat{\boldsymbol{y}}$ are orthogonal to every feature column, and the weights solve $\boldsymbol{X}^\top \boldsymbol{X}\hat{\boldsymbol{w}} = \boldsymbol{X}^\top \boldsymbol{y}$. In practice, libraries solve this with the QR or singular value decomposition (for example `np.linalg.lstsq`), which is more accurate than forming $\boldsymbol{X}^\top \boldsymbol{X}$. We will derive the same equations with calculus in Chapter 8.

Exercises for Section 5.6

Exercise 5.19. Project $\boldsymbol{b} = (3, 1)$ onto the line spanned by $\boldsymbol{a} = (1, 1)$, and verify that the error is orthogonal to $\boldsymbol{a}$.

Solution on p. 250 →

Exercise 5.20. Project $\boldsymbol{b} = (1, 2, 3)$ onto the line spanned by $\boldsymbol{a} = (1, 0, 1)$. How far is $\boldsymbol{b}$ from the line?

Solution on p. 250 →

Exercise 5.21. Find the least-squares line through $(0, 1)$, $(1, 3)$ and $(2, 4)$.

Solution on p. 250 →

Exercise 5.22. Fit a constant model $\hat{y} = c$ to the data $y = (2, 4, 9)$ by least squares, using the normal equations with $\boldsymbol{X}$ a single column of ones. Compare with Section 2.4.

Solution on p. 250 →

Exercise 5.23. Show that the projection matrix $\boldsymbol{P} = \frac{\boldsymbol{a}\boldsymbol{a}^\top}{\boldsymbol{a}^\top\boldsymbol{a}}$ satisfies $\boldsymbol{P}^2 = \boldsymbol{P}$ and $\boldsymbol{P}^\top = \boldsymbol{P}$. Explain $\boldsymbol{P}^2 = \boldsymbol{P}$ geometrically.

Solution on p. 250 →

5.7 Eigenvalues and eigenvectors

Motivation. A matrix typically turns vectors in all sorts of directions. But for most square matrices there are a few special directions that are left alone: vectors along them are only stretched or shrunk. These *eigenvectors* (from the German *eigen*, "own" or "characteristic") reveal the skeleton of a transformation. They tell us the main axes of variation in data (principal component analysis), whether repeated multiplication explodes or dies out (recurrent networks, Markov chains, PageRank), and how curved a loss surface is (Chapters 7 and 8).

5.7.1 The idea

Consider $\boldsymbol{A} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and apply it to a few vectors (Figure 5.7):

$$\boldsymbol{A}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}2\\1\end{pmatrix}, \qquad \boldsymbol{A}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix} = 3\begin{pmatrix}1\\1\end{pmatrix}, \qquad \boldsymbol{A}\begin{pmatrix}1\\-1\end{pmatrix} = \begin{pmatrix}1\\-1\end{pmatrix} = 1\begin{pmatrix}1\\-1\end{pmatrix}.$$

The vector $(1, 0)$ is knocked off its line. But $(1, 1)$ stays on its line and is stretched by a factor 3, and $(1, -1)$ stays exactly where it is (factor 1).

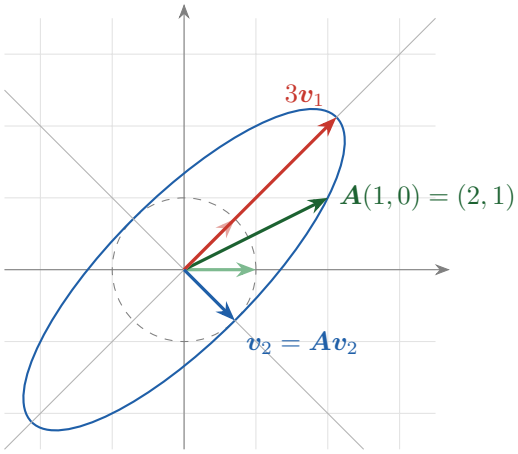


Figure 5.7: The matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ maps the unit circle (dashed) to the ellipse. Vectors along the eigenvector directions (grey lines) stay on their lines: the unit vector $\boldsymbol{v}_1$ along $(1, 1)$ is stretched by 3, and $\boldsymbol{v}_2$ along $(1, -1)$ is unchanged. Other vectors, such as $(1, 0)$ (green), change direction.

Definition 5.25 (Eigenvalue and eigenvector)**.** Let $\boldsymbol{A}$ be a square matrix. A non-zero vector $\boldsymbol{v}$ is an **eigenvector** of $\boldsymbol{A}$, with **eigenvalue** λ ("lambda"), if

$$\boxed{\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}}$$

Any non-zero multiple of an eigenvector is again an eigenvector with the same eigenvalue, so really an eigenvector stands for a whole direction. The eigenvalue says what happens

along that direction: $\lambda > 1$ stretches, $0 < \lambda < 1$ shrinks, $\lambda < 0$ flips, and $\lambda = 0$ squashes the direction to zero (so $\boldsymbol{A}$ is singular).

Example 5.26 (Eigenvectors you can see)**.** (a) For the diagonal matrix $\left(\begin{smallmatrix} 3 & 0 \\ 0 & 0.5 \end{smallmatrix}\right)$, the eigenvectors are $\boldsymbol{e}_1$ (eigenvalue 3) and $\boldsymbol{e}_2$ (eigenvalue 0.5): it stretches the x_1-axis and shrinks the x_2-axis. (b) The projection onto the x_1-axis has eigenvalue 1 for $\boldsymbol{e}_1$ (left alone) and 0 for $\boldsymbol{e}_2$ (squashed). (c) A rotation by 90° has no real eigenvectors at all: it moves every non-zero vector off its line.

5.7.2 Finding eigenvalues

$\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}$ can be rewritten as $(\boldsymbol{A} - \lambda\boldsymbol{I})\boldsymbol{v} = \boldsymbol{0}$. A non-zero solution $\boldsymbol{v}$ exists exactly when $\boldsymbol{A} - \lambda\boldsymbol{I}$ is singular, i.e. when its determinant is zero.

Theorem 5.27 (Characteristic equation)**.** The eigenvalues of $\boldsymbol{A}$ are the solutions λ of

$$\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0.$$

For a 2×2 matrix $\boldsymbol{A} = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ this is the quadratic equation

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0,$$

i.e. $\lambda^2 - \operatorname{tr}(\boldsymbol{A})\,\lambda + \det(\boldsymbol{A}) = 0$, where the **trace** $\operatorname{tr}(\boldsymbol{A}) = a + d$ is the sum of the diagonal entries.

Once an eigenvalue λ is known, its eigenvectors are the non-zero solutions of $(\boldsymbol{A} - \lambda\boldsymbol{I})\boldsymbol{v} = \boldsymbol{0}$, found by elimination.

Example 5.28 (Computing eigenvalues and eigenvectors)**.** Find the eigenvalues and eigenvectors of $\boldsymbol{A} = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$.

Solution. $\operatorname{tr}\boldsymbol{A} = 7$ and $\det\boldsymbol{A} = 12 - 2 = 10$, so the characteristic equation is $\lambda^2 - 7\lambda + 10 = 0$, i.e. $(\lambda - 5)(\lambda - 2) = 0$. The eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = 2$.

For $\lambda_1 = 5$: $\boldsymbol{A} - 5\boldsymbol{I} = \left(\begin{smallmatrix} -1 & 1 \\ 2 & -2 \end{smallmatrix}\right)$; both rows say $v_1 = v_2$, so $\boldsymbol{v}_1 = (1, 1)$.

For $\lambda_2 = 2$: $\boldsymbol{A} - 2\boldsymbol{I} = \left(\begin{smallmatrix} 2 & 1 \\ 2 & 1 \end{smallmatrix}\right)$; the rows say $2v_1 + v_2 = 0$, so $\boldsymbol{v}_2 = (1, -2)$.

Check: $\boldsymbol{A}(1, -2) = (4 - 2, 2 - 6) = (2, -4) = 2(1, -2)$. ✓

Proposition 5.29 (Trace and determinant)**.** The sum of the eigenvalues of a matrix equals its trace, and their product equals its determinant. (Above: $5 + 2 = 7$ and $5 \times 2 = 10$.)

5.7.3 Diagonalisation and powers

If an $n \times n$ matrix has n independent eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ with eigenvalues $\lambda_1, \ldots, \lambda_n$, put the eigenvectors as the columns of $\boldsymbol{V}$ and the eigenvalues on the diagonal of $\boldsymbol{\Lambda}$ ("capital lambda"). The equations $\boldsymbol{A}\boldsymbol{v}_i = \lambda_i\boldsymbol{v}_i$, written side by side, say $\boldsymbol{A}\boldsymbol{V} = \boldsymbol{V}\boldsymbol{\Lambda}$, i.e.

$$\boxed{\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1}}$$

This **eigendecomposition** says: in the coordinate system of the eigenvectors, $\boldsymbol{A}$ is just a diagonal matrix, a pure stretch along each axis.

It makes powers easy: $\boldsymbol{A}^k = \boldsymbol{V}\boldsymbol{\Lambda}^k\boldsymbol{V}^{-1}$, and $\boldsymbol{\Lambda}^k$ is diagonal with entries λ_i^k. In particular, applying $\boldsymbol{A}$ many times to a vector amplifies its component along the eigenvector with the largest $|\lambda|$ and suppresses the others.

Example 5.30 (Repeated multiplication)**.** For $\boldsymbol{A} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, compute $\boldsymbol{A}^k\boldsymbol{x}$ for $\boldsymbol{x} = (1, 0)$.

Solution. Write $\boldsymbol{x}$ in the eigenvector basis: $(1, 0) = \frac{1}{2}(1, 1) + \frac{1}{2}(1, -1)$. Then

$$\boldsymbol{A}^k\boldsymbol{x} = \tfrac{1}{2} \cdot 3^k(1, 1) + \tfrac{1}{2} \cdot 1^k(1, -1).$$

For large k the first term dominates, so $\boldsymbol{A}^k\boldsymbol{x}$ points almost exactly along the eigenvector $(1, 1)$ and grows like 3^k. Repeatedly multiplying and normalising is a practical algorithm (*power iteration*) for finding the dominant eigenvector.

5.7.4 Symmetric matrices

Symmetric matrices, which are everywhere in machine learning (covariance matrices, $\boldsymbol{X}^\top\boldsymbol{X}$, Hessians), have particularly nice eigenvectors.

Theorem 5.31 (Spectral theorem)**.** A real symmetric matrix $\boldsymbol{A}$ has only real eigenvalues, and it has an orthonormal basis of eigenvectors. Hence

$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top,$$

where $\boldsymbol{Q}$ is orthogonal (its columns are the orthonormal eigenvectors) and $\boldsymbol{\Lambda}$ is diagonal (the eigenvalues).

Geometrically: a symmetric matrix rotates into its eigenvector coordinates ($\boldsymbol{Q}^\top$), stretches along the axes ($\boldsymbol{\Lambda}$), and rotates back ($\boldsymbol{Q}$). This is exactly what Figure 5.7 shows: perpendicular eigenvector directions, stretched by 3 and by 1, turning the circle into an ellipse.

Why this matters for ML (Where eigenvalues show up)**.** Eigenvalues appear across machine learning.

- **Principal component analysis:** the eigenvectors of a data set's covariance matrix are its main directions of variation (Section 5.10).
- **Exploding and vanishing gradients:** a recurrent network multiplies its state by the same matrix at every time step. Components along eigenvectors with $|\lambda| > 1$ explode, and those with $|\lambda| < 1$ vanish, exactly as in Example 5.30.
- **Curvature:** the eigenvalues of the Hessian matrix of a loss measure how sharply it curves in each direction; they decide whether a point is a minimum or a saddle and how large the learning rate may be (Chapters 7 and 8).
- **PageRank and Markov chains:** the long-run importance of web pages is the dominant eigenvector of a matrix of link probabilities.

Exercises for Section 5.7

Exercise 5.24. Find the eigenvalues and eigenvectors of (a) $\left(\begin{smallmatrix} 3 & 0 \\ 0 & -2 \end{smallmatrix}\right)$; (b) $\left(\begin{smallmatrix} 5 & 2 \\ 2 & 2 \end{smallmatrix}\right)$. In (b), check that the eigenvectors are orthogonal.

Solution on p. 250 →

Exercise 5.25. Check that $(1, -1)$ is an eigenvector of $\left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ and find its eigenvalue. Then find the other eigenvalue using the trace.

Solution on p. 250 →

Exercise 5.26. Show that the rotation by 90°, $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, has no real eigenvalues.

Solution on p. 250 →

Exercise 5.27. Find the eigenvalues of $\left(\begin{smallmatrix} 1 & 4 \\ 2 & 3 \end{smallmatrix}\right)$ and check Proposition 5.29.

Solution on p. 250 →

Exercise 5.28. A recurrent network multiplies its hidden state by $\boldsymbol{W} = \left(\begin{smallmatrix} 0.5 & 0 \\ 0 & 1.2 \end{smallmatrix}\right)$ at every time step. Starting from $\boldsymbol{h} = (1, 1)$, compute the state after 10 steps and comment.

Solution on p. 250 →

5.8 The singular value decomposition

Motivation. Eigendecomposition only works for square matrices, and not even for all of them. The *singular value decomposition* (SVD) works for *every* matrix, of any shape. It is arguably the most useful single result of applied linear algebra: it underlies data compression, principal component analysis, recommender systems, least squares and the analysis of neural network weights.

Theorem 5.32 (Singular value decomposition)**.** Every $m \times n$ matrix $\boldsymbol{A}$ can be written as

$$\boxed{\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{\top}}$$

where $\boldsymbol{U}$ $(m \times m)$ and $\boldsymbol{V}$ $(n \times n)$ are orthogonal matrices and $\boldsymbol{\Sigma}$ $(m \times n)$ is zero except for the numbers $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$ on its main diagonal. The σ_i are the **singular values** of $\boldsymbol{A}$, and the columns $\boldsymbol{u}_i$ of $\boldsymbol{U}$ and $\boldsymbol{v}_i$ of $\boldsymbol{V}$ are its left and right *singular vectors*. They satisfy $\boldsymbol{A}\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$.

Intuition (Rotate, stretch, rotate)**.** Reading $\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{\top}\boldsymbol{x}$ from right to left: $\boldsymbol{V}^{\top}$ rotates (or reflects) the input, $\boldsymbol{\Sigma}$ stretches along the coordinate axes by the factors σ_i, and $\boldsymbol{U}$ rotates the result into place. So *every* linear transformation is a rotation, an axis-aligned stretch and another rotation. In two dimensions it maps the unit circle to an ellipse whose semi-axes have lengths σ_1 and σ_2, pointing along $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ (Figure 5.8).

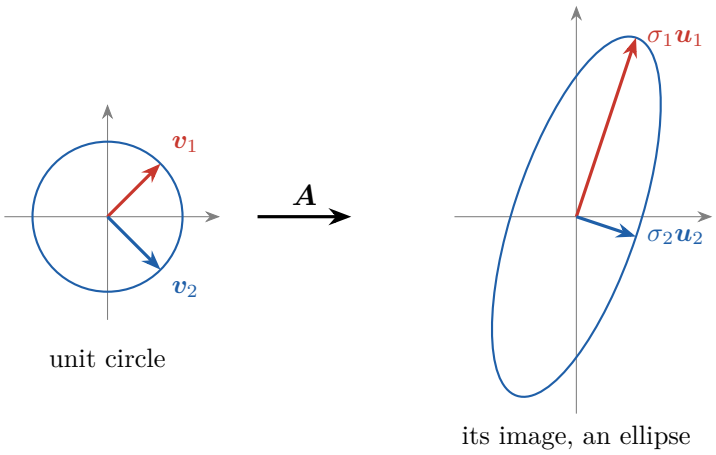


Figure 5.8: The SVD of $\boldsymbol{A} = \left(\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right)$. The perpendicular directions $\boldsymbol{v}_1$, $\boldsymbol{v}_2$ are mapped to the perpendicular axes of an ellipse, of lengths $\sigma_1 = \sqrt{45} \approx 6.71$ and $\sigma_2 = \sqrt{5} \approx 2.24$. (The two panels are drawn at different scales.)

5.8.1 How to find the SVD

From $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$ we get $\boldsymbol{A}^\top\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Sigma}^\top\boldsymbol{U}^\top\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top = \boldsymbol{V}(\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma})\boldsymbol{V}^\top$. This is an eigendecomposition of the symmetric matrix $\boldsymbol{A}^\top\boldsymbol{A}$, so:

- the right singular vectors $\boldsymbol{v}_i$ are the eigenvectors of $\boldsymbol{A}^\top\boldsymbol{A}$;
- the singular values are the square roots of its eigenvalues, $\sigma_i = \sqrt{\lambda_i(\boldsymbol{A}^\top\boldsymbol{A})}$;
- the left singular vectors are $\boldsymbol{u}_i = \boldsymbol{A}\boldsymbol{v}_i/\sigma_i$ (for $\sigma_i > 0$).

Example 5.33 (An SVD by hand)**.** Find the SVD of $\boldsymbol{A} = \left(\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right)$.

Solution. $\boldsymbol{A}^\top\boldsymbol{A} = \left(\begin{smallmatrix} 3 & 4 \\ 0 & 5 \end{smallmatrix}\right)\left(\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 25 & 20 \\ 20 & 25 \end{smallmatrix}\right)$, with trace 50 and determinant $625-400 = 225$. Its eigenvalues solve $\lambda^2 - 50\lambda + 225 = 0$: $\lambda = 45$ and $\lambda = 5$, with eigenvectors $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)$ and $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$. So $\sigma_1 = \sqrt{45} = 3\sqrt{5}$ and $\sigma_2 = \sqrt{5}$. Then

$$\boldsymbol{u}_1 = \frac{\boldsymbol{A}\boldsymbol{v}_1}{\sigma_1} = \frac{(3,9)/\sqrt{2}}{3\sqrt{5}} = \frac{1}{\sqrt{10}}(1,3), \qquad \boldsymbol{u}_2 = \frac{\boldsymbol{A}\boldsymbol{v}_2}{\sigma_2} = \frac{(3,-1)/\sqrt{2}}{\sqrt{5}} = \frac{1}{\sqrt{10}}(3,-1).$$

As a check, $\boldsymbol{u}_1 \cdot \boldsymbol{u}_2 = 0$, and $\sigma_1\sigma_2 = 15 = |\det \boldsymbol{A}|$.

5.8.2 Low-rank approximation

Multiplying out $\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$ column by row writes $\boldsymbol{A}$ as a sum of rank-one matrices, ordered by importance:

$$\boldsymbol{A} = \sigma_1\boldsymbol{u}_1\boldsymbol{v}_1^\top + \sigma_2\boldsymbol{u}_2\boldsymbol{v}_2^\top + \cdots + \sigma_r\boldsymbol{u}_r\boldsymbol{v}_r^\top,$$

where r is the rank (the number of non-zero singular values).

Theorem 5.34 (Best low-rank approximation (Eckart–Young))**.** Keeping only the first k terms, $\boldsymbol{A}_k = \sigma_1\boldsymbol{u}_1\boldsymbol{v}_1^\top + \cdots + \sigma_k\boldsymbol{u}_k\boldsymbol{v}_k^\top$, gives the best approximation of $\boldsymbol{A}$ by a matrix of rank k, in the sense that it minimises $\|\boldsymbol{A} - \boldsymbol{B}\|_F$ over all matrices $\boldsymbol{B}$ of rank at most k. The error is $\|\boldsymbol{A} - \boldsymbol{A}_k\|_F = \sqrt{\sigma_{k+1}^2 + \cdots + \sigma_r^2}$.

For the matrix of Example 5.33, the best rank-one approximation is $\boldsymbol{A}_1 = \sigma_1\boldsymbol{u}_1\boldsymbol{v}_1^\top =$

$\left(\begin{smallmatrix}1.5 & 1.5\\ 4.5 & 4.5\end{smallmatrix}\right)$, and the error $\boldsymbol{A} - \boldsymbol{A}_1 = \left(\begin{smallmatrix}1.5 & -1.5\\ -0.5 & 0.5\end{smallmatrix}\right)$ has squared Frobenius norm $5 = \sigma_2^2$, as the theorem predicts.

Why this matters for ML (The SVD at work)**.** The SVD is used throughout machine learning.

- **Compression.** A grayscale image is a matrix. If its singular values decay quickly, a rank-50 approximation of a 1000×1000 image needs only $50\times(1000+1000+1) \approx 10^5$ numbers instead of 10^6, and often looks almost identical.
- **Recommender systems.** A matrix of ratings (users $\times$ films) is mostly empty. Assuming it is approximately of low rank, i.e. that tastes are described by a few hidden factors, lets us fill in the missing ratings. This idea was central to the famous Netflix Prize.
- **Conditioning.** The ratio σ_1/σ_n of the largest to the smallest singular value, the **condition number**, measures how much a matrix can amplify errors. Badly conditioned problems are hard to solve accurately and slow to optimise.
- **Principal component analysis** can be computed from the SVD of the centred data matrix (Section 5.10).

Exercises for Section 5.8

Exercise 5.29. What are the singular values of $\left(\begin{smallmatrix}3 & 0\\ 0 & -2\end{smallmatrix}\right)$? Why are they not simply the diagonal entries?

Solution on p. 250 →

Exercise 5.30. Find the singular values of $\left(\begin{smallmatrix}1 & 1\\ 1 & 1\end{smallmatrix}\right)$ and write the matrix as $\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^\top$.

Solution on p. 251 →

Exercise 5.31. An 800×1000 image is approximated with rank 40. How many numbers must be stored, and what fraction of the original is this?

Solution on p. 251 →

Exercise 5.32. A matrix has singular values $10, 5, 1, 0.1$. What is the Frobenius error of its best rank-2 approximation, and what fraction of $\|\boldsymbol{A}\|_F^2$ does the approximation capture?

Solution on p. 251 →

5.9 Quadratic forms and positive definite matrices

Motivation. Near a minimum, almost every smooth loss function looks like a bowl described by a quadratic expression in the parameters (Chapter 7). Whether it is really a bowl, rather than a saddle or a flat trough, is decided by a symmetric matrix. Positive definite matrices are the matrices of bowls. They also appear as covariance matrices of data and of Gaussian distributions.

Definition 5.35 (Quadratic form)**.** For a symmetric $n \times n$ matrix $\boldsymbol{A}$, the function $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} = \sum_{i,j} a_{ij} x_i x_j$ is called a **quadratic form**.

In two dimensions, with $\boldsymbol{A} = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$,

$$\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} = ax_1^2 + 2bx_1x_2 + cx_2^2.$$

For example, $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ gives $2x_1^2 + 2x_1x_2 + 2x_2^2$.

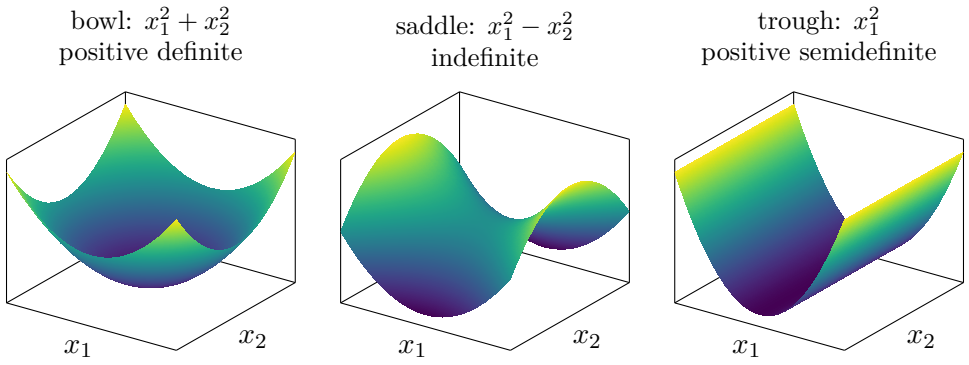


Figure 5.9: The three basic shapes of quadratic forms in two variables.

Definition 5.36 (Definiteness)**.** A symmetric matrix $\boldsymbol{A}$ is

- **positive definite** if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} > 0$ for every $\boldsymbol{x} \neq \boldsymbol{0}$ (a bowl);
- **positive semidefinite** if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \geq 0$ for every $\boldsymbol{x}$ (a bowl or a trough);
- *negative definite* if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} < 0$ for every $\boldsymbol{x} \neq \boldsymbol{0}$ (an upside-down bowl);
- **indefinite** if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}$ takes both positive and negative values (a saddle).

The definition involves infinitely many vectors $\boldsymbol{x}$, so we need practical tests. The spectral theorem provides one: in the coordinates $\boldsymbol{y} = \boldsymbol{Q}^\top \boldsymbol{x}$ of the orthonormal eigenvectors,

$$\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} = \boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top \boldsymbol{x} = \boldsymbol{y}^\top \boldsymbol{\Lambda}\boldsymbol{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2,$$

a sum of squares weighted by the eigenvalues.

Theorem 5.37 (Tests for definiteness)**.** A symmetric matrix is positive definite exactly when all its eigenvalues are positive, and positive semidefinite exactly when all are ≥ 0; it is indefinite when it has eigenvalues of both signs. A symmetric 2×2 matrix $\left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$ is positive definite exactly when $a > 0$ and $ac - b^2 > 0$.

Example 5.38 (Classifying quadratic forms)**.** (a) $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has eigenvalues 3 and 1: positive definite. Indeed $2x_1^2 + 2x_1x_2 + 2x_2^2 = x_1^2 + x_2^2 + (x_1 + x_2)^2 > 0$ for $\boldsymbol{x} \neq \boldsymbol{0}$.

(b) $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ has eigenvalues 3 and -1: indefinite. At $\boldsymbol{x} = (1, -1)$ the form is $1 - 4 + 1 = -2 < 0$, while at $(1, 1)$ it is $6 > 0$.

(c) $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has eigenvalues 2 and 0: positive semidefinite but not definite. The form is $(x_1 + x_2)^2$, which is zero along the whole line $x_2 = -x_1$.

Intuition (Level curves are ellipses)**.** For a positive definite $\boldsymbol{A}$, the curves $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} = c$ are ellipses whose axes point along the eigenvectors. Along an eigenvector with eigenvalue λ, the form grows like λt^2, so a large eigenvalue means a steep, narrow

direction and a small eigenvalue a shallow, wide direction (Figure 5.10). These elongated ellipses will explain, in Chapter 8, why gradient descent zig-zags on badly conditioned problems.

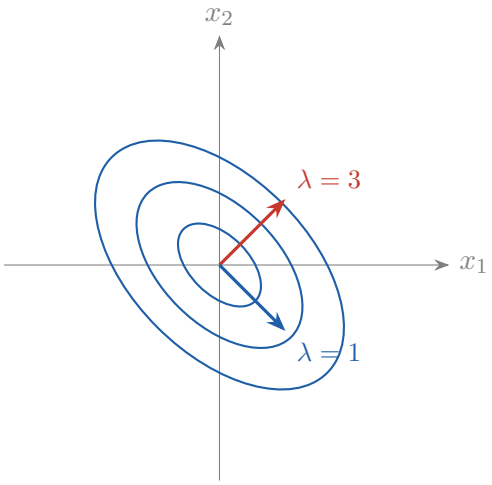


Figure 5.10: Level curves $2x_1^2 + 2x_1x_2 + 2x_2^2 = c$ for $c = 0.5, 2, 4.5$. They are ellipses with axes along the eigenvectors $(1, 1)$ and $(1, -1)$; the form rises three times as fast along the first.

Proposition 5.39 (Gram matrices are positive semidefinite)**.** For any matrix $\boldsymbol{A}$, the matrix $\boldsymbol{A}^\top \boldsymbol{A}$ is positive semidefinite, and it is positive definite if the columns of $\boldsymbol{A}$ are independent.

Proof. $\boldsymbol{x}^\top(\boldsymbol{A}^\top \boldsymbol{A})\boldsymbol{x} = (\boldsymbol{A}\boldsymbol{x})^\top(\boldsymbol{A}\boldsymbol{x}) = \|\boldsymbol{A}\boldsymbol{x}\|^2 \geq 0$. It is zero only if $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$, which for independent columns forces $\boldsymbol{x} = \boldsymbol{0}$. ∎

Why this matters for ML. Positive (semi)definite matrices are the mathematical signature of "bowls" and of "spread".

- The least-squares loss $\|\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}\|^2$ is a quadratic in $\boldsymbol{w}$ whose matrix is $\boldsymbol{X}^\top \boldsymbol{X}$, which is positive semidefinite. That is why linear regression has no bad local minima (Chapter 8).
- A critical point of a loss is a minimum if its Hessian matrix is positive definite and a saddle if it is indefinite (Chapter 7).
- Covariance matrices are always positive semidefinite (Chapter 9), and the multivariate Gaussian distribution needs a positive definite one.
- Kernel methods require the matrix of similarities between data points to be positive semidefinite.

Exercises for Section 5.9

Exercise 5.33. Write out the quadratic form of $\left(\begin{smallmatrix} 3 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ and decide whether the matrix is positive definite.

Solution on p. 251 →

Exercise 5.34. Classify each matrix as positive definite, positive semidefinite, negative definite or indefinite: (a) $\left(\begin{smallmatrix}1&3\\3&1\end{smallmatrix}\right)$; (b) $\left(\begin{smallmatrix}4&2\\2&1\end{smallmatrix}\right)$; (c) $\begin{pmatrix}-2&0\\0&-1\end{pmatrix}$.

Solution on p. 251 →

Exercise 5.35. For which values of c is $\left(\begin{smallmatrix}1&c\\c&4\end{smallmatrix}\right)$ positive definite?

Solution on p. 251 →

5.10 Putting it together: principal component analysis

Motivation. Real data sets have many features, and many of them are correlated: in a data set of people, height and arm span carry nearly the same information. *Principal component analysis* (PCA) finds new axes, ordered by how much the data varies along them. Keeping only the first few axes compresses the data with little loss, removes noise, and makes high-dimensional data visualisable. PCA uses almost every idea of this chapter.

5.10.1 The recipe

Let the data be n points $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n \in \mathbb{R}^d$.

Algorithm 5.40 (Principal component analysis)**.** To find the principal components of a data set:

Step 1. **Centre** the data: subtract the mean $\bar{\boldsymbol{x}} = \frac{1}{n}\sum_i \boldsymbol{x}_i$ from every point, and stack the centred points as the rows of an $n \times d$ matrix $\boldsymbol{X}_c$.

Step 2. Form the $d \times d$ **covariance matrix** $\boldsymbol{C} = \frac{1}{n}\boldsymbol{X}_c^\top \boldsymbol{X}_c$. (Its diagonal entries are the average squared deviations of each feature, i.e. how spread out it is; the off-diagonal entries measure how pairs of features vary together. Chapter 9 explains these "variances" and "covariances" properly.)

Step 3. Compute the **eigenvalues** $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$ and orthonormal **eigenvectors** $\boldsymbol{q}_1, \dots, \boldsymbol{q}_d$ of $\boldsymbol{C}$ (the spectral theorem applies, since $\boldsymbol{C}$ is symmetric, and the eigenvalues are ≥ 0 by Proposition 5.39). The eigenvectors are the **principal components**, and λ_j is the spread of the data along $\boldsymbol{q}_j$.

Step 4. **Project:** to reduce the data to k dimensions, replace each centred point $\boldsymbol{x}_i - \bar{\boldsymbol{x}}$ by its k coordinates $((\boldsymbol{x}_i - \bar{\boldsymbol{x}}) \cdot \boldsymbol{q}_1, \dots, (\boldsymbol{x}_i - \bar{\boldsymbol{x}}) \cdot \boldsymbol{q}_k)$. The fraction of the total spread that is kept is $\frac{\lambda_1 + \cdots + \lambda_k}{\lambda_1 + \cdots + \lambda_d}$.

Example 5.41 (PCA by hand)**.** Apply PCA to the five points $(1,2)$, $(2,1)$, $(3,4)$, $(4,3)$, $(5,5)$.

Solution. **Centre.** The mean is $(3,3)$, so the centred points are $(-2,-1)$, $(-1,-2)$, $(0,1)$, $(1,0)$, $(2,2)$.

Covariance. The average of the squared first coordinates is $\frac{4+1+0+1+4}{5} = 2$; of the squared second coordinates also 2; and of the products of the two coordinates $\frac{2+2+0+0+4}{5} = 1.6$. So

$$\boldsymbol{C} = \begin{pmatrix} 2 & 1.6 \\ 1.6 & 2 \end{pmatrix}.$$

Eigenvectors. $\operatorname{tr} \boldsymbol{C} = 4$ and $\det \boldsymbol{C} = 4 - 2.56 = 1.44$, so $\lambda^2 - 4\lambda + 1.44 = 0$, giving $\lambda_1 = 3.6$ and $\lambda_2 = 0.4$, with eigenvectors $\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$ and $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$.

Interpretation. The data is spread mostly along the diagonal direction $(1, 1)$: the first principal component carries $\frac{3.6}{3.6+0.4} = 90\%$ of the total spread. Projecting onto it reduces each point to a single number, $(\boldsymbol{x}_i - \bar{\boldsymbol{x}}) \cdot \boldsymbol{q}_1$:

$$-\tfrac{3}{\sqrt{2}},\ -\tfrac{3}{\sqrt{2}},\ \tfrac{1}{\sqrt{2}},\ \tfrac{1}{\sqrt{2}},\ \tfrac{4}{\sqrt{2}} \approx -2.12,\ -2.12,\ 0.71,\ 0.71,\ 2.83.$$

We have compressed two-dimensional data to one dimension while keeping 90% of its variation (Figure 5.11).

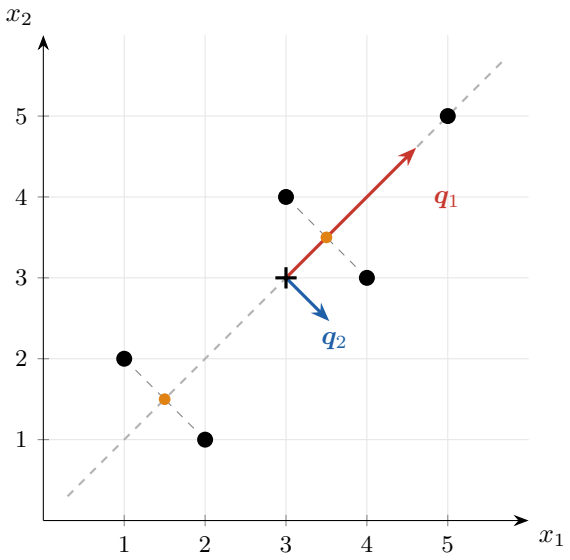


Figure 5.11: PCA of five points. The mean is marked +. The first principal direction $\boldsymbol{q}_1$ (red) points along the diagonal, where the data varies most; the second $\boldsymbol{q}_2$ (blue) is perpendicular. The arrows are drawn with lengths proportional to $\sqrt{\lambda_j}$. Projecting each point onto the dashed line through the mean (orange dots) keeps 90% of the variation.

Remark (PCA and the SVD)**.** Instead of forming $\boldsymbol{C}$, one can compute the SVD of the centred data matrix, $\boldsymbol{X}_c = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$. Then $\boldsymbol{C} = \frac{1}{n}\boldsymbol{X}_c^\top \boldsymbol{X}_c = \boldsymbol{V}\frac{\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma}}{n}\boldsymbol{V}^\top$, so the principal components are the right singular vectors and $\lambda_j = \sigma_j^2/n$. This is how PCA is computed in practice, because it is more accurate.

Why this matters for ML (PCA in practice)**.** PCA is used to visualise high-dimensional data (project onto the first two or three components and plot), to compress features before training another model, and to remove noise (the last components often contain mostly noise). For data such as images of handwritten digits, a few dozen components out of hundreds of pixels already capture most of the variation. Two cautions: always centre the data, and put features on comparable scales first, because PCA is driven by spread and a feature measured in larger units would otherwise dominate.

Exercises for Section 5.10

Exercise 5.36. Apply PCA to the four points $(2, 0)$, $(0, 2)$, $(4, 6)$, $(6, 4)$: find the mean, the covariance matrix, the principal components and the fraction of spread explained by the first one.

Solution on p. 251 →

Exercise 5.37. Why must the data be centred before PCA? What would the "first principal component" tend to capture if we skipped this step?

Solution on p. 251 →

Exercise 5.38. The eigenvalues of a covariance matrix are 6, 3, 0.6, 0.3 and 0.1. How many principal components are needed to keep at least 90% of the total spread?

Solution on p. 251 →

5.11 Chapter summary

Key takeaways

- The span of vectors is the set of their linear combinations; the column space $C(\boldsymbol{A})$ is the set of outputs $\boldsymbol{Ax}$, and $\boldsymbol{Ax} = \boldsymbol{b}$ is solvable exactly when $\boldsymbol{b} \in C(\boldsymbol{A})$.
- Independent vectors have no redundancy; collinear features make $\boldsymbol{X}^\top \boldsymbol{X}$ singular.
- A basis gives unique coordinates; the dimension counts basis vectors; the rank counts independent columns; rank + nullity = number of columns.
- Norms measure size: L^1 (sum of absolute values), L^2 (length), L^∞ (largest entry). Ridge (L^2) shrinks weights, lasso (L^1) makes them sparse.
- In an orthonormal basis, coordinates are dot products; orthogonal matrices ($\boldsymbol{Q}^\top \boldsymbol{Q} = \boldsymbol{I}$) preserve lengths and angles.
- Projection onto a line: $\frac{\boldsymbol{a}\cdot\boldsymbol{b}}{\boldsymbol{a}\cdot\boldsymbol{a}}\boldsymbol{a}$. Least squares solves $\boldsymbol{X}^\top \boldsymbol{X}\hat{\boldsymbol{w}} = \boldsymbol{X}^\top \boldsymbol{y}$: linear regression is a projection.
- Eigenvectors satisfy $\boldsymbol{Av} = \lambda \boldsymbol{v}$; eigenvalues solve $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$. Symmetric matrices have real eigenvalues and orthonormal eigenvectors: $\boldsymbol{A} = \boldsymbol{Q\Lambda Q}^\top$.
- Every matrix has an SVD $\boldsymbol{A} = \boldsymbol{U\Sigma V}^\top$ (rotate, stretch, rotate); truncating it gives the best low-rank approximation.
- Positive definite matrices ($\boldsymbol{x}^\top \boldsymbol{Ax} > 0$, all eigenvalues positive) describe bowls; $\boldsymbol{X}^\top \boldsymbol{X}$ is always positive semidefinite.
- PCA: centre, form the covariance matrix, take its top eigenvectors, project.

This completes our tour of linear algebra. The next three chapters develop the other pillar of machine learning: calculus, the mathematics of change, which tells us how to adjust all these vectors and matrices to make a model better.

Chapter 6

Calculus of One Variable

Motivation. Training a machine learning model means adjusting its parameters to make a loss function smaller. Suppose the loss depends on a single weight w. If we nudge w up a little, does the loss go up or down, and how fast? The answer is given by the *derivative* of the loss with respect to w. Knowing it for every weight tells us which way to move all of them, and this is exactly what gradient descent does (Chapter 8). The tool that computes these derivatives through the many layers of a neural network, *backpropagation*, is nothing but the *chain rule* of this chapter applied systematically.

Calculus also has a second half, *integration*, which adds up infinitely many infinitely small pieces. We need it because probabilities of continuous quantities are areas under curves (Chapter 9).

In this chapter you will learn to...

- explain average and instantaneous rates of change;
- understand limits intuitively and evaluate simple limits;
- define the derivative and interpret it as a slope, a rate and a sensitivity;
- differentiate functions using the power, product, quotient and chain rules and the derivatives of e^x, $\ln x$, $\sin x$, $\cos x$;
- compute the derivatives of the sigmoid and other functions used in ML;
- find and classify maxima and minima, and solve optimisation problems;
- approximate functions by Taylor polynomials;
- compute integrals and interpret them as areas and accumulated change;
- estimate derivatives numerically.

6.1 Rates of change

A ball is dropped from a tower. Physics tells us that after t seconds it has fallen $d(t) = 4.9t^2$ metres. How fast is it moving at exactly $t = 1$ second?

Speed is distance divided by time, but at a single instant no distance is covered and no time passes, so "$\frac{0}{0}$" is no help. What we *can* compute is the *average* speed over a time interval:

$$\text{average speed between } t = 1 \text{ and } t = 1 + h = \frac{d(1+h) - d(1)}{h}.$$

interval length h	1	0.1	0.01	0.001
average speed (m/s)	14.7	10.29	9.849	9.8049

As the interval shrinks, the average speeds settle down towards 9.8 m/s. This number, the value the averages approach, is what we mean by the *instantaneous* speed at $t = 1$.

Definition 6.1 (Average rate of change)**.** The **average rate of change** of a function

f between $x = a$ and $x = b$ is

$$\frac{f(b) - f(a)}{b - a} = \frac{\Delta f}{\Delta x}.$$

(The Greek letter Δ, "delta", is the standard symbol for a change in a quantity.) Geometrically, it is the slope of the **secant line** through the points $(a, f(a))$ and $(b, f(b))$ on the graph.

The instantaneous rate of change is what the average rate approaches as the interval shrinks to nothing. To make "approaches" precise we need the idea of a limit.

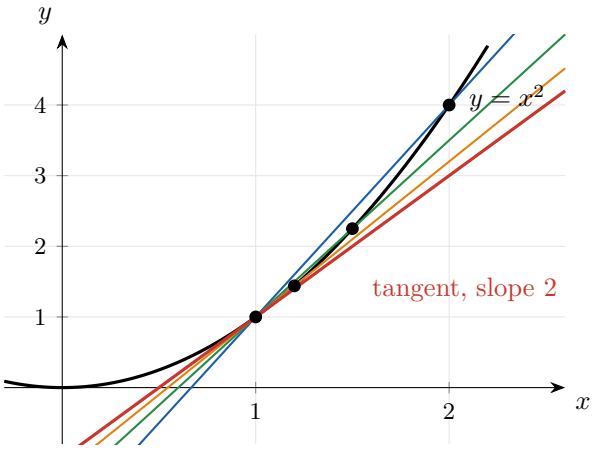


Figure 6.1: Secant lines of $y = x^2$ through $(1, 1)$ and $(1 + h, (1 + h)^2)$ for $h = 1$ (blue, slope 3), $h = 0.5$ (green, slope 2.5) and $h = 0.2$ (orange, slope 2.2). As h shrinks they approach the tangent line (red), whose slope 2 is the derivative at $x = 1$.

Exercises for Section 6.1

Exercise 6.1. Find the average rate of change of $f(x) = x^2 + 1$ on the interval $[1, 3]$ and on the interval $[1, 1.1]$.

Solution on p. 251 →

Exercise 6.2. A model's training loss is 2.3 at step 0 and 0.5 at step 1000. What is the average rate of change of the loss per step? What does the sign tell you?

Solution on p. 251 →

Exercise 6.3. A population (in thousands) is $P(t) = 50 \cdot 1.02^t$ after t years. What is its average growth rate between $t = 0$ and $t = 10$?

Solution on p. 251 →

6.2 Limits

6.2.1 The idea of a limit

Definition 6.2 (Limit, informally)**.** We write

$$\lim_{x \to a} f(x) = L$$

(read "the **limit** of $f(x)$ as x approaches a is L") if $f(x)$ can be made as close to L as we like by taking x close enough to a, but not equal to a.

The words "not equal to a" matter: a limit describes what happens *near* a, not *at* a. The function need not even be defined at a.

Example 6.3 (A limit at a hole)**.** Find $\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$.

Solution. At $x = 1$ the expression is $\frac{0}{0}$, which is meaningless. But for $x \neq 1$ we can factor and cancel: $\frac{(x-1)(x+1)}{x-1} = x + 1$. As x approaches 1, this approaches 2. A table confirms it:

x	0.9	0.99	0.999	1.001	1.01	1.1
$\frac{x^2-1}{x-1}$	1.9	1.99	1.999	2.001	2.01	2.1

So the limit is 2, even though the function has a "hole" at $x = 1$.

Example 6.4 (A famous limit)**.** For x in radians, $\frac{\sin x}{x}$ takes the values 0.8415, 0.9589, 0.99833, 0.99998 at $x = 1, 0.5, 0.1, 0.01$. So $\lim_{x \to 0} \frac{\sin x}{x} = 1$: for small angles, $\sin x \approx x$.

6.2.2 Limits that do not exist, and one-sided limits

The step function of Section 2.10 is 0 just to the left of 0 and 1 just to the right. Its *left-hand limit* at 0, written $\lim_{x \to 0^-}$, is 0 and its *right-hand limit* $\lim_{x \to 0^+}$ is 1. Since they disagree, the limit $\lim_{x \to 0}$ does not exist. Another way for a limit to fail is to blow up: as $x \to 0^+$, $\frac{1}{x}$ grows without bound. We write $\lim_{x \to 0^+} \frac{1}{x} = \infty$, which is shorthand for "no limit; it becomes arbitrarily large".

6.2.3 Limits at infinity

We also ask what happens as x grows without bound:

$$\lim_{x\to\infty}\frac{1}{x}=0, \qquad \lim_{x\to\infty}e^{-x}=0, \qquad \lim_{x\to\infty}\sigma(x)=1, \qquad \lim_{x\to-\infty}\sigma(x)=0.$$

The number e itself was defined in Section 2.5 as a limit: $e=\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$.

6.2.4 Rules and continuity

Limits behave as you would hope: the limit of a sum is the sum of the limits, and likewise for differences, products, quotients (if the denominator's limit is not zero) and constant multiples.

Definition 6.5 (Continuity). A function f is **continuous** at a if $\lim_{x\to a}f(x)=f(a)$. It is continuous if it is continuous at every point of its domain.

Informally, a continuous function has a graph you can draw without lifting your pen: no jumps, no holes, no sudden blow-ups. Polynomials, e^x, $\ln x$ (for $x>0$), $\sin x$, $\cos x$, the sigmoid and the ReLU are all continuous; the step function is not continuous at 0.

Remark (The precise definition). Mathematicians make "as close as we like" precise as follows: $\lim_{x\to a}f(x)=L$ means that for every tolerance $\varepsilon>0$, there is a distance $\delta>0$ such that $|f(x)-L|<\varepsilon$ whenever $0<|x-a|<\delta$. You will not need this "ε–δ" definition in these notes, but it is the foundation on which all of calculus rests.

Why this matters for ML. Limits describe the long-run behaviour of algorithms ("as the number of training steps goes to infinity, does the loss converge?"), the behaviour of learning-rate schedules that shrink towards zero, and the *saturation* of activation functions: since $\sigma(x)\to 1$ as $x\to\infty$, a sigmoid neuron with a very large input is "stuck" near 1 and barely reacts to changes, an effect we will quantify with derivatives below.

Exercises for Section 6.2

Exercise 6.4. Evaluate (a) $\lim_{x\to 2}\frac{x^2-4}{x-2}$; (b) $\lim_{x\to 3}(2x+1)$; (c) $\lim_{x\to 0}\frac{x^2+3x}{x}$.

Solution on p. 251 →

Exercise 6.5. Evaluate (a) $\lim_{x\to\infty}\frac{3x+1}{x-2}$; (b) $\lim_{x\to\infty}e^{-2x}$; (c) $\lim_{x\to-\infty}\sigma(x)$. *Hint for (a):* divide the top and bottom by x.

Solution on p. 251 →

Exercise 6.6. Is $\mathrm{ReLU}(x)=\max(0,x)$ continuous at 0? Is the step function?

Solution on p. 251 →

6.3 The derivative

Definition 6.6 (Derivative)**.** The **derivative** of f at the point a is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists; then f is called **differentiable** at a. The line through $(a, f(a))$ with slope $f'(a)$ is the **tangent line** to the graph at a.

The fraction is the slope of a secant line over an interval of width h (Figure 6.1), so the derivative is the slope of the tangent line: the slope of the graph *at a single point*.

Example 6.7 (Derivatives from the definition)**.** (a) For $f(x) = x^2$:

$$\frac{(a+h)^2 - a^2}{h} = \frac{a^2 + 2ah + h^2 - a^2}{h} = \frac{2ah + h^2}{h} = 2a + h \longrightarrow 2a \quad \text{as } h \to 0.$$

So $f'(a) = 2a$. At $a = 1$ the tangent has slope 2, as in Figure 6.1.

(b) For the falling ball, $d(t) = 4.9t^2$, the same computation gives $d'(t) = 9.8t$, so the speed at $t = 1$ is $9.8\,\text{m/s}$, as our table suggested.

6.3.1 Notation and interpretation

Computing $f'(a)$ for every a gives a new function f', the derivative of f. There are several notations, all common:

$$f'(x), \qquad \frac{\mathrm{d}f}{\mathrm{d}x}, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} \ \text{(when } y = f(x)\text{)}, \qquad \frac{\mathrm{d}}{\mathrm{d}x} f(x).$$

The second is due to Leibniz: think of $\mathrm{d}x$ as a tiny change in x and $\mathrm{d}f$ as the corresponding tiny change in f; the derivative is their ratio. (For us, "$\frac{\mathrm{d}}{\mathrm{d}x}$" is just a symbol for "the derivative with respect to x of".)

Intuition (Three ways to read a derivative)**.** The single number $f'(a)$ can be read in three equivalent ways.

- **Slope:** $f'(a)$ is the steepness of the graph at a.
- **Rate:** $f'(a)$ is how fast f changes per unit change of the input, at that moment (speed, growth rate, marginal cost).
- **Sensitivity:** for a small change h of the input,

$$f(a+h) \approx f(a) + f'(a)\,h.$$

  "If I nudge the input by h, the output moves by about $f'(a)$ times h." This is the reading machine learning cares about most.

The sign of f' tells us the direction: where $f' > 0$ the function is increasing, where $f' < 0$ it is decreasing, and where $f' = 0$ the tangent is horizontal.

6.3.2 Where derivatives do not exist

A function is not differentiable where its graph has a jump (such as the step function at 0), a corner, or a vertical tangent. The absolute value $|x|$ has a corner at 0: the slope is -1 to the left and $+1$ to the right, so the difference quotient has no single limit. Being differentiable is stronger than being continuous: $|x|$ is continuous everywhere but not differentiable at 0.

Why this matters for ML (Corners and jumps in neural networks)**.** The ReLU has a corner at 0, just like $|x|$: its slope is 0 for $x < 0$ and 1 for $x > 0$. In practice this causes no trouble: an input of exactly $0.000\ldots$ is rare, and software simply uses the slope 0 there. (The general notion behind such choices is the *subgradient*.) The step function is a different story: its slope is 0 everywhere except at the jump, where it does not exist. A slope of 0 says "changing this weight changes nothing", which gives gradient-based training no information at all. This is why the step function of the early perceptrons was replaced by smooth functions such as the sigmoid when networks began to be trained by gradient descent.

Exercises for Section 6.3

Exercise 6.7. Use the definition of the derivative to find $f'(x)$ for (a) $f(x) = 3x + 2$; (b) $f(x) = x^2 + x$.

Solution on p. 252 →

Exercise 6.8. Use the definition to show that the derivative of $f(x) = \frac{1}{x}$ is $-\frac{1}{x^2}$.

Solution on p. 252 →

Exercise 6.9. Find the equation of the tangent line to $y = x^2$ at $x = 3$.

Solution on p. 252 →

Exercise 6.10. Where is $f(x) = |x - 2|$ not differentiable? Where is $\mathrm{ReLU}(x - 1)$ not differentiable?

Solution on p. 252 →

6.4 Rules of differentiation

Computing every derivative from the definition would be exhausting. Instead we derive a few rules once and then combine them.

6.4.1 Basic rules

Theorem 6.8 (Basic differentiation rules)**.** For differentiable functions f, g and constants c, n:

constant rule:	$\frac{\mathrm{d}}{\mathrm{d}x}\, c = 0,$
power rule:	$\frac{\mathrm{d}}{\mathrm{d}x}\, x^n = n\, x^{n-1}$ (for every real number n),
constant multiple rule:	$\frac{\mathrm{d}}{\mathrm{d}x}\, c\, f(x) = c\, f'(x),$
sum rule:	$\frac{\mathrm{d}}{\mathrm{d}x}\, \big(f(x) + g(x)\big) = f'(x) + g'(x).$

A constant does not change, so its rate of change is zero. For whole-number powers, the power rule can be checked from the definition, as we did for x^2; it also holds for negative and fractional powers.

Example 6.9 (Using the power rule)**.**

$\frac{\mathrm{d}}{\mathrm{d}x}x^5 = 5x^4$, $\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{x} = \frac{\mathrm{d}}{\mathrm{d}x}x^{1/2} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$, $\frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{x^2} = \frac{\mathrm{d}}{\mathrm{d}x}x^{-2} = -2x^{-3} = -\frac{2}{x^3}$.

Combining with the sum and constant multiple rules, $\frac{\mathrm{d}}{\mathrm{d}x}(4x^3 - 2x + 7) = 12x^2 - 2$.

6.4.2 Exponentials, logarithms and trigonometric functions

Theorem 6.10 (Derivatives of special functions)**.**

$$(e^x)' = e^x, \qquad (\ln x)' = \frac{1}{x}, \qquad (a^x)' = a^x \ln a,$$
$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x.$$

(The trigonometric formulas require x in radians.)

The first formula is the special property of e promised in Section 2.5: *the slope of e^x equals its value*. Here is why. From the definition,

$$\frac{e^{x+h} - e^x}{h} = e^x \cdot \frac{e^h - 1}{h},$$

and numerically $\frac{e^h-1}{h}$ equals 1.0517, 1.00502, 1.0005 for $h = 0.1, 0.01, 0.001$: it tends to 1. (That is precisely what singles out e: for any other base a the corresponding limit is $\ln a$ instead of 1.) Similarly, the derivative of $\sin x$ comes down to the limit $\frac{\sin h}{h} \to 1$ of Example 6.4.

6.4.3 The product and quotient rules

Theorem 6.11 (Product and quotient rules)**.**

$$(fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad (\text{where } g \neq 0).$$

Intuition (Why the product rule looks like that)**.** Think of $f(x)g(x)$ as the area of a rectangle with sides f and g. When x increases by a tiny $\mathrm{d}x$, the sides grow by $\mathrm{d}f$ and $\mathrm{d}g$. The area grows by two thin strips, $\mathrm{d}f \cdot g$ and $f \cdot \mathrm{d}g$, plus a tiny corner $\mathrm{d}f\,\mathrm{d}g$ that is negligible in comparison. Dividing by $\mathrm{d}x$ gives $f'g + fg'$. Note that $(fg)'$ is *not* $f'g'$.

Example 6.12 (Product and quotient rules)**.** (a) $\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2e^x\right) = 2x\,e^x + x^2e^x = (x^2 + 2x)e^x$.

(b) $\frac{\mathrm{d}}{\mathrm{d}x}\frac{x}{1+x^2} = \frac{1\cdot(1+x^2) - x\cdot 2x}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$.

6.4.4 The chain rule

The most important rule for machine learning tells us how to differentiate a composition $f(g(x))$ (Section 2.7).

Theorem 6.13 (Chain rule). If $y = f(u)$ and $u = g(x)$, then

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = f'(g(x)) \cdot g'(x)} \qquad \text{or, in Leibniz notation,} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \cdot \frac{\mathrm{d}u}{\mathrm{d}x}.$$

Intuition (Rates multiply). If u changes 3 times as fast as x, and y changes 2 times as fast as u, then y changes $2 \times 3 = 6$ times as fast as x. Think of gears: if gear A turns twice for each turn of gear B, and gear B turns three times for each turn of gear C, then A turns six times for each turn of C. Sensitivities along a chain of functions multiply. A neural network is a long chain of functions, and backpropagation (Chapter 11) multiplies the local sensitivities of all its links.

Example 6.14 (The chain rule in action). In each case identify the outer and inner functions.

(a) $\frac{\mathrm{d}}{\mathrm{d}x}(3x+1)^5$: outer u^5, inner $u = 3x+1$. Result: $5(3x+1)^4 \cdot 3 = 15(3x+1)^4$.
(b) $\frac{\mathrm{d}}{\mathrm{d}x} e^{-x^2}$: outer e^u, inner $u = -x^2$. Result: $e^{-x^2} \cdot (-2x) = -2x\, e^{-x^2}$.
(c) $\frac{\mathrm{d}}{\mathrm{d}x} \ln(1+x^2)$: outer $\ln u$, inner $u = 1+x^2$. Result: $\frac{1}{1+x^2} \cdot 2x = \frac{2x}{1+x^2}$.
(d) $\frac{\mathrm{d}}{\mathrm{d}x} \sin(e^{2x})$: three layers. Working from the outside in: $\cos(e^{2x}) \cdot e^{2x} \cdot 2$.

Common pitfall (Forgetting the inner derivative). The most common mistake is to differentiate the outer function and stop: $\frac{\mathrm{d}}{\mathrm{d}x} e^{3x}$ is $3e^{3x}$, not e^{3x}. Always ask: "what is inside, and have I multiplied by its derivative?"

Table of derivatives

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
c	0	e^x	e^x
x^n	nx^{n-1}	a^x	$a^x \ln a$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$\ln x$	$\frac{1}{x}$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$\sin x$	$\cos x$
$\sigma(x)$	$\sigma(x)(1-\sigma(x))$	$\cos x$	$-\sin x$
$\tanh x$	$1 - \tanh^2 x$	$\mathrm{ReLU}(x)$	0 for $x<0$, 1 for $x>0$

$(cf)' = cf'$, $(f+g)' = f'+g'$, $(fg)' = f'g + fg'$,
$(f/g)' = (f'g - fg')/g^2$, $(f(g(x)))' = f'(g(x))\, g'(x)$

6.4.5 Derivatives that machine learning uses every day

Example 6.15 (The derivative of the sigmoid)**.** Show that $\sigma'(x) = \sigma(x)(1-\sigma(x))$, where $\sigma(x) = \frac{1}{1+e^{-x}}$.

Solution. Write $\sigma(x) = (1+e^{-x})^{-1}$ and use the chain rule with outer function u^{-1} and inner function $u = 1+e^{-x}$, whose derivative is $-e^{-x}$:

$$\sigma'(x) = -(1+e^{-x})^{-2}\cdot(-e^{-x}) = \frac{e^{-x}}{(1+e^{-x})^2} = \frac{1}{1+e^{-x}}\cdot\frac{e^{-x}}{1+e^{-x}}.$$

The first factor is $\sigma(x)$. The second is $\frac{(1+e^{-x})-1}{1+e^{-x}} = 1-\sigma(x)$.

Why this matters for ML (Saturation and vanishing gradients)**.** Since $\sigma(x)(1-\sigma(x))$ is a product of two numbers that add up to 1, it is largest when both equal $\frac{1}{2}$, i.e. at $x=0$, where $\sigma'(0) = \frac{1}{4}$. For large $|x|$ the sigmoid is flat and $\sigma'(x) \approx 0$: the neuron is *saturated*. By the chain rule, a signal passing backwards through ten sigmoid layers is multiplied by at most $\left(\frac{1}{4}\right)^{10} \approx 10^{-6}$. This *vanishing gradient* problem made deep sigmoid networks very hard to train, and is one reason ReLU, whose slope is 1 for all positive inputs, took over.

Example 6.16 (Differentiating a tiny model's loss)**.** A one-weight model predicts $\hat{y} = \sigma(wx)$, and its loss on one data point is $L(w) = (y-\sigma(wx))^2$. Find $L'(w)$, and evaluate it for $x=2$, $y=1$, $w=0.5$.

Solution. Three nested layers: $w \mapsto z = wx \mapsto \hat{y} = \sigma(z) \mapsto L = (y-\hat{y})^2$. The chain rule multiplies their derivatives:

$$L'(w) = \underbrace{-2(y-\hat{y})}_{\mathrm{d}L/\mathrm{d}\hat{y}}\cdot\underbrace{\sigma(z)\big(1-\sigma(z)\big)}_{\mathrm{d}\hat{y}/\mathrm{d}z}\cdot\underbrace{x}_{\mathrm{d}z/\mathrm{d}w}.$$

With the given numbers, $z=1$, $\hat{y} = \sigma(1) \approx 0.731$ and $\sigma'(1) \approx 0.197$, so $L'(0.5) \approx -2(0.269)(0.197)(2) \approx -0.212$. The derivative is negative: increasing w will decrease the loss, so training should increase w. This little computation is backpropagation in miniature.

Exercises for Section 6.4

Exercise 6.11. Differentiate: (a) $4x^3 - 2x + 7$; (b) $\sqrt{x} + \frac{1}{x}$; (c) $5e^x - 3\ln x$; (d) $x^{3/2}$.

Solution on p. 252 →

Exercise 6.12. Differentiate: (a) x^2e^x; (b) $x\ln x$; (c) $\frac{x^2}{x+1}$.

Solution on p. 252 →

Exercise 6.13. Differentiate: (a) $(2x+1)^4$; (b) e^{3x}; (c) $\ln(x^2+1)$; (d) $\sin(x^2)$; (e) $e^{-x^2/2}$.

Solution on p. 252 →

Exercise 6.14. Show that the derivative of the softplus function $\ln(1+e^x)$ is the sigmoid $\sigma(x)$.

Solution on p. 252 →

Exercise 6.15. For the loss $L(w) = (y - wx)^2$ of a linear model with one data point $x = 3$, $y = 5$, find $L'(w)$ and evaluate it at $w = 1$. Should w be increased or decreased to reduce the loss?

Solution on p. 252 →

Exercise 6.16. Show that $\tanh'(x) = 1 - \tanh^2(x)$, using the quotient rule on $\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

Solution on p. 252 →

6.5 Maxima, minima and the shape of a graph

6.5.1 Critical points

At the top of a hill or the bottom of a valley, the tangent line is horizontal. This simple observation is the basis of optimisation.

Definition 6.17 (Local extrema and critical points)**.** A function f has a **local minimum** at c if $f(c) \le f(x)$ for all x near c, and a **local maximum** if $f(c) \ge f(x)$ for all x near c. It has a **global minimum** at c if $f(c) \le f(x)$ for *all* x in the domain. A point c where $f'(c) = 0$ is called a **critical point** (or stationary point).

Theorem 6.18 (Extrema occur at critical points)**.** If f is differentiable and has a local minimum or maximum at an interior point c of its domain, then $f'(c) = 0$.

The converse is false: $f(x) = x^3$ has $f'(0) = 0$ but neither a minimum nor a maximum at 0 (it keeps increasing). So critical points are only *candidates*, and we need tests to classify them.

6.5.2 The second derivative and concavity

The derivative of f' is the **second derivative** f'' (also written $\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}$). It measures how the slope itself changes:

- where $f'' > 0$, the slope increases: the graph bends upwards like a cup and is called *convex* (or concave up);
- where $f'' < 0$, the slope decreases: the graph bends downwards like a cap and is called *concave* (or concave down);
- a point where the bending changes direction is an **inflection point**.

Theorem 6.19 (Second derivative test)**.** Let $f'(c) = 0$.

- If $f''(c) > 0$, then f has a local minimum at c (bottom of a cup).
- If $f''(c) < 0$, then f has a local maximum at c (top of a cap).
- If $f''(c) = 0$, the test is inconclusive.

Alternatively, the *first derivative test* looks at the sign of f' on either side of c: a change from $-$ to $+$ (falling, then rising) means a minimum, and a change from $+$ to $-$ means a maximum.

Example 6.20 (Classifying critical points)**.** Find and classify the critical points of $f(x) = x^3 - 3x$ (the cubic of Figure 2.6).

Solution. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$, so the critical points are $x = \pm 1$. Next, $f''(x) = 6x$. At $x = 1$: $f''(1) = 6 > 0$, a local minimum with value $f(1) = -2$. At $x = -1$: $f''(-1) = -6 < 0$, a local maximum with value $f(-1) = 2$. Since $f''(x) = 6x$ changes sign at 0, the origin is an inflection point. Neither extremum is global: $f(x) \to \pm\infty$ as $x \to \pm\infty$.

Example 6.21 (The mean minimises the squared error)**.** In Section 2.4 we saw that the number c that minimises the total squared error $L(c) = \sum_{i=1}^{n} (y_i - c)^2$ is the mean of the y_i. Prove it with calculus.

Solution. By the sum and chain rules, $L'(c) = \sum_{i=1}^{n} 2(y_i - c)\cdot(-1) = -2\left(\sum_{i=1}^{n} y_i - nc\right)$. Setting $L'(c) = 0$ gives $nc = \sum_i y_i$, i.e. $c = \frac{1}{n}\sum_i y_i = \bar{y}$. And $L''(c) = 2n > 0$, so this critical point is a minimum (in fact the global one, since L is a parabola opening upwards).

Remark (The median minimises the absolute error)**.** If instead we minimise $\sum_i |y_i - c|$, the derivative of each term is -1 when $c < y_i$ and $+1$ when $c > y_i$. The total derivative is (number of data points below c) $-$ (number above c), which is zero when c is in the middle of the data: the *median*. So the mean squared error is minimised by the mean and the mean absolute error by the median, which is why models trained with absolute errors are less sensitive to outliers.

Example 6.22 (An optimisation problem)**.** A rectangle has perimeter 20 m. Which shape has the largest area?

Solution. If one side is x, the other is $10 - x$, and the area is $A(x) = x(10-x) = 10x - x^2$ for $0 \le x \le 10$. Then $A'(x) = 10 - 2x = 0$ gives $x = 5$, and $A''(x) = -2 < 0$, a maximum. The best rectangle is the 5×5 square, with area $25\,\text{m}^2$. (At the endpoints $x = 0$ and $x = 10$ the area is 0.)

6.5.3 Local versus global

A function can have many local minima (Figure 6.2), and derivatives only see the neighbourhood of a point. To find a *global* minimum we must compare the values at all critical points (and at the ends of the domain, if there are any). For special functions called *convex* functions, every local minimum is automatically global; we study them in Chapter 8.

Why this matters for ML (Loss landscapes)**.** The loss of a neural network, viewed as a function of its weights, is a landscape with many hills and valleys, so it has many critical points. One might fear that training would constantly get stuck in poor local minima. In practice, for large networks, this is rarely the main problem; in many dimensions most critical points turn out to be *saddle points* (Chapter 7) rather than

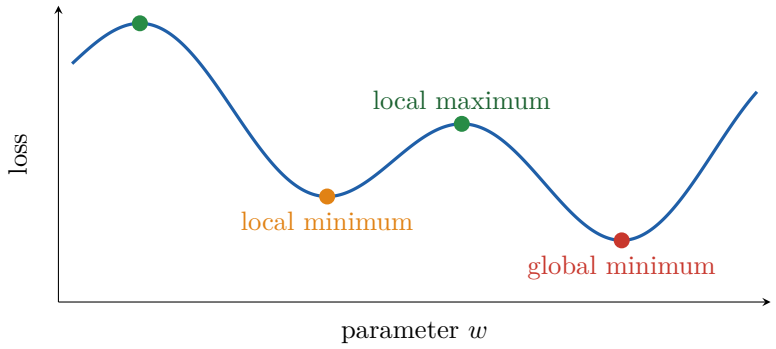


Figure 6.2: A loss function of one parameter with several critical points (two local maxima in green, two local minima). All have $f' = 0$, but only one is the global minimum. Gradient descent started at the left, say at $w = 2$, would slide into the local minimum and stay there.

minima, and many of the minima found by gradient descent are almost equally good.

Exercises for Section 6.5

Exercise 6.17. Find and classify the critical points of $f(x) = x^3 - 6x^2 + 9x + 1$.

Solution on p. 252 →

Exercise 6.18. Use derivatives to find the minimum of $f(x) = x^2 - 4x + 7$, and compare with completing the square.

Solution on p. 252 →

Exercise 6.19. Find the value of c that minimises the weighted squared error $L(c) = 2(c-1)^2 + (c-4)^2$. Interpret the answer.

Solution on p. 252 →

Exercise 6.20. Find the maximum of $f(x) = x\,e^{-x}$ for $x \geq 0$.

Solution on p. 252 →

Exercise 6.21. Show that the second derivative test is inconclusive for $f(x) = x^4$ at 0, and decide by another argument whether 0 is a minimum.

Solution on p. 252 →

6.6 Linear and quadratic approximation

6.6.1 The tangent line approximation

Zoom in far enough on the graph of a differentiable function and it looks like a straight line: its tangent line. This gives the **linear approximation**

$$f(a+h) \approx f(a) + f'(a)\,h$$

valid for small h. For example, with $f(x) = \sqrt{x}$, $a = 4$ and $h = 0.1$: $f(4) = 2$ and $f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$, so $\sqrt{4.1} \approx 2 + \frac{1}{4}(0.1) = 2.025$. The true value is $2.02485\ldots$.

6.6.2 The quadratic approximation

A straight line cannot follow the bending of a curve. Adding a term with the second derivative fixes this:

$$f(a+h) \approx f(a) + f'(a)\,h + \tfrac{1}{2} f''(a)\,h^2$$

For $\sqrt{x}$ at 4: $f''(x) = -\frac{1}{4}x^{-3/2}$, so $f''(4) = -\frac{1}{32}$ and $\sqrt{4.1} \approx 2.025 - \frac{1}{2}\cdot\frac{1}{32}\cdot 0.01 \approx 2.024844$, correct to six decimal places. The factor $\frac{1}{2}$ makes the second derivative of the approximation equal to $f''(a)$.

6.6.3 Taylor polynomials

Continuing the pattern gives the **Taylor polynomial** of degree n:

$$f(a+h) \approx f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \frac{f'''(a)}{3!}h^3 + \cdots + \frac{f^{(n)}(a)}{n!}h^n,$$

where $f^{(k)}$ is the k-th derivative. For e^x at $a = 0$, all derivatives equal 1, so

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots + \frac{x^n}{n!},$$

and the approximation improves as n grows (Figure 6.3). For example, $e^{0.1} \approx 1 + 0.1 + 0.005 = 1.105$, while the true value is $1.10517\ldots$.

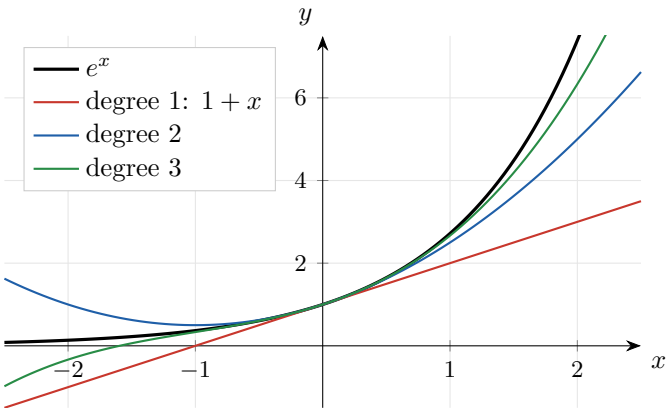


Figure 6.3: Taylor polynomials of e^x around 0. Each extra term makes the approximation good over a wider range around 0.

Intuition (Every smooth minimum looks like a parabola)**.** At a minimum a we have $f'(a) = 0$, so the quadratic approximation becomes

$$f(a+h) \approx f(a) + \tfrac{1}{2} f''(a)\,h^2.$$

Close to the bottom, every smooth valley is a parabola, and its width is controlled by the curvature $f''(a)$: a large $f''(a)$ means a narrow, steep valley; a small one means a wide, flat valley. In Chapter 8 this explains how large the steps of gradient descent may be.

Why this matters for ML. Optimisation algorithms are built on these approximations. Gradient descent uses the linear approximation: it trusts the tangent to tell it which way is downhill. Newton's method uses the quadratic approximation and jumps straight to the bottom of the approximating parabola (Chapter 8). The multivariable versions of both are developed in Chapter 7.

Exercises for Section 6.6

Exercise 6.22. Find the linear approximation of $\ln(1+x)$ near $x=0$ and use it to estimate $\ln 1.05$.

Solution on p. 252 →

Exercise 6.23. Find the quadratic approximation of $\cos x$ near 0 and estimate $\cos(0.2)$.

Solution on p. 252 →

Exercise 6.24. Find the quadratic approximation of $f(x)=\frac{1}{x}$ around $a=1$ and use it to estimate $\frac{1}{1.1}$.

Solution on p. 252 →

Exercise 6.25. A loss function has a minimum at $w=a$ with $f(a)=2$ and $f''(a)=8$. Estimate $f(a+0.1)$ and $f(a-0.1)$.

Solution on p. 253 →

6.7 Integration

Motivation. Differentiation turns a quantity into its rate of change. Integration goes the other way: from a rate of change it recovers the accumulated total (from speed to distance travelled), and more generally it computes areas under curves. In machine learning, integrals turn up whenever continuous quantities are involved: the probability that a continuous random variable falls in an interval is an area under its density curve, and averages (expected values) of continuous quantities are integrals (Chapter 9).

6.7.1 Area as a limit of sums

How large is the area between the graph of $f(x)=x^2$ and the x-axis, for x from 0 to 1? Approximate it by rectangles (Figure 6.4). Using n rectangles of width $\frac{1}{n}$ whose heights are the function values at their right ends gives the estimate

$$\sum_{k=1}^{n} f\left(\frac{k}{n}\right)\frac{1}{n}.$$

For $n=4,8,10,100$ these sums are 0.469, 0.398, 0.385, 0.338; as the rectangles get thinner, the sums approach $\frac{1}{3}$.

Definition 6.23 (Definite integral). The **definite integral** of f from a to b,

$$\int_a^b f(x)\,\mathrm{d}x,$$

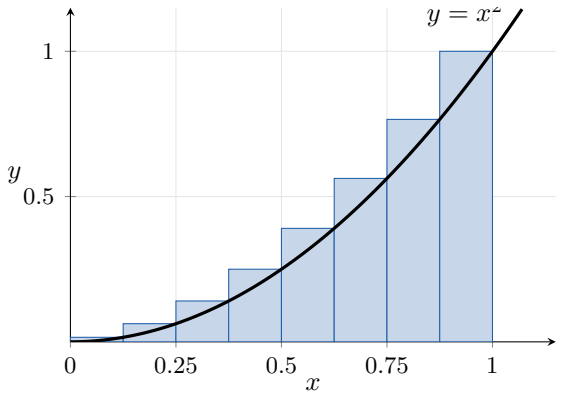


Figure 6.4: Approximating the area under $y = x^2$ on $[0, 1]$ by 8 rectangles (total area 0.398). With thinner rectangles the total approaches the exact area $\frac{1}{3}$.

is the limit of such sums of rectangle areas as the width of the rectangles goes to zero. It is the *signed* area between the graph and the x-axis: area above the axis counts positively and area below counts negatively.

Notation. The integral sign $\int$ is an elongated S, for "sum". The symbol $\mathrm{d}x$ recalls the width of the infinitely thin rectangles, and $f(x)\,\mathrm{d}x$ the area of one of them. The numbers a and b are the *limits of integration*, and f is the *integrand*. The variable x is a dummy variable, like the index of a sum: $\int_0^1 x^2\,\mathrm{d}x = \int_0^1 t^2\,\mathrm{d}t$.

6.7.2 The fundamental theorem of calculus

Computing limits of sums directly is laborious. Fortunately there is a shortcut, which connects integration to differentiation.

Definition 6.24 (Antiderivative). A function F with $F' = f$ is called an **antiderivative** of f. If F is one antiderivative, all others are $F + C$ for a constant C, and we write the **indefinite integral** $\int f(x)\,\mathrm{d}x = F(x) + C$.

Theorem 6.25 (Fundamental theorem of calculus). If f is continuous on $[a, b]$ and F is any antiderivative of f, then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$$

The right-hand side is often written $\big[F(x)\big]_a^b$.

Intuition (Accumulating a rate gives the total change). Let $F(t)$ be the position of a car and $f(t) = F'(t)$ its speed. In a tiny time interval $\mathrm{d}t$, the car moves about $f(t)\,\mathrm{d}t$. Adding up all these tiny moves from time a to time b (that is, integrating the speed) gives the total distance moved, which is the change of position $F(b) - F(a)$. Integration and differentiation are inverse operations.

6.7.3 Computing integrals

Reading the table of derivatives backwards gives a table of antiderivatives:

$f(x)$	$\int f(x)\,dx$	$f(x)$	$\int f(x)\,dx$
$x^n \quad (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$	e^x	$e^x + C$
$\frac{1}{x}$	$\ln\lvert x\rvert + C$	e^{kx}	$\frac{1}{k}e^{kx} + C$
$\cos x$	$\sin x + C$	$\sin x$	$-\cos x + C$

Integrals are also linear: $\int (cf + g) = c\int f + \int g$.

Example 6.26 (Definite integrals). (a) $\int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3} - 0 = \frac{1}{3}$, confirming the rectangle computation.

(b) $\int_1^3 (2x+1)\,dx = \left[x^2 + x\right]_1^3 = (9+3) - (1+1) = 10$.

(c) $\int_0^1 e^x\,dx = \left[e^x\right]_0^1 = e - 1 \approx 1.718$.

(d) $\int_0^\pi \sin x\,dx = \left[-\cos x\right]_0^\pi = -(-1) - (-1) = 2$.

(e) $\int_{-1}^1 x\,dx = \left[\frac{x^2}{2}\right]_{-1}^1 = \frac{1}{2} - \frac{1}{2} = 0$: the area above the axis (for $x > 0$) cancels the area below.

The chain rule, read backwards, gives the technique of *substitution*: for instance, since $\frac{d}{dx}e^{x^2} = 2x\,e^{x^2}$, we have $\int 2x\,e^{x^2}\,dx = e^{x^2} + C$.

6.7.4 Integrals over infinite intervals

An integral can extend to infinity, as a limit:

$$\int_0^\infty e^{-x}\,dx = \lim_{b\to\infty}\left[-e^{-x}\right]_0^b = \lim_{b\to\infty}\left(1 - e^{-b}\right) = 1.$$

Although the region under e^{-x} stretches out forever, its area is finite. Such integrals are essential in probability. A famous one, which we state without proof, is the area under the bell curve:

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.$$

Why this matters for ML (Integrals in machine learning). For a continuous random variable with density $p(x)$, the probability of landing between a and b is $\int_a^b p(x)\,dx$, the total area $\int_{-\infty}^{\infty} p(x)\,dx$ must equal 1, and the average value is $\int x\,p(x)\,dx$ (Chapter 9). The constant $\sqrt{\pi}$ above is why $\sqrt{2\pi}$ appears in the formula of the normal distribution. A popular measure of classifier quality, the *area under the ROC curve* (AUC), is literally an integral. In practice, integrals that cannot be done by hand are estimated numerically or by random sampling (Chapter 10).

Exercises for Section 6.7

Exercise 6.26. Compute: (a) $\int_0^2 3x^2\,dx$; (b) $\int_1^4 \sqrt{x}\,dx$; (c) $\int_0^1 (e^x + 1)\,dx$; (d) $\int_1^2 \frac{1}{x}\,dx$.

Solution on p. 253 →

Exercise 6.27. Find antiderivatives of x^5, $\frac{1}{x^2}$, e^{2x} and $\cos x$.

Solution on p. 253 →

Exercise 6.28. Approximate $\int_0^2 x\,\mathrm{d}x$ using 4 rectangles whose heights are taken at their *left* ends, and compare with the exact value.

Solution on p. 253 →

Exercise 6.29. The function $p(x) = 2x$ on $[0, 1]$ (and 0 elsewhere) is a probability density. Check that its total area is 1, and compute $\int_0^1 x\,p(x)\,\mathrm{d}x$, its average value.

Solution on p. 253 →

Exercise 6.30. Compute $\int_0^\infty e^{-2x}\,\mathrm{d}x$.

Solution on p. 253 →

6.8 Derivatives on a computer

The definition of the derivative suggests a way to estimate it numerically: pick a small h and compute a difference quotient. The *forward difference* $\frac{f(x+h)-f(x)}{h}$ works, but the **central difference**

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

is much more accurate, because the errors on the two sides largely cancel. For $f(x) = x^3$ at $x = 2$, where the true derivative is 12:

h	forward difference	central difference
0.1	12.61	12.01
0.01	12.0601	12.0001

Common pitfall (Do not make h too small)**.** It is tempting to take h extremely small, say 10^{-15}. But computers store numbers with limited precision (Section 1.1), and subtracting two almost equal numbers $f(x+h) - f(x)$ loses most of the significant digits. In double precision, values of h around 10^{-5} to 10^{-7} usually work best.

Why this matters for ML (Gradient checking and automatic differentiation)**.** Deep learning libraries do not differentiate numerically; numerical differentiation needs a separate evaluation of the loss for every one of millions of weights, and it is inexact. Instead they use *automatic differentiation*, which applies the chain rule exactly and efficiently to the sequence of operations the program performed (Chapter 11). Numerical derivatives remain a valuable debugging tool: *gradient checking* compares the derivatives computed by hand-written code against central differences on a few weights. If they disagree, there is a bug.

Exercises for Section 6.8

Exercise 6.31. Estimate the derivative of e^x at $x = 1$ with $h = 0.01$ using the forward and the central difference. Compare with the exact value.

Solution on p. 253 →

Exercise 6.32. Use Example 6.16 to explain how you would check the formula for $L'(w)$ numerically.

Solution on p. 253 →

6.9 Chapter summary

Key takeaways

- The average rate of change $\frac{f(b)-f(a)}{b-a}$ is the slope of a secant; the derivative $f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ is the slope of the tangent, an instantaneous rate, and a sensitivity: $f(a+h) \approx f(a) + f'(a)h$.
- Limits describe what a function approaches; continuous functions have no jumps; differentiable functions have no corners either.
- Rules: power, constant, sum, product $(fg)' = f'g + fg'$, quotient, and the chain rule $\big(f(g(x))\big)' = f'(g(x))g'(x)$: rates multiply.
- $(e^x)' = e^x$, $(\ln x)' = 1/x$, $\sigma' = \sigma(1-\sigma) \le \frac{1}{4}$, $\mathrm{ReLU}' \in \{0, 1\}$.
- At interior extrema $f' = 0$; $f'' > 0$ means a minimum (cup), $f'' < 0$ a maximum (cap). The mean minimises squared error.
- Taylor: $f(a+h) \approx f(a) + f'(a)h + \frac{1}{2}f''(a)h^2$; near a minimum every smooth function looks like a parabola.
- $\int_a^b f(x)\,\mathrm{d}x$ is a signed area, computed as $F(b) - F(a)$ for an antiderivative F.
- Central differences estimate derivatives numerically, useful for gradient checking.

Machine learning models have not one parameter but millions. The next chapter extends every idea of this chapter to functions of many variables, leading to the gradient, the most important vector in machine learning.

Chapter 7

Multivariable Calculus

Motivation. The loss of a real model depends not on one parameter but on thousands, millions or billions of them. To train it, we need to know how the loss responds to a small change in *each* parameter, and then which *combined* change of all parameters decreases the loss fastest. The answer is a single vector, the *gradient*, and computing it efficiently through a deep network is the job of the multivariable *chain rule*. Second derivatives also generalise, to a matrix called the *Hessian*, which describes the curvature of the loss landscape and tells minima apart from saddle points. This chapter extends every idea of Chapter 6 to functions of many variables, using the linear algebra of Chapters 4 and 5 as its language.

In this chapter you will learn to...

- visualise functions of two variables with surfaces and contour plots;
- compute partial derivatives, including second-order ones;
- compute gradients and explain why the gradient points uphill and is perpendicular to contour lines;
- compute directional derivatives;
- apply the multivariable chain rule, including on computation graphs;
- compute Jacobian matrices and use the chain rule in matrix form;
- compute Hessians and classify critical points as minima, maxima or saddles;
- write first- and second-order Taylor approximations and use them to understand gradient descent;
- compute simple double integrals.

7.1 Functions of several variables

A function of several variables takes a vector as input and returns a number: $f\colon \mathbb{R}^n \to \mathbb{R}$. Examples:

- $f(x, y) = x^2 + y^2$, the squared distance from the origin;
- $f(\boldsymbol{x}) = \boldsymbol{w}^\top \boldsymbol{x} + b$, a linear model with n inputs;
- the mean squared error of a linear regression model, viewed as a function of its parameters w and b: $L(w, b) = \frac{1}{n}\sum_{i=1}^{n}\big(y_i - (wx_i + b)\big)^2$.

In the last example the data (x_i, y_i) are fixed numbers and the variables are the parameters. This change of viewpoint is typical of machine learning: during training, the *parameters* are the variables.

7.1.1 Pictures: surfaces and contour plots

A function of two variables can be drawn as a *surface*: above each point (x, y) of the plane, plot the height $z = f(x, y)$ (Figure 7.1, left). Surfaces are attractive but hard to draw and read. A **contour plot** is often more useful: it draws the **level curves** $f(x, y) = c$ for several values of c, exactly like the contour lines of a hiking map, which connect points

of equal height (Figure 7.1, right). Where contour lines are close together, the terrain is steep.

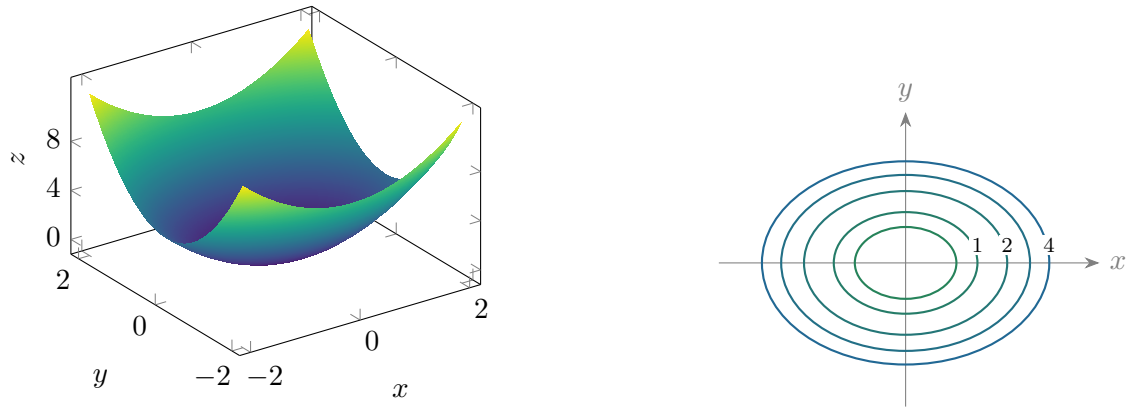


Figure 7.1: Two pictures of $f(x, y) = x^2 + 2y^2$. Left: its graph, a bowl-shaped surface. Right: its contour plot, the ellipses $x^2 + 2y^2 = c$ for $c = 0.5, 1, 2, 3, 4$ (some labelled). The contours are closer together in the y-direction, where the bowl is steeper.

Why this matters for ML (Loss landscapes). The graph of the loss as a function of the parameters is called the *loss landscape*. For a model with two parameters we can draw it, and training looks like a ball rolling down into a valley. For a real network the landscape lives in millions of dimensions, but the intuition from two dimensions (valleys, ridges, saddles, flat plateaus) remains the main way practitioners think about training.

Exercises for Section 7.1

Exercise 7.1. Let $f(x, y) = x^2 - y + 3$. Evaluate $f(2, 1)$, and describe the level curve $f(x, y) = 3$.

Solution on p. 253 →

Exercise 7.2. Describe the contour lines of $f(x, y) = x + 2y$ and of $g(x, y) = x^2 + y^2$.

Solution on p. 253 →

Exercise 7.3. For the two data points $(x, y) = (1, 2)$ and $(2, 4)$, the mean squared error of the model $\hat{y} = wx + b$ is $L(w, b) = \frac{1}{2}\left[(2 - w - b)^2 + (4 - 2w - b)^2\right]$. Evaluate $L(2, 0)$ and $L(1, 1)$. Which parameters fit better?

Solution on p. 253 →

7.2 Partial derivatives

The simplest way to measure how a function of several variables changes is to change *one* variable at a time, keeping all the others fixed.

Definition 7.1 (Partial derivative). The **partial derivative** of $f(x, y)$ with respect to x is

$$\frac{\partial f}{\partial x}(x, y) = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h},$$

the derivative of f with respect to x when y is held constant. The partial derivative

$\frac{\partial f}{\partial y}$ is defined in the same way, holding x constant. For a function of n variables there is one partial derivative $\frac{\partial f}{\partial x_i}$ for each variable.

Notation. The curly ∂ ("partial" or "del") signals that other variables are being held fixed. Common alternatives are f_x, $\partial_x f$ and $\partial f/\partial x$. So $\frac{\partial f}{\partial x}$ is read "partial f by partial x" or "the partial derivative of f with respect to x".

Picture this. Slice the surface $z = f(x, y)$ with the vertical plane $y = y_0$. The cut is a curve, the graph of the one-variable function $x \mapsto f(x, y_0)$. The partial derivative $\frac{\partial f}{\partial x}(x_0, y_0)$ is the slope of that curve at x_0: the steepness of the surface if you walk in the x-direction (Figure 7.2).

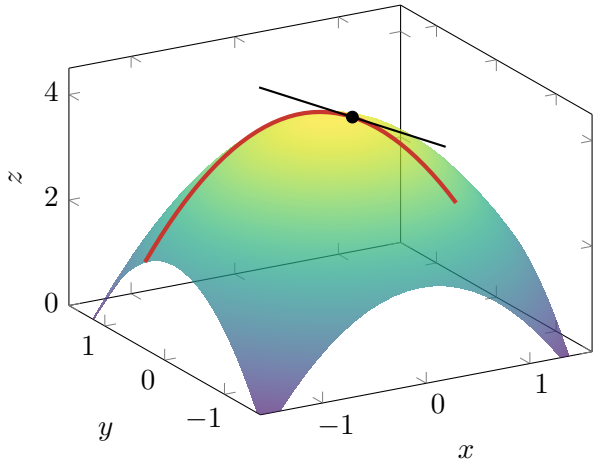


Figure 7.2: The surface $z = 4 - x^2 - y^2$ cut by the plane $y = 0.5$ (red curve). The partial derivative $\frac{\partial f}{\partial x}(0.5, 0.5) = -2(0.5) = -1$ is the slope of the black tangent line to the red curve.

7.2.1 Computing partial derivatives

There is nothing new to learn: treat every other variable as if it were a number, and use the rules of Chapter 6.

Example 7.2 (Partial derivatives)**.** Compute the partial derivatives of (a) $f(x, y) = x^2y + 3y^3$; (b) $g(x, y) = e^{xy}$; (c) $h(x, y) = \ln(x^2 + y^2)$.

Solution. (a) With y constant, x^2y behaves like "constant times x^2" and $3y^3$ is a constant: $\frac{\partial f}{\partial x} = 2xy$. With x constant: $\frac{\partial f}{\partial y} = x^2 + 9y^2$.

(b) By the chain rule, $\frac{\partial g}{\partial x} = e^{xy} \cdot y = y\,e^{xy}$ and $\frac{\partial g}{\partial y} = x\,e^{xy}$.

(c) $\frac{\partial h}{\partial x} = \frac{2x}{x^2+y^2}$ and $\frac{\partial h}{\partial y} = \frac{2y}{x^2+y^2}$.

7.2.2 Higher-order partial derivatives

Partial derivatives are themselves functions of x and y, so we can differentiate them again. There are four second-order partial derivatives of $f(x, y)$:

$$\frac{\partial^2 f}{\partial x^2}, \qquad \frac{\partial^2 f}{\partial y^2}, \qquad \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right), \qquad \frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right).$$

For $f = x^2y + 3y^3$: $f_{xx} = 2y$, $f_{yy} = 18y$, and both mixed partials equal $2x$. That the mixed partials agree is no accident.

Theorem 7.3 (Symmetry of mixed partials)**.** If the second partial derivatives of f are continuous, then $\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x}$: the order of differentiation does not matter.

Why this matters for ML (One weight at a time)**.** For a model with weights $w_1, \ldots, w_n$, the partial derivative $\frac{\partial L}{\partial w_i}$ answers: "if I change only w_i a tiny bit, how fast does the loss change?" For example, for a single data point (x, y) and the model $\hat{y} = wx + b$ with squared error $L(w, b) = (y - wx - b)^2$, the chain rule gives

$$\frac{\partial L}{\partial w} = -2x(y - wx - b), \qquad \frac{\partial L}{\partial b} = -2(y - wx - b).$$

Both are proportional to the error $y - \hat{y}$: parameters are adjusted more when the prediction is worse, and the weight w more when the input x is larger.

Exercises for Section 7.2

Exercise 7.4. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for $f(x, y) = 3x^2y - 2xy^3 + 5$.

Solution on p. 253 →

Exercise 7.5. Find the partial derivatives of $f(x, y) = x\,e^y + \ln x$.

Solution on p. 253 →

Exercise 7.6. For $f(x, y) = \sin(xy)$, compute f_x, f_y, and both mixed second partial derivatives, and check that they agree.

Solution on p. 253 →

Exercise 7.7. For the single-point loss $L(w, b) = (y - wx - b)^2$ with $x = 2$ and $y = 7$, compute both partial derivatives at $w = 1$, $b = 1$.

Solution on p. 253 →

7.3 The gradient

Definition 7.4 (Gradient)**.** The **gradient** of $f\colon \mathbb{R}^n \to \mathbb{R}$ at $\boldsymbol{x}$ is the vector of all its partial derivatives:

$$\nabla f(\boldsymbol{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(\boldsymbol{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\boldsymbol{x}) \end{pmatrix}$$

The symbol ∇ is called "nabla" or "del", and ∇f is read "grad f". When it is important to say which variables we differentiate with respect to, we write, for example, $\nabla_{\boldsymbol{w}} L$.

The gradient packs n numbers into one vector that lives in the same space as the input $\boldsymbol{x}$. For $f(x, y) = x^2 + y^2$, $\nabla f = (2x, 2y)$; at the point $(1, 2)$ the gradient is $(2, 4)$.

Example 7.5 (Gradients of common functions of vectors)**.** For $\boldsymbol{x} \in \mathbb{R}^n$:

(a) $f(\boldsymbol{x}) = \boldsymbol{a}^\top \boldsymbol{x} = a_1 x_1 + \cdots + a_n x_n$: each partial derivative is $\frac{\partial f}{\partial x_i} = a_i$, so $\nabla f = \boldsymbol{a}$. (The analogue of $\frac{\mathrm{d}}{\mathrm{d}x} ax = a$.)

(b) $f(\boldsymbol{x}) = \|\boldsymbol{x}\|^2 = x_1^2 + \cdots + x_n^2$: $\frac{\partial f}{\partial x_i} = 2x_i$, so $\nabla f = 2\boldsymbol{x}$. (The analogue of $\frac{\mathrm{d}}{\mathrm{d}x} x^2 = 2x$.)

(c) $f(\boldsymbol{x}) = \|\boldsymbol{x} - \boldsymbol{a}\|^2$: by the chain rule, $\nabla f = 2(\boldsymbol{x} - \boldsymbol{a})$.

7.3.1 The geometry of the gradient

Theorem 7.6 (The gradient points uphill)**.** At any point where $\nabla f \neq \boldsymbol{0}$:

(a) ∇f points in the direction in which f increases fastest (*steepest ascent*), and $-\nabla f$ in the direction of steepest descent;

(b) the length $\|\nabla f\|$ is the rate of increase in that direction;

(c) ∇f is perpendicular to the level curve (contour) through the point.

We prove this in Section 7.4. Figure 7.3 shows the picture: gradient arrows cross the contour lines at right angles, pointing uphill, and are longest where the contours are most crowded.

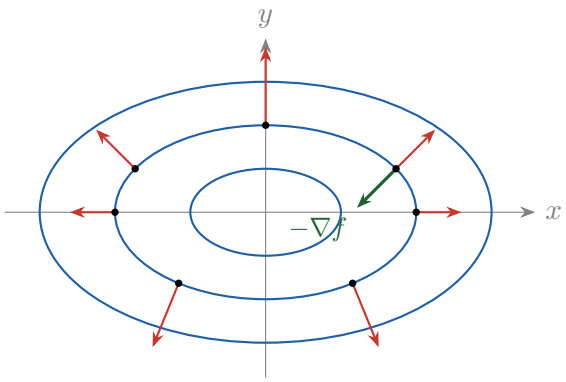


Figure 7.3: Contours of $f(x, y) = x^2 + 3y^2$ (blue) and gradient vectors $\nabla f = (2x, 6y)$ at points of the middle contour (red, scaled down). Each gradient is perpendicular to the contour and points uphill; the green arrow is the direction of steepest descent.

At a minimum or maximum of a differentiable function, there is no uphill direction, so the gradient is zero. Points with $\nabla f = \boldsymbol{0}$ are called *critical points*, as in one variable.

7.3.2 Linear approximation and the tangent plane

The one-variable approximation $f(a + h) \approx f(a) + f'(a)h$ becomes

$$\boxed{f(\boldsymbol{x} + \boldsymbol{h}) \approx f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top \boldsymbol{h}}$$

for a small vector $\boldsymbol{h}$: each variable contributes its partial derivative times its own change, and the contributions add up. For two variables, the graph of the right-hand side is the *tangent plane* to the surface. This formula is the reason the gradient governs optimisation.

Example 7.7 (The gradient of a regression loss)**.** For the data $(1,2)$, $(2,3)$, $(3,5)$ and the model $\hat{y} = wx + b$, the mean squared error is $L(w,b) = \frac{1}{3}\sum_{i=1}^{3}(y_i - wx_i - b)^2$. Compute ∇L at $(w,b) = (0,0)$ and find where $\nabla L = \mathbf{0}$.

Solution. Write $r_i = y_i - wx_i - b$ for the residuals. By the chain rule,

$$\frac{\partial L}{\partial w} = -\frac{2}{3}\sum_i x_i r_i, \qquad \frac{\partial L}{\partial b} = -\frac{2}{3}\sum_i r_i.$$

At $(0,0)$ the residuals are just $(2,3,5)$, so $\frac{\partial L}{\partial w} = -\frac{2}{3}(2+6+15) \approx -15.3$ and $\frac{\partial L}{\partial b} = -\frac{2}{3}(10) \approx -6.67$. The gradient $(-15.3, -6.67)$ points towards smaller w and b; the loss *decreases* fastest in the opposite direction, so training would increase both parameters, and w faster than b.

Setting both partial derivatives to zero gives $\sum_i x_i r_i = 0$ and $\sum_i r_i = 0$: the residuals must be orthogonal to the column of inputs and to the column of ones. These are exactly the normal equations of Section 5.6, with solution $w = 1.5$, $b = \frac{1}{3}$. Calculus and geometry give the same answer.

Exercises for Section 7.3

Exercise 7.8. Compute the gradient of $f(x,y) = x^2 y + y^3$ at the point $(1,2)$.

Solution on p. 253 →

Exercise 7.9. Find the critical point of $f(x,y) = x^2 + y^2 - 2x + 4y$.

Solution on p. 254 →

Exercise 7.10. Use the linear approximation at $(1,1)$ to estimate $f(1.1, 0.9)$ for $f(x,y) = x^2 + y^2$, and compare with the exact value.

Solution on p. 254 →

Exercise 7.11. For the data $(1,1)$ and $(2,3)$ and the model $\hat{y} = wx + b$ with mean squared error, compute ∇L at $(w,b) = (1,0)$.

Solution on p. 254 →

7.4 Directional derivatives

Partial derivatives measure the rate of change along the coordinate axes. What about other directions?

Definition 7.8 (Directional derivative)**.** The **directional derivative** of f at $\boldsymbol{x}$ in the direction of a unit vector $\boldsymbol{u}$ is

$$D_{\boldsymbol{u}} f(\boldsymbol{x}) = \lim_{h \to 0} \frac{f(\boldsymbol{x} + h\boldsymbol{u}) - f(\boldsymbol{x})}{h},$$

the rate of change of f as we move from $\boldsymbol{x}$ in the direction $\boldsymbol{u}$.

By the linear approximation, $f(\boldsymbol{x}+h\boldsymbol{u}) \approx f(\boldsymbol{x}) + h\,\nabla f(\boldsymbol{x})^\top \boldsymbol{u}$, so

Theorem 7.9 (Directional derivatives from the gradient)**.**

$$D_{\boldsymbol{u}} f(\boldsymbol{x}) = \nabla f(\boldsymbol{x}) \cdot \boldsymbol{u}.$$

Now the geometry of the gradient follows from the geometry of the dot product. By Theorem 4.8, $\nabla f \cdot \boldsymbol{u} = \|\nabla f\| \cos\theta$, where θ is the angle between $\boldsymbol{u}$ and the gradient. This is largest (namely $\|\nabla f\|$) when $\theta = 0$, i.e. when $\boldsymbol{u}$ points along ∇f; smallest (namely $-\|\nabla f\|$) when $\boldsymbol{u}$ points along $-\nabla f$; and zero when $\boldsymbol{u}$ is perpendicular to ∇f. Moving along a contour line does not change f, so the direction of the contour has zero directional derivative, which means it is perpendicular to the gradient. This proves Theorem 7.6.

Example 7.10 (Directional derivatives)**.** Let $f(x,y) = x^2 + 3y^2$ and $\boldsymbol{x} = (1,1)$, so $\nabla f = (2,6)$.

- In the direction $\boldsymbol{u} = (1,0)$: $D_{\boldsymbol{u}} f = 2$ (this is just $\frac{\partial f}{\partial x}$).
- In the direction $\boldsymbol{u} = (0.6, 0.8)$: $D_{\boldsymbol{u}} f = 1.2 + 4.8 = 6$.
- The steepest direction is $\frac{(2,6)}{\|(2,6)\|}$, with rate $\|(2,6)\| = \sqrt{40} \approx 6.32$; no direction gives a faster increase.

Why this matters for ML (Steepest descent)**.** To decrease a loss as quickly as possible from the current parameters, move in the direction $-\nabla L$. This is the idea of *gradient descent*: repeatedly replace the parameters $\boldsymbol{\theta}$ by $\boldsymbol{\theta} - \eta \nabla L(\boldsymbol{\theta})$ for a small step size η (Chapter 8). The steepest direction is only steepest *locally*, which is why the steps must be small and repeated many times.

Exercises for Section 7.4

Exercise 7.12. For $f(x,y) = x^2 + 3y^2$ at $(1,1)$, compute the directional derivative in the direction of $(1,1)$. (Remember to normalise.)

Solution on p. 254 →

Exercise 7.13. For $f(x,y) = xy$ at the point $(2,3)$: in which direction does f increase fastest, and at what rate? In which directions does f not change at all?

Solution on p. 254 →

Exercise 7.14. What is the directional derivative of f at a minimum, in any direction?

Solution on p. 254 →

7.5 The chain rule for several variables

7.5.1 One input, several paths

Suppose $z = f(x,y)$ and both x and y depend on a single variable t. When t changes a little, it changes x, which changes z; and it changes y, which also changes z. The total effect is the sum of the two effects.

Theorem 7.11 (Chain rule, two intermediate variables)**.** If $z = f(x, y)$ with $x = x(t)$ and $y = y(t)$, then

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\partial z}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial z}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}.$$

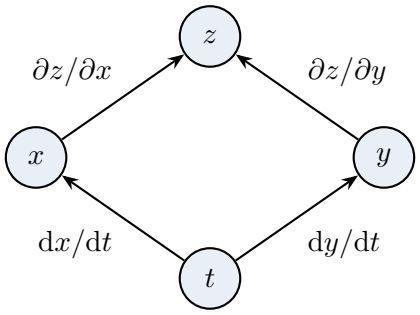


Figure 7.4: Dependency diagram for $z = f(x(t), y(t))$. The derivative $\frac{\mathrm{d}z}{\mathrm{d}t}$ is the sum, over the two paths from t to z, of the products of the derivatives along each path.

Example 7.12 (Chain rule along two paths)**.** Let $z = x^2 y$ with $x = \cos t$ and $y = \sin t$. Find $\frac{\mathrm{d}z}{\mathrm{d}t}$ at $t = 0$.

Solution. $\frac{\partial z}{\partial x} = 2xy$, $\frac{\partial z}{\partial y} = x^2$, $\frac{\mathrm{d}x}{\mathrm{d}t} = -\sin t$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = \cos t$. At $t = 0$: $x = 1$, $y = 0$, so $\frac{\mathrm{d}z}{\mathrm{d}t} = 2 \cdot 1 \cdot 0 \cdot 0 + 1 \cdot 1 = 1$. (Check: $z = \cos^2 t \sin t$, whose derivative at 0 is $-2\cos t \sin^2 t + \cos^3 t = 1$.)

7.5.2 The general rule: sum over paths

The same principle works for any number of variables and any number of layers.

Theorem 7.13 (General chain rule)**.** If z depends on intermediate variables $u_1, \ldots, u_m$, each of which depends on x, then

$$\boxed{\frac{\partial z}{\partial x} = \sum_{k=1}^{m} \frac{\partial z}{\partial u_k}\frac{\partial u_k}{\partial x}}$$

More generally, in any network of dependencies, $\frac{\partial z}{\partial x}$ is the sum, over all paths from x to z, of the product of the local derivatives along the path.

7.5.3 Computation graphs

Any formula can be broken into elementary steps and drawn as a **computation graph**: each node is an intermediate value, and arrows show which values are used to compute which. Derivatives can then be computed mechanically, one node at a time.

Example 7.14 (Gradients on a computation graph)**.** Let $f(x, y) = (x + y)(y + 1)$. Introduce $a = x + y$ and $b = y + 1$, so that $f = ab$. Compute all partial derivatives at $x = 2$, $y = 3$.

Solution. Forward pass (compute values): $a = 5$, $b = 4$, $f = 20$.

Backward pass (compute derivatives, from the output back to the inputs). The local derivatives of $f = ab$ are $\frac{\partial f}{\partial a} = b = 4$ and $\frac{\partial f}{\partial b} = a = 5$. Next, x influences f only through a, with $\frac{\partial a}{\partial x} = 1$, so $\frac{\partial f}{\partial x} = 4 \cdot 1 = 4$. But y influences f along two paths, through a and through b, so we add:

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial a}\frac{\partial a}{\partial y} + \frac{\partial f}{\partial b}\frac{\partial b}{\partial y} = 4 \cdot 1 + 5 \cdot 1 = 9.$$

Check by expanding: $f = xy + x + y^2 + y$, so $f_x = y + 1 = 4$ and $f_y = x + 2y + 1 = 9$. ✓

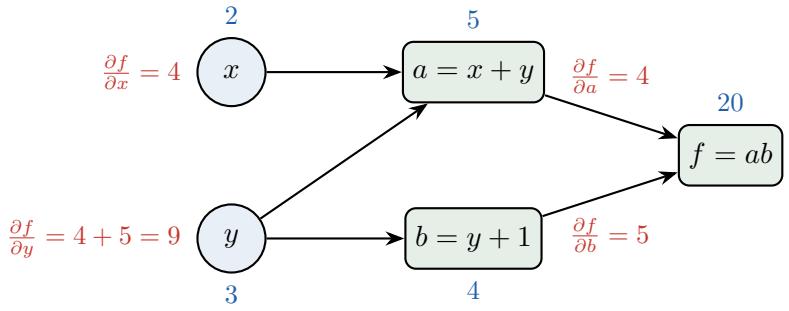


Figure 7.5: The computation graph of $f = (x + y)(y + 1)$ at $x = 2$, $y = 3$. Forward values are in blue; derivatives of the output, computed backwards, are in red. The two paths from y to f contribute 4 and 5.

Why this matters for ML (Backpropagation is the chain rule on a computation graph). A neural network and its loss form one large computation graph, with the weights as inputs and the loss as output. *Backpropagation* computes $\frac{\partial L}{\partial w}$ for every weight by exactly the procedure of Example 7.14: one forward pass to compute all intermediate values, then one backward pass that multiplies local derivatives and adds them up wherever paths merge. Its great virtue is efficiency: the derivatives of all the weights are obtained for roughly the cost of two evaluations of the network, because the products along shared parts of the paths are computed once and reused. We will carry this out in detail in Chapter 11.

Exercises for Section 7.5

Exercise 7.15. Let $z = x^2 + y^2$ with $x = 2t$ and $y = t^2$. Find $\frac{\mathrm{d}z}{\mathrm{d}t}$ with the chain rule, and check by substituting first.

Solution on p. 254 →

Exercise 7.16. Let $f(x, y) = xy$ with $x = 3t + 1$ and $y = t^2$. Find $\frac{\mathrm{d}f}{\mathrm{d}t}$ at $t = 1$.

Solution on p. 254 →

Exercise 7.17. Draw the computation graph of $f(x, y, z) = xy + z$ (with $q = xy$) and compute the forward value and all three partial derivatives at $x = -2$, $y = 5$, $z = -4$.

Solution on p. 254 →

Exercise 7.18. A neuron computes $\hat{y} = \sigma(w_1 x_1 + w_2 x_2 + b)$ with squared error $L = (\hat{y} - y)^2$. Use the chain rule to write $\frac{\partial L}{\partial w_1}$, $\frac{\partial L}{\partial w_2}$ and $\frac{\partial L}{\partial b}$, and evaluate them for $\boldsymbol{x} = (1, 2)$, $\boldsymbol{w} = (0.5, -0.5)$, $b = 0$, $y = 1$.

Solution on p. 254 →

7.6 Vector-valued functions and the Jacobian

A neural network layer takes a vector and returns a vector. Its derivative must describe how *each* output depends on *each* input, which calls for a matrix.

Definition 7.15 (Jacobian matrix)**.** Let $\boldsymbol{f}\colon \mathbb{R}^n \to \mathbb{R}^m$ have component functions $f_1, \dots, f_m$. Its **Jacobian matrix** at $\boldsymbol{x}$ is the $m \times n$ matrix of all first partial derivatives:

$$\boldsymbol{J_f}(\boldsymbol{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}, \qquad (\boldsymbol{J_f})_{ij} = \frac{\partial f_i}{\partial x_j}.$$

Row i is the gradient of f_i, written as a row.

Example 7.16 (Jacobians)**.** (a) For $\boldsymbol{f}(x, y) = (x^2 y,\ 5x + \sin y)$: $\boldsymbol{J} = \begin{pmatrix} 2xy & x^2 \\ 5 & \cos y \end{pmatrix}$.

(b) For a linear map $\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}$, each $f_i = \sum_j a_{ij} x_j + b_i$ has $\frac{\partial f_i}{\partial x_j} = a_{ij}$, so $\boldsymbol{J} = \boldsymbol{A}$: the derivative of a linear map is the matrix itself.

(c) For an activation applied to each component, $\boldsymbol{f}(\boldsymbol{z}) = (\sigma(z_1), \dots, \sigma(z_n))$, output i depends only on input i, so the Jacobian is diagonal: $\boldsymbol{J} = \operatorname{diag}(\sigma'(z_1), \dots, \sigma'(z_n))$.

The Jacobian gives the best linear approximation of $\boldsymbol{f}$ near $\boldsymbol{x}$:

$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{h}) \approx \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{J_f}(\boldsymbol{x})\, \boldsymbol{h}.$$

In the language of Chapter 4: zoomed in close enough, any smooth map looks like a linear transformation, and its matrix is the Jacobian.

Theorem 7.17 (Chain rule in matrix form)**.** If $\boldsymbol{g}\colon \mathbb{R}^n \to \mathbb{R}^m$ and $\boldsymbol{f}\colon \mathbb{R}^m \to \mathbb{R}^p$ are differentiable, then

$$\boxed{\boldsymbol{J}_{\boldsymbol{f}\circ\boldsymbol{g}}(\boldsymbol{x}) = \boldsymbol{J_f}(\boldsymbol{g}(\boldsymbol{x}))\, \boldsymbol{J_g}(\boldsymbol{x})}$$

The Jacobian of a composition is the product of the Jacobians.

This is the "sum over paths" rule written with matrices: entry (i, j) of the product is $\sum_k \frac{\partial f_i}{\partial g_k} \frac{\partial g_k}{\partial x_j}$. It also matches Section 4.8: composing linear maps multiplies their matrices, and a composition of smooth maps is, locally, a composition of their linear approximations.

Example 7.18 (The Jacobian of a layer)**.** A layer computes $\boldsymbol{h} = \mathrm{ReLU}(\boldsymbol{W}\boldsymbol{x} + \boldsymbol{b})$. Writing $\boldsymbol{z} = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{b}$, the chain rule gives

$$\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}} = \operatorname{diag}(\mathrm{ReLU}'(z_1), \dots, \mathrm{ReLU}'(z_m))\, \boldsymbol{W},$$

i.e. $\boldsymbol{W}$ with the rows of the "inactive" neurons (those with $z_i < 0$) set to zero. For

instance, with the layer of Section 4.6, where $\boldsymbol{z} = (5.5, -2)$, the second neuron is inactive and the Jacobian is $\left(\begin{smallmatrix} 1 & -1 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$.

Remark (Jacobian determinants change areas). For a map from $\mathbb{R}^2$ to $\mathbb{R}^2$, the determinant of the Jacobian is the local area-scaling factor (Section 4.10). For polar coordinates, $(r, \theta) \mapsto (r\cos\theta, r\sin\theta)$, the Jacobian is $\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}$, with determinant $r\cos^2\theta + r\sin^2\theta = r$: small rectangles in the (r, θ)-plane become patches of area r times larger. We use this in Section 7.9.

Exercises for Section 7.6

Exercise 7.19. Find the Jacobian of $\boldsymbol{f}(x, y) = (xy,\ x + y,\ x^2)$. What is its size?

Solution on p. 254 →

Exercise 7.20. What is the Jacobian of $\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}$ with $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$? Does it depend on $\boldsymbol{x}$ or $\boldsymbol{b}$?

Solution on p. 254 →

Exercise 7.21. Compute the Jacobian of the function that applies tanh to each component, $\boldsymbol{f}(\boldsymbol{z}) = (\tanh z_1, \tanh z_2)$, at the point $\boldsymbol{z} = (0, 1)$.

Solution on p. 254 →

7.7 The Hessian and the classification of critical points

Definition 7.19 (Hessian matrix). The **Hessian matrix** of $f\colon \mathbb{R}^n \to \mathbb{R}$ at $\boldsymbol{x}$ is the $n \times n$ matrix of second partial derivatives:

$$\boldsymbol{H}(\boldsymbol{x}) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1\, \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n\, \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}.$$

By Theorem 7.3 it is symmetric. It is the Jacobian of the gradient.

Example 7.20 (Hessians). (a) $f = x^2 + 3y^2$: $\boldsymbol{H} = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 6 \end{smallmatrix}\right)$ (the same at every point). (b) $f = x^2 - y^2$: $\boldsymbol{H} = \left(\begin{smallmatrix} 2 & 0 \\ 0 & -2 \end{smallmatrix}\right)$. (c) $f = x^3 - 3x + y^2$: $\boldsymbol{H} = \left(\begin{smallmatrix} 6x & 0 \\ 0 & 2 \end{smallmatrix}\right)$, which depends on the point. (d) For a quadratic function $f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}^\top \boldsymbol{x} + c$ with $\boldsymbol{A}$ symmetric: $\nabla f = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}$ and $\boldsymbol{H} = \boldsymbol{A}$.

7.7.1 Curvature in every direction

The second derivative of f along the line $\boldsymbol{x} + t\boldsymbol{u}$ (for a unit vector $\boldsymbol{u}$) is $\boldsymbol{u}^\top \boldsymbol{H}\boldsymbol{u}$, a quadratic form. By Section 5.9, its values are governed by the eigenvalues of $\boldsymbol{H}$: the eigenvectors of the Hessian are the directions of extreme curvature, and the eigenvalues are the curvatures in those directions. A large eigenvalue means a steep, narrow valley wall; a small one a gentle slope.

7.7.2 The second derivative test in several variables

Theorem 7.21 (Second derivative test)**.** Let $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$, with Hessian $\boldsymbol{H} = \boldsymbol{H}(\boldsymbol{x}^*)$.

- If $\boldsymbol{H}$ is positive definite (all eigenvalues > 0), $\boldsymbol{x}^*$ is a local minimum: the function curves upwards in every direction.
- If $\boldsymbol{H}$ is negative definite (all eigenvalues < 0), $\boldsymbol{x}^*$ is a local maximum.
- If $\boldsymbol{H}$ has both positive and negative eigenvalues, $\boldsymbol{x}^*$ is a **saddle point**: a minimum along some directions and a maximum along others.
- If $\boldsymbol{H}$ is only semidefinite (some eigenvalue is 0), the test is inconclusive.

For two variables: with $D = f_{xx}f_{yy} - f_{xy}^2 = \det \boldsymbol{H}$, the point is a minimum if $D > 0$ and $f_{xx} > 0$, a maximum if $D > 0$ and $f_{xx} < 0$, and a saddle if $D < 0$.

The shapes are those of Figure 5.9: near a critical point, $f(\boldsymbol{x}^* + \boldsymbol{h}) \approx f(\boldsymbol{x}^*) + \frac{1}{2}\boldsymbol{h}^\top \boldsymbol{H}\boldsymbol{h}$, so the function looks like the quadratic form of its Hessian.

Example 7.22 (Classifying critical points)**.** Find and classify the critical points of $f(x, y) = x^3 - 3x + y^2$.

Solution. $\nabla f = (3x^2 - 3,\ 2y) = \mathbf{0}$ gives $x = \pm 1$, $y = 0$. The Hessian is $\operatorname{diag}(6x, 2)$.

- At $(1, 0)$: $\boldsymbol{H} = \operatorname{diag}(6, 2)$, both eigenvalues positive: a local minimum, with $f(1, 0) = -2$.
- At $(-1, 0)$: $\boldsymbol{H} = \operatorname{diag}(-6, 2)$, eigenvalues of opposite signs: a saddle point, with $f(-1, 0) = 2$. Along the x-direction it is a maximum (of $x^3 - 3x$), along the y-direction a minimum (of y^2).

Example 7.23 (A minimum with a non-diagonal Hessian)**.** Find the minimum of $f(x, y) = x^2 + xy + y^2 - 3x$.

Solution. $\nabla f = (2x + y - 3,\ x + 2y) = \mathbf{0}$. The second equation gives $x = -2y$; substituting, $-4y + y - 3 = 0$, so $y = -1$ and $x = 2$. The Hessian $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has eigenvalues 3 and 1, so it is positive definite and $(2, -1)$ is a minimum, with $f(2, -1) = 4 - 2 + 1 - 6 = -3$.

Why this matters for ML (Saddles, curvature and the size of Hessians)**.** Three facts about Hessians shape how neural networks are trained.

- **Saddle points are everywhere in high dimensions.** A critical point is a minimum only if *all* eigenvalues of the Hessian are positive. With millions of parameters, it is far more common for at least a few to be negative, so most critical points of neural network losses are saddles, not minima. Gradient descent tends to escape them, because they have downhill directions.
- **Curvature limits the learning rate.** In a direction of curvature λ, a gradient step of size η overshoots unless $\eta < 2/\lambda$ (Chapter 8). A Hessian whose largest and smallest eigenvalues differ greatly (a large condition number) produces long, narrow valleys in which gradient descent zig-zags.
- **Hessians are too big to compute.** For a model with 10^9 parameters, the Hessian has 10^{18} entries. Deep learning therefore uses *first-order* methods, which

need only gradients, and approximates curvature information cheaply when it needs it (as the Adam optimiser does, Chapter 8).

Exercises for Section 7.7

Exercise 7.22. Compute the Hessian of $f(x, y) = x^3 + 2xy + y^2$.

Solution on p. 254 →

Exercise 7.23. Find and classify the critical point of $f(x, y) = x^2 + y^2 - 4x + 2y$.

Solution on p. 254 →

Exercise 7.24. Show that $(0, 0)$ is a saddle point of $f(x, y) = xy$.

Solution on p. 254 →

Exercise 7.25. Find and classify all critical points of $f(x, y) = x^3 - 12x + y^3 - 3y$.

Solution on p. 254 →

Exercise 7.26. For $f(x, y) = x^4 + y^4$, show that the second derivative test is inconclusive at the origin, but that the origin is a minimum.

Solution on p. 255 →

7.8 Taylor approximations in several variables

The one-variable approximations of Section 6.6 generalise directly, with the gradient in place of f' and the Hessian in place of f''.

Theorem 7.24 (Taylor approximations)**.** For a small vector $\boldsymbol{h}$,

$$f(\boldsymbol{x}+\boldsymbol{h}) \approx f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^{\top}\boldsymbol{h} \qquad \text{(first order, the tangent plane)},$$
$$f(\boldsymbol{x}+\boldsymbol{h}) \approx f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^{\top}\boldsymbol{h} + \tfrac{1}{2}\,\boldsymbol{h}^{\top}\boldsymbol{H}(\boldsymbol{x})\,\boldsymbol{h} \qquad \text{(second order)}.$$

Example 7.25 (A second-order approximation)**.** Approximate $f(x, y) = x^2 y + e^y$ near $(1, 0)$ and estimate $f(1.1, 0.1)$.

Solution. $f(1, 0) = 1$. $\nabla f = (2xy,\ x^2 + e^y)$, which is $(0, 2)$ at $(1, 0)$. The Hessian is $\left(\begin{smallmatrix} 2y & 2x \\ 2x & e^y \end{smallmatrix}\right)$, which is $\left(\begin{smallmatrix} 0 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ at $(1, 0)$. With $\boldsymbol{h} = (0.1, 0.1)$:

$$\begin{aligned} \text{first order:} &\quad 1 + (0, 2)\cdot(0.1, 0.1) = 1.2, \\ \text{second order:} &\quad 1.2 + \tfrac{1}{2}\,(0 \cdot 0.01 + 2 \cdot 2 \cdot 0.01 + 1 \cdot 0.01) = 1.2 + 0.025 = 1.225. \end{aligned}$$

The exact value is $1.21 \times 0.1 + e^{0.1} \approx 1.2262$.

7.8.1 Why gradient descent works

Take a step from $\boldsymbol{x}$ in the direction of the negative gradient, $\boldsymbol{h} = -\eta\,\boldsymbol{g}$ with $\boldsymbol{g} = \nabla f(\boldsymbol{x})$ and a small $\eta > 0$. The first-order approximation gives

$$f(\boldsymbol{x} - \eta\boldsymbol{g}) \approx f(\boldsymbol{x}) - \eta\,\boldsymbol{g}^{\top}\boldsymbol{g} = f(\boldsymbol{x}) - \eta\|\boldsymbol{g}\|^2,$$

which is *smaller* than $f(\boldsymbol{x})$ whenever $\boldsymbol{g} \neq \boldsymbol{0}$. So a small enough step against the gradient always decreases the function. The second-order approximation shows what "small enough" means:

$$f(\boldsymbol{x} - \eta \boldsymbol{g}) \approx f(\boldsymbol{x}) - \eta \|\boldsymbol{g}\|^2 + \tfrac{1}{2}\eta^2 \, \boldsymbol{g}^\top \boldsymbol{H} \boldsymbol{g}.$$

The last term, due to curvature, grows like η^2; if η is too large it wins, and the step makes things worse. Choosing η well is the art of the learning rate (Chapter 8). Finally, minimising the quadratic approximation exactly (set its gradient $\boldsymbol{g} + \boldsymbol{H}\boldsymbol{h}$ to zero) gives the *Newton step* $\boldsymbol{h} = -\boldsymbol{H}^{-1}\boldsymbol{g}$, the multivariable version of Newton's method.

Exercises for Section 7.8

Exercise 7.27. Write the second-order Taylor approximation of $f(x, y) = e^{x+y}$ around $(0, 0)$, and use it to estimate $f(0.1, 0.05)$.

Solution on p. 255 →

Exercise 7.28. Let $f(\boldsymbol{x}) = \tfrac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{x}$ with $\boldsymbol{A} = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 4 \end{smallmatrix}\right)$ and $\boldsymbol{b} = (2, 4)$. Find the gradient, the Hessian and the minimiser.

Solution on p. 255 →

Exercise 7.29. For $f(x, y) = x^2 + 3y^2$, start at $(1, 1)$ and take one gradient step $\boldsymbol{x} - \eta \nabla f(\boldsymbol{x})$ with $\eta = 0.1$, and then with $\eta = 0.5$. Compare the values of f before and after, and explain the result.

Solution on p. 255 →

7.9 Integrals over several variables

Integrals also extend to several variables, and we will need them for probabilities of two or more continuous quantities (Chapter 9). We only sketch the ideas.

The *double integral* $\iint_R f(x, y)\,\mathrm{d}A$ of a function over a region R of the plane is the volume between the region and the surface $z = f(x, y)$ (counting volume below the plane negatively), defined as a limit of sums of thin columns. Over a rectangle it can be computed as an *iterated integral*: integrate over one variable while holding the other fixed, then integrate the result over the other variable.

Example 7.26 (An iterated integral)**.** Compute $\int_0^1 \int_0^2 (x + y)\,\mathrm{d}y\,\mathrm{d}x$.

Solution. Inner integral, with x held fixed: $\int_0^2 (x + y)\,\mathrm{d}y = \left[xy + \frac{y^2}{2}\right]_0^2 = 2x + 2$. Outer integral: $\int_0^1 (2x + 2)\,\mathrm{d}x = \left[x^2 + 2x\right]_0^1 = 3$. (Integrating in the other order gives the same answer.)

Remark (Where $\sqrt{\pi}$ comes from)**.** Here is a beautiful use of the polar Jacobian of Section 7.6. Let $I = \int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x$. Then

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,\mathrm{d}x\,\mathrm{d}y = \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\, r\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi \cdot \tfrac{1}{2} = \pi,$$

where we switched to polar coordinates ($x^2 + y^2 = r^2$, with the area factor r) and used $\int_0^\infty r e^{-r^2}\mathrm{d}r = \left[-\frac{1}{2}e^{-r^2}\right]_0^\infty = \frac{1}{2}$. So $I = \sqrt{\pi}$, the constant behind the normal distribution.

Exercises for Section 7.9

Exercise 7.30. Compute $\int_0^2 \int_0^1 (x^2 + y)\,\mathrm{d}x\,\mathrm{d}y$.

Solution on p. 255 →

Exercise 7.31. Check that $p(x, y) = 4xy$ on the unit square $[0, 1] \times [0, 1]$ (and 0 elsewhere) has total integral 1, so that it is a valid joint probability density.

Solution on p. 255 →

7.10 Chapter summary

Key takeaways

- Functions $f\colon \mathbb{R}^n \to \mathbb{R}$ can be pictured as surfaces or contour plots; in ML the parameters are the variables of the loss.
- The partial derivative $\frac{\partial f}{\partial x_i}$ is the derivative in one variable with the others held fixed; mixed partials commute.
- The gradient ∇f collects all partial derivatives. It points in the direction of steepest ascent, its length is the steepest rate, and it is perpendicular to contours. $\nabla(\boldsymbol{a}^\top \boldsymbol{x}) = \boldsymbol{a}$, $\nabla\|\boldsymbol{x}\|^2 = 2\boldsymbol{x}$.
- Directional derivative: $D_{\boldsymbol{u}} f = \nabla f \cdot \boldsymbol{u}$; steepest descent is along $-\nabla f$.
- Chain rule: sum over paths of products of local derivatives. On a computation graph this is backpropagation.
- The Jacobian of $\boldsymbol{f}\colon \mathbb{R}^n \to \mathbb{R}^m$ is the $m \times n$ matrix of partial derivatives; Jacobians of compositions multiply.
- The Hessian $\boldsymbol{H}$ is the symmetric matrix of second partials. At a critical point: positive definite means minimum, negative definite maximum, indefinite saddle.
- $f(\boldsymbol{x}+\boldsymbol{h}) \approx f(\boldsymbol{x}) + \nabla f^\top \boldsymbol{h} + \frac{1}{2}\boldsymbol{h}^\top \boldsymbol{H}\boldsymbol{h}$; a small step against the gradient decreases f.

We now have all the tools to state and solve the central problem of machine learning training: finding parameters that minimise a loss. That is the subject of the next chapter.

Chapter 8

Optimization

Motivation. Almost every machine learning method can be summarised in one sentence: *choose the parameters that minimise a loss.* Linear regression, logistic regression, support vector machines, neural networks and large language models all fit this pattern; they differ in the model, the loss and the data, but the training is an *optimisation problem.* This chapter puts the calculus of Chapters 6 and 7 to work. We learn when a problem can be solved with a formula, and, since most cannot, how to solve it iteratively with *gradient descent* and its descendants (stochastic gradient descent, momentum, Adam), which train essentially every neural network in use today. We also learn why the learning rate matters so much, why *convex* problems are easy, and how to handle constraints.

In this chapter you will learn to...

- formulate optimisation problems and read the min and arg min notation;
- solve small problems analytically, including least squares and ridge regression;
- run gradient descent by hand and in code;
- explain how the learning rate and the curvature determine whether and how fast gradient descent converges;
- recognise convex sets and functions and explain why convexity matters;
- describe stochastic and mini-batch gradient descent;
- describe momentum, RMSProp and Adam;
- apply Newton's method;
- solve equality-constrained problems with Lagrange multipliers.

8.1 Optimisation problems

An **optimisation problem** asks for the input that makes a function as small (or as large) as possible:

$$\text{minimise } f(\boldsymbol{x}) \quad \text{over } \boldsymbol{x} \in \mathbb{R}^n \qquad \text{(possibly subject to constraints on } \boldsymbol{x}\text{)}.$$

The function f is the **objective function** (also called the cost or, in machine learning, the loss), and $\boldsymbol{x}$ contains the *decision variables.* Maximising f is the same as minimising $-f$, so we can always talk about minimisation.

Notation. $\min_{\boldsymbol{x}} f(\boldsymbol{x})$ denotes the smallest *value* of f, while $\arg\min_{\boldsymbol{x}} f(\boldsymbol{x})$ ("the argument of the minimum") denotes the *input* at which it is attained, the **minimiser**, often written $\boldsymbol{x}^*$. For example, for $f(x) = (x-3)^2 + 1$:

$$\min_x f(x) = 1, \qquad \arg\min_x f(x) = 3.$$

8.1.1 Training as optimisation

Definition 8.1 (Empirical risk minimisation)**.** Given training data $(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)$, a model $\hat{y} = f(\boldsymbol{x}; \boldsymbol{\theta})$ with parameters $\boldsymbol{\theta}$, and a **loss function** $\ell(\hat{y}, y)$ that measures how bad a prediction is, training means solving

$$\min_{\boldsymbol{\theta}} \; L(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \ell(f(\boldsymbol{x}_i; \boldsymbol{\theta}),\, y_i)$$

The average loss L over the training set is called the **empirical risk**.

Typical choices of ℓ are the squared error $(\hat{y} - y)^2$ for predicting numbers, the absolute error $|\hat{y} - y|$, and the cross-entropy for predicting classes (Chapter 11). Often a regularisation term such as $\lambda \|\boldsymbol{\theta}\|^2$ is added (Section 5.4).

Exercises for Section 8.1

Exercise 8.1. Give min and arg min of $f(x) = (x + 2)^2 + 1$. Write the problem of maximising $g(x) = -(x - 1)^2 + 4$ as a minimisation problem, and solve it.

Solution on p. 255 →

Exercise 8.2. For the data $(x, y) = (1, 1), (2, 2), (3, 2)$ and the model $\hat{y} = wx$, write the empirical risk with the absolute-error loss, and evaluate it at $w = 0.5$ and $w = 1$.

Solution on p. 255 →

8.2 Solving with a formula

When the objective is differentiable, every interior minimiser satisfies $\nabla f(\boldsymbol{x}^*) = \boldsymbol{0}$ (Section 7.3). Sometimes these equations can be solved exactly, and the Hessian then confirms that we have a minimum.

8.2.1 Least squares by calculus

Linear regression with data matrix $\boldsymbol{X}$ (one row per example, including a column of ones for the intercept) and targets $\boldsymbol{y}$ minimises the sum of squared errors

$$L(\boldsymbol{w}) = \|\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}\|^2 = (\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y})^\top (\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}) = \boldsymbol{w}^\top \boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{w} - 2\boldsymbol{y}^\top \boldsymbol{X}\boldsymbol{w} + \boldsymbol{y}^\top \boldsymbol{y}.$$

Using the gradients of Examples 7.5 and 7.20 (the first term is a quadratic form with symmetric matrix $\boldsymbol{X}^\top \boldsymbol{X}$, the second is linear in $\boldsymbol{w}$):

$$\nabla L(\boldsymbol{w}) = 2\boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{w} - 2\boldsymbol{X}^\top \boldsymbol{y} = 2\boldsymbol{X}^\top (\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}), \qquad \boldsymbol{H} = 2\boldsymbol{X}^\top \boldsymbol{X}.$$

Setting the gradient to zero gives the normal equations $\boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{w} = \boldsymbol{X}^\top \boldsymbol{y}$, exactly as the geometry of projections did in Section 5.6. The Hessian $2\boldsymbol{X}^\top \boldsymbol{X}$ is positive semidefinite (Proposition 5.39), so any solution is a global minimum.

8.2.2 Ridge regression

Adding the penalty $\lambda \|\boldsymbol{w}\|^2$ with $\lambda > 0$ gives $L(\boldsymbol{w}) = \|\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}\|^2 + \lambda \|\boldsymbol{w}\|^2$, with gradient $2\boldsymbol{X}^\top (\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}) + 2\lambda \boldsymbol{w}$. Setting it to zero:

$$(\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I})\, \boldsymbol{w} = \boldsymbol{X}^\top \boldsymbol{y}.$$

The matrix $\boldsymbol{X}^\top\boldsymbol{X} + \lambda\boldsymbol{I}$ has all eigenvalues at least $\lambda > 0$, so it is always invertible, even when features are collinear: ridge regularisation also cures the numerical problems of Section 5.2.

Example 8.2 (Least squares and ridge in one dimension)**.** Fit $\hat{y} = wx$ (no intercept) to the data $(1, 2)$, $(2, 3)$, $(3, 5)$, with and without the ridge penalty $\lambda = 1$.

Solution. Here $\boldsymbol{X}^\top\boldsymbol{X} = \sum x_i^2 = 14$ and $\boldsymbol{X}^\top\boldsymbol{y} = \sum x_i y_i = 23$. Least squares: $w = \frac{23}{14} \approx 1.643$. Ridge: $w = \frac{23}{14+1} \approx 1.533$. The penalty shrinks the weight towards zero.

Remark (When formulas run out)**.** Closed-form solutions exist for linear regression and a few other problems, but not for logistic regression, and certainly not for neural networks, whose equations $\nabla L = \mathbf{0}$ are hopelessly complicated. Even when a formula exists, solving a linear system with millions of unknowns may be too slow. In these cases we optimise *iteratively*: start somewhere and improve step by step.

Exercises for Section 8.2

Exercise 8.3. Minimise $f(x, y) = (x - 1)^2 + 2(y + 3)^2 + 5$.

Solution on p. 255 →

Exercise 8.4. For the model $\hat{y} = wx$ and data (x_i, y_i), show by differentiating $L(w) = \sum_i (y_i - wx_i)^2$ that the least-squares weight is $w = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2}$.

Solution on p. 255 →

Exercise 8.5. Show that, for the one-dimensional ridge problem $\min_w \sum_i (y_i - wx_i)^2 + \lambda w^2$, the solution is $w = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}$. What happens as $\lambda \to \infty$?

Solution on p. 255 →

8.3 Gradient descent

Picture this. You are standing on a mountainside in thick fog and want to reach the valley. You cannot see where the valley is, but you can feel the slope under your feet. A sensible strategy: take a step in the steepest downhill direction, feel the slope again, take another step, and repeat. That is gradient descent. The gradient tells you the steepest uphill direction (Theorem 7.6), so you step the opposite way.

Algorithm 8.3 (Gradient descent)**.** To minimise a differentiable function $L(\boldsymbol{\theta})$:

Step 1. Choose a starting point $\boldsymbol{\theta}_0$ and a **learning rate** (step size) $\eta > 0$.

Step 2. For $k = 0, 1, 2, \ldots$, update

$$\boxed{\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta\,\nabla L(\boldsymbol{\theta}_k)}$$

Step 3. Stop when the gradient is very small, when L stops improving, or after a fixed number of steps.

By the first-order Taylor approximation (Section 7.8), each step changes the loss by about $-\eta\|\nabla L\|^2$, which is negative: for a small enough learning rate, every step goes downhill. As the minimum is approached the gradient shrinks, so the steps automatically become shorter.

Example 8.4 (Gradient descent in one dimension)**.** Minimise $f(x) = (x-3)^2$ by gradient descent with $x_0 = 0$ and $\eta = 0.1$.

Solution. $f'(x) = 2(x-3)$, so the update is $x_{k+1} = x_k - 0.2(x_k - 3)$.

k	0	1	2	3	4	5	10
x_k	0	0.6	1.08	1.464	1.771	2.017	2.678

Subtracting 3 from both sides of the update gives $x_{k+1} - 3 = 0.8(x_k - 3)$: the distance to the minimiser shrinks by the factor 0.8 at every step, so $x_k = 3 - 3 \cdot 0.8^k \to 3$.

Example 8.5 (Gradient descent for linear regression)**.** Run gradient descent on the mean squared error of Example 7.7 (data $(1,2)$, $(2,3)$, $(3,5)$, model $\hat{y} = wx + b$), starting from $(w, b) = (0, 0)$ with $\eta = 0.05$.

Solution. The gradient is $\nabla L = -\frac{2}{3}\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})$. At the start it is $(-15.33, -6.67)$ (Example 7.7), so the first step gives $\boldsymbol{\theta}_1 = (0,0) - 0.05(-15.33, -6.67) = (0.767, 0.333)$. Continuing:

step k	0	1	2	3	100	500
w_k	0	0.767	1.109	1.262	1.464	1.500
b_k	0	0.333	0.480	0.544	0.415	0.334
loss	12.67	2.57	0.561	0.164	0.0565	0.0556

The loss drops quickly at first and then creeps towards its minimum $\frac{1}{18} \approx 0.0556$, reached at the least-squares solution $(1.5, \frac{1}{3})$ of Example 5.24.

Remark (Gradient descent in code)**.** The whole of Example 8.5 takes a few lines of Python:

```
import numpy as np

X = np.array([[1, 1], [2, 1], [3, 1]], dtype=float)  # x and ones
y = np.array([2, 3, 5], dtype=float)
theta = np.zeros(2)                  # parameters (w, b)
eta = 0.05                           # learning rate

for step in range(500):
    residuals = y - X @ theta
    grad = -2 / len(y) * X.T @ residuals   # gradient of the MSE
    theta = theta - eta * grad             # gradient descent update

print(theta)                         # approximately [1.5, 0.333]
```

Training a neural network has exactly the same structure; only the computation of the gradient is more involved (Chapter 11).

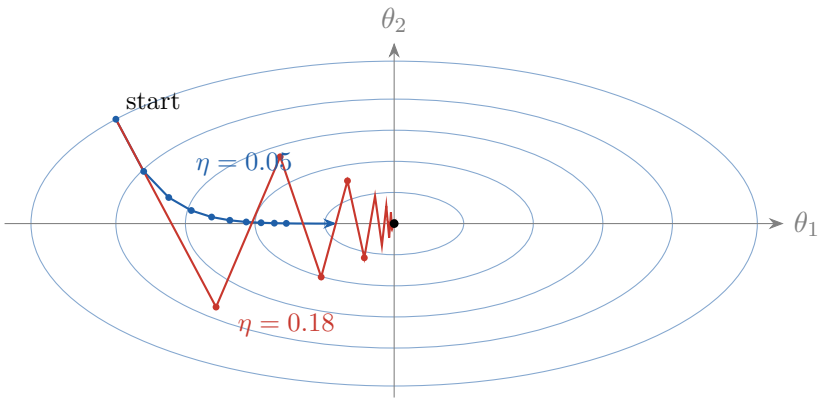


Figure 8.1: Gradient descent on $f(\boldsymbol{\theta}) = \theta_1^2 + 5\theta_2^2$ (contours in grey-blue) from $(-4, 1.5)$, 15 steps each. With $\eta = 0.05$ (blue) the path is smooth but crawls along the flat θ_1-direction. With $\eta = 0.18$ (red) it makes faster progress along θ_1 but zig-zags across the steep θ_2-direction.

Exercises for Section 8.3

Exercise 8.6. Run three steps of gradient descent on $f(x) = x^2 - 4x$ starting from $x_0 = 0$ with $\eta = 0.25$. Where is it heading?

Solution on p. 255 →

Exercise 8.7. Run two steps of gradient descent on $f(x, y) = x^2 + 2y^2$ from $(2, 1)$ with $\eta = 0.1$, and compute f after each step.

Solution on p. 255 →

Exercise 8.8. In Example 8.4, how many steps are needed before x_k is within 0.01 of the minimiser?

Solution on p. 255 →

8.4 The learning rate

The learning rate is the single most important setting (*hyperparameter*) of gradient-based training. Figure 8.2 shows the four things that can happen on the simplest possible function, $f(x) = x^2$, for which the update is $x_{k+1} = x_k - 2\eta x_k = (1 - 2\eta)x_k$.

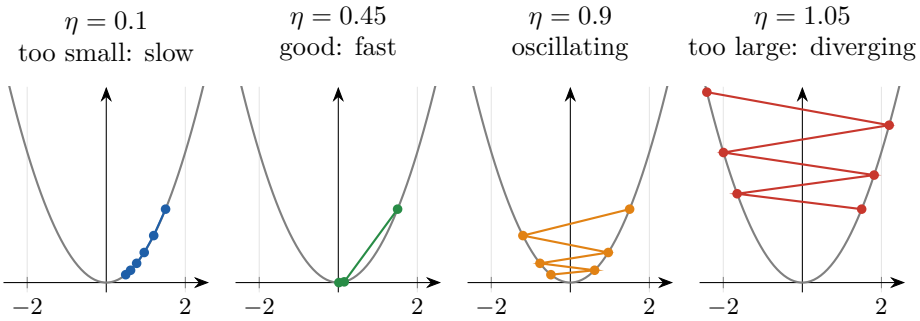


Figure 8.2: Gradient descent on $f(x) = x^2$ from $x_0 = 1.5$ with four learning rates. The update multiplies x by $1 - 2\eta$, which is 0.8, 0.1, -0.8 and -1.1 respectively.

8.4.1 What the curvature allows

For a one-dimensional quadratic $f(x) = \frac{a}{2}x^2$ with $a > 0$ (curvature $f'' = a$), the update is $x_{k+1} = (1 - \eta a)\, x_k$. The iterates converge to the minimum exactly when $|1 - \eta a| < 1$, i.e.

$$\boxed{0 < \eta < \frac{2}{a}}$$

The fastest choice is $\eta = 1/a$, which jumps to the minimum in one step; with $\frac{1}{a} < \eta < \frac{2}{a}$ the iterates overshoot and oscillate but still converge; with $\eta > \frac{2}{a}$ they diverge. Steep (highly curved) functions need small learning rates.

In several dimensions, near a minimum the loss is approximately a quadratic $\frac{1}{2}\boldsymbol{h}^\top \boldsymbol{H}\boldsymbol{h}$ (Section 7.8). Along each eigenvector of the Hessian, gradient descent behaves like the one-dimensional case with a replaced by the corresponding eigenvalue λ_i. Stability requires $\eta < 2/\lambda_{\max}$, but progress along the flattest direction proceeds only by the factor $1 - \eta\lambda_{\min}$ per step. When the **condition number** $\kappa = \lambda_{\max}/\lambda_{\min}$ is large, no single learning rate is good for all directions: a small one crawls along the flat directions, and a large one zig-zags across the steep ones (Figure 8.1, where $\kappa = 10/2 = 5$).

Why this matters for ML (Choosing and scheduling the learning rate)**.** In practice the curvature is unknown, so learning rates are found by experiment: try values on a logarithmic grid such as $10^{-1}, 3 \times 10^{-2}, 10^{-2}, \ldots$, and watch the training curve. A loss that explodes or oscillates wildly means the rate is too large; a loss that decreases painfully slowly means it is too small. Most training runs also change the learning rate over time, following a **learning-rate schedule** (Figure 8.3): large steps early to make fast progress, small steps later to settle into a minimum. Common schedules are step decay (divide by a factor every few epochs), exponential decay, and cosine decay, often preceded by a short *warm-up* in which the rate grows from zero.

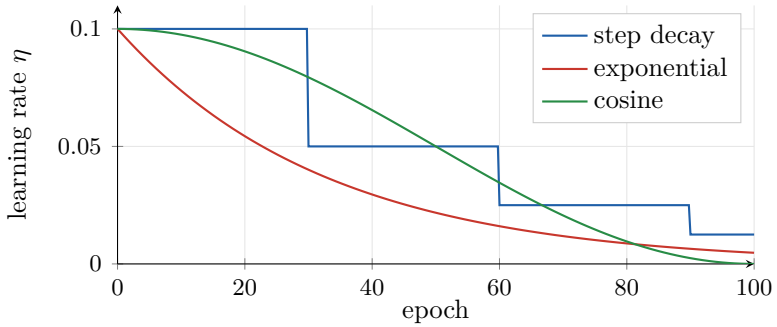


Figure 8.3: Three learning-rate schedules starting at $\eta = 0.1$: halving every 30 epochs, multiplying by 0.97 every epoch, and a cosine curve that reaches zero after 100 epochs.

Exercises for Section 8.4

Exercise 8.9. For $f(x) = 5x^2$, which learning rates make gradient descent converge? Which learning rate reaches the minimum in a single step?

Solution on p. 255 →

Exercise 8.10. For $f(x, y) = x^2 + 50y^2$, what is the largest learning rate for which gradient descent converges? With $\eta = 0.019$, by what factor does the x-coordinate shrink per step? What does this say about badly conditioned problems?

Solution on p. 256 →

Exercise 8.11. A learning rate starts at 0.1 and is halved every 30 epochs. What is it at epoch 75? For the cosine schedule $\eta_t = 0.05\,(1 + \cos \frac{\pi t}{100})$, what is it at $t = 50$ and $t = 100$?

Solution on p. 256 →

8.5 Convexity

Motivation. Gradient descent only sees the local slope, so in general it can get stuck in a local minimum (Figure 6.2). For a special and important class of functions this cannot happen: for *convex* functions, every local minimum is a global minimum. Many classical machine learning problems are convex, which is why they can be solved reliably.

Definition 8.6 (Convex set)**.** A set C is **convex** if for any two points $\boldsymbol{x}, \boldsymbol{y} \in C$, the whole line segment between them lies in C: $\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y} \in C$ for all $\lambda \in [0, 1]$.

Discs, squares, half-planes, lines and the unit balls of all norms (Figure 5.3) are convex. A crescent moon, a doughnut or an L-shape is not (Figure 8.4).

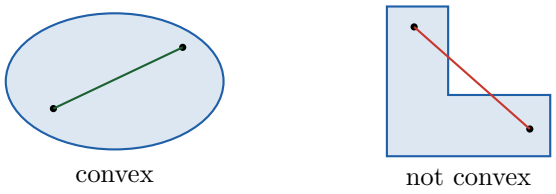


Figure 8.4: A convex set contains the segment between any two of its points. The L-shaped set is not convex: the red segment between two of its points leaves the set.

Definition 8.7 (Convex function)**.** A function f is **convex** if, for all $\boldsymbol{x}, \boldsymbol{y}$ and all $\lambda \in [0, 1]$,

$$f(\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y}) \leq \lambda f(\boldsymbol{x}) + (1 - \lambda) f(\boldsymbol{y}).$$

Geometrically: the chord between any two points of the graph lies on or above the graph (Figure 8.5).

Theorem 8.8 (Tests for convexity)**.** For a twice-differentiable function f:

(a) in one variable, f is convex if and only if $f''(x) \geq 0$ everywhere;
(b) in several variables, f is convex if and only if its Hessian is positive semidefinite everywhere;
(c) equivalently, f is convex if and only if it lies above all its tangent lines (planes): $f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x})$.

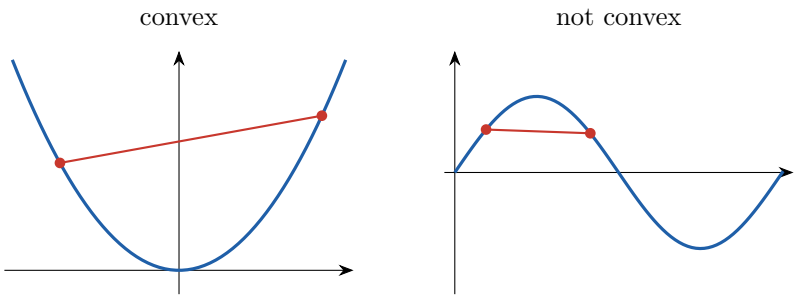


Figure 8.5: Left: for the convex function x^2, every chord (red) lies above the graph. Right: for $\sin x$, the chord shown lies below the graph, so $\sin x$ is not convex.

Example 8.9 (Convex and non-convex functions)**.** *Convex:* x^2, e^x, $-\ln x$ (for $x > 0$), $|x|$, $\mathrm{ReLU}(x)$, every norm $\|\boldsymbol{x}\|$, every linear function, and every positive semidefinite quadratic form. Sums of convex functions, and maxima of convex functions, are convex. Therefore the least-squares loss, the ridge loss and the lasso loss are all convex. *Not convex:* x^3, $\sin x$, $-x^2$, and the losses of neural networks with hidden layers.

Theorem 8.10 (Local minima of convex functions are global)**.** If f is convex, every local minimum is a global minimum. If f is *strictly* convex (the inequality in Definition 8.7 is strict for $\boldsymbol{x} \neq \boldsymbol{y}$ and $0 < \lambda < 1$), there is at most one minimiser.

Why. Suppose $\boldsymbol{x}$ is a local minimum but some $\boldsymbol{y}$ has $f(\boldsymbol{y}) < f(\boldsymbol{x})$. Points $\boldsymbol{z} = \lambda\boldsymbol{x} + (1-\lambda)\boldsymbol{y}$ on the segment, with λ close to 1, are arbitrarily close to $\boldsymbol{x}$, and by convexity $f(\boldsymbol{z}) \leq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}) < f(\boldsymbol{x})$. So there are points arbitrarily close to $\boldsymbol{x}$ with smaller values, contradicting that $\boldsymbol{x}$ is a local minimum. ∎

Why this matters for ML (Convex and non-convex learning)**.** For convex problems (linear and ridge regression, logistic regression, support vector machines), gradient descent with a suitable learning rate converges to a global minimum, and the answer does not depend on where we start. Neural network losses are not convex: they have many local minima and saddle points, and different starting points lead to different solutions. It is one of the surprises of deep learning that gradient descent nevertheless works so well on these problems; understanding why is an active research area.

Exercises for Section 8.5

Exercise 8.12. Which of these functions are convex on all of $\mathbb{R}$? x^4, $-x^2$, e^{-x}, x^3, $|x-2|$, $\max(0, x)$.

Solution on p. 256 →

Exercise 8.13. Show that $f(x, y) = x^2 + xy + y^2$ is convex.

Solution on p. 256 →

Exercise 8.14. Is the set $\{x \in \mathbb{R} : |x| \geq 1\}$ convex? Is $\{x \in \mathbb{R} : |x| \leq 1\}$?

Solution on p. 256 →

8.6 Stochastic gradient descent

The empirical risk is an average over the training set, $L(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} \ell_i(\boldsymbol{\theta})$, where ℓ_i is the loss on example i. Its gradient is therefore also an average,

$$\nabla L(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} \nabla \ell_i(\boldsymbol{\theta}),$$

and computing it requires a pass over *all* n examples. With millions of training examples, one exact gradient step is very expensive.

Intuition (Estimating the gradient from a sample)**.** To estimate the average height of a country's population you do not measure everyone; a random sample of a few hundred people gives a good estimate. In the same way, the average of the gradients over a small random **mini-batch** of examples is a good, cheap estimate of the full gradient. It is noisy, but on average it points in the right direction (in the language of Chapter 9, it is an *unbiased estimate*).

Algorithm 8.11 (Mini-batch stochastic gradient descent)**.** To minimise $L(\boldsymbol{\theta}) = \frac{1}{n}\sum_i \ell_i(\boldsymbol{\theta})$:

Step 1. Choose $\boldsymbol{\theta}_0$, a learning rate η and a batch size b.

Step 2. Repeat for each **epoch** (a full pass over the data): shuffle the examples randomly and split them into mini-batches of b examples.

Step 3. For each mini-batch B, update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \cdot \frac{1}{b}\sum_{i\in B} \nabla \ell_i(\boldsymbol{\theta})$.

With $b = 1$ this is **stochastic gradient descent** (SGD) in its purest form; with $b = n$ it is ordinary ("full-batch") gradient descent. Typical batch sizes in deep learning range from tens to thousands of examples.

Example 8.12 (SGD for the mean)**.** The mean of the data $1, 3, 5, 7$ minimises $L(c) = \frac{1}{4}\sum_i (c - y_i)^2$, whose minimiser is 4. The loss on a single example is $(c - y_i)^2$, with gradient $2(c - y_i)$. Run SGD with $b = 1$, $\eta = 0.1$, $c_0 = 0$, visiting the examples in the order $5, 1, 7, 3, 5, 7$.

Solution. $c_1 = 0 - 0.2(0 - 5) = 1$; $c_2 = 1 - 0.2(1 - 1) = 1$; $c_3 = 1 - 0.2(1 - 7) = 2.2$; $c_4 = 2.2 - 0.2(2.2 - 3) = 2.36$; $c_5 = 2.36 - 0.2(2.36 - 5) = 2.888$; $c_6 = 2.888 - 0.2(2.888 - 7) \approx 3.710$. Each step pulls c towards the current example rather than towards the mean, so the path is erratic, but it drifts towards 4. With a constant learning rate it keeps jiggling around 4; a decreasing learning rate lets it settle.

Why this matters for ML (Why SGD is the workhorse of deep learning)**.** For the same amount of computation, SGD takes many cheap, noisy steps instead of a few expensive, exact ones, and on large data sets this wins by a wide margin. The noise is not only a nuisance: it helps the iterates escape saddle points and is believed to steer them towards "flat" minima that generalise better. Mini-batches of a few hundred examples also map well onto parallel hardware. Essentially every modern neural network is trained with mini-batch SGD or one of the refinements of the next section.

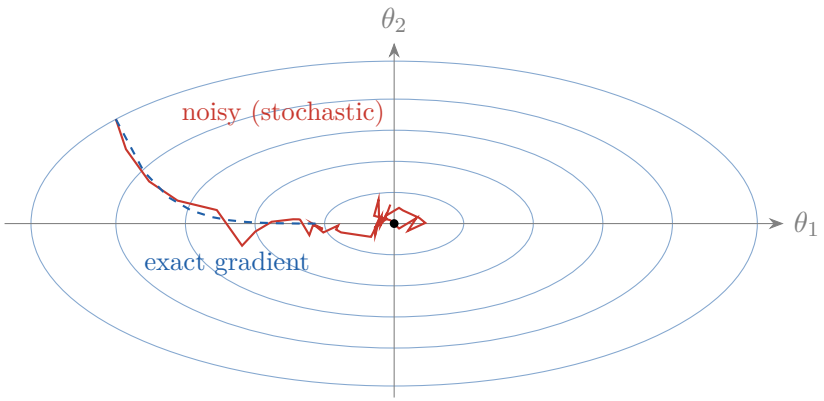


Figure 8.6: Gradient descent with exact gradients (blue, dashed) and with noisy gradient estimates (red), both with $\eta = 0.05$, on $\theta_1^2 + 5\theta_2^2$. The noisy path wanders but makes progress towards the minimum, and then keeps jittering around it.

Exercises for Section 8.6

Exercise 8.15. A training set contains 50 000 images and the batch size is 100. How many parameter updates are made per epoch? How many in 20 epochs?

Solution on p. 256 →

Exercise 8.16. Run three steps of SGD ($b = 1$, $\eta = 0.25$, $c_0 = 0$) for the mean of the data $2, 4, 6, 8$, visiting the examples 6, 2, 8.

Solution on p. 256 →

Exercise 8.17. Suppose an epoch visits every example exactly once, in mini-batches of equal size. Show that the average of the mini-batch gradients (all evaluated at the same $\boldsymbol{\theta}$) equals the full gradient.

Solution on p. 256 →

8.7 Momentum and adaptive methods

Plain gradient descent has two weaknesses that we have already seen: it zig-zags in narrow valleys, and a single learning rate must serve all parameters, even if they need very different step sizes. Two ideas address them.

8.7.1 Momentum

Picture this. Replace the cautious hiker by a heavy ball rolling down the landscape. The ball builds up speed in directions where the slope is consistent, and its inertia smooths out back-and-forth wobbles, because pushes in opposite directions cancel.

Algorithm 8.13 (Gradient descent with momentum)**.** Choose a momentum coefficient $\beta \in [0, 1)$, typically 0.9, and start with the "velocity" $\boldsymbol{v}_0 = \mathbf{0}$. At each step, with $\boldsymbol{g}_k = \nabla L(\boldsymbol{\theta}_k)$:

Step 1. $\boldsymbol{v}_{k+1} = \beta\, \boldsymbol{v}_k + \boldsymbol{g}_k$ (a running sum of past gradients, with old ones fading);
Step 2. $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta\, \boldsymbol{v}_{k+1}$.

If the gradient keeps pointing the same way, the velocity grows towards $\frac{1}{1-\beta}$ times the gradient (ten times for $\beta = 0.9$), so the ball accelerates along consistent directions. In a zig-zag, successive gradients in the steep direction have opposite signs and largely cancel.

8.7.2 Adaptive learning rates: RMSProp and Adam

RMSProp gives each parameter its own effective learning rate. It keeps a running average of the *squared* gradient of each parameter, $s_{k+1} = \rho s_k + (1-\rho) g_k^2$ (all operations component by component), and divides the step by its square root: $\theta_{k+1} = \theta_k - \eta\, g_k / (\sqrt{s_{k+1}} + \epsilon)$. Parameters with consistently large gradients get smaller steps, and parameters with small gradients get larger ones. The tiny constant ϵ (such as 10^{-8}) only prevents division by zero.

Adam ("adaptive moment estimation") combines momentum with RMSProp, and is the default optimiser for training most deep networks, including large language models.

Algorithm 8.14 (Adam). With default values $\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, start with $\boldsymbol{m}_0 = \boldsymbol{s}_0 = \mathbf{0}$. At step $t = 1, 2, \ldots$, with gradient $\boldsymbol{g}_t$ (all operations component by component):

Step 1. $\boldsymbol{m}_t = \beta_1 \boldsymbol{m}_{t-1} + (1-\beta_1)\, \boldsymbol{g}_t$ (average gradient: momentum);
Step 2. $\boldsymbol{s}_t = \beta_2 \boldsymbol{s}_{t-1} + (1-\beta_2)\, \boldsymbol{g}_t^2$ (average squared gradient);
Step 3. $\hat{\boldsymbol{m}}_t = \boldsymbol{m}_t / (1-\beta_1^t)$ and $\hat{\boldsymbol{s}}_t = \boldsymbol{s}_t / (1-\beta_2^t)$ (corrections for starting at zero);
Step 4. $\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} - \eta\, \hat{\boldsymbol{m}}_t / (\sqrt{\hat{\boldsymbol{s}}_t} + \epsilon)$.

Intuition (What Adam's step looks like). At the first step, $\hat{\boldsymbol{m}}_1 = \boldsymbol{g}_1$ and $\hat{\boldsymbol{s}}_1 = \boldsymbol{g}_1^2$, so the update of each parameter is about $\eta\, g/|g| = \pm\eta$: *every parameter moves by roughly η, whatever the size of its gradient.* Later, the ratio $\hat{m}/\sqrt{\hat{s}}$ is close to ± 1 when a parameter's gradient is consistent and closer to 0 when it keeps changing sign. So Adam's steps have a predictable size of about η per parameter, which makes its learning rate much easier to choose than that of plain SGD.

Remark (AdamW). Regularising with $\lambda\|\boldsymbol{\theta}\|^2$ adds $2\lambda\boldsymbol{\theta}$ to the gradient, which Adam would then rescale along with everything else. The widely used variant *AdamW* instead applies the weight decay directly, $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\lambda\boldsymbol{\theta}$, separately from the adaptive step; this tends to work better.

Exercises for Section 8.7

Exercise 8.18. Gradient descent with momentum ($\beta = 0.9$, $\eta = 0.1$) is applied in a region where the gradient is constantly $g = 1$. Compute the first three steps taken, and the step size the method approaches in the long run.

Solution on p. 256 →

Exercise 8.19. In Adam's first step, the gradient is $\boldsymbol{g}_1 = (0.002, -30)$ and $\eta = 0.001$. Compute (approximately) the change in each parameter.

Solution on p. 256 →

8.8 Newton's method

Gradient descent trusts only the slope. **Newton's method** also uses the curvature: at each step it fits the quadratic Taylor approximation (Sections 6.6 and 7.8) and jumps straight to its minimum. In one variable, the parabola $f(x_k) + f'(x_k)h + \frac{1}{2}f''(x_k)h^2$ is minimised at $h = -f'(x_k)/f''(x_k)$, giving

$$\boxed{x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}} \qquad \text{and, in several variables,} \qquad \boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \boldsymbol{H}^{-1}\nabla L(\boldsymbol{\theta}_k).$$

For a quadratic function the approximation is exact, so Newton's method finds the minimum in a single step.

Example 8.15 (Newton's method)**.** Minimise $f(x) = e^x - 2x$, starting from $x_0 = 0$.

Solution. $f'(x) = e^x - 2$ and $f''(x) = e^x$, so $x_{k+1} = x_k - \frac{e^{x_k}-2}{e^{x_k}}$. The iterates are

$$x_0 = 0, \quad x_1 = 1, \quad x_2 \approx 0.7358, \quad x_3 \approx 0.69404, \quad x_4 \approx 0.6931476,$$

approaching the exact minimiser $\ln 2 \approx 0.6931472$ (where $f' = 0$). The errors are about 0.31, 0.043, 0.00090, 0.0000004: the number of correct digits roughly doubles at every step. This *quadratic convergence* is far faster than gradient descent, whose error shrinks by a constant factor per step.

Why this matters for ML (Why deep learning does not use Newton's method)**.** For a model with n parameters, the Hessian has n^2 entries and solving with it costs on the order of n^3 operations. For $n = 10^8$ this is out of the question. Moreover, near saddle points the Hessian is indefinite and Newton's method happily jumps *towards* the saddle instead of away from it. For problems with up to a few thousand parameters, quasi-Newton methods such as L-BFGS, which build a cheap approximation of the Hessian from successive gradients, are excellent. For deep learning, adaptive first-order methods such as Adam, which use only cheap per-parameter curvature estimates, are the practical compromise.

Exercises for Section 8.8

Exercise 8.20. Apply one step of Newton's method to $f(x) = x^2 - 6x + 2$ from an arbitrary starting point x_0. Explain the result.

Solution on p. 256 →

Exercise 8.21. Compare one step of Newton's method with one step of gradient descent ($\eta = 0.1$) for $f(x) = e^x - 2x$ starting at $x_0 = 1$. Which ends up closer to $\ln 2 \approx 0.693$?

Solution on p. 256 →

8.9 Constrained optimisation and Lagrange multipliers

Sometimes the variables are not free. Probabilities must add up to one; a budget cannot be exceeded; in Figure 5.4 the weights were confined to a ball. We consider problems of the form

$$\text{minimise } f(\boldsymbol{x}) \quad \text{subject to} \quad g(\boldsymbol{x}) = c.$$

Intuition (Tangency). Walk along the constraint curve $g(\boldsymbol{x}) = c$ while watching the contour lines of f. As long as you cross contour lines, you can still decrease f by walking on. At the constrained minimum you stop crossing them: the constraint curve just *touches* a contour line (Figure 8.7). Two curves that touch have the same perpendicular direction, so the gradients ∇f and ∇g (each perpendicular to its own contour) are parallel there:

$$\nabla f(\boldsymbol{x}^*) = \lambda \nabla g(\boldsymbol{x}^*)$$

for some number λ, called the **Lagrange multiplier**.

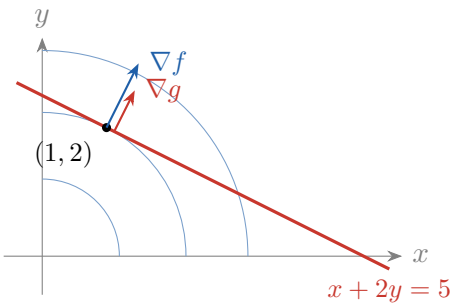


Figure 8.7: Minimising $f = x^2 + y^2$ (circular contours) on the line $g = x + 2y = 5$. At the solution $(1, 2)$ the line touches a contour, and $\nabla f = (2, 4)$ is parallel to $\nabla g = (1, 2)$.

Theorem 8.16 (Method of Lagrange multipliers). If $\boldsymbol{x}^*$ minimises (or maximises) $f(\boldsymbol{x})$ subject to $g(\boldsymbol{x}) = c$, and $\nabla g(\boldsymbol{x}^*) \neq \mathbf{0}$, then there is a number λ with

$$\nabla f(\boldsymbol{x}^*) = \lambda \nabla g(\boldsymbol{x}^*) \qquad \text{and} \qquad g(\boldsymbol{x}^*) = c.$$

Equivalently, $\boldsymbol{x}^*$ and λ make the gradient of the **Lagrangian** $\mathcal{L}(\boldsymbol{x}, \lambda) = f(\boldsymbol{x}) - \lambda(g(\boldsymbol{x}) - c)$ equal to zero.

The recipe: write down the equations $\nabla f = \lambda \nabla g$ (one per variable) together with the constraint, solve them, and compare the values of f at the solutions.

Example 8.17 (The closest point on a line). Minimise $f(x, y) = x^2 + y^2$ subject to $x + 2y = 5$.

Solution. $\nabla f = (2x, 2y)$ and $\nabla g = (1, 2)$. The equations are $2x = \lambda$, $2y = 2\lambda$, $x + 2y = 5$. The first two give $y = 2x$; the constraint then gives $5x = 5$, so $(x, y) = (1, 2)$, with $\lambda = 2$ and $f = 5$. The point $(1, 2)$ is the point of the line closest to the origin, at distance $\sqrt{5}$ (compare the exercise on distances to lines in Section 4.4).

Example 8.18 (Maximising a product). Maximise $f(x, y) = xy$ subject to $x + y = 10$.

Solution. $\nabla f = (y, x) = \lambda(1, 1)$ gives $x = y$; with the constraint, $x = y = 5$ and $f = 25$. This is the rectangle problem of Section 6.5, solved a second way.

Example 8.19 (Maximum likelihood for categories). A die-like experiment with three outcomes produced the counts n_1, n_2, n_3 in $N = n_1 + n_2 + n_3$ trials. Which

probabilities p_1, p_2, p_3 make the observations most likely? (In Chapter 10 we will see that this means maximising $f(\boldsymbol{p}) = n_1 \ln p_1 + n_2 \ln p_2 + n_3 \ln p_3$.) The constraint is $p_1 + p_2 + p_3 = 1$.

Solution. $\frac{\partial f}{\partial p_i} = \frac{n_i}{p_i}$ and $\nabla g = (1, 1, 1)$, so $\frac{n_i}{p_i} = \lambda$, i.e. $p_i = \frac{n_i}{\lambda}$. Summing, $1 = \frac{N}{\lambda}$, so $\lambda = N$ and

$$p_i = \frac{n_i}{N}.$$

The best probabilities are the observed frequencies, for example $(0.3, 0.5, 0.2)$ for counts $(3, 5, 2)$. Reassuringly, mathematics confirms common sense.

Remark (Inequality constraints and projections)**.** Constraints of the form $g(\boldsymbol{x}) \le c$ lead to the *Karush–Kuhn–Tucker (KKT) conditions*: at the optimum, either the constraint is not active ($g < c$, and then $\lambda = 0$ and the problem is locally unconstrained), or it is active ($g = c$, and the Lagrange condition holds with λ of the appropriate sign). The theory of support vector machines is built on these conditions. A simpler practical approach is *projected gradient descent*: take an ordinary gradient step, then move the result back to the nearest point satisfying the constraints, for example by clipping negative weights to zero when weights must be non-negative.

Exercises for Section 8.9

Exercise 8.22. Minimise $x^2 + y^2$ subject to $x + y = 4$.

Solution on p. 256 →

Exercise 8.23. Find the maximum and minimum of $f(x, y) = x + y$ on the circle $x^2 + y^2 = 2$.

Solution on p. 256 →

Exercise 8.24. In Example 8.19, what are the maximum-likelihood probabilities for the counts $(12, 6, 2)$?

Solution on p. 256 →

Exercise 8.25. Weights must satisfy $w_i \ge 0$. After a gradient step the weights are $(0.4, -0.1, 0.0, -2.3)$. What does projected gradient descent do next?

Solution on p. 256 →

8.10 Chapter summary

Key takeaways

- Training is empirical risk minimisation: $\min_{\boldsymbol{\theta}} \frac{1}{n} \sum_i \ell(f(\boldsymbol{x}_i; \boldsymbol{\theta}), y_i)$, possibly plus a regulariser.
- Some problems have formulas: least squares solves $\boldsymbol{X}^\top \boldsymbol{X} \boldsymbol{w} = \boldsymbol{X}^\top \boldsymbol{y}$; ridge solves $(\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}) \boldsymbol{w} = \boldsymbol{X}^\top \boldsymbol{y}$.
- Gradient descent: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla L(\boldsymbol{\theta})$. Each small step goes downhill.
- On a quadratic with curvature a, gradient descent converges iff $0 < \eta < 2/a$. A large condition number causes zig-zagging and slow progress. Learning rates are

tuned on a log scale and decayed by schedules.

- Convex functions (chords above the graph, Hessian PSD) have no bad local minima. Linear and logistic regression are convex; neural networks are not.
- Mini-batch SGD uses cheap, noisy gradient estimates; one epoch is one pass over the data.
- Momentum accelerates consistent directions; Adam adapts the step size of each parameter and is the default for deep learning.
- Newton's method uses the Hessian and converges very fast, but is too expensive for large models.
- Lagrange multipliers: at a constrained optimum, $\nabla f = \lambda \nabla g$.

We now know how to fit models. But how much should we trust a fitted model, and will it work on data it has not seen? Answering such questions requires the language of uncertainty: probability, which is the subject of the next chapter.

Chapter 9

Probability

Motivation. Machine learning lives with uncertainty. Measurements are noisy, labels are sometimes wrong, and the future is never fully determined by the past. A good model does not just say "cat"; it says "cat with probability 0.97", and it knows when it is unsure. Probability is the mathematics of uncertainty, and it is woven through the whole field:

- classifiers output probability distributions over classes, and a language model outputs a probability distribution over the next word;
- the standard loss functions are derived from probability (maximum likelihood, Chapter 10; cross-entropy, Chapter 11);
- generative models (which create images, text or music) are learned probability distributions from which we can sample;
- training itself is random: random initialisation, random mini-batches, random dropout.

This chapter develops probability from scratch: events and their probabilities, conditional probability and Bayes' theorem, random variables, expectation and variance, the most important distributions, and how several random quantities relate to each other.

In this chapter you will learn to...

- describe random experiments with sample spaces and events;
- apply the axioms and rules of probability, and count outcomes;
- compute conditional probabilities and use the law of total probability;
- recognise independent and conditionally independent events;
- apply Bayes' theorem, and avoid the base-rate fallacy;
- work with discrete and continuous random variables, PMFs, PDFs and CDFs;
- compute expectations, variances and standard deviations;
- use the Bernoulli, binomial, categorical, uniform and normal distributions;
- work with joint, marginal and conditional distributions;
- compute covariance, correlation and covariance matrices.

9.1 Random experiments, outcomes and events

A **random experiment** is a process whose result we cannot predict with certainty: tossing a coin, rolling a die, measuring tomorrow's temperature, drawing a random mini-batch of training data.

Definition 9.1 (Sample space and events)**.** The **sample space** Ω (capital omega) of a random experiment is the set of all its possible **outcomes**. An **event** is a subset of the sample space: a collection of outcomes. We say the event A *occurs* if the outcome

of the experiment lies in A.

Example 9.2 (Sample spaces)**.** Here are four random experiments and their sample spaces.

- One coin toss: $\Omega = \{H, T\}$.
- One die: $\Omega = \{1, 2, 3, 4, 5, 6\}$. The event "an even number" is $A = \{2, 4, 6\}$.
- Two dice: Ω consists of the 36 ordered pairs $(1, 1), (1, 2), \ldots, (6, 6)$.
- A person's height in metres: $\Omega = (0, \infty)$, an infinite, continuous set.

9.1.1 Combining events

Events are sets (Section 1.1), and there are three basic ways to combine sets, which we now introduce (Figure 9.1):

- $A \cup B$ ("A *or* B", the **union**): at least one of A, B occurs;
- $A \cap B$ ("A *and* B", the **intersection**): both occur;
- A^c ("*not* A", the **complement**): A does not occur;
- A and B are **disjoint** (or *mutually exclusive*) if they cannot both occur: $A \cap B = \emptyset$, where $\emptyset$ is the empty set.

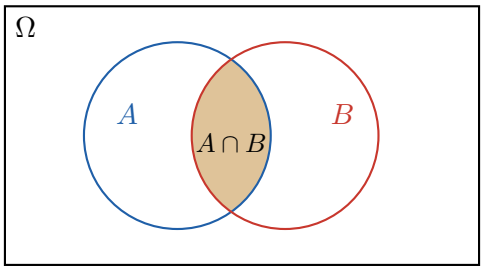


Figure 9.1: A Venn diagram. The rectangle is the sample space, the circles are events A and B, and the shaded overlap is $A \cap B$. Everything inside either circle is $A \cup B$; everything outside A is A^c.

9.1.2 What does a probability mean?

The *probability* $P(A)$ of an event is a number between 0 and 1 that measures how likely it is. There are two complementary interpretations.

- **Frequentist:** $P(A)$ is the long-run fraction of repetitions of the experiment in which A occurs. A fair coin lands heads in about half of many tosses.
- **Bayesian:** $P(A)$ is a degree of belief, which can apply to one-off events: "the probability that this email is spam", "the probability that it rains tomorrow".

Both interpretations obey exactly the same mathematical rules, which is all we need.

Exercises for Section 9.1

Exercise 9.1. Write down the sample space for tossing a coin twice, and the event "at least one head".

Solution on p. 257 →

Exercise 9.2. For one roll of a die let $A = \{2, 4, 6\}$ (even) and $B = \{4, 5, 6\}$ (at least 4). Find $A \cup B$, $A \cap B$ and A^c. Are A and B disjoint?

Solution on p. 257 →

9.2 The rules of probability

All of probability theory rests on three simple rules.

Definition 9.3 (Axioms of probability)**.** A **probability** P assigns to each event A a number $P(A)$ such that:

(a) $P(A) \geq 0$ for every event A;
(b) $P(\Omega) = 1$ (something certainly happens);
(c) if A and B are disjoint, then $P(A \cup B) = P(A) + P(B)$ (and the same for any sequence of pairwise disjoint events).

Theorem 9.4 (Basic rules)**.** For all events A and B:

(a) $P(A^c) = 1 - P(A)$ (complement rule);
(b) $P(\emptyset) = 0$ and $0 \leq P(A) \leq 1$;
(c) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (inclusion–exclusion).

Proof. A and A^c are disjoint with union Ω, so $P(A)+P(A^c) = 1$. This gives the complement rule, and with $A = \Omega$ it gives $P(\emptyset) = 0$. For inclusion–exclusion, adding $P(A)$ and $P(B)$ counts the overlap $A \cap B$ twice (Figure 9.1), so it must be subtracted once. ∎

9.2.1 Equally likely outcomes

When the sample space is finite and all outcomes are equally likely (a fair coin, a fair die, a well-shuffled deck), probabilities are just proportions:

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega} = \frac{|A|}{|\Omega|}.$$

(For a finite set, $|A|$ denotes the number of its elements.)

Example 9.5 (Two dice)**.** Roll two fair dice. Find the probability that (a) the sum is 7; (b) at least one of the dice shows a six.

Solution. There are 36 equally likely outcomes. (a) The sum is 7 for the six outcomes $(1,6)$, $(2,5)$, $(3,4)$, $(4,3)$, $(5,2)$ and $(6,1)$, so $P = \frac{6}{36} = \frac{1}{6}$. (b) It is easier to count the complement, "no six": $5 \times 5 = 25$ outcomes. So $P(\text{at least one six}) = 1 - \frac{25}{36} = \frac{11}{36} \approx 0.306$.

9.2.2 Counting

Computing proportions requires counting, and counting can be surprisingly tricky. Three tools suffice for most purposes.

- **Multiplication principle.** If a choice is made in two stages, with m options at the first and n at the second, there are mn possibilities in total. (Two dice: $6 \times 6 = 36$.)

- **Arrangements.** n distinct objects can be put in order in $n! = n \times (n-1) \times \cdots \times 1$ ways (Section 1.7): n choices for the first position, $n-1$ for the second, and so on.
- **Combinations.** The number of ways to *choose* k objects out of n, when the order does not matter, is the **binomial coefficient**

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!},$$

read "n choose k". For example, $\binom{4}{2} = \frac{24}{2 \cdot 2} = 6$: from $\{a, b, c, d\}$ we can choose ab, ac, ad, bc, bd, cd.

Example 9.6 (Counting with combinations)**.** (a) How many different 5-card hands can be dealt from a 52-card deck? (b) What is the probability of exactly 2 heads in 4 fair coin tosses?

Solution. (a) $\binom{52}{5} = 2\,598\,960$. (b) There are $2^4 = 16$ equally likely sequences of heads and tails. A sequence with exactly 2 heads is determined by *which* 2 of the 4 tosses are heads: $\binom{4}{2} = 6$ sequences. So $P = \frac{6}{16} = 0.375$.

Exercises for Section 9.2

Exercise 9.3. $P(A) = 0.3$, $P(B) = 0.5$ and $P(A \cap B) = 0.1$. Find $P(A \cup B)$, $P(A^c)$, and the probability that neither A nor B occurs.

Solution on p. 257 →

Exercise 9.4. Two fair dice are rolled. Find the probability that (a) the sum is 8; (b) both show the same number; (c) the sum is at least 10.

Solution on p. 257 →

Exercise 9.5. What is the probability of at least one head in 5 tosses of a fair coin?

Solution on p. 257 →

Exercise 9.6. A committee of 3 is chosen at random from 8 people, one of whom is Alice. How many committees are possible, and what is the probability that Alice is on the committee?

Solution on p. 257 →

9.3 Conditional probability

New information changes probabilities. If you learn that a die roll was even, the chance that it was a 6 goes up from $\frac{1}{6}$ to $\frac{1}{3}$: there are now only three possibilities, $2, 4, 6$, and 6 is one of them.

Definition 9.7 (Conditional probability)**.** If $P(B) > 0$, the **conditional probability** of A given B is

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}}$$

read "the probability of A given B".

Intuition (Shrinking the universe)**.** Conditioning on B means: we now know that B happened, so throw away all outcomes outside B, and rescale the probabilities of what is left so that they add up to 1 again. In the Venn diagram, B becomes the new sample space and $P(A \mid B)$ is the fraction of B occupied by $A \cap B$.

Example 9.8 (Conditional probabilities from a table)**.** Of 100 emails, 40 are spam. The word "free" appears in 30 of the spam emails and in 5 of the 60 others:

	contains "free"	no "free"	total
spam	30	10	40
not spam	5	55	60
total	35	65	100

Find $P(\text{spam})$, $P(\text{spam} \mid \text{free})$ and $P(\text{free} \mid \text{spam})$.

Solution. $P(\text{spam}) = \frac{40}{100} = 0.4$. Restricting attention to the 35 emails that contain "free", and then to the 40 spam emails, we find

$$P(\text{spam} \mid \text{free}) = \frac{30}{35} \approx 0.857, \qquad P(\text{free} \mid \text{spam}) = \frac{30}{40} = 0.75.$$

Common pitfall ($P(A \mid B)$ is not $P(B \mid A)$)**.** In the example, $P(\text{spam} \mid \text{free}) \approx 0.86$ but $P(\text{free} \mid \text{spam}) = 0.75$. Confusing the two is one of the most common and most consequential errors in reasoning about probability. The probability that a person who has a disease shows a symptom is not the probability that a person with the symptom has the disease. Bayes' theorem (Section 9.5) is the correct way to turn one into the other.

9.3.1 The multiplication rule

Rearranging the definition gives the **multiplication rule**

$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A),$$

and, for several events, the *chain rule of probability*

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)\, P(A_2 \mid A_1)\, P(A_3 \mid A_1 \cap A_2) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

Why this matters for ML (How language models assign probabilities to text)**.** A language model assigns a probability to a whole sentence $w_1 w_2 \dots w_n$ by the chain rule:

$$P(w_1, \dots, w_n) = P(w_1)\, P(w_2 \mid w_1)\, P(w_3 \mid w_1, w_2) \cdots P(w_n \mid w_1, \dots, w_{n-1}).$$

Each factor is a prediction of the next word given everything before it, and this is exactly what models such as GPT are trained to produce. Generating text means repeatedly sampling the next word from these conditional probabilities.

9.3.2 The law of total probability

Often a probability is easiest to compute by splitting into cases. Events $B_1, \dots, B_k$ form a *partition* of Ω if they are pairwise disjoint and together cover Ω (exactly one of them occurs).

Theorem 9.9 (Law of total probability)**.** If $B_1, \dots, B_k$ partition Ω and each $P(B_i) > 0$, then for any event A,

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i)\, P(B_i).$$

Example 9.10 (Defective items)**.** Machine 1 makes 60% of a factory's items, of which 2% are defective; machine 2 makes the other 40%, of which 5% are defective. What fraction of all items is defective?

Solution. $P(D) = P(D \mid M_1)P(M_1) + P(D \mid M_2)P(M_2) = 0.02 \cdot 0.6 + 0.05 \cdot 0.4 = 0.012 + 0.020 = 0.032$. The tree diagram in Figure 9.2 shows the computation: multiply along the branches, add over the paths that end in "defective".

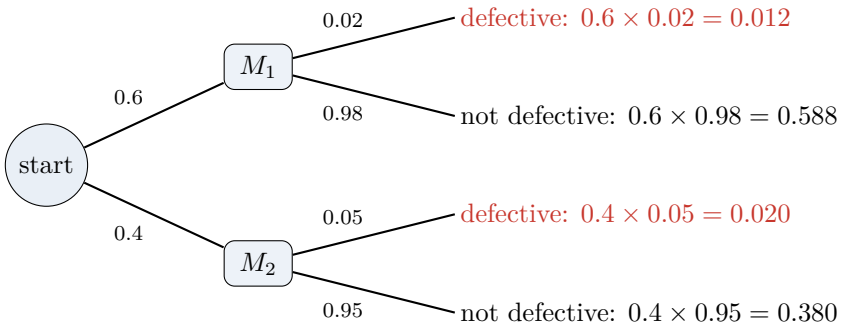


Figure 9.2: A probability tree for Example 9.10. Probabilities along a path are multiplied; the two red paths together give $P(\text{defective}) = 0.032$.

Exercises for Section 9.3

Exercise 9.7. Using the table of Example 9.8, find

$$P(\text{free} \mid \text{not spam}) \quad \text{and} \quad P(\text{not spam} \mid \text{no "free"}).$$

Solution on p. 257 →

Exercise 9.8. Machine A makes 70% of items with a 1% defect rate, and machine B makes 30% with a 4% defect rate. What fraction of items is defective?

Solution on p. 257 →

Exercise 9.9. Two cards are drawn from a deck without replacement. What is the probability that both are aces?

Solution on p. 257 →

Exercise 9.10. A language model gives $P(\text{the}) = 0.05$, $P(\text{cat} \mid \text{the}) = 0.001$ and $P(\text{sat} \mid \text{the cat}) = 0.1$. What probability does it assign to "the cat sat", and what is the natural logarithm of that probability?

Solution on p. 257 →

9.4 Independence

Definition 9.11 (Independent events)**.** Events A and B are **independent** if

$$P(A \cap B) = P(A)\,P(B).$$

Equivalently (when $P(B) > 0$), $P(A \mid B) = P(A)$: learning that B occurred does not change the probability of A.

Two tosses of a coin, two rolls of a die and, ideally, two randomly sampled training examples are independent. For independent events the probability that all occur is the product of their probabilities: ten heads in a row has probability $\left(\frac{1}{2}\right)^{10} = \frac{1}{1024}$.

Common pitfall (Independent is not the same as disjoint)**.** Disjoint events are as *dependent* as events can be: if A and B cannot both happen, then knowing that B happened tells you that A did not. Formally, if $P(A), P(B) > 0$ and $A \cap B = \emptyset$, then $P(A \cap B) = 0 \neq P(A)P(B)$.

Definition 9.12 (Conditional independence)**.** A and B are **conditionally independent** given C if $P(A \cap B \mid C) = P(A \mid C)\,P(B \mid C)$.

Conditional independence is common: the symptoms "cough" and "fever" are correlated in the population (both are caused by flu), but among people known to have flu they may be roughly independent.

Why this matters for ML (Independence assumptions in machine learning)**.** Most of statistical learning assumes that the training examples are **i.i.d.**: *independent and identically distributed*, i.e. drawn independently from the same distribution. This is what makes the probability of a whole data set a product of individual probabilities (Chapter 10). The *naive Bayes* classifier makes a much stronger assumption: that the words of an email are conditionally independent given the class (spam or not). This is certainly false, but it makes the model simple and it often works surprisingly well (Section 9.5).

Exercises for Section 9.4

Exercise 9.11. Roll a die. Let $A = \{2, 4, 6\}$, $B = \{1, 2\}$ and $C = \{1, 3, 5\}$. Are A and B independent? Are A and C?

Solution on p. 257 →

Exercise 9.12. A classifier is correct on each test example with probability 0.9, independently. What is the probability that it gets all of 10 examples right?

Solution on p. 257 →

9.5 Bayes' theorem

The multiplication rule says $P(A \mid B)P(B) = P(B \mid A)P(A)$. Dividing by $P(B)$ gives one of the most important formulas in all of science.

Theorem 9.13 (Bayes' theorem)**.** If $P(B) > 0$,

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

where, by the law of total probability, $P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$.

Notation. When A is a hypothesis H and B is observed data D, the terms have names:

$$\underbrace{P(H \mid D)}_{\text{posterior}} = \frac{\overbrace{P(D \mid H)}^{\text{likelihood}}\ \overbrace{P(H)}^{\text{prior}}}{\underbrace{P(D)}_{\text{evidence}}}.$$

The **prior** is our belief before seeing the data; the **likelihood** says how well the hypothesis predicts the data; the **posterior** is the updated belief. In words: *posterior is proportional to likelihood times prior.*

Example 9.14 (A medical test)**.** A disease affects 1% of the population. A test detects it in 99% of sick people, but also gives a (false) positive result for 5% of healthy people. You test positive. What is the probability that you have the disease?

Solution. With D = "has the disease" and $+$ = "tests positive": $P(D) = 0.01$, $P(+ \mid D) = 0.99$ and $P(+ \mid D^c) = 0.05$. By Bayes,

$$P(D \mid +) = \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99} = \frac{0.0099}{0.0099 + 0.0495} \approx 0.167.$$

Despite the positive result from a "99% accurate" test, the chance of having the disease is only about 17%.

Intuition (Think in natural frequencies)**.** Imagine 10 000 people (Figure 9.3). About 100 are sick, and 99 of them test positive. Of the 9900 healthy people, 5%, i.e. 495, also test positive. So among the $99 + 495 = 594$ positives, only 99 are sick: $\frac{99}{594} \approx 17\%$. The rare disease is swamped by false alarms from the large healthy majority.

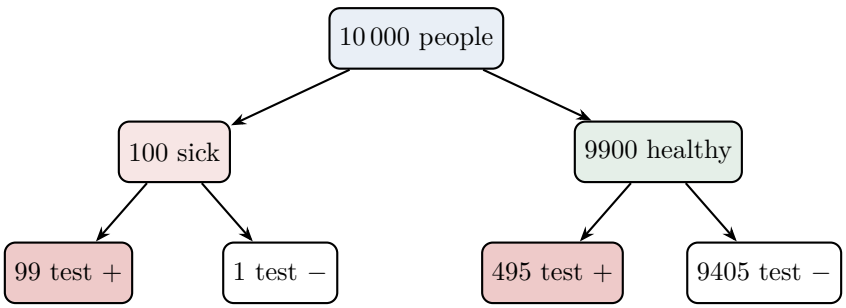


Figure 9.3: The medical test in natural frequencies. Of the 594 people who test positive (shaded), only 99 are actually sick.

Common pitfall (The base-rate fallacy)**.** Ignoring the prior (the *base rate* of 1%) and answering "99%" is the *base-rate fallacy*. It affects doctors, juries and machine learning practitioners alike: a fraud detector that is "99% accurate" may still raise mostly false alarms if fraud is rare. Always ask how common the thing you are looking for is to begin with.

Example 9.15 (A naive Bayes spam filter)**.** From the table of Example 9.8 we know that $P(\text{spam}) = 0.4$ and

$$P(\text{free} \mid \text{spam}) = 0.75, \qquad P(\text{free} \mid \text{ham}) = \tfrac{5}{60} \approx 0.083,$$

where "ham" means "not spam". Suppose also that the word "meeting" appears in 5% of spam and 30% of ham. An email contains both "free" and "meeting". Is it spam?

Solution. Assume, as naive Bayes does, that the words are conditionally independent given the class. Then

$$P(\text{spam})\,P(\text{free} \mid \text{spam})\,P(\text{meeting} \mid \text{spam}) = 0.4 \times 0.75 \times 0.05 = 0.015,$$
$$P(\text{ham})\,P(\text{free} \mid \text{ham})\,P(\text{meeting} \mid \text{ham}) = 0.6 \times 0.0833 \times 0.30 = 0.015.$$

By Bayes' theorem the posterior probabilities are proportional to these numbers, so $P(\text{spam} \mid \text{both words}) = \frac{0.015}{0.015+0.015} = 0.5$. The word "free" pushes towards spam and "meeting" pushes back equally hard.

Why this matters for ML (Bayesian thinking in machine learning)**.** Bayes' theorem underlies classifiers such as naive Bayes, the interpretation of regularisation as a prior belief about the weights (Chapter 10), and the whole field of Bayesian machine learning, which maintains probability distributions over model parameters instead of single best values. Note also that a posterior can serve as the prior for the next piece of evidence: Bayes' theorem describes learning from data step by step.

Exercises for Section 9.5

Exercise 9.13. A disease has prevalence 2%. A test detects it in 90% of sick people and gives false positives for 10% of healthy people. What is the probability of disease after a positive test?

Solution on p. 257 →

Exercise 9.14. In the previous exercise, the patient takes a second, independent test, which is also positive. Using 0.155 as the new prior, what is the probability of disease now?

Solution on p. 257 →

Exercise 9.15. Urn 1 contains 3 red and 1 blue ball; urn 2 contains 1 red and 3 blue balls. An urn is chosen by a fair coin toss and a ball is drawn: it is red. What is the probability that it came from urn 1?

Solution on p. 257 →

Exercise 9.16. With the numbers of Example 9.15, find the probability that an email containing "meeting" (and we ignore all other words) is spam.

Solution on p. 257 →

9.6 Random variables and their distributions

Often we care about a *number* computed from the outcome: the sum of two dice, the number of heads, the height of a randomly chosen person, the loss on a randomly chosen training example.

Definition 9.16 (Random variable)**.** A **random variable** is a function that assigns a number to each outcome of a random experiment.

Notation. Random variables are written with capital letters (X, Y, S) and the values they take with lower-case letters (x, y, s). So $P(X = 3)$ is "the probability that X takes the value 3" and $P(X \leq x)$ "the probability that X is at most x".

9.6.1 Discrete random variables

A random variable is *discrete* if it takes a finite (or countable) list of values. Its distribution is described by its **probability mass function** (PMF)

$$p(x) = P(X = x), \qquad \text{with } p(x) \geq 0 \text{ and } \sum_x p(x) = 1.$$

Example 9.17 (The sum of two dice)**.** Let S be the sum of two fair dice. Counting outcomes as in Example 9.5:

s	2	3	4	5	6	7	8	9	10	11	12
$36\,p(s)$	1	2	3	4	5	6	5	4	3	2	1

The PMF is drawn as a bar chart in Figure 9.4. The probabilities add up to $\frac{36}{36} = 1$.

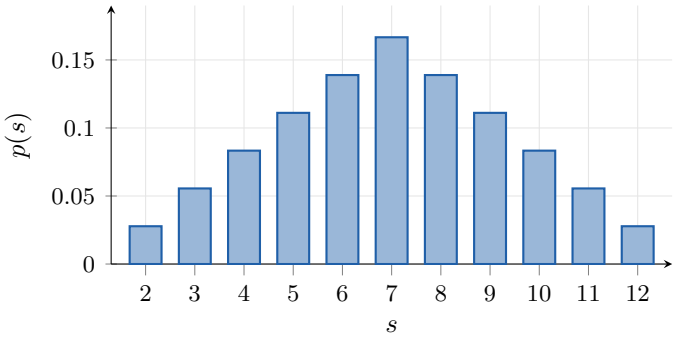


Figure 9.4: The probability mass function of the sum of two fair dice.

9.6.2 Continuous random variables

A height, a temperature or a model weight can take any value in an interval. For such *continuous* random variables, the probability of any single exact value is zero (what is the chance that a person is *exactly* 1.7500000… metres tall?), so instead we describe probabilities of intervals with a density.

Definition 9.18 (Probability density function)**.** A continuous random variable X has **probability density function** (PDF) $p(x)$ if, for all $a \le b$,

$$P(a \le X \le b) = \int_a^b p(x)\,dx.$$

A density satisfies $p(x) \ge 0$ and $\int_{-\infty}^{\infty} p(x)\,dx = 1$.

So probabilities are *areas under the density curve* (Figure 9.6 below shows an example).

Common pitfall (A density is not a probability)**.** $p(x)$ itself is not the probability of anything: $P(X = x) = 0$. A density measures probability *per unit length*, and it can be larger than 1. For example, if X is uniform on $[0, 0.5]$, then $p(x) = 2$ on that interval (so that the total area is $2 \times 0.5 = 1$).

Definition 9.19 (Cumulative distribution function)**.** The **cumulative distribution function** (CDF) of any random variable X is $F(x) = P(X \le x)$. For a continuous random variable $F(x) = \int_{-\infty}^{x} p(t)\,dt$, and so, by the fundamental theorem of calculus, $p = F'$.

Example 9.20 (Two continuous distributions)**.** (a) X is *uniform* on $[0, 2]$, with $p(x) = \frac{1}{2}$ there (and 0 elsewhere). Then $P(0.5 \le X \le 1.2) = \int_{0.5}^{1.2} \frac{1}{2}\,dx = 0.35$.

(b) The waiting time T until the next event in a random stream of events (the next customer, the next radioactive decay) is often modelled by the *exponential distribution* $p(t) = \lambda e^{-\lambda t}$ for $t \ge 0$. With rate $\lambda = 2$ per minute, the probability of waiting more than a minute is

$$P(T > 1) = \int_1^{\infty} 2e^{-2t}\,dt = \left[-e^{-2t}\right]_1^{\infty} = e^{-2} \approx 0.135.$$

Exercises for Section 9.6

Exercise 9.17. Let X be the number of heads in 3 tosses of a fair coin. Write down its PMF and compute $F(1) = P(X \le 1)$.

Solution on p. 257 →

Exercise 9.18. X is uniform on $[0, 4]$. What is its density? Find $P(1 \le X \le 2.5)$ and $P(X = 2)$.

Solution on p. 257 →

Exercise 9.19. Show that $p(x) = 3x^2$ on $[0, 1]$ (and 0 elsewhere) is a valid density, and compute $P(X \le 0.5)$. Is it a problem that $p(1) = 3 > 1$?

Solution on p. 257 →

Exercise 9.20. For an exponential waiting time with rate $\lambda = 0.5$, find $P(T > 2)$.

Solution on p. 258 →

9.7 Expectation

Definition 9.21 (Expected value)**.** The **expected value** (or **mean**, or *expectation*) of a random variable X is

$$\boxed{\mathbb{E}[X] = \sum_x x\, p(x)} \qquad \text{or} \qquad \mathbb{E}[X] = \int_{-\infty}^{\infty} x\, p(x)\, \mathrm{d}x$$

for discrete and continuous X respectively. It is often written μ.

The expected value is a probability-weighted average of the possible values. For a fair die, $\mathbb{E}[X] = \frac{1}{6}(1+2+3+4+5+6) = 3.5$. Note that the expected value need not be a possible value.

Intuition (Two pictures of the mean)**.** **Long-run average:** if you repeat the experiment many times, the average of the observed values approaches $\mathbb{E}[X]$ (the *law of large numbers*, Chapter 10). Roll a die a thousand times and the average will be close to 3.5. **Centre of mass:** if you place weights $p(x)$ at the positions x on a ruler, the ruler balances at $\mathbb{E}[X]$.

9.7.1 Expectations of functions

To compute the expected value of a function $g(X)$, weight the values of g by the probabilities of X:

$$\mathbb{E}[g(X)] = \sum_x g(x)\, p(x) \qquad \text{or} \qquad \int g(x)\, p(x)\, \mathrm{d}x.$$

For a die, $\mathbb{E}[X^2] = \frac{1}{6}(1+4+9+16+25+36) = \frac{91}{6} \approx 15.17$.

9.7.2 Linearity of expectation

Theorem 9.22 (Linearity of expectation)**.** For any random variables X, Y and constants a, b:

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b, \qquad \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

The second rule holds even if X and Y are dependent.

Linearity is enormously useful. The expected sum of 100 dice is $100 \times 3.5 = 350$, with no need to work out the distribution of the sum. The expected number of heads in n tosses of a coin with $P(\text{heads}) = p$ is np, because it is a sum of n terms, each 1 (heads) or 0 (tails) with expected value p.

Common pitfall (Expectation does not pass through non-linear functions)**.** In general $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$. For a die, $\mathbb{E}[X^2] \approx 15.17$, but $(\mathbb{E}[X])^2 = 12.25$. Likewise $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ holds for independent X and Y, but not in general.

Why this matters for ML (Risk, and why stochastic gradients work)**.** If the data $(\boldsymbol{X}, Y)$ come from some distribution, the quality of a model is its *expected* loss, the **risk** $R(\boldsymbol{\theta}) = \mathbb{E}[\ell(f(\boldsymbol{X}; \boldsymbol{\theta}), Y)]$. We cannot compute it, because we do not know the distribution, so we minimise its estimate from the data, the empirical risk of

Definition 8.1. Expectation also explains stochastic gradient descent (Section 8.6): if I is an index chosen uniformly at random from $1, \dots, n$, then $\mathbb{E}[\nabla \ell_I] = \sum_{i=1}^{n} \frac{1}{n} \nabla \ell_i = \nabla L$. The gradient of a random example is, on average, exactly the full gradient: an *unbiased* estimate.

Exercises for Section 9.7

Exercise 9.21. X takes the values 0, 1, 2 with probabilities 0.2, 0.5, 0.3. Find $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

Solution on p. 258 →

Exercise 9.22. A game costs nothing to play: you win 10 with probability 0.1 and lose 2 otherwise. What is the expected gain? Would you play many times?

Solution on p. 258 →

Exercise 9.23. What is the expected number of sixes in 30 rolls of a die? What is the expected number of correct predictions of a classifier that is right with probability 0.8 on each of 200 test examples?

Solution on p. 258 →

Exercise 9.24. Compute $\mathbb{E}[X]$ for the density $p(x) = 3x^2$ on $[0, 1]$.

Solution on p. 258 →

9.8 Variance and standard deviation

Two random variables can have the same mean but very different spreads: a fair coin bet of ± 1 and a bet of ± 1000 both have mean 0. Variance measures spread.

Definition 9.23 (Variance and standard deviation)**.** The **variance** of X, with mean $\mu = \mathbb{E}[X]$, is the expected squared distance from the mean,

$$\boxed{\operatorname{Var}(X) = \mathbb{E}[(X - \mu)^2]}$$

and the **standard deviation** is its square root, $\sigma = \sqrt{\operatorname{Var}(X)}$.

Squaring makes all deviations positive and emphasises large ones; the square root brings the units back to those of X (if X is in metres, $\operatorname{Var}(X)$ is in square metres and σ in metres). For calculations, the following shortcut is usually easier.

Proposition 9.24 (Computational formula)**.** $\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Proof. Expand the square and use linearity: $\mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2$. ∎

For a die: $\operatorname{Var}(X) = \frac{91}{6} - 3.5^2 = \frac{35}{12} \approx 2.92$ and $\sigma \approx 1.71$.

Theorem 9.25 (Properties of variance)**.** For constants a, b: $\operatorname{Var}(aX+b) = a^2 \operatorname{Var}(X)$, so the standard deviation scales by $|a|$ and is unaffected by shifts. If X and Y are independent, $\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)$.

9.8.1 Standardisation

Subtracting the mean and dividing by the standard deviation gives the **standardised** variable or **z-score**

$$Z = \frac{X - \mu}{\sigma},$$

which has mean 0 and variance 1 (by Theorem 9.25). Its value says how many standard deviations X is above or below its mean.

Remark (Chebyshev's inequality)**.** For any random variable with mean μ and standard deviation σ, and any $k > 0$,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

So at most $\frac{1}{4}$ of the probability lies 2 or more standard deviations away from the mean, whatever the distribution. This makes precise the idea that the standard deviation is the typical size of the fluctuations.

Why this matters for ML (Variance in neural networks)**.** Controlling variances is a recurring theme in deep learning.

- **Feature standardisation.** Features are usually standardised (z-scored) before training, so that all of them have mean 0 and variance 1; this fixes the scale problem of Section 3.2 and makes gradient descent better conditioned.
- **Weight initialisation.** A neuron computes $z = \sum_{i=1}^{n} w_i x_i$. If the inputs and weights are independent with mean 0, then $\operatorname{Var}(z) = n \operatorname{Var}(w) \operatorname{Var}(x)$. To keep signals from growing or shrinking from layer to layer, initialisation schemes choose $\operatorname{Var}(w) \approx \frac{1}{n}$ (*Xavier/Glorot* initialisation) or $\frac{2}{n}$ for ReLU networks (*He* initialisation).
- **Normalisation layers** (batch norm, layer norm) standardise activations inside the network during training.

Exercises for Section 9.8

Exercise 9.25. For the random variable of the first exercise of Section 9.7 (values $0, 1, 2$ with probabilities $0.2, 0.5, 0.3$), compute the variance and standard deviation.

Solution on p. 258 →

Exercise 9.26. A Bernoulli random variable is 1 with probability p and 0 otherwise. Show that its variance is $p(1-p)$. For which p is it largest?

Solution on p. 258 →

Exercise 9.27. If $\operatorname{Var}(X) = 4$, what are $\operatorname{Var}(3X - 5)$ and the standard deviation of $3X - 5$?

Solution on p. 258 →

Exercise 9.28. Test scores have mean 70 and standard deviation 8. What is the z-score of a score of 82? What score has z-score -0.5?

Solution on p. 258 →

Exercise 9.29. A neuron sums 100 independent inputs, each with mean 0 and variance 1, multiplied by independent weights with mean 0 and variance v. What value of v makes the variance of the sum equal to 1?

Solution on p. 258 →

9.9 Common distributions

A handful of distributions appear again and again. Learning their shapes, means and variances pays off.

9.9.1 Bernoulli and binomial

A **Bernoulli distribution** describes a single yes/no trial: $X = 1$ ("success") with probability p and $X = 0$ with probability $1 - p$. Its mean is p and its variance $p(1 - p)$.

The **binomial distribution** $\mathrm{Bin}(n, p)$ counts the successes in n independent Bernoulli trials with the same p. A particular sequence with k successes and $n - k$ failures has probability $p^k(1 - p)^{n-k}$, and there are $\binom{n}{k}$ such sequences, so

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Its mean is np and its variance $np(1 - p)$ (it is a sum of n independent Bernoulli variables).

Example 9.26 (A binomial probability)**.** A free-throw shooter scores with probability 0.3. In 10 independent attempts, what is the probability of exactly 3 successes?

Solution. $P(X = 3) = \binom{10}{3}(0.3)^3(0.7)^7 = 120 \times 0.027 \times 0.0824 \approx 0.267$. The whole distribution is shown in Figure 9.5; its mean is $np = 3$ and its standard deviation $\sqrt{10 \times 0.3 \times 0.7} \approx 1.45$.

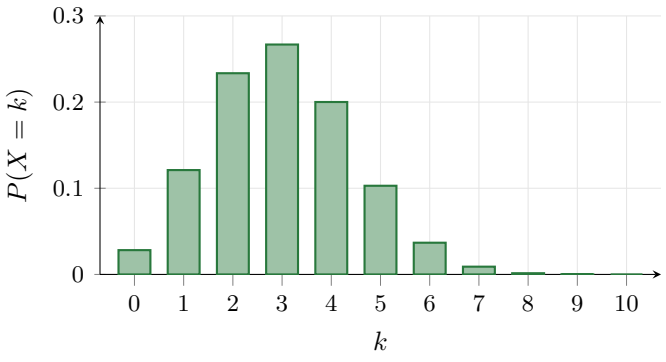


Figure 9.5: The binomial distribution with $n = 10$ and $p = 0.3$.

9.9.2 Categorical

The **categorical distribution** generalises the Bernoulli distribution to K outcomes: $P(X = k) = p_k$ for $k = 1, \ldots, K$, with $p_k \geq 0$ and $\sum_k p_k = 1$. A fair die is categorical with $p_k = \frac{1}{6}$. The outcome is often encoded as a one-hot vector (Section 4.1).

Why this matters for ML. Classifiers with K classes output the parameters $(p_1, \ldots, p_K)$ of a categorical distribution, usually produced by the softmax function (Chapter 11). A language model's next-word prediction is a categorical distribution over a vocabulary of tens of thousands of tokens. Binary classifiers output a single Bernoulli parameter p, usually via a sigmoid.

9.9.3 Uniform

The continuous **uniform distribution** on $[a, b]$ has constant density $\frac{1}{b-a}$ on the interval: every value is equally likely. Its mean is the midpoint $\frac{a+b}{2}$ and its variance $\frac{(b-a)^2}{12}$. Random number generators produce uniform numbers on $[0, 1]$, from which all other distributions can be simulated.

9.9.4 The normal (Gaussian) distribution

The most important distribution of all is the bell-shaped **normal** or **Gaussian distribution** $\mathcal{N}(\mu, \sigma^2)$, with density

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

It has mean μ (the centre of the bell) and variance σ^2 (its width). The constant in front makes the total area 1; it comes from the integral $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$ of Section 6.7. The case $\mu = 0$, $\sigma = 1$ is the *standard normal distribution*, whose random variables are usually called Z.

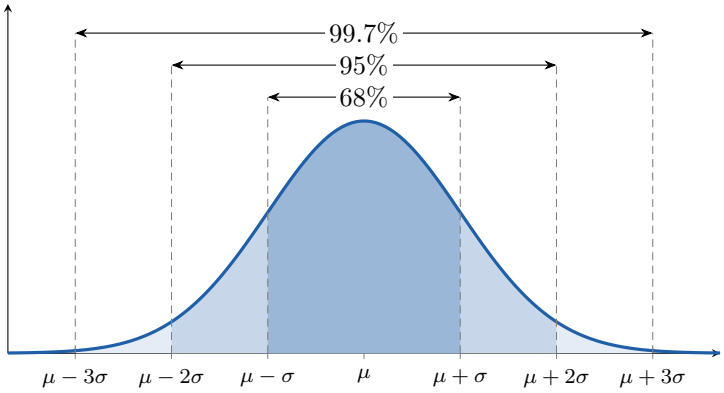


Figure 9.6: The normal distribution. About 68% of the probability lies within one standard deviation of the mean, 95% within two, and 99.7% within three.

Theorem 9.27 (Properties of the normal distribution)**.** Let $X \sim \mathcal{N}(\mu, \sigma^2)$, where the symbol $\sim$ means "is distributed as".

(a) **The 68–95–99.7 rule:** a normal random variable lies within 1, 2 and 3 standard deviations of its mean with probability about 68.3%, 95.4% and 99.7% (Figure 9.6).

(b) **Linear transformations:** for constants $a \neq 0$ and b, $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$. In particular $Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0, 1)$.

(c) **Sums:** the sum of independent normal random variables is normal, with the means and the variances adding up.

Example 9.28 (Using the normal distribution)**.** IQ scores are designed to follow $\mathcal{N}(100, 15^2)$. What fraction of people have an IQ above 130?

Solution. Standardise: 130 is $\frac{130-100}{15} = 2$ standard deviations above the mean. By the 95% rule, 5% lies outside $\pm 2\sigma$, half of it above, so about 2.5%. (Tables or software give $P(Z > 2) \approx 0.0228$.)

Intuition (Why the normal distribution is everywhere)**.** The *central limit theorem* (Chapter 10) says that a sum of many small independent random effects is approximately normal, whatever the distributions of the individual effects. Heights, measurement errors and the noise in many systems arise this way, so the bell curve appears throughout nature, and in machine learning it is the default model for noise. It is also the distribution with the most uncertainty (in the sense of entropy, Chapter 11) for a given mean and variance.

9.9.5 Two more distributions

The **Poisson distribution** with rate λ counts how many times a rare event happens in a fixed period (emails per hour, typos per page): $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \ldots$, with mean and variance both equal to λ. The **exponential distribution** of Example 9.20 describes the waiting time between such events.

Distribution	Values	PMF / PDF	Mean	Variance
Bernoulli(p)	$\{0, 1\}$	$p^x(1-p)^{1-x}$	p	$p(1-p)$
Binomial(n, p)	$\{0, \ldots, n\}$	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Categorical($\boldsymbol{p}$)	$\{1, \ldots, K\}$	p_k	—	—
Poisson(λ)	$\{0, 1, \ldots\}$	$\lambda^k e^{-\lambda}/k!$	λ	λ
Uniform(a, b)	$[a, b]$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$[0, \infty)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal(μ, σ^2)	$\mathbb{R}$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2

Exercises for Section 9.9

Exercise 9.30. $X \sim \text{Bin}(5, 0.5)$. Find $P(X = 2)$, the mean and the variance.

Solution on p. 258 →

Exercise 9.31. $X \sim \mathcal{N}(50, 10^2)$. Use the 68–95–99.7 rule to estimate $P(X > 70)$ and $P(40 < X < 60)$.

Solution on p. 258 →

Exercise 9.32. If $Z \sim \mathcal{N}(0, 1)$, what is the distribution of $3Z + 2$? Find the mean and variance of the uniform distribution on $[2, 8]$.

Solution on p. 258 →

Exercise 9.33. A model with true accuracy 0.9 is evaluated on 1000 independent test examples. What are the mean and standard deviation of the number of correct predictions, and of the measured accuracy?

Solution on p. 258 →

Exercise 9.34. Emails arrive at an average rate of 3 per hour, modelled by a Poisson distribution. What is the probability that no email arrives in a given hour?

Solution on p. 258 →

9.10 Joint, marginal and conditional distributions

Machine learning is about relationships between quantities (features and labels, pixels and objects), so we need to describe several random variables at once.

Definition 9.29 (Joint distribution)**.** The **joint distribution** of two discrete random variables X and Y is given by the joint PMF $p(x, y) = P(X = x \text{ and } Y = y)$. For continuous random variables, a joint density $p(x, y)$ gives probabilities of regions R of the plane as double integrals $\iint_R p(x, y)\,\mathrm{d}x\,\mathrm{d}y$ (Section 7.9).

Example 9.30 (Weather and umbrellas)**.** Let W be the weather (sun or rain) and U whether a person carries an umbrella. Their joint distribution is

	$U = \text{no}$	$U = \text{yes}$	$P(W)$
$W = \text{sun}$	0.50	0.10	0.60
$W = \text{rain}$	0.05	0.35	0.40
$P(U)$	0.55	0.45	1

The entries add up to 1. The row and column totals in the margins are the *marginal distributions* of W and U.

Definition 9.31 (Marginal and conditional distributions)**.** The **marginal distribution** of X is obtained by summing (or integrating) out the other variable:

$$p(x) = \sum_y p(x, y) \qquad \text{or} \qquad p(x) = \int p(x, y)\,\mathrm{d}y.$$

The **conditional distribution** of Y given $X = x$ is $p(y \mid x) = \frac{p(x,y)}{p(x)}$.

In Example 9.30: $P(\text{rain}) = 0.05 + 0.35 = 0.40$, and $P(\text{rain} \mid \text{umbrella}) = \frac{0.35}{0.45} \approx 0.78$. Seeing an umbrella makes rain much more likely than its base rate of 40%.

The two rules of probabilistic modelling

Almost every computation in probabilistic machine learning uses just two rules:

sum rule: $p(x) = \sum_y p(x, y)$, **product rule:** $p(x, y) = p(y \mid x)\,p(x)$.

Bayes' theorem follows from applying the product rule both ways round.

Definition 9.32 (Independent random variables)**.** X and Y are **independent** if $p(x, y) = p(x)\,p(y)$ for all x and y: the joint distribution is the product of the marginals.

In Example 9.30, $P(\text{rain})P(\text{yes}) = 0.4 \times 0.45 = 0.18 \neq 0.35$, so weather and umbrellas are dependent, as one would hope.

Exercises for Section 9.10

Exercise 9.35. Using Example 9.30, find $P(\text{sun} \mid U = \text{no})$ and $P(U = \text{yes} \mid \text{rain})$.

Solution on p. 258 →

Exercise 9.36. Write down the joint distribution table of two independent tosses of a fair coin, X_1 and X_2 (each H or T), and check that it factorises into the marginals.

Solution on p. 258 →

Exercise 9.37. The joint density $p(x, y) = x + y$ on the unit square $[0, 1]^2$ (and 0 elsewhere). Check that it integrates to 1, and find the marginal density of X.

Solution on p. 258 →

9.11 Covariance and correlation

Definition 9.33 (Covariance)**.** The **covariance** of X and Y, with means μ_X and μ_Y, is

$$\operatorname{Cov}(X, Y) = \mathbb{E}\big[(X - \mu_X)(Y - \mu_Y)\big] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y].$$

If X and Y tend to be above their means at the same time (and below them at the same time), the products $(X - \mu_X)(Y - \mu_Y)$ are mostly positive and the covariance is positive. If one tends to be high when the other is low, it is negative. Note that $\operatorname{Cov}(X, X) = \operatorname{Var}(X)$.

The size of the covariance depends on the units, so it is hard to interpret. Its normalised version is easier.

Definition 9.34 (Correlation)**.** The **correlation** of X and Y is

$$\rho_{XY} = \operatorname{Corr}(X, Y) = \frac{\operatorname{Cov}(X, Y)}{\sigma_X\,\sigma_Y},$$

a number between -1 and 1.

Intuition (Correlation is a cosine)**.** For data, centre each variable by subtracting its mean, and collect the centred values into vectors $\boldsymbol{x}$ and $\boldsymbol{y}$. Then the sample covariance is $\frac{1}{n}\boldsymbol{x} \cdot \boldsymbol{y}$ and the sample correlation is

$$\frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\|\boldsymbol{x}\|\,\|\boldsymbol{y}\|} = \cos\theta,$$

the cosine of the angle between the two centred vectors (Theorem 4.8). This is why correlation always lies in $[-1, 1]$ (Cauchy–Schwarz), why $\rho = \pm 1$ means a perfect linear relationship (the vectors are parallel), and why $\rho = 0$ means the centred vectors are orthogonal.

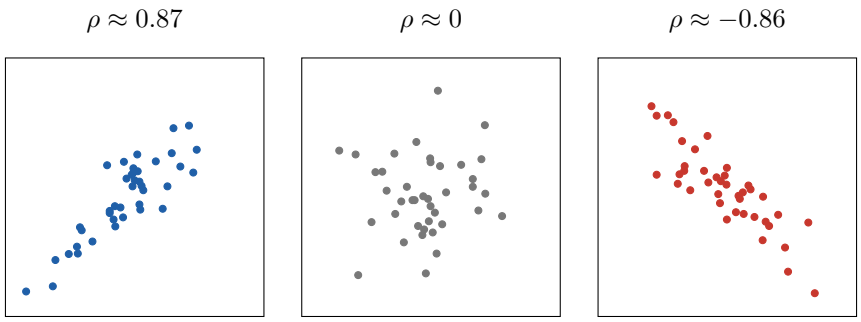


Figure 9.7: Samples of pairs (X, Y) with strong positive correlation (left), no correlation (middle) and strong negative correlation (right).

Example 9.35 (Covariance from a table)**.** Encode Example 9.30 with numbers: $X = 1$ for rain (0 for sun) and $Y = 1$ for an umbrella (0 otherwise). Then $\mathbb{E}[X] = 0.4$, $\mathbb{E}[Y] = 0.45$ and $\mathbb{E}[XY] = P(X = 1, Y = 1) = 0.35$, so $\mathrm{Cov}(X, Y) = 0.35 - 0.4 \times 0.45 = 0.17$. The variances are $0.4 \times 0.6 = 0.24$ and $0.45 \times 0.55 = 0.2475$ (Bernoulli), so $\rho = \frac{0.17}{\sqrt{0.24 \times 0.2475}} \approx 0.70$: a strong positive correlation.

Common pitfall (Two warnings about correlation)**.** Correlation is easy to compute and easy to over-interpret.

- **Correlation is not causation.** Ice-cream sales and drownings are positively correlated, because both rise in summer, not because one causes the other. A model that predicts well using correlated features has not necessarily discovered cause and effect.
- **Zero correlation does not imply independence.** Correlation measures only *linear* relationships. If X is -1, 0 or 1 with equal probabilities and $Y = X^2$, then Y is completely determined by X, yet $\mathrm{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0 - 0 = 0$. (Independent variables do, however, always have zero covariance.)

Theorem 9.36 (Variance of a sum)**.** For any random variables, $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y)$.

9.11.1 Covariance matrices

For a random vector $\boldsymbol{X} = (X_1, \ldots, X_d)$, all the variances and covariances are collected in the $d \times d$ **covariance matrix**

$$\boldsymbol{\Sigma} = \mathbb{E}[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top], \qquad \Sigma_{ij} = \mathrm{Cov}(X_i, X_j),$$

where $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{X}]$. Its diagonal holds the variances. It is symmetric, and it is positive semidefinite: for any fixed vector $\boldsymbol{v}$, $\boldsymbol{v}^\top \boldsymbol{\Sigma} \boldsymbol{v} = \mathrm{Var}(\boldsymbol{v}^\top \boldsymbol{X}) \geq 0$. This is the matrix whose eigenvectors gave the principal components in Section 5.10: there we computed its estimate from data, $\frac{1}{n} \boldsymbol{X}_c^\top \boldsymbol{X}_c$.

9.11.2 The multivariate normal distribution

The **multivariate normal distribution** $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ in $\mathbb{R}^d$, with mean vector $\boldsymbol{\mu}$ and positive definite covariance matrix $\boldsymbol{\Sigma}$, has density

$$p(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^d \det \boldsymbol{\Sigma}}} \exp\!\Big(-\tfrac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^{\top}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\Big).$$

The exponent is a positive definite quadratic form (Section 5.9), so the contours of the density are ellipses (ellipsoids in higher dimensions) centred at $\boldsymbol{\mu}$, with axes along the eigenvectors of $\boldsymbol{\Sigma}$ and widths proportional to the square roots of its eigenvalues (Figure 9.8). The determinant measures the volume of the cloud (Section 4.10).

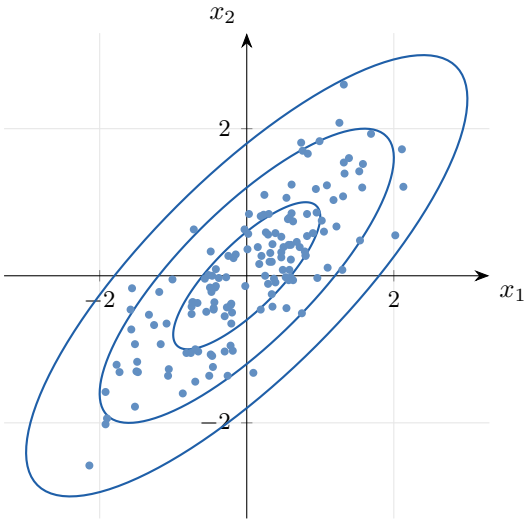


Figure 9.8: 150 samples from a two-dimensional normal distribution with $\boldsymbol{\mu} = \mathbf{0}$ and $\boldsymbol{\Sigma} = \left(\begin{smallmatrix} 1 & 0.8 \\ 0.8 & 1 \end{smallmatrix}\right)$, with contour ellipses of the density. The ellipses point along the eigenvectors $(1,1)$ and $(1,-1)$ of $\boldsymbol{\Sigma}$, whose eigenvalues are 1.8 and 0.2.

Why this matters for ML. The multivariate normal is the workhorse model for continuous data in machine learning: Gaussian noise in regression, Gaussian mixture models for clustering, the latent space of variational autoencoders, and the noise that diffusion models (which generate images) gradually add and then learn to remove. Covariance matrices describe how features vary together and are the input to PCA.

Exercises for Section 9.11

Exercise 9.38. For the data $x = (1, 2, 3, 4)$ and $y = (2, 4, 5, 9)$, compute the covariance (dividing by n) and the correlation.

Solution on p. 258 →

Exercise 9.39. $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$ and $\mathrm{Cov}(X, Y) = -2$. Find $\mathrm{Var}(X + Y)$ and the correlation of X and Y.

Solution on p. 258 →

Exercise 9.40. Let X take the values -1, 0, 1 with probability $\frac{1}{3}$ each, and let $Y = X^2$. Show that $\mathrm{Cov}(X, Y) = 0$, and that X and Y are not independent.

Solution on p. 259 →

Exercise 9.41. Can $\left(\begin{smallmatrix} 4 & 3 \\ 3 & 2 \end{smallmatrix}\right)$ be a covariance matrix? What about $\left(\begin{smallmatrix} 4 & 2 \\ 2 & 2 \end{smallmatrix}\right)$? For the valid one, find the correlation.

Solution on p. 259 →

9.12 Chapter summary

Key takeaways

- Events are subsets of the sample space; probabilities obey $P \geq 0$, $P(\Omega) = 1$ and additivity. $P(A^c) = 1 - P(A)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- $P(A \mid B) = P(A \cap B)/P(B)$; $P(A \mid B) \neq P(B \mid A)$. Law of total probability: $P(A) = \sum_i P(A \mid B_i)P(B_i)$.
- Independence: $P(A \cap B) = P(A)P(B)$. Disjoint is not independent.
- Bayes: posterior $\propto$ likelihood $\times$ prior. Mind the base rate.
- Random variables have PMFs (discrete) or PDFs (continuous, probability = area); CDF $F(x) = P(X \leq x)$.
- $\mathbb{E}[X] = \sum xp(x)$ is linear; $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$; $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$; z-scores standardise.
- Key distributions: Bernoulli, binomial, categorical, uniform, Poisson, exponential and the normal $\mathcal{N}(\mu, \sigma^2)$ (68–95–99.7).
- Joint distributions; marginals by summing out; sum and product rules.
- Covariance and correlation ($\rho \in [-1, 1]$, a cosine) measure linear association; correlation is not causation. Covariance matrices are symmetric and PSD; the multivariate normal has elliptical contours.

Probability goes from a known distribution to predictions about data. Statistics goes the other way, from data to conclusions about the unknown distribution. That is what learning from data means, and it is the subject of the next chapter.

Chapter 10

Statistics for Machine Learning

Motivation. Probability starts from a known distribution and predicts what data it will produce. Statistics runs in the opposite direction: it starts from data and draws conclusions about the unknown distribution that produced them. That is exactly the situation of machine learning. We never know the true relationship between inputs and outputs; we only have a finite, noisy sample of examples, and from it we must

- *estimate* the parameters of a model (training);
- judge how reliable a number computed from a finite sample is (is a test accuracy of 91.2% really better than 90.8%?);
- understand why the standard loss functions, squared error and cross-entropy, are what they are;
- understand why a model that fits its training data perfectly can fail badly on new data, and what to do about it.

This chapter covers these ideas: estimators and their errors, the law of large numbers and the central limit theorem, maximum likelihood and Bayesian estimation, hypothesis testing, and generalisation.

In this chapter you will learn to…

- summarise data with means, medians, variances and percentiles;
- describe estimators by their bias, variance and standard error;
- state and use the law of large numbers and Monte Carlo estimation;
- state the central limit theorem and compute confidence intervals;
- derive maximum likelihood estimates;
- derive the squared-error and cross-entropy losses from likelihoods;
- connect priors with regularisation through MAP estimation;
- carry out and correctly interpret a hypothesis test;
- explain overfitting, underfitting and the bias–variance trade-off, and use validation sets and cross-validation.

10.1 Data, samples and summary statistics

Statistics views a data set $x_1, \dots, x_n$ as the observed values of random variables $X_1, \dots, X_n$. Usually they are assumed to be *i.i.d.* (Section 9.4): drawn independently from one and the same distribution, called the **population**. The data are a **sample** from the population.

Numbers that describe the population, such as its mean μ and variance σ^2, are called **parameters**; they are fixed but unknown. Numbers computed from the sample are called **statistics**; they are known, but they would come out differently for a different sample.

Definition 10.1 (Sample mean, variance and standard deviation)**.** For data $x_1, \dots, x_n$, the **sample mean** and the **sample variance** are

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2,$$

and the *sample standard deviation* is $s = \sqrt{s^2}$.

The division by $n-1$ instead of n looks strange; Section 10.2 explains it. For large n the difference is negligible.

Example 10.2 (Computing summary statistics)**.** Find the sample mean, variance and standard deviation of the data $2, 4, 4, 4, 5, 5, 7, 9$.

Solution. The mean is $\bar{x} = \frac{40}{8} = 5$. The deviations from the mean are -3, -1, -1, -1, 0, 0, 2 and 4, and their squares add up to $9+1+1+1+0+0+4+16 = 32$. So $s^2 = \frac{32}{7} \approx 4.57$ and $s \approx 2.14$.

10.1.1 Medians and percentiles

Definition 10.3 (Median and percentiles)**.** The **median** of a data set is its middle value after sorting (the average of the two middle values if n is even). More generally, the p-th **percentile** is a value below which $p\%$ of the data lie; the median is the 50th percentile, and the 25th and 75th percentiles are called the lower and upper **quartiles**.

Example 10.4 (Mean versus median)**.** The annual salaries (in thousands) of eight employees are

$$32,\ 35,\ 38,\ 40,\ 41,\ 44,\ 47,\ 250.$$

The mean is $\frac{527}{8} \approx 65.9$, more than all but one of the salaries. The median is $\frac{40+41}{2} = 40.5$, a much better description of a typical salary.

Intuition (Two kinds of centre)**.** The mean is the balance point of the data, so one extreme value (an **outlier**) can drag it far away. The median only cares about the order of the values and is *robust* to outliers. There is a deeper connection: the mean is the number c that minimises the total *squared* distance $\sum_i (x_i - c)^2$ (Example 6.21), while the median minimises the total *absolute* distance $\sum_i |x_i - c|$ (an exercise below). This is the same contrast as between the squared and the absolute error (Section 1.10).

Why this matters for ML (Look at your data)**.** Before training anything, practitioners compute summary statistics of every feature: means and standard deviations (for standardisation, Section 9.8), minima and maxima (to find impossible values), and percentiles (to see skew and outliers). Percentiles are also how systems are measured: the "p99 latency" of a model server is the 99th percentile of its response times.

Exercises for Section 10.1

Exercise 10.1. For the data $4, 8, 6, 5, 3, 10$, compute the mean, the median, the sample variance and the sample standard deviation.

Solution on p. 259 →

Exercise 10.2. In the previous exercise, the value 10 is replaced by 100 (a typo). What are the new mean and median?

Solution on p. 259 →

Exercise 10.3. Every value of a data set is transformed by $y_i = 2x_i + 3$. How do the mean, the median and the standard deviation change?

Solution on p. 259 →

Exercise 10.4. Let $g(c) = \sum_{i=1}^{n} |x_i - c|$. Explain why, if c is not a data value, the slope of g at c equals (number of data values below c) minus (number of data values above c). Deduce that g is smallest at the median.

Solution on p. 259 →

10.2 Estimators: bias, variance and standard error

Definition 10.5 (Estimator)**.** An **estimator** is a rule that computes, from the data, a guess $\hat{\theta}$ (read "theta hat") of an unknown parameter θ. Because the data are random, an estimator is a random variable, and its distribution is called its **sampling distribution**.

The sample mean $\bar{X}$ is an estimator of μ; a model's accuracy on a test set is an estimator of its accuracy on all possible inputs; the weights found by training are estimators of the "best" weights. How good is an estimator? Two numbers answer this.

Definition 10.6 (Bias and unbiasedness)**.** The **bias** of an estimator $\hat{\theta}$ is $\mathrm{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$. The estimator is **unbiased** if its bias is 0: on average, it hits the true value.

Theorem 10.7 (Mean and variance of the sample mean)**.** Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2. Then

$$\mathbb{E}[\bar{X}] = \mu \qquad \text{and} \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Proof. By linearity of expectation, $\mathbb{E}[\bar{X}] = \frac{1}{n}\sum_i \mathbb{E}[X_i] = \frac{1}{n} \cdot n\mu = \mu$. By independence the variances of the X_i add up, so $\mathrm{Var}\left(\sum_i X_i\right) = n\sigma^2$, and by Theorem 9.25 $\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\, n\sigma^2 = \frac{\sigma^2}{n}$. ∎

Definition 10.8 (Standard error)**.** The **standard error** of an estimator is its standard deviation. For the sample mean it is $\mathrm{SE}(\bar{X}) = \sigma/\sqrt{n}$, which we estimate by $s/\sqrt{n}$.

Intuition (The square-root law)**.** Averaging n independent measurements makes the error $\sqrt{n}$ times smaller, not n times. To halve the uncertainty you need four times as much data; to gain one more correct decimal digit, a hundred times as much. This law explains a great deal about machine learning, from the noise in mini-batch gradients to the size of test sets.

Example 10.9 (How precise is a test accuracy?)**.** A classifier is correct on 850 of 1000 test examples. How precise is the accuracy estimate 0.85?

Solution. Each test example gives a Bernoulli variable, 1 if the prediction is correct and 0 otherwise, with some unknown mean p (the true accuracy) and variance $p(1-p)$. The test accuracy is their sample mean, so its standard error is

$$\sqrt{\frac{p(1-p)}{n}} \approx \sqrt{\frac{0.85 \times 0.15}{1000}} \approx 0.011.$$

The accuracy is uncertain by about one percentage point, so a difference between 85.3% and 85.7% on this test set means very little.

10.2.1 Why divide by $n-1$?

Theorem 10.10 (An unbiased estimator of the variance)**.** For i.i.d. $X_1, \dots, X_n$ with variance σ^2,

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = (n-1)\,\sigma^2.$$

So dividing by n gives an estimator that is too small on average, by the factor $\frac{n-1}{n}$, whereas s^2, which divides by $n-1$, is unbiased.

Proof. Write $X_i - \bar{X} = (X_i - \mu) - (\bar{X} - \mu)$ and expand. Since $\sum_i (X_i - \mu) = n(\bar{X} - \mu)$,

$$\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - 2n(\bar{X} - \mu)^2 + n(\bar{X} - \mu)^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Taking expectations, using $\mathbb{E}[(X_i - \mu)^2] = \sigma^2$ and $\mathbb{E}[(\bar{X} - \mu)^2] = \operatorname{Var}(\bar{X}) = \sigma^2/n$, gives $n\sigma^2 - \sigma^2 = (n-1)\sigma^2$. ∎

The intuition: the data are on average closer to their own mean $\bar{x}$ (which was computed from them) than to the true mean μ, so squared deviations from $\bar{x}$ underestimate the spread. Dividing by $n-1$, a correction known as **Bessel's correction**, compensates exactly.

10.2.2 Mean squared error

Bias is not everything: an unbiased estimator that is wildly variable is useless. The overall quality of an estimator combines both.

Theorem 10.11 (Bias–variance decomposition of the MSE)**.** The **mean squared error** of an estimator, $\operatorname{MSE}(\hat{\theta}) = \mathbb{E}\big[(\hat{\theta} - \theta)^2\big]$, satisfies

$$\boxed{\operatorname{MSE}(\hat{\theta}) = \operatorname{Bias}(\hat{\theta})^2 + \operatorname{Var}(\hat{\theta})}$$

Proof. Let $m = \mathbb{E}[\hat{\theta}]$. Then $\hat{\theta} - \theta = (\hat{\theta} - m) + (m - \theta)$, where the second part is a constant. Squaring and taking expectations,

$$\mathbb{E}\big[(\hat{\theta} - \theta)^2\big] = \mathbb{E}\big[(\hat{\theta} - m)^2\big] + 2(m - \theta)\,\mathbb{E}[\hat{\theta} - m] + (m - \theta)^2.$$

The middle term vanishes because $\mathbb{E}[\hat{\theta} - m] = 0$; the other two are the variance and the squared bias. ∎

Why this matters for ML (A little bias can help)**.** A biased estimator can have a smaller mean squared error than an unbiased one, if it has much smaller variance. Shrinking an estimate towards zero introduces bias but reduces variance, and this is exactly what regularisation does to the weights of a model (Example 8.2). We meet the same trade-off for whole models in Section 10.9.

Exercises for Section 10.2

Exercise 10.5. Measurements have standard deviation $\sigma = 10$. How many must be averaged for the standard error of the mean to be 1? To be 0.5?

Solution on p. 259 →

Exercise 10.6. From i.i.d. data $X_1, \dots, X_{25}$ with mean μ and variance σ^2, someone estimates μ by the first observation X_1 alone. Is this estimator unbiased? Compare its variance with that of $\bar{X}$.

Solution on p. 259 →

Exercise 10.7. With $n = 5$ observations, what is the expected value of the variance estimator $\frac{1}{n}\sum_i (X_i - \bar{X})^2$, and what is its bias?

Solution on p. 259 →

Exercise 10.8. Estimator A is unbiased with variance 4. Estimator B has bias 1 and variance 1. Which has the smaller mean squared error?

Solution on p. 259 →

10.3 The law of large numbers and Monte Carlo

Why should averages of data tell us anything about the population? Because of the following theorem, which turns the intuition "averages settle down" into mathematics.

Theorem 10.12 (Law of large numbers)**.** Let $X_1, X_2, \dots$ be i.i.d. with mean μ and finite variance σ^2, and let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then for every $\varepsilon > 0$,

$$P\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \to 0 \qquad \text{as } n \to \infty.$$

Proof. Apply Chebyshev's inequality (Section 9.8) to $\bar{X}_n$, whose mean is μ and variance $\frac{\sigma^2}{n}$ (Theorem 10.7): $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}$, which tends to 0. ∎

Common pitfall (The gambler's fallacy)**.** The law of large numbers does *not* say that after a run of heads, tails become "due". A coin has no memory: after ten heads in a row, the next toss is still heads with probability $\frac{1}{2}$. Early imbalances are not

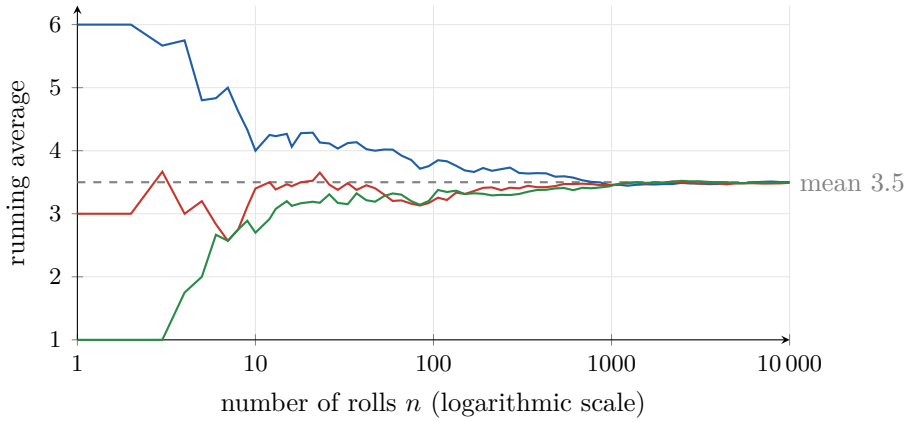


Figure 10.1: The law of large numbers. Each curve shows the average of the first n rolls of a die, for three independent sequences of 10 000 rolls. The averages wander at first and then settle down to the mean 3.5.

compensated; they are *diluted* by the many tosses that follow.

10.3.1 Monte Carlo estimation

The law of large numbers gives a universal recipe for computing expectations, probabilities and integrals: *simulate and average*.

Definition 10.13 (Monte Carlo estimation)**.** To estimate $\mathbb{E}[g(X)]$, draw independent samples $x_1, \dots, x_n$ from the distribution of X and compute

$$\mathbb{E}[g(X)] \approx \frac{1}{n} \sum_{i=1}^{n} g(x_i).$$

This is the **Monte Carlo method**. Its standard error is $\sigma_g/\sqrt{n}$, where σ_g is the standard deviation of $g(X)$, *whatever the dimension of* X.

Example 10.14 (Estimating π by throwing darts)**.** Choose points uniformly at random in the unit square $[0,1]^2$. The probability that a point lands inside the quarter disc $x^2 + y^2 \le 1$ is its area, $\frac{\pi}{4}$. In a simulation with 400 points (Figure 10.2), 318 landed inside, giving the estimate

$$\pi \approx 4 \times \frac{318}{400} = 3.18.$$

The standard error is about $4\sqrt{0.785 \times 0.215/400} \approx 0.08$, and the estimate is indeed within 0.04 of π.

Monte Carlo also computes integrals: $\int_0^1 e^{-x^2}\,\mathrm{d}x$, which has no formula in terms of elementary functions, equals $\mathbb{E}[e^{-U^2}]$ for U uniform on $[0,1]$. Averaging e^{-u^2} over 10 000 random numbers u gave 0.7460, with standard error 0.002; the true value is 0.7468. The method gets more accurate only slowly (the square-root law again), but it works the same way in a thousand dimensions, where other methods are hopeless.

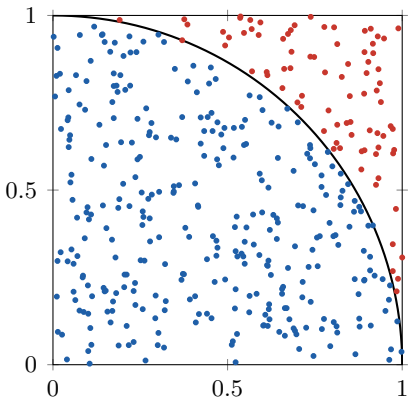


Figure 10.2: Monte Carlo estimation of π: 318 of 400 random points (blue) lie inside the quarter circle.

Why this matters for ML (Monte Carlo is everywhere in machine learning)**.** Whenever an expectation is too expensive to compute exactly, machine learning replaces it by an average over random samples.

- A **mini-batch gradient** is a Monte Carlo estimate of the full gradient (Section 8.6): unbiased, with a standard error proportional to $1/\sqrt{B}$ for batch size B. Quadrupling the batch only halves the noise.
- A **test-set evaluation** is a Monte Carlo estimate of a model's expected loss on new data.
- Generative models, reinforcement learning and Bayesian methods all rely on averages over random samples to approximate expectations that cannot be computed exactly.

Exercises for Section 10.3

Exercise 10.9. Use Chebyshev's inequality to find a number of die rolls n that guarantees $P(|\bar{X}_n - 3.5| \geq 0.1) \leq 0.05$. (The variance of one roll is $\frac{35}{12}$.)

Solution on p. 259 →

Exercise 10.10. The five numbers $0.1, 0.4, 0.5, 0.8, 0.9$ are random samples from the uniform distribution on $[0, 1]$. Use them to give a Monte Carlo estimate of $\mathbb{E}[X^2]$, and compare with the exact value.

Solution on p. 259 →

Exercise 10.11. With a batch size of 32, the standard deviation of a mini-batch gradient component is 0.4. Roughly which batch size would bring it down to 0.1?

Solution on p. 259 →

Exercise 10.12. A fair coin has landed heads ten times in a row. What is the probability of heads on the next toss? What was the probability, before any toss, of eleven heads in a row?

Solution on p. 259 →

10.4 The central limit theorem and confidence intervals

The law of large numbers says that $\bar{X}_n$ approaches μ. The central limit theorem says something much more detailed: *how* $\bar{X}_n$ fluctuates around μ. The answer is always the same: like a normal distribution.

Theorem 10.15 (Central limit theorem)**.** Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and finite variance $\sigma^2 > 0$. For large n, the sample mean is approximately normal:

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Precisely: for every number z,

$$P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \le z\right) \to P(Z \le z) \qquad \text{as } n \to \infty,$$

where $Z \sim \mathcal{N}(0,1)$.

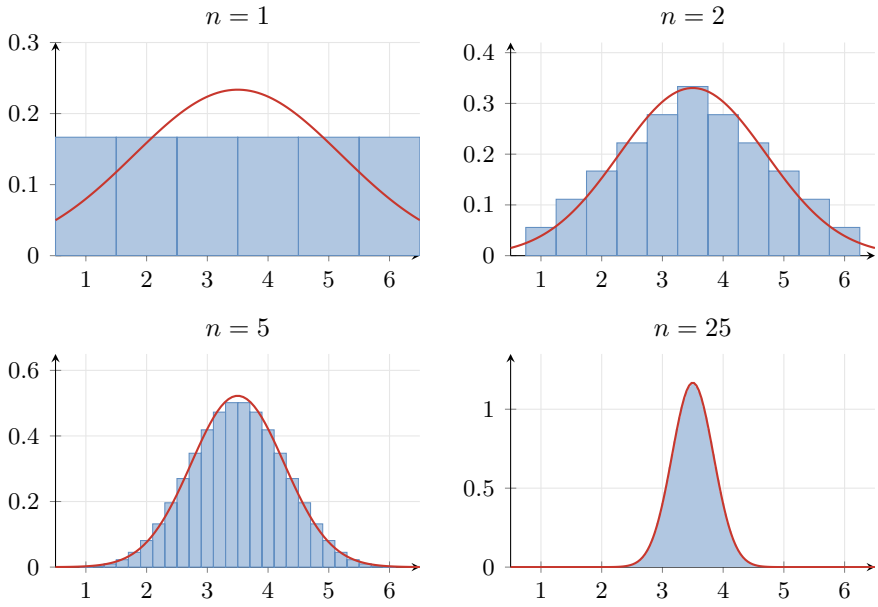


Figure 10.3: The central limit theorem in action. The bars show the exact distribution of the average of n rolls of a fair die, drawn as a density (bar heights are probabilities divided by the bar width $\frac{1}{n}$). The red curves are the normal densities $\mathcal{N}(3.5, \frac{35}{12n})$ predicted by the theorem. A flat distribution turns into a bell curve that becomes narrower as n grows.

Intuition (Why averages become bell-shaped)**.** An average of many independent pieces has many ways to come out near the middle and few ways to come out at the extremes: to average 6 over 25 dice, every single die must show a 6. Remarkably, the limiting shape does not depend on the shape of the individual distribution (flat, skewed, two-humped): only its mean and variance survive. This is why the normal distribution appears whenever many small independent effects add up.

10.4.1 Confidence intervals

The central limit theorem lets us attach error bars to an estimate. Since $\bar{X}$ is approximately normal with mean μ and standard deviation $\mathrm{SE} = \sigma/\sqrt{n}$, and a normal random variable lies within 1.96 standard deviations of its mean with probability 0.95,

$$P(\bar{X} - 1.96\,\mathrm{SE} \le \mu \le \bar{X} + 1.96\,\mathrm{SE}) \approx 0.95.$$

Definition 10.16 (Confidence interval)**.** An approximate 95% **confidence interval** for a mean μ is

$$\bar{x} \pm 1.96\,\frac{s}{\sqrt{n}} = \left[\bar{x} - 1.96\,\frac{s}{\sqrt{n}},\ \bar{x} + 1.96\,\frac{s}{\sqrt{n}}\right].$$

Example 10.17 (Error bars for an accuracy)**.** For the classifier of Example 10.9, with accuracy 0.85 on 1000 test examples, the standard error is 0.0113, and an approximate 95% confidence interval for the true accuracy is

$$0.85 \pm 1.96 \times 0.0113 = 0.85 \pm 0.022 = [0.828,\ 0.872].$$

Common pitfall (What "95% confident" means)**.** The true accuracy is a fixed number; it is the interval that is random. The 95% describes the *procedure*: if we repeated the whole experiment many times, with new test sets, about 95% of the intervals computed this way would contain the true value. It does not mean that the true value lies in our particular interval with probability 95% (in the Bayesian approach of Section 10.7 one can make such statements, using a prior).

Why this matters for ML (Error bars in machine learning)**.** Results computed from finite samples should come with error bars: report the mean and standard error over several training runs with different random seeds, or a confidence interval for a test metric. The central limit theorem also explains why the input $\sum_i w_i x_i$ to a neuron with many inputs is approximately normal, which is the starting point of the analysis of weight initialisation (Section 9.8).

Exercises for Section 10.4

Exercise 10.13. A die is rolled 100 times. What is the approximate distribution of the average roll? Estimate the probability that the average exceeds 3.84.

Solution on p. 259 →

Exercise 10.14. A model is correct on 360 of 400 test examples. Give an approximate 95% confidence interval for its accuracy.

Solution on p. 260 →

Exercise 10.15. About how many test examples are needed to measure an accuracy of about 90% to within ± 1 percentage point (as a 95% confidence interval)?

Solution on p. 260 →

Exercise 10.16. Let $U_1, \dots, U_{12}$ be independent and uniform on $[0,1]$ (mean $\frac{1}{2}$, variance $\frac{1}{12}$). Find the mean and variance of $S = U_1 + \cdots + U_{12} - 6$. What does the central limit theorem suggest about its distribution?

Solution on p. 260 →

10.5 Maximum likelihood estimation

Most models in machine learning are *parametric*: a family of distributions $p(x \mid \theta)$, one for each value of a parameter θ (which may be a vector of millions of weights). Training means choosing θ. The most important principle for doing this is maximum likelihood.

Picture this (Which coin produced these tosses?)**.** A coin is tossed 10 times and shows 7 heads and 3 tails. Suppose the probability of heads is θ. How probable was this particular sequence of tosses?

θ	0.5	0.7	0.9
$P(\text{data} \mid \theta) = \theta^7(1-\theta)^3$	0.00098	0.00222	0.00048

Of these values, $\theta = 0.7$ makes the observed data most probable, more than twice as probable as a fair coin does. Maximum likelihood says: *choose the parameter value under which the observed data would have been most probable.*

Definition 10.18 (Likelihood)**.** For data $x_1, \dots, x_n$ and a model $p(x \mid \theta)$, the **likelihood** is the probability (or density) of the data, viewed as a function of the parameter:

$$\mathcal{L}(\theta) = p(x_1, \dots, x_n \mid \theta) = \prod_{i=1}^{n} p(x_i \mid \theta),$$

where the product form holds for i.i.d. data. Its logarithm $\ln \mathcal{L}(\theta) = \sum_{i=1}^{n} \ln p(x_i \mid \theta)$ is the **log-likelihood**.

Definition 10.19 (Maximum likelihood estimate)**.** The **maximum likelihood estimate** (MLE) is the parameter value that maximises the likelihood:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \ln \mathcal{L}(\theta) = \arg\min_{\theta} \Big(-\sum_{i=1}^{n} \ln p(x_i \mid \theta) \Big)$$

The quantity being minimised on the right is the **negative log-likelihood** (NLL).

Maximising $\ln \mathcal{L}$ instead of $\mathcal{L}$ gives the same answer, because ln is increasing, and it is far more convenient: sums are easier to differentiate than products, and a product of thousands of probabilities would underflow to 0 on a computer (Section 2.6).

Example 10.20 (Maximum likelihood for a coin)**.** Find the MLE of $\theta = P(\text{heads})$ after h heads in n tosses; in particular for 7 heads in 10 tosses.

Solution. The log-likelihood is $\ln \mathcal{L}(\theta) = h \ln \theta + (n-h) \ln(1-\theta)$. Setting its derivative

to zero,

$$\frac{h}{\theta} - \frac{n-h}{1-\theta} = 0 \quad \Longrightarrow \quad h(1-\theta) = (n-h)\,\theta \quad \Longrightarrow \quad \hat{\theta} = \frac{h}{n}.$$

The second derivative $-\frac{h}{\theta^2} - \frac{n-h}{(1-\theta)^2}$ is negative, so this is a maximum. For 7 heads in 10 tosses, $\hat{\theta} = 0.7$ (Figure 10.4): the MLE is the observed frequency, as common sense suggests.

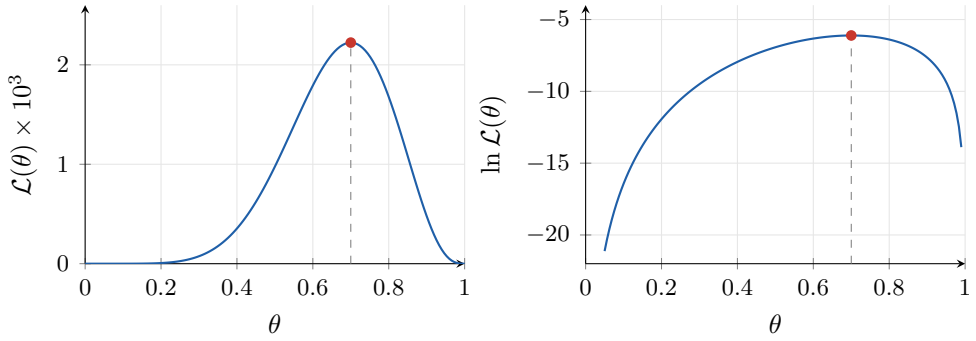


Figure 10.4: The likelihood (left) and log-likelihood (right) of θ after 7 heads in 10 tosses. Both are maximised at $\hat{\theta} = 0.7$; the logarithm changes the shape of the curve but not the location of its maximum.

Example 10.21 (Maximum likelihood for a normal distribution)**.** Data $x_1, \ldots, x_n$ are modelled as i.i.d. from $\mathcal{N}(\mu, \sigma^2)$. Find the MLEs of μ and σ^2.

Solution. Taking the logarithm of the normal density (Section 9.9) and adding over the data,

$$\ln \mathcal{L}(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

For any fixed σ^2, this is largest when $\sum_i (x_i - \mu)^2$ is smallest, which happens at $\hat{\mu} = \bar{x}$ (Example 6.21). Next, writing $v = \sigma^2$ and differentiating with respect to v,

$$-\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(x_i - \bar{x})^2 = 0 \quad \Longrightarrow \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Note that the MLE of the variance divides by n, so it is slightly biased (Theorem 10.10). Maximum likelihood estimates need not be unbiased.

For a categorical distribution with K outcomes, observed $n_1, \ldots, n_K$ times in N trials, the MLE is $\hat{p}_k = \frac{n_k}{N}$, the observed frequencies; we derived this with a Lagrange multiplier in Example 8.19.

Remark (How good is maximum likelihood?)**.** If the model is correct, then as the amount of data grows the MLE converges to the true parameter, its sampling distribution becomes approximately normal, and no other (approximately unbiased) estimator has a smaller variance. With little data, however, it can be badly overconfident: after 3 heads in 3 tosses the MLE is $\hat{\theta} = 1$, which declares tails impossible. Priors

(Section 10.7) fix this.

Why this matters for ML (Training is maximum likelihood)**.** Most of machine learning is maximum likelihood in disguise. A language model is trained by maximising the log-likelihood of its training text: by the chain rule of probability (Section 9.3), this is the sum of the log-probabilities that it assigns to each actual next token. Minimising the average negative log-likelihood with stochastic gradient descent is the training procedure of nearly every modern neural network, as the next section shows.

Exercises for Section 10.5

Exercise 10.17. A coin shows 12 heads in 40 tosses. What is the maximum likelihood estimate of the probability of heads?

Solution on p. 260 →

Exercise 10.18. Counts $x_1, \dots, x_n$ are modelled as Poisson with $p(x \mid \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$. Show that the MLE is $\hat{\lambda} = \bar{x}$, and compute it for the data $2, 0, 3, 1, 4$.

Solution on p. 260 →

Exercise 10.19. Waiting times $0.5, 1.5, 2.0, 1.0$ are modelled as exponential with density $\lambda e^{-\lambda t}$. Find the MLE of λ.

Solution on p. 260 →

Exercise 10.20. Find the MLEs of μ and σ^2 for the data of Example 10.2, assuming a normal model, and compare $\hat{\sigma}^2$ with s^2.

Solution on p. 260 →

Exercise 10.21. The log-likelihood of a data set is -120.5 under model A and -118.2 under model B. Which model explains the data better, and how many times larger is its likelihood?

Solution on p. 260 →

10.6 Loss functions from likelihoods

In supervised learning the model describes the *conditional* distribution $p(y \mid \boldsymbol{x}, \boldsymbol{\theta})$ of a label y given an input $\boldsymbol{x}$. Maximum likelihood then says: minimise $-\sum_i \ln p(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})$. Choosing a distribution for the labels determines the loss function.

10.6.1 Regression: Gaussian noise gives squared error

Suppose that the targets are a deterministic function plus Gaussian noise,

$$y = f(\boldsymbol{x}; \boldsymbol{\theta}) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

so that $p(y \mid \boldsymbol{x}, \boldsymbol{\theta})$ is the normal density with mean $f(\boldsymbol{x}; \boldsymbol{\theta})$ and variance σ^2 (Figure 10.5). The negative log-likelihood is

$$-\ln \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2\sigma^2} \sum_{i=1}^{n} \big(y_i - f(\boldsymbol{x}_i; \boldsymbol{\theta})\big)^2 + n \ln\!\left(\sigma \sqrt{2\pi}\right).$$

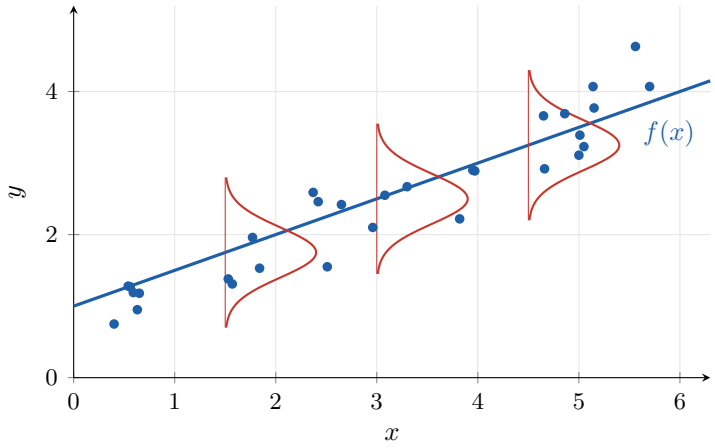


Figure 10.5: The probabilistic view of regression. For each input x, the target y is drawn from a normal distribution (red) centred on the line $f(x)$. Maximum likelihood for this model is least squares.

The second term does not depend on $\boldsymbol{\theta}$, and the first is a positive multiple of the sum of squared errors. Therefore:

Theorem 10.22 (Least squares is maximum likelihood)**.** If the noise is Gaussian with constant variance, maximum likelihood estimation of $\boldsymbol{\theta}$ is the same as minimising the sum of squared errors (equivalently, the mean squared error) $\sum_i (y_i - f(\boldsymbol{x}_i;\boldsymbol{\theta}))^2$.

This is the statistical reason why squared error is the default loss for regression. It also explains its weakness: the normal distribution has very thin tails, so the model regards a large residual as nearly impossible and bends itself to reduce it, which makes least squares sensitive to outliers. If the noise is instead modelled by the heavier-tailed *Laplace distribution*, with density $\frac{1}{2b}e^{-|\varepsilon|/b}$, maximum likelihood gives the more robust absolute error (an exercise below).

10.6.2 Classification: Bernoulli labels give cross-entropy

For binary classification, $y \in \{0,1\}$, and the model outputs the probability $q = P(y=1 \mid \boldsymbol{x},\boldsymbol{\theta})$, for example $q = \sigma(\boldsymbol{w}^\top \boldsymbol{x} + b)$ with the sigmoid function of Chapter 2; this model is called **logistic regression**. The Bernoulli probability of the observed label can be written in one formula as $q^y(1-q)^{1-y}$ (which is q if $y=1$ and $1-q$ if $y=0$). So the negative log-likelihood is

$$-\ln \mathcal{L}(\boldsymbol{\theta}) = -\sum_{i=1}^{n} \Big[y_i \ln q_i + (1-y_i)\ln(1-q_i) \Big]$$

where q_i is the predicted probability for example i. This loss is the **binary cross-entropy** or *log loss*.

Example 10.23 (Computing the binary cross-entropy)**.** Three examples have labels $1, 0, 1$, and a model predicts $q = 0.9, 0.2, 0.6$. Compute the loss.

Solution. For each example the loss is $-\ln$ of the probability given to the true label: $-\ln 0.9 \approx 0.105$, $-\ln(1-0.2) = -\ln 0.8 \approx 0.223$ and $-\ln 0.6 \approx 0.511$. The total is 0.839 and the average 0.280.

Intuition (Cross-entropy punishes confident mistakes)**.** The loss on an example is $-\ln$(probability assigned to the truth). It is 0 for a confident correct prediction, about 0.69 for an undecided prediction of 0.5, and it grows without limit as the probability of the truth goes to 0: predicting $q = 0.99$ for an example whose label is 0 costs $-\ln 0.01 \approx 4.6$. A model trained with cross-entropy learns to be confident only when it has reason to be.

For K classes the labels are categorical, the model outputs probabilities $q_1, \ldots, q_K$ (usually through a softmax function), and the negative log-likelihood is $-\sum_i \ln q_{i,y_i}$, the *cross-entropy* loss. Its connection with information theory, and its strikingly simple gradient, are the subject of Chapter 11.

Choosing a loss means choosing a noise model

Target	Model for y given $\boldsymbol{x}$	Negative log-likelihood
real number	normal, mean $f(\boldsymbol{x})$	squared error
real number	Laplace, centre $f(\boldsymbol{x})$	absolute error
binary label	Bernoulli$(q(\boldsymbol{x}))$	binary cross-entropy
one of K classes	categorical$(\boldsymbol{q}(\boldsymbol{x}))$	cross-entropy
count	Poisson$(\lambda(\boldsymbol{x}))$	$\lambda(\boldsymbol{x}) - y \ln \lambda(\boldsymbol{x})$ (Poisson loss)

Exercises for Section 10.6

Exercise 10.22. Four examples have labels $0, 1, 1, 0$, and a model predicts the probabilities $q = 0.1, 0.7, 0.4, 0.3$ that the label is 1. Compute the average binary cross-entropy. Which example contributes most?

Solution on p. 260 →

Exercise 10.23. Suppose the noise is Laplace: $p(y \mid \boldsymbol{x}) = \frac{1}{2b}\exp(-|y - f(\boldsymbol{x})|/b)$. Show that maximum likelihood minimises the sum of absolute errors $\sum_i |y_i - f(\boldsymbol{x}_i)|$.

Solution on p. 260 →

Exercise 10.24. A "model" predicts the same probability q for every example of a data set with h positive and $n - h$ negative labels. Which q minimises the binary cross-entropy?

Solution on p. 260 →

Exercise 10.25. For a label $y = 1$, compare the binary cross-entropy of the predictions $q = 0.9$, $q = 0.5$, $q = 0.1$ and $q = 0.001$.

Solution on p. 260 →

10.7 Bayesian inference and MAP estimation

Maximum likelihood uses only the data. Often we also have prior knowledge: coins are usually close to fair; the weights of a good model are usually not huge. Bayes' theorem (Section 9.5) combines prior knowledge with data. Treating the parameter θ itself as a

random variable with a **prior** distribution $p(\theta)$,

$$\underbrace{p(\theta \mid \mathcal{D})}_{\text{posterior}} = \frac{p(\mathcal{D} \mid \theta)\, p(\theta)}{p(\mathcal{D})} \;\propto\; \underbrace{p(\mathcal{D} \mid \theta)}_{\text{likelihood}}\, \underbrace{p(\theta)}_{\text{prior}},$$

where $\mathcal{D}$ denotes the data and $\propto$ means "is proportional to" (the denominator does not depend on θ).

Definition 10.24 (MAP estimate)**.** The **maximum a posteriori** (MAP) estimate is the parameter value with the highest posterior density:

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta}\, p(\theta \mid \mathcal{D}) = \arg\min_{\theta} \Big(\underbrace{-\ln p(\mathcal{D} \mid \theta)}_{\text{NLL}}\; \underbrace{-\ln p(\theta)}_{\text{penalty}} \Big).$$

So MAP estimation is maximum likelihood plus a penalty term coming from the prior.

Example 10.25 (A coin with a prior)**.** Take the prior density $p(\theta) = 6\theta(1-\theta)$ on $[0,1]$, which favours values near $\frac{1}{2}$ without ruling anything out. (It is a valid density: $\int_0^1 6\theta(1-\theta)\,\mathrm{d}\theta = 6\left(\frac{1}{2} - \frac{1}{3}\right) = 1$.) After h heads and t tails, find the MAP estimate.

Solution. The posterior is proportional to $\theta^h(1-\theta)^t \cdot \theta(1-\theta) = \theta^{h+1}(1-\theta)^{t+1}$. This has the same form as the likelihood in Example 10.20, with $h+1$ "heads" and $t+1$ "tails", so it is maximised at

$$\hat{\theta}_{\text{MAP}} = \frac{h+1}{n+2}, \qquad n = h + t.$$

After 3 heads in 3 tosses, the MLE is 1 but the MAP estimate is $\frac{4}{5} = 0.8$ (Figure 10.6). After 700 heads in 1000 tosses, the two are almost identical (0.7 and 0.6996): with lots of data, the prior's influence fades away.

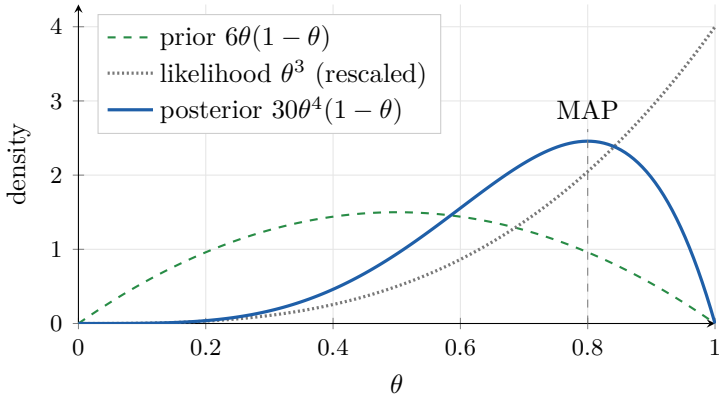


Figure 10.6: Bayesian updating for a coin that shows 3 heads in 3 tosses. The likelihood alone is maximised at $\theta = 1$; combined with the prior, the posterior peaks at $\theta = 0.8$.

The estimate $\frac{h+1}{n+2}$ is known as **Laplace smoothing** or *add-one smoothing*. In a naive Bayes spam filter (Example 9.15) it prevents a word that never appeared in the spam training emails from getting probability 0, which would make any email containing it impossible to classify as spam, whatever its other words.

10.7.1 Regularisation is a prior

Consider linear regression with Gaussian noise of variance σ^2, and give each weight a normal prior, $w_j \sim \mathcal{N}(0, \tau^2)$, independently: we believe that weights are usually small. Then $-\ln p(\boldsymbol{w}) = \frac{1}{2\tau^2}\|\boldsymbol{w}\|^2 + \text{constant}$, and the MAP objective becomes

$$\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i)^2 + \frac{1}{2\tau^2}\|\boldsymbol{w}\|^2.$$

Multiplying by $2\sigma^2$ (which does not change the minimiser) gives

$$\sum_{i=1}^{n}(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i)^2 + \lambda\|\boldsymbol{w}\|^2, \qquad \lambda = \frac{\sigma^2}{\tau^2},$$

which is exactly ridge regression (Section 8.2).

Theorem 10.26 (Priors and penalties)**.** MAP estimation with a Gaussian prior on the weights is L^2 regularisation (ridge regression, weight decay). With a Laplace prior, $p(w_j) \propto e^{-|w_j|/b}$, it is L^1 regularisation (the lasso), which favours sparse weight vectors (Section 5.4).

The stronger the prior belief that the weights are small (the smaller τ), the larger λ. A flat prior ($\tau \to \infty$) gives $\lambda \to 0$: MAP becomes maximum likelihood.

Remark (Full Bayesian inference)**.** MAP keeps only the peak of the posterior. A fully Bayesian approach keeps the whole posterior and averages the predictions of all parameter values, weighted by their posterior probabilities: $p(y \mid \boldsymbol{x}, \mathcal{D}) = \int p(y \mid \boldsymbol{x}, \boldsymbol{\theta})\, p(\boldsymbol{\theta} \mid \mathcal{D})\, \mathrm{d}\boldsymbol{\theta}$. This gives honest uncertainty estimates, but the integral is usually intractable for neural networks; approximations include ensembles of independently trained networks, Monte Carlo dropout and variational inference.

Exercises for Section 10.7

Exercise 10.26. With the prior $6\theta(1-\theta)$ of Example 10.25, find the MAP estimate after 7 heads in 10 tosses, and compare it with the MLE.

Solution on p. 260 →

Exercise 10.27. In ridge regression derived as MAP estimation, the noise variance is $\sigma^2 = 1$ and the prior variance of each weight is $\tau^2 = 0.1$. What is λ? What happens to λ as τ^2 grows?

Solution on p. 260 →

Exercise 10.28. Show that with the uniform prior $p(\theta) = 1$ on $[0, 1]$, the MAP estimate equals the MLE.

Solution on p. 260 →

Exercise 10.29. In 50 spam training emails a certain word never occurs. Estimate the probability that a spam email contains the word, by maximum likelihood and with Laplace smoothing. Why is the first answer dangerous for a naive Bayes classifier?

Solution on p. 260 →

10.8 Hypothesis testing and p-values

A new model scores better than the old one on a test set. Is it really better, or was it lucky? Hypothesis testing is a framework for answering such questions.

Definition 10.27 (Hypothesis test)**.** A hypothesis test has the following ingredients:

- a **null hypothesis** H_0, the sceptical default ("the new model is no better", "the coin is fair"), and an **alternative hypothesis** H_1;
- a **test statistic**: a number computed from the data whose distribution is known if H_0 is true;
- the **p-value**: the probability, *assuming* H_0 *is true*, of obtaining a test statistic at least as extreme as the one observed;
- a **significance level** α, chosen in advance (often 0.05): if $p \le \alpha$, we *reject* H_0 and call the result *statistically significant*.

A small p-value means: if H_0 were true, data like ours would be surprising. So either something surprising happened, or H_0 is false.

Example 10.28 (Is the new model better?)**.** A baseline classifier has accuracy 80%. A new model is correct on 340 of 400 test examples (85%). Is this convincing evidence that its accuracy is higher than 80%?

Solution. Take H_0: the new model's accuracy is $p = 0.8$, and H_1: $p > 0.8$. Under H_0 the number of correct predictions is binomial with mean $400 \times 0.8 = 320$ and standard deviation $\sqrt{400 \times 0.8 \times 0.2} = 8$. The observed 340 is

$$z = \frac{340 - 320}{8} = 2.5$$

standard deviations above the mean. By the central limit theorem, the p-value is approximately $P(Z \ge 2.5) \approx 0.006$ (Figure 10.7); an exact binomial computation gives the same. Since $0.006 < 0.05$, we reject H_0: the improvement is very unlikely to be luck.

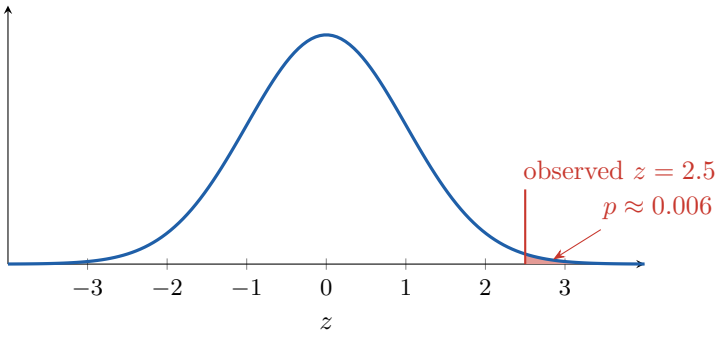


Figure 10.7: The p-value of Example 10.28 is the area of the tail of the standard normal distribution beyond the observed test statistic.

If we had asked whether the new model is *different* (better or worse), we would count extreme results in both directions; this *two-sided* p-value is $P(|Z| \ge 2.5) \approx 0.012$.

10.8.1 Two kinds of error

A test can go wrong in two ways:

	H_0 true	H_0 false
reject H_0	**Type I error** (false positive)	correct
do not reject	correct	**Type II error** (false negative)

The significance level α is the probability of a Type I error. The probability of correctly rejecting a false H_0 is the **power** of the test; it grows with the amount of data. This is the same trade-off as between false positives and false negatives of a classifier, or of the medical test of Example 9.14.

Common pitfall (Misreading p-values)**.** p-values are among the most misinterpreted numbers in science. Keep four points in mind.

- **A p-value is not the probability that H_0 is true.** It is $P(\text{data this extreme} \mid H_0)$, not $P(H_0 \mid \text{data})$; confusing the two is the error of Section 9.3.
- **"Not significant" does not mean "no effect".** With little data, a test has low power and may miss a real effect.
- **Significant does not mean important.** With a million test examples, an improvement of 0.1 percentage points can be highly significant and practically irrelevant. Report the size of the effect, with a confidence interval, not just the p-value.
- **Multiple comparisons.** Test 20 useless ideas at $\alpha = 0.05$ and, on average, one will look significant: the probability of at least one false positive is $1 - 0.95^{20} \approx 0.64$. Trying many models or hyperparameters and reporting the best test score has the same effect.

Remark (Tests and confidence intervals)**.** A two-sided test at level 5% rejects $H_0 : \mu = \mu_0$ exactly when (up to the approximations involved) μ_0 lies outside the 95% confidence interval. Confidence intervals carry more information, because they also show the size of the effect and the precision with which it is known.

Why this matters for ML (Testing in machine learning)**.** Statistical tests are used to compare models (preferably *paired* tests that evaluate both models on the same test examples), to run A/B tests of deployed systems, and to check whether an apparent improvement exceeds the noise between random seeds. The multiple-comparisons problem is the reason for keeping a final test set that is used only once, after all decisions have been made (Section 10.9).

Exercises for Section 10.8

Exercise 10.30. A coin shows 61 heads in 100 tosses. Test the hypothesis that it is fair, using a two-sided test at the 5% level and the normal approximation.

Solution on p. 261 →

Exercise 10.31. A study reports $p = 0.03$. Which statements are correct? (a) There is a 3% probability that H_0 is true. (b) If H_0 were true, results at least this extreme would occur about 3% of the time. (c) The effect is large. (d) At the 5% level, H_0 is rejected.

Solution on p. 261 →

Exercise 10.32. You compare 20 hyperparameter settings, none of which is actually better than the baseline, each with a test at $\alpha = 0.05$ (assume the tests are independent). What is the probability that at least one comparison looks significant? What per-test level would keep this probability below about 0.05?

Solution on p. 261 →

Exercise 10.33. A model has accuracy 0.9 on 400 test examples. Is it significantly different from a competitor whose accuracy is known to be 0.88? Use the confidence interval of the second exercise of Section 10.4.

Solution on p. 261 →

10.9 Generalisation, overfitting and the bias–variance trade-off

The goal of learning is not to do well on the training data, which we already have, but on *new* data from the same distribution: to **generalise**. The training error is an optimistic estimate of the error on new data, because the parameters were chosen to fit precisely those examples.

Definition 10.29 (Training error, test error, generalisation gap)**.** The **training error** is the average loss of a model on the data it was trained on. Its **test error** (or *generalisation error*) is its expected loss on new examples from the same distribution, estimated on a held-out **test set**. The difference between the two is the *generalisation gap*.

Example 10.30 (Fitting polynomials)**.** Ten points are generated from $y = \sin(2\pi x)$ plus Gaussian noise with standard deviation 0.25. Polynomials of degree d are fitted by least squares (Section 5.6), and their root mean squared errors (RMSE) are measured on the training points and on 2000 fresh test points:

degree d	0	1	3	9
training RMSE	0.68	0.55	0.35	0.00
test RMSE	0.75	0.55	0.30	1.05

The degree-9 polynomial has 10 coefficients, enough to pass exactly through all 10 points: zero training error. But it achieves this by wild oscillations between the points (Figure 10.8), and its test error is the worst of all. The cubic captures the true shape and comes close to the noise level 0.25, the best any model can do.

Definition 10.31 (Underfitting and overfitting)**.** A model **underfits** when it is too simple to capture the pattern in the data: both its training and its test error are high. It **overfits** when it is so flexible that it fits the noise in the training data: its training

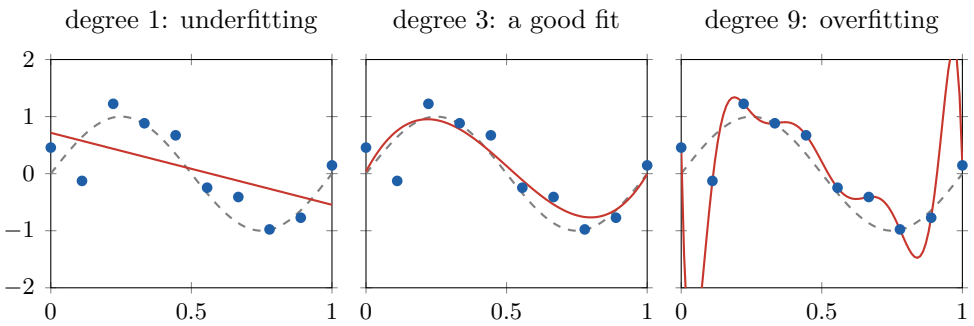


Figure 10.8: Polynomial fits (red) to ten noisy samples (blue) of $\sin(2\pi x)$ (dashed). The straight line is too simple, the cubic is about right, and the degree-9 polynomial passes through every point but oscillates wildly between them.

error is low but its test error is high.

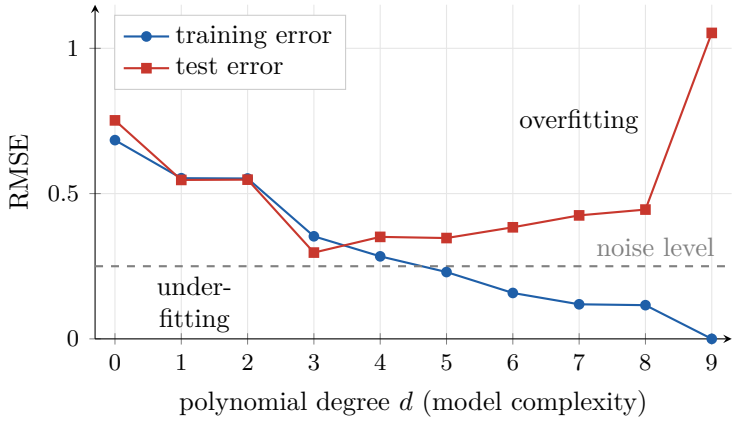


Figure 10.9: Training and test error of the polynomial fits of Example 10.30 for every degree. The training error always decreases with complexity; the test error first decreases, then increases.

10.9.1 The bias–variance trade-off

Why does the test error first fall and then rise? Imagine repeating the whole experiment many times, each time with a fresh training set, and look at the prediction $\hat{f}(x)$ at one fixed input x. Its error has three sources.

Theorem 10.32 (Bias–variance decomposition)**.** Suppose $y = f(x) + \varepsilon$, where the noise ε has mean 0 and variance σ^2 and is independent of the training set, and let $\hat{f}$ be learned from a random training set. Then the expected squared error of the prediction at x is

$$\mathbb{E}\big[(y - \hat{f}(x))^2\big] = \underbrace{\big(f(x) - \mathbb{E}[\hat{f}(x)]\big)^2}_{\text{bias}^2} + \underbrace{\operatorname{Var}\big(\hat{f}(x)\big)}_{\text{variance}} + \underbrace{\sigma^2}_{\text{noise}}.$$

Proof. Write $y - \hat{f}(x) = \varepsilon + (f(x) - \hat{f}(x))$. The noise is independent of $\hat{f}(x)$ and has mean 0, so the cross term has expectation 0 and $\mathbb{E}\big[(y - \hat{f}(x))^2\big] = \sigma^2 + \mathbb{E}\big[(f(x) - \hat{f}(x))^2\big]$. The last expectation is the mean squared error of $\hat{f}(x)$ as an estimator of $f(x)$, which is bias

squared plus variance by Theorem 10.11. ∎

- **Bias** is the systematic error of the model class: even averaged over many training sets, a straight line cannot follow a sine curve.
- **Variance** measures how much the learned function changes from one training set to another. A degree-9 polynomial through ten noisy points changes completely with every new sample.
- **Noise** is the irreducible error: no model can predict the random part of y.

Simple models have high bias and low variance; flexible models have low bias and high variance. The best test error lies in between (Figure 10.10).

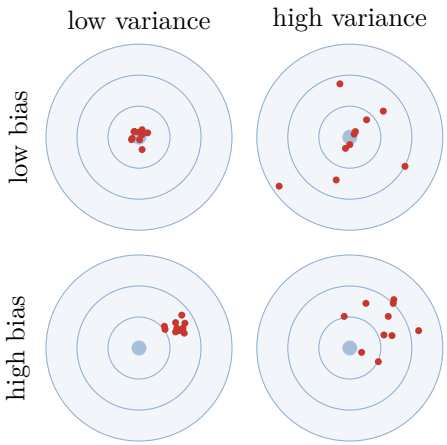


Figure 10.10: Bias and variance as a game of darts. Each dot is the prediction of a model trained on a different training set; the centre is the truth. Bias is the distance from the centre of the cloud to the bull's-eye; variance is the spread of the cloud.

10.9.2 Fighting overfitting

- **More data.** Variance shrinks as the training set grows; a flexible model that overfits 10 points may be perfect for 10 000.
- **Regularisation.** Penalise large weights (L^2 or L^1, Section 10.7), randomly drop units during training (*dropout*), or stop training when the validation error starts to rise (*early stopping*).
- **A simpler model**, or fewer features.
- **Data augmentation**: create new training examples by transformations that do not change the label (rotating or cropping images).
- **Ensembles**: averaging several models reduces variance.

Underfitting calls for the opposite: a bigger model, better features, longer training or less regularisation.

10.9.3 Validation sets and cross-validation

Choices such as the polynomial degree, the regularisation strength λ or the learning rate are called *hyperparameters*. They cannot be chosen by training error (which always favours complexity), nor by test error (which would then no longer be an honest estimate: this is the multiple-comparisons problem of Section 10.8). The standard solution is to split the data three ways:

- a **training set**, used to fit the parameters;

- a **validation set**, used to choose hyperparameters and compare models;
- a **test set**, used once, at the very end, to estimate the performance of the final model.

When data are scarce, **cross-validation** uses them more efficiently. In *k-fold* cross-validation, the data are split into k equal parts (*folds*); the model is trained k times, each time on $k-1$ folds and validated on the remaining one, and the k validation scores are averaged (Figure 10.11).

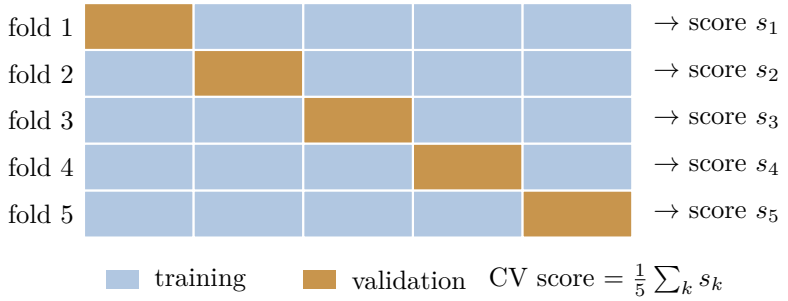


Figure 10.11: Five-fold cross-validation. Every example is used for validation exactly once and for training four times.

Common pitfall (Data leakage)**.** Any information from the validation or test data that leaks into training makes the error estimates optimistic. Common leaks: standardising features with the mean and standard deviation of the *whole* data set instead of the training set; duplicates or near-duplicates of training examples in the test set; and choosing a model because of its test score.

Remark (Overparameterised models and double descent)**.** Modern neural networks often have far more parameters than training examples and can fit even randomly shuffled labels perfectly, yet they generalise well on real data. Beyond the point where a model can interpolate the training data, the test error can *decrease again* as the model grows, a phenomenon called **double descent**. Explaining this (through the implicit preferences of gradient descent, the architecture, and the structure of real data) is an active area of research. The practical tools of this section, held-out data and regularisation, remain as important as ever.

Exercises for Section 10.9

Exercise 10.34. Diagnose each situation and suggest a remedy. (a) Training error 1%, validation error 30%. (b) Training error 25%, validation error 26%, while humans achieve 5%.

Solution on p. 261 →

Exercise 10.35. In 5-fold cross-validation on 1000 examples, how many times is the model trained, and on how many examples is each trained and validated?

Solution on p. 261 →

Exercise 10.36. At some input, a model has squared bias 0.04 and variance 0.10, and the noise variance is 0.25. What is the expected squared prediction error? Which parts could a better method reduce?

Solution on p. 261 →

Exercise 10.37. You train 100 models and report the test error of the one with the lowest test error. Why is this number likely to be too optimistic?

Solution on p. 261 →

Exercise 10.38. Explain, using the bias–variance decomposition, why collecting more training data usually helps an overfitting model much more than an underfitting one.

Solution on p. 261 →

10.10 Chapter summary

Key takeaways

- Data are modelled as i.i.d. samples from a population. The mean is sensitive to outliers; the median is robust. The sample variance divides by $n-1$ to be unbiased.
- An estimator has a bias and a variance; $\text{MSE} = \text{Bias}^2 + \text{Var}$. The sample mean has standard error $\sigma/\sqrt{n}$: four times the data halves the error.
- Law of large numbers: averages converge to expectations, which justifies Monte Carlo estimation (mini-batch gradients, test-set evaluation).
- Central limit theorem: $\bar{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$; approximate 95% confidence interval $\bar{x} \pm 1.96\, s/\sqrt{n}$.
- Maximum likelihood: choose θ to maximise $\sum_i \ln p(x_i \mid \theta)$; for a coin, $\hat{\theta} = h/n$; for a normal distribution, $\hat{\mu} = \bar{x}$.
- Gaussian noise $\Rightarrow$ squared error; Bernoulli labels $\Rightarrow$ binary cross-entropy; categorical labels $\Rightarrow$ cross-entropy.
- MAP = maximum likelihood + prior penalty; a Gaussian prior gives L^2 regularisation, a Laplace prior L^1.
- A p-value is $P(\text{data this extreme} \mid H_0)$, not the probability that H_0 is true. Beware of multiple comparisons.
- Test error = bias^2 + variance + noise. Underfitting is high bias; overfitting is high variance. Use validation sets or cross-validation to choose hyperparameters, and touch the test set only once.

We now have all the ingredients of modern deep learning: functions and their compositions, vectors and matrices, gradients and optimisation, probability and likelihood. The final chapter puts them together: softmax and cross-entropy, information theory, and the backpropagation algorithm that trains neural networks.

Chapter 11

Mathematics for Deep Learning

Motivation. A deep neural network is a long composition of simple functions: matrix multiplications followed by simple non-linear functions. It turns its input into probabilities, is scored by a loss derived from probability theory, and is trained by gradient descent, with gradients computed by the chain rule. Every one of these ingredients has appeared in this book:

- layers are matrix–vector products (Chapters 4 and 5);
- activation functions, exponentials and logarithms come from Chapter 2;
- gradients and the chain rule come from Chapters 6 and 7;
- training is optimisation (Chapter 8);
- outputs are probability distributions, losses are negative log-likelihoods, and generalisation is a statistical question (Chapters 9 and 10).

This final technical chapter puts them together. We study softmax outputs, the information theory behind the cross-entropy loss, the matrix calculus that backpropagation needs, backpropagation itself, the problem of vanishing gradients, automatic differentiation and attention, and we end with a complete neural network trained from scratch.

In this chapter you will learn to...

- describe a feed-forward network as a composition of layers and count its parameters;
- compute softmax probabilities, and do so in a numerically stable way;
- compute entropy, cross-entropy and KL divergence, and explain why cross-entropy is the loss for classification;
- use the basic identities of matrix calculus and check them with shapes;
- derive the gradient $\boldsymbol{q} - \boldsymbol{y}$ of the softmax cross-entropy loss;
- carry out backpropagation by hand on a small network;
- explain vanishing and exploding gradients and their remedies;
- explain how automatic differentiation works;
- compute scaled dot-product attention;
- assemble the complete training loop of a neural network.

11.1 Neural networks: compositions of simple functions

A linear model $\hat{y} = \boldsymbol{w}^\top \boldsymbol{x} + b$ can only separate classes by a straight line (or a plane, in higher dimensions). Neural networks get their flexibility by stacking many simple non-linear steps.

Definition 11.1 (Neuron and layer)**.** A **neuron** computes a weighted sum of its inputs plus a bias, the **pre-activation** $z = \boldsymbol{w}^\top \boldsymbol{x} + b$, and outputs $a = \varphi(z)$, where φ is an **activation function** such as ReLU, sigmoid or tanh (Section 2.10). A *layer* of m neurons with n inputs computes

$$\boldsymbol{z} = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{b}, \qquad \boldsymbol{a} = \varphi(\boldsymbol{z}),$$

where $\boldsymbol{W}$ is an $m \times n$ matrix (one row of weights per neuron), $\boldsymbol{b} \in \mathbb{R}^m$, and φ is applied to each component.

Definition 11.2 (Feed-forward network)**.** A **feed-forward neural network** (or *multilayer perceptron*) with L layers computes, starting from $\boldsymbol{a}^{(0)} = \boldsymbol{x}$,

$$\boxed{\boldsymbol{z}^{(l)} = \boldsymbol{W}^{(l)}\boldsymbol{a}^{(l-1)} + \boldsymbol{b}^{(l)}, \qquad \boldsymbol{a}^{(l)} = \varphi(\boldsymbol{z}^{(l)}), \qquad l = 1, \dots, L}$$

The superscript (l) numbers the layers. The layers between input and output are called **hidden layers**. The output layer usually has no activation, or a special one (Section 11.2), and its outputs $\boldsymbol{z}^{(L)}$ are called **logits**.

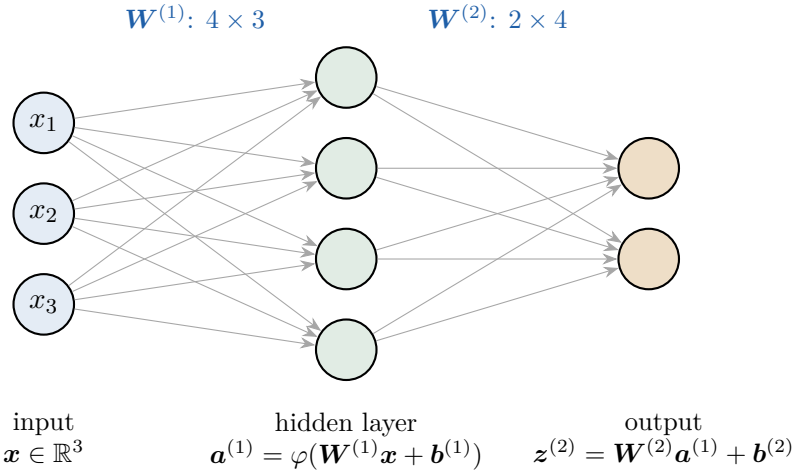


Figure 11.1: A network with 3 inputs, one hidden layer of 4 neurons and 2 outputs. Each arrow carries one weight; each neuron also has a bias.

Example 11.3 (A two-layer network)**.** Take the layer of Section 4.6: $\boldsymbol{W}^{(1)} = \left(\begin{smallmatrix} 1 & -1 & 2 \\ 0 & -2 & 1 \end{smallmatrix}\right)$, $\boldsymbol{b}^{(1)} = (0.5, -1)$, ReLU activation and input $\boldsymbol{x} = (1, 2, 3)$, which gave $\boldsymbol{a}^{(1)} = (5.5, 0)$. Add an output layer with $\boldsymbol{W}^{(2)} = (1 \;\; 3)$ and $b^{(2)} = -2$. The network's output is $z^{(2)} = 1 \cdot 5.5 + 3 \cdot 0 - 2 = 3.5$. It has $2 \cdot 3 + 2 = 8$ parameters in the first layer and $2 + 1 = 3$ in the second.

In general a layer with n inputs and m outputs has $mn + m$ parameters. A classic network for recognising handwritten digits, with 784 inputs (the pixels of a 28×28 image), 128 hidden units and 10 outputs, has $784 \cdot 128 + 128 + 128 \cdot 10 + 10 = 101\,770$ parameters; large language models have hundreds of billions.

Proposition 11.4 (Depth needs non-linearity)**.** Without activation functions, a network collapses to a single linear layer:

$$\boldsymbol{W}^{(2)}(\boldsymbol{W}^{(1)}\boldsymbol{x}+\boldsymbol{b}^{(1)})+\boldsymbol{b}^{(2)}=(\boldsymbol{W}^{(2)}\boldsymbol{W}^{(1)})\boldsymbol{x}+(\boldsymbol{W}^{(2)}\boldsymbol{b}^{(1)}+\boldsymbol{b}^{(2)}).$$

This is Proposition 2.20 again, in matrix form: compositions of linear functions are linear, so stacking linear layers adds nothing. The activation functions are what make depth useful. A network of ReLU layers computes a function that is linear on each of many small regions, with kinks between them; with enough neurons, such functions can follow any shape.

Remark (Universal approximation)**.** A network with a single hidden layer can approximate any continuous function on a bounded region as accurately as desired, provided it has enough hidden neurons. This *universal approximation theorem* guarantees that neural networks are flexible enough; it says nothing about how many neurons are needed or how to find the weights. In practice, deep networks represent complicated functions far more economically than wide, shallow ones.

Exercises for Section 11.1

Exercise 11.1. How many parameters does a network with 2 inputs, one hidden layer of 3 neurons and 1 output have? And one with layers of sizes 10, 20, 20, 5?

Solution on p. 261 →

Exercise 11.2. Two linear layers have $\boldsymbol{W}^{(1)}=\left(\begin{smallmatrix}1&2\\0&1\end{smallmatrix}\right)$, $\boldsymbol{b}^{(1)}=(1,0)$, $\boldsymbol{W}^{(2)}=(1\ \ -1)$ and $b^{(2)}=2$, and no activation. Write the network as a single linear function $w_1x_1+w_2x_2+c$.

Solution on p. 261 →

Exercise 11.3. In Example 11.3, change the input to $\boldsymbol{x}=(0,1,0)$ and compute the output.

Solution on p. 261 →

11.2 From scores to probabilities: the softmax function

A classifier's last layer produces arbitrary real numbers, the logits. To turn them into probabilities we need positive numbers that add up to 1.

For **two classes**, one logit suffices: $q=\sigma(z)=\frac{1}{1+e^{-z}}$ is the probability of class 1, as in logistic regression (Section 10.6). The logit is then the log-odds, $z=\ln\frac{q}{1-q}$, the inverse of the sigmoid (Section 2.8).

For K **classes**, the network outputs K logits, and we exponentiate them (making them positive, Section 2.5) and normalise.

Definition 11.5 (Softmax)**.** The **softmax** of a vector $\boldsymbol{z}\in\mathbb{R}^K$ is the vector $\boldsymbol{q}\in\mathbb{R}^K$ with

$$q_k=\operatorname{softmax}(\boldsymbol{z})_k=\frac{e^{z_k}}{\sum_{j=1}^{K}e^{z_j}}$$

Example 11.6 (Computing a softmax). For the logits $\boldsymbol{z} = (2, 1, 0.1)$: $e^2 \approx 7.389$, $e^1 \approx 2.718$ and $e^{0.1} \approx 1.105$, with sum 11.212. So $\text{softmax}(\boldsymbol{z}) \approx (0.659, 0.242, 0.099)$. The largest logit gets the largest probability, but the others are not ignored.

Theorem 11.7 (Properties of softmax). For every $\boldsymbol{z} \in \mathbb{R}^K$:

(a) the outputs are positive and add up to 1, so they form a categorical distribution (Section 9.9);
(b) larger logits get larger probabilities;
(c) **shift invariance:** $\text{softmax}(\boldsymbol{z} + c\mathbf{1}) = \text{softmax}(\boldsymbol{z})$ for any number c, where $\mathbf{1} = (1, \ldots, 1)$;
(d) for $K = 2$, $\text{softmax}(z_1, z_2)_1 = \sigma(z_1 - z_2)$: softmax generalises the sigmoid.

Proof. (1) and (2) hold because e^x is positive and increasing. (3): $e^{z_k + c} = e^c e^{z_k}$, and the factor e^c cancels between numerator and denominator. (4): dividing numerator and denominator by e^{z_1},

$$\frac{e^{z_1}}{e^{z_1} + e^{z_2}} = \frac{1}{1 + e^{-(z_1 - z_2)}} = \sigma(z_1 - z_2).$$

■

Intuition (A soft maximum). If the logits are multiplied by a large number, the largest one dominates and softmax approaches the one-hot vector of the largest logit, a "hard" maximum. Softmax is a smooth, differentiable version of picking the maximum, which is exactly what gradient-based training needs.

11.2.1 Temperature

Dividing the logits by a **temperature** $T > 0$ before the softmax, $\boldsymbol{q} = \text{softmax}(\boldsymbol{z}/T)$, controls how peaked the distribution is (Figure 11.2). A low temperature makes the model more decisive; a high temperature makes it more uniform. Language models use temperature when generating text: low temperatures give predictable text, high temperatures give varied (and eventually incoherent) text.

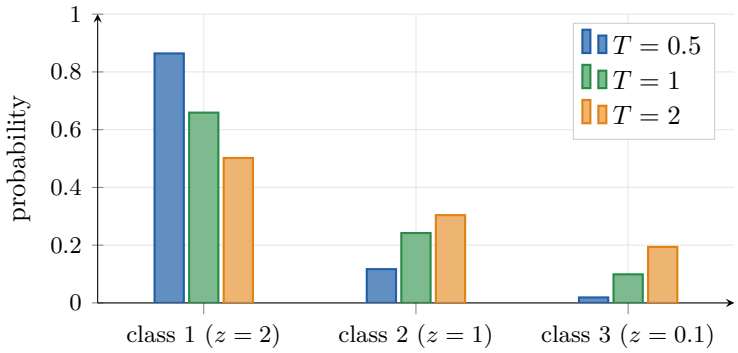


Figure 11.2: The softmax of the logits $(2, 1, 0.1)$ at three temperatures. Lower temperatures sharpen the distribution; higher ones flatten it.

11.2.2 Computing softmax safely

A computer cannot evaluate e^{1000}: it overflows (double-precision numbers stop at about $1.8 \times 10^{308} \approx e^{709.8}$). Shift invariance is the cure: subtract the largest logit first. For

example, $\mathrm{softmax}(1000, 999) = \mathrm{softmax}(1, 0) \approx (0.731, 0.269)$. The same trick evaluates the **log-sum-exp** function, which appears in the logarithm of the softmax:

$$\ln \sum_{j=1}^{K} e^{z_j} = m + \ln \sum_{j=1}^{K} e^{z_j - m}, \qquad m = \max_j z_j, \qquad \text{and} \qquad \ln q_k = z_k - \ln \sum_{j=1}^{K} e^{z_j}.$$

Exercises for Section 11.2

Exercise 11.4. Compute $\mathrm{softmax}(0, \ln 2, \ln 3)$ exactly.

Solution on p. 261 →

Exercise 11.5. Compute $\mathrm{softmax}(1000, 1001, 1002)$ by hand, using shift invariance.

Solution on p. 261 →

Exercise 11.6. What does $\mathrm{softmax}(\boldsymbol{z}/T)$ approach as $T \to \infty$? As $T \to 0$ (assuming the largest logit is unique)?

Solution on p. 262 →

Exercise 11.7. Show that $\max_j z_j \le \ln \sum_{j=1}^{K} e^{z_j} \le \max_j z_j + \ln K$. (This is why log-sum-exp is called a "soft maximum".)

Solution on p. 262 →

11.3 Information theory: entropy, cross-entropy and KL divergence

Why is $-\ln q$ the right loss for classification? Chapter 10 gave one answer, maximum likelihood. Information theory, founded by Claude Shannon in 1948 to study communication, gives a second one, and with it a vocabulary (entropy, cross-entropy, KL divergence) used throughout machine learning.

11.3.1 Surprise and entropy

Learning that an unlikely event happened is more informative than learning that a likely one did. The **information content** or *surprise* of an event of probability p is

$$I = -\log_2 p \quad \text{bits}.$$

A fair coin toss carries $-\log_2 \frac{1}{2} = 1$ bit of information; a roll of a fair die $\log_2 6 \approx 2.585$ bits; an event of probability $\frac{1}{1024}$, 10 bits. A certain event carries none. For independent events probabilities multiply, so surprises add (Theorem 2.16).

Definition 11.8 (Entropy). The **entropy** of a discrete distribution $\boldsymbol{p} = (p_1, \ldots, p_K)$ is its expected surprise,

$$\boxed{H(\boldsymbol{p}) = -\sum_{k=1}^{K} p_k \log p_k}$$

with the convention $0 \log 0 = 0$. With $\log_2$ it is measured in *bits*; with the natural logarithm, as is usual in machine learning, in *nats*.

Example 11.9 (Entropies). A fair coin has entropy 1 bit. A coin with $P(\text{heads}) = 0.9$ has $-(0.9\log_2 0.9 + 0.1\log_2 0.1) \approx 0.469$ bits: it is much more predictable. A fair die has $\log_2 6 \approx 2.585$ bits. The distribution $\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$ has $\frac{1}{2}\cdot 1 + \frac{1}{4}\cdot 2 + \frac{1}{8}\cdot 3 + \frac{1}{8}\cdot 3 = 1.75$ bits.

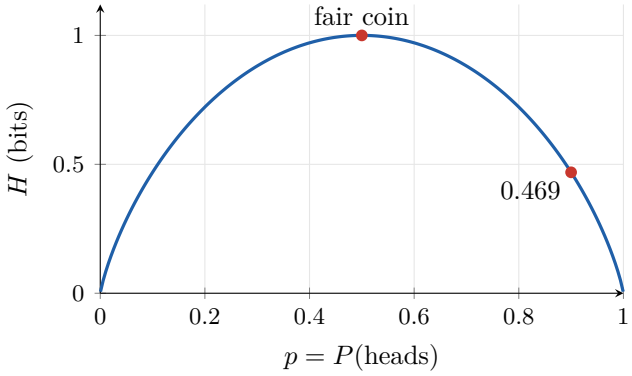


Figure 11.3: The entropy of a coin with $P(\text{heads}) = p$. It is largest, 1 bit, for a fair coin and 0 when the outcome is certain.

Intuition (Entropy as the length of a code). Entropy is the average number of yes/no questions (bits) needed to communicate an outcome, using the best possible code. For $\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$ the code $\texttt{0}, \texttt{10}, \texttt{110}, \texttt{111}$ gives frequent outcomes short codewords, with average length $\frac{1}{2}\cdot 1 + \frac{1}{4}\cdot 2 + \frac{1}{8}\cdot 3 + \frac{1}{8}\cdot 3 = 1.75$ bits, exactly the entropy. Shannon proved that no code can do better on average. In general, the best code spends $-\log_2 p_k$ bits on an outcome of probability p_k.

11.3.2 Cross-entropy and KL divergence

Suppose the outcomes follow $\boldsymbol{p}$, but we use a code designed for a different distribution $\boldsymbol{q}$, spending $-\log q_k$ on outcome k.

Definition 11.10 (Cross-entropy and KL divergence). The **cross-entropy** of $\boldsymbol{q}$ relative to $\boldsymbol{p}$ and the **Kullback–Leibler divergence** (KL divergence) of $\boldsymbol{q}$ from $\boldsymbol{p}$ are

$$H(\boldsymbol{p}, \boldsymbol{q}) = -\sum_{k=1}^{K} p_k \log q_k, \qquad \boxed{D_{\mathrm{KL}}(\boldsymbol{p} \,\|\, \boldsymbol{q}) = \sum_{k=1}^{K} p_k \log \frac{p_k}{q_k} = H(\boldsymbol{p}, \boldsymbol{q}) - H(\boldsymbol{p})}$$

The cross-entropy is the average code length when we are wrong about the distribution, and the KL divergence is the price of being wrong: the number of extra bits (or nats) per outcome.

Theorem 11.11 (Gibbs' inequality). $D_{\mathrm{KL}}(\boldsymbol{p} \,\|\, \boldsymbol{q}) \geq 0$, with equality only if $\boldsymbol{p} = \boldsymbol{q}$. Consequently $H(\boldsymbol{p}, \boldsymbol{q}) \geq H(\boldsymbol{p})$.

Proof. We use natural logarithms and the inequality $\ln t \leq t - 1$, which says that the concave function ln lies below its tangent line at $t = 1$, touching it only at $t = 1$ (Section 8.5).

Summing over the k with $p_k > 0$,

$$-D_{\mathrm{KL}}(\boldsymbol{p} \,\|\, \boldsymbol{q}) = \sum_k p_k \ln \frac{q_k}{p_k} \le \sum_k p_k \left(\frac{q_k}{p_k} - 1 \right) = \sum_k q_k - \sum_k p_k \le 1 - 1 = 0.$$

Equality requires $\frac{q_k}{p_k} = 1$ for all these k, i.e. $\boldsymbol{q} = \boldsymbol{p}$. ■

Example 11.12 (KL divergence is not symmetric)**.** Let $\boldsymbol{p} = (0.5, 0.5)$ and $\boldsymbol{q} = (0.9, 0.1)$. In nats,

$$D_{\mathrm{KL}}(\boldsymbol{p} \,\|\, \boldsymbol{q}) = 0.5 \ln \tfrac{0.5}{0.9} + 0.5 \ln \tfrac{0.5}{0.1} \approx 0.511,$$
$$D_{\mathrm{KL}}(\boldsymbol{q} \,\|\, \boldsymbol{p}) = 0.9 \ln \tfrac{0.9}{0.5} + 0.1 \ln \tfrac{0.1}{0.5} \approx 0.368.$$

The KL divergence measures how different two distributions are, but it is not a distance in the sense of Section 3.2: it is not symmetric.

11.3.3 Cross-entropy as a loss function

For one training example of class c, the "true" distribution is the one-hot vector $\boldsymbol{y}$ (with $y_c = 1$), and the model predicts $\boldsymbol{q} = \mathrm{softmax}(\boldsymbol{z})$. Then

$$H(\boldsymbol{y}, \boldsymbol{q}) = -\sum_k y_k \ln q_k = -\ln q_c,$$

the negative log-probability of the correct class: precisely the negative log-likelihood of Section 10.6. Since a one-hot vector has entropy $H(\boldsymbol{y}) = 0$, this is also $D_{\mathrm{KL}}(\boldsymbol{y} \,\|\, \boldsymbol{q})$.

Three names for one loss

Averaged over a training set, the following are the same thing:

- the negative log-likelihood of the data (Section 10.5);
- the cross-entropy between the labels and the predictions;
- the KL divergence from the labels to the predictions, up to a constant that does not depend on the model.

Training a classifier by minimising cross-entropy means making its predicted distribution as close as possible to the observed one.

Why this matters for ML (Information theory in machine learning)**.** Entropy, cross-entropy and KL divergence turn up all over machine learning.

- **Perplexity.** Language models report $e^{\text{cross-entropy}}$, the **perplexity**. A model with an average cross-entropy of 3 nats per token has perplexity $e^3 \approx 20$: it is as uncertain as if it were choosing uniformly among 20 tokens. Guessing uniformly from a vocabulary of 50 000 tokens gives a cross-entropy of $\ln 50\,000 \approx 10.8$ nats.
- **KL divergence** appears in variational autoencoders, in *knowledge distillation* (training a small model to match the softened output distribution of a large one), and in fine-tuning language models with human feedback, where a KL penalty keeps the new model close to the original.
- **Maximum entropy.** For a continuous distribution with density p, the entropy is defined as $-\int p(x) \ln p(x)\, \mathrm{d}x$. Among all distributions with a given variance, the normal distribution has the largest entropy, which is one more reason why it

is the default model of noise (Section 9.9).

Exercises for Section 11.3

Exercise 11.8. Compute the entropy in bits of $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$ and of $\left(\frac{1}{2}, \frac{1}{2}, 0, 0\right)$.

Solution on p. 262 →

Exercise 11.9. A model predicts $\boldsymbol{q} = (0.7, 0.2, 0.1)$. What is the cross-entropy loss if the true class is 1? If it is 2?

Solution on p. 262 →

Exercise 11.10. Compute $D_{\mathrm{KL}}(\boldsymbol{p} \,\|\, \boldsymbol{q})$ in bits for $\boldsymbol{p} = \left(\frac{1}{2}, \frac{1}{2}\right)$ and $\boldsymbol{q} = \left(\frac{1}{4}, \frac{3}{4}\right)$.

Solution on p. 262 →

Exercise 11.11. Use Gibbs' inequality with $\boldsymbol{q}$ the uniform distribution on K outcomes to show that $H(\boldsymbol{p}) \le \log K$ for every distribution $\boldsymbol{p}$ on K outcomes.

Solution on p. 262 →

Exercise 11.12. A language model has an average cross-entropy of 2.3 nats per token on a test text. What is its perplexity?

Solution on p. 262 →

11.4 Matrix calculus

Backpropagation needs derivatives of a scalar loss with respect to vectors and matrices. We follow one simple convention.

Notation. For a scalar L and a vector $\boldsymbol{x} \in \mathbb{R}^n$, $\frac{\partial L}{\partial \boldsymbol{x}} = \nabla_{\boldsymbol{x}} L$ is the vector of partial derivatives $\frac{\partial L}{\partial x_i}$ (Section 7.3). For a matrix $\boldsymbol{W}$, $\frac{\partial L}{\partial \boldsymbol{W}}$ is the matrix with entries $\frac{\partial L}{\partial W_{ij}}$. In both cases, *the derivative has the same shape as the variable.*

The reliable method for finding such derivatives is to write everything in components, differentiate with respect to one entry, and reassemble the result into vectors and matrices.

Theorem 11.13 (Basic gradients)**.** For a constant vector $\boldsymbol{a}$, a constant matrix $\boldsymbol{A}$ and a constant vector $\boldsymbol{b}$:

(a) $\nabla_{\boldsymbol{x}}(\boldsymbol{a}^\top \boldsymbol{x}) = \boldsymbol{a}$;
(b) $\nabla_{\boldsymbol{x}}(\boldsymbol{x}^\top \boldsymbol{A} \boldsymbol{x}) = (\boldsymbol{A} + \boldsymbol{A}^\top)\boldsymbol{x}$, which equals $2\boldsymbol{A}\boldsymbol{x}$ if $\boldsymbol{A}$ is symmetric;
(c) $\nabla_{\boldsymbol{x}} \|\boldsymbol{x}\|^2 = 2\boldsymbol{x}$;
(d) $\nabla_{\boldsymbol{x}} \|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|^2 = 2\boldsymbol{A}^\top(\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b})$.

Proof. (1) $\boldsymbol{a}^\top \boldsymbol{x} = \sum_i a_i x_i$, whose derivative with respect to x_k is a_k. (2) $\boldsymbol{x}^\top \boldsymbol{A} \boldsymbol{x} = \sum_i \sum_j x_i A_{ij} x_j$. The variable x_k appears when $i = k$ and when $j = k$, so $\frac{\partial}{\partial x_k} = \sum_j A_{kj} x_j + \sum_i x_i A_{ik} = (\boldsymbol{A}\boldsymbol{x})_k + (\boldsymbol{A}^\top \boldsymbol{x})_k$. (3) is (2) with $\boldsymbol{A} = \boldsymbol{I}$. (4) follows from (3) and the chain rule: with $\boldsymbol{r} = \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}$, Theorem 11.15 below gives $\nabla_{\boldsymbol{x}} = \boldsymbol{A}^\top \nabla_{\boldsymbol{r}} \|\boldsymbol{r}\|^2 = 2\boldsymbol{A}^\top \boldsymbol{r}$. ∎

Setting the gradient in (4) to zero gives $\boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^\top \boldsymbol{b}$, the normal equations of least squares (Theorem 5.23), found this time with calculus instead of geometry.

Example 11.14 (Checking an identity). Let $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} = x_1^2 + 5x_1x_2 + 4x_2^2$. At $\boldsymbol{x} = (1, 1)$ the formula gives $(\boldsymbol{A} + \boldsymbol{A}^\top)\boldsymbol{x} = \left(\begin{smallmatrix} 2 & 5 \\ 5 & 8 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right) = (7, 13)$. Directly: $\frac{\partial f}{\partial x_1} = 2x_1 + 5x_2 = 7$ and $\frac{\partial f}{\partial x_2} = 5x_1 + 8x_2 = 13$. ✓

11.4.1 Gradients through a layer

The following two rules are all that backpropagation needs.

Theorem 11.15 (Gradients through a linear layer). Let $\boldsymbol{z} = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{b}$, with $\boldsymbol{W}$ an $m \times n$ matrix, and let L be a scalar that depends on $\boldsymbol{z}$, with gradient $\boldsymbol{\delta} = \frac{\partial L}{\partial \boldsymbol{z}} \in \mathbb{R}^m$. Then

$$\boxed{\frac{\partial L}{\partial \boldsymbol{W}} = \boldsymbol{\delta}\,\boldsymbol{x}^\top, \qquad \frac{\partial L}{\partial \boldsymbol{x}} = \boldsymbol{W}^\top \boldsymbol{\delta}, \qquad \frac{\partial L}{\partial \boldsymbol{b}} = \boldsymbol{\delta}}$$

Proof. In components, $z_i = \sum_j W_{ij}x_j + b_i$. The weight W_{ij} appears only in z_i, with $\frac{\partial z_i}{\partial W_{ij}} = x_j$, so $\frac{\partial L}{\partial W_{ij}} = \delta_i x_j$: the matrix of these numbers is the outer product $\boldsymbol{\delta}\boldsymbol{x}^\top$ (Section 4.9). The input x_j appears in every z_i, with $\frac{\partial z_i}{\partial x_j} = W_{ij}$, so by the multivariable chain rule $\frac{\partial L}{\partial x_j} = \sum_i \delta_i W_{ij} = (\boldsymbol{W}^\top \boldsymbol{\delta})_j$. Finally, $\frac{\partial z_i}{\partial b_i} = 1$. ∎

Intuition (Reading the weight gradient). The weight W_{ij} connects input j to output i, and its gradient is $\delta_i x_j$: (how much the loss cares about output i) times (how active input j was). A weight whose input was 0 gets no gradient, since it had no influence on this example.

For the activation step we need a new product. The **Hadamard product** of two vectors of the same length multiplies them component by component: $\boldsymbol{u} \odot \boldsymbol{v} = (u_1v_1, \ldots, u_nv_n)$.

Theorem 11.16 (Gradients through an activation). If $\boldsymbol{a} = \varphi(\boldsymbol{z})$ componentwise and L depends on $\boldsymbol{a}$, then

$$\boxed{\frac{\partial L}{\partial \boldsymbol{z}} = \frac{\partial L}{\partial \boldsymbol{a}} \odot \varphi'(\boldsymbol{z})}$$

Proof. Each $a_i = \varphi(z_i)$ depends only on z_i, so $\frac{\partial L}{\partial z_i} = \frac{\partial L}{\partial a_i}\varphi'(z_i)$. (In matrix language, the Jacobian of $\boldsymbol{a}$ is diagonal, as in Example 7.18.) ∎

Remark (Let the shapes guide you). Shapes catch most mistakes in matrix calculus. If $\boldsymbol{W}$ is $m \times n$, then $\frac{\partial L}{\partial \boldsymbol{W}}$ must be $m \times n$ too, and the only way to build such a matrix from $\boldsymbol{\delta} \in \mathbb{R}^m$ and $\boldsymbol{x} \in \mathbb{R}^n$ is $\boldsymbol{\delta}\boldsymbol{x}^\top$. Likewise $\frac{\partial L}{\partial \boldsymbol{x}} \in \mathbb{R}^n$ can only be $\boldsymbol{W}^\top \boldsymbol{\delta}$. For a mini-batch stored as a matrix $\boldsymbol{X}$ with one example per row, the forward pass is $\boldsymbol{Z} = \boldsymbol{X}\boldsymbol{W}^\top + \boldsymbol{1}\boldsymbol{b}^\top$, and with $\boldsymbol{\Delta} = \frac{\partial L}{\partial \boldsymbol{Z}}$ (one row per example) the gradients become $\frac{\partial L}{\partial \boldsymbol{W}} = \boldsymbol{\Delta}^\top \boldsymbol{X}$, the sum of the outer products of all the examples, and $\frac{\partial L}{\partial \boldsymbol{X}} = \boldsymbol{\Delta}\boldsymbol{W}$.

Exercises for Section 11.4

Exercise 11.13. Find $\nabla_{\boldsymbol{x}} \left(\boldsymbol{x}^\top \boldsymbol{x} + 3\boldsymbol{a}^\top \boldsymbol{x} \right)$.

Solution on p. 262 →

Exercise 11.14. Show that the gradient of the mean squared error $L(\boldsymbol{w}) = \frac{1}{n}\|\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}\|^2$ is $\frac{2}{n}\boldsymbol{X}^\top(\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y})$.

Solution on p. 262 →

Exercise 11.15. For a layer $\boldsymbol{z} = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{b}$ with $\boldsymbol{W} = \left(\begin{smallmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{smallmatrix}\right)$ and $\boldsymbol{x} = (3, 0, 1)$, suppose $\boldsymbol{\delta} = \frac{\partial L}{\partial \boldsymbol{z}} = (1, -2)$. Compute $\frac{\partial L}{\partial \boldsymbol{W}}$, $\frac{\partial L}{\partial \boldsymbol{x}}$ and $\frac{\partial L}{\partial \boldsymbol{b}}$.

Solution on p. 262 →

Exercise 11.16. A layer has a 128×784 weight matrix. What is the shape of the gradient of the loss with respect to it, and of the vector $\boldsymbol{\delta}$?

Solution on p. 262 →

11.5 The gradient of the cross-entropy loss

We now differentiate the most important loss in deep learning: softmax followed by cross-entropy. The result is remarkably simple.

Theorem 11.17 (Gradient of softmax with cross-entropy)**.** Let $\boldsymbol{q} = \mathrm{softmax}(\boldsymbol{z})$ and $L = -\ln q_c$, where c is the correct class, and let $\boldsymbol{y}$ be the one-hot vector of c. Then

$$\boxed{\frac{\partial L}{\partial \boldsymbol{z}} = \boldsymbol{q} - \boldsymbol{y}}$$

Proof. By the definition of softmax,

$$L = -\ln \frac{e^{z_c}}{\sum_j e^{z_j}} = -z_c + \ln \sum_j e^{z_j}.$$

The derivative of $-z_c$ with respect to z_k is -1 if $k = c$ and 0 otherwise, that is, $-y_k$. By the chain rule, the derivative of $\ln \sum_j e^{z_j}$ is $\frac{e^{z_k}}{\sum_j e^{z_j}} = q_k$. Together, $\frac{\partial L}{\partial z_k} = q_k - y_k$. ∎

Intuition (Prediction minus truth)**.** The gradient is simply "what the model predicted minus what it should have predicted". For the correct class, $q_c - 1 < 0$, so gradient descent raises its logit; every other logit is lowered in proportion to the probability it wrongly received. A perfect prediction gives a zero gradient, and the gradient can never be larger than 1 in any component.

Example 11.18 (One step on the logits)**.** In Example 11.6, $\boldsymbol{z} = (2, 1, 0.1)$ and $\boldsymbol{q} \approx (0.659, 0.242, 0.099)$. If the correct class is the first, the loss is $-\ln 0.659 \approx 0.417$ and the gradient is

$$\boldsymbol{q} - \boldsymbol{y} \approx (-0.341,\ 0.242,\ 0.099).$$

A gradient step of size 1 on the logits gives $\boldsymbol{z} \approx (2.341, 0.758, 0.001)$, for which

$q_1 \approx 0.768$ and the loss has dropped to 0.264.

The same pattern holds for the other standard pairs of output and loss. For a sigmoid output $q = \sigma(z)$ with binary cross-entropy, the chain rule and $\sigma' = \sigma(1-\sigma)$ (Example 6.15) give

$$\frac{\partial L}{\partial z} = \left(-\frac{y}{q} + \frac{1-y}{1-q}\right) q(1-q) = q - y,$$

and for a linear output $\hat{y} = z$ with the squared error $\frac{1}{2}(\hat{y}-y)^2$, $\frac{\partial L}{\partial z} = \hat{y} - y$. In each case: prediction minus target. Combined with Theorem 11.15, logistic regression ($z = \boldsymbol{w}^\top \boldsymbol{x} + b$) has the gradient

$$\nabla_{\boldsymbol{w}} L = (q-y)\,\boldsymbol{x}, \qquad \frac{\partial L}{\partial b} = q - y.$$

Remark (Why not squared error for classification?)**.** Suppose we used the squared error $\frac{1}{2}(q-y)^2$ on a sigmoid output. Its gradient with respect to the logit is $(q-y)\,q(1-q)$. When the model is confidently wrong, say $z = -5$ for an example with $y = 1$, we get $q = \sigma(-5) \approx 0.0067$ and a gradient of only about -0.0066, while the cross-entropy gives $q - y \approx -0.993$, about 150 times larger. With squared error the factor $q(1-q)$ silences exactly the examples that most need correcting; cross-entropy cancels it.

Common pitfall (Compute the loss from the logits)**.** Never compute $\mathrm{softmax}(\boldsymbol{z})$ and then take its logarithm: a tiny probability can underflow to 0, and $-\ln 0 = \infty$. Use $-\ln q_c = -z_c + \ln \sum_j e^{z_j}$ with the stable log-sum-exp of Section 11.2. Deep learning libraries offer "cross-entropy with logits" functions for exactly this reason.

Exercises for Section 11.5

Exercise 11.17. The logits are $\boldsymbol{z} = (0, 0, 0)$ and the correct class is 3. Compute the probabilities, the loss and the gradient.

Solution on p. 262 →

Exercise 11.18. Show that the components of $\boldsymbol{q} - \boldsymbol{y}$ always add up to 0. Explain this using the shift invariance of softmax.

Solution on p. 262 →

Exercise 11.19. A binary classifier outputs the logit $z = 2$ for an example with label $y = 0$. Find q, the loss and $\frac{\partial L}{\partial z}$.

Solution on p. 262 →

Exercise 11.20. Logistic regression has $\boldsymbol{w} = (1, -1)$ and $b = 0$. For $\boldsymbol{x} = (2, 1)$ and $y = 1$, compute the gradients of the cross-entropy loss with respect to $\boldsymbol{w}$ and b.

Solution on p. 262 →

11.6 Backpropagation

We now have everything needed to compute the gradient of the loss with respect to every weight of a network. Write $\boldsymbol{\delta}^{(l)} = \frac{\partial L}{\partial \boldsymbol{z}^{(l)}}$ for the gradient with respect to the pre-activations of layer l, sometimes called the *error signal* of that layer.

Theorem 11.19 (The backpropagation equations)**.** For the network of Definition 11.2 and a loss L computed from its output:

$$\begin{aligned}
\boldsymbol{\delta}^{(L)} &= \frac{\partial L}{\partial \boldsymbol{z}^{(L)}} && \text{(for example } \boldsymbol{q}-\boldsymbol{y}\text{, by Theorem 11.17)},\\
\boldsymbol{\delta}^{(l)} &= (\boldsymbol{W}^{(l+1)\top}\boldsymbol{\delta}^{(l+1)}) \odot \varphi'(\boldsymbol{z}^{(l)}) && \text{for } l = L-1,\dots,1,\\
\frac{\partial L}{\partial \boldsymbol{W}^{(l)}} &= \boldsymbol{\delta}^{(l)}\,\boldsymbol{a}^{(l-1)\top}, \qquad \frac{\partial L}{\partial \boldsymbol{b}^{(l)}} = \boldsymbol{\delta}^{(l)} && \text{for } l = 1,\dots,L.
\end{aligned}$$

Proof. Layer $l+1$ computes $\boldsymbol{z}^{(l+1)} = \boldsymbol{W}^{(l+1)}\boldsymbol{a}^{(l)} + \boldsymbol{b}^{(l+1)}$, so by Theorem 11.15, $\frac{\partial L}{\partial \boldsymbol{a}^{(l)}} = \boldsymbol{W}^{(l+1)\top}\boldsymbol{\delta}^{(l+1)}$; then $\boldsymbol{a}^{(l)} = \varphi(\boldsymbol{z}^{(l)})$ and Theorem 11.16 give the second equation. The weight and bias gradients follow from Theorem 11.15 applied to layer l, whose input is $\boldsymbol{a}^{(l-1)}$. ∎

Algorithm 11.20 (Backpropagation)**.** To compute the gradient of the loss for one example $(\boldsymbol{x}, \boldsymbol{y})$:

Step 1. Forward pass. Set $\boldsymbol{a}^{(0)} = \boldsymbol{x}$ and compute, for $l = 1,\dots,L$, the vectors $\boldsymbol{z}^{(l)}$ and $\boldsymbol{a}^{(l)}$, storing them all. Compute the loss.

Step 2. Output layer. Compute $\boldsymbol{\delta}^{(L)} = \frac{\partial L}{\partial \boldsymbol{z}^{(L)}}$.

Step 3. Backward pass. For $l = L, L-1,\dots,1$: record the gradients $\boldsymbol{\delta}^{(l)}\boldsymbol{a}^{(l-1)\top}$ and $\boldsymbol{\delta}^{(l)}$, and if $l > 1$ compute $\boldsymbol{\delta}^{(l-1)} = (\boldsymbol{W}^{(l)\top}\boldsymbol{\delta}^{(l)}) \odot \varphi'(\boldsymbol{z}^{(l-1)})$.

For a mini-batch, average the gradients of its examples (or use the matrix form of the remark in Section 11.4).

Example 11.21 (Backpropagation by hand)**.** A network has 2 inputs, a hidden layer of 2 ReLU neurons and 1 linear output, with

$$\boldsymbol{W}^{(1)} = \begin{pmatrix} 1 & -1 \\ 0.5 & 1 \end{pmatrix}, \quad \boldsymbol{b}^{(1)} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \boldsymbol{W}^{(2)} = \begin{pmatrix} 2 & -1 \end{pmatrix}, \quad b^{(2)} = 0.5,$$

and the loss $L = \frac{1}{2}(\hat{y} - y)^2$. Compute all gradients for the example $\boldsymbol{x} = (1, 2)$, $y = 1$, and take one gradient descent step with $\eta = 0.1$.

Solution. Forward pass. $\boldsymbol{z}^{(1)} = (1 - 2 + 0,\ 0.5 + 2 - 1) = (-1, 1.5)$, so $\boldsymbol{a}^{(1)} = \mathrm{ReLU}(\boldsymbol{z}^{(1)}) = (0, 1.5)$. Then $\hat{y} = z^{(2)} = 2\cdot 0 - 1\cdot 1.5 + 0.5 = -1$, and $L = \frac{1}{2}(-1-1)^2 = 2$.

Output layer. $\delta^{(2)} = \hat{y} - y = -2$, so

$$\frac{\partial L}{\partial \boldsymbol{W}^{(2)}} = \delta^{(2)}\boldsymbol{a}^{(1)\top} = (0,\ -3), \qquad \frac{\partial L}{\partial b^{(2)}} = -2.$$

Hidden layer. $\boldsymbol{W}^{(2)\top}\delta^{(2)} = (-4, 2)$, and $\mathrm{ReLU}'(\boldsymbol{z}^{(1)}) = (0, 1)$ because the first neuron is inactive. So $\boldsymbol{\delta}^{(1)} = (-4, 2) \odot (0, 1) = (0, 2)$ and

$$\frac{\partial L}{\partial \boldsymbol{W}^{(1)}} = \boldsymbol{\delta}^{(1)}\boldsymbol{x}^\top = \begin{pmatrix} 0 \\ 2 \end{pmatrix}\begin{pmatrix} 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 2 & 4 \end{pmatrix}, \qquad \frac{\partial L}{\partial \boldsymbol{b}^{(1)}} = (0, 2).$$

Update. Subtracting 0.1 times each gradient gives $\boldsymbol{W}^{(1)} = \left(\begin{smallmatrix} 1 & -1 \\ 0.3 & 0.6 \end{smallmatrix}\right)$, $\boldsymbol{b}^{(1)} = (0, -1.2)$, $\boldsymbol{W}^{(2)} = (2 \quad -0.7)$ and $b^{(2)} = 0.7$. Running the forward pass again: $\boldsymbol{z}^{(1)} = (-1, 0.3)$,

$\boldsymbol{a}^{(1)} = (0, 0.3)$, $\hat{y} = -0.21 + 0.7 = 0.49$, and the loss has fallen from 2 to $\frac{1}{2}(0.49 - 1)^2 \approx 0.13$.

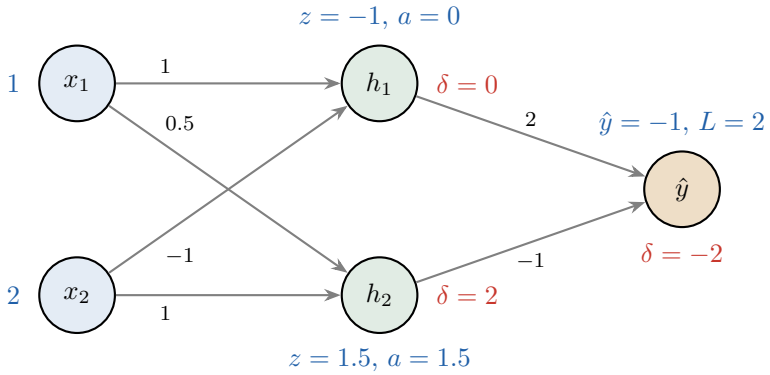


Figure 11.4: The network of Example 11.21, with the weights in black, the forward values in blue and the error signals $\delta = \frac{\partial L}{\partial z}$ in red. The inactive neuron h_1 passes no gradient back.

Intuition (Why backpropagation is fast)**.** The backward pass costs about as much as the forward pass: one multiplication by $\boldsymbol{W}^{(l)\top}$ per layer, plus an outer product. So all the partial derivatives, for millions or billions of weights, cost only a few forward passes. Numerical differentiation (Section 6.8) would need a separate forward pass for every single weight. The price is memory: every $\boldsymbol{z}^{(l)}$ and $\boldsymbol{a}^{(l)}$ of the forward pass must be stored until the backward pass uses it, which is why training needs much more memory than using a trained network.

Exercises for Section 11.6

Exercise 11.21. In Example 11.21, what are the gradients if the target is $y = -1$ instead of 1?

Solution on p. 262 →

Exercise 11.22. Why is the first row of $\frac{\partial L}{\partial \boldsymbol{W}^{(1)}}$ zero in Example 11.21? What would happen if that neuron were inactive for every training example?

Solution on p. 262 →

Exercise 11.23. A network has one input, one hidden ReLU neuron and one linear output: $h = \mathrm{ReLU}(w_1 x + b_1)$, $\hat{y} = w_2 h + b_2$, with loss $\frac{1}{2}(\hat{y} - y)^2$. For $x = 1$, $w_1 = 2$, $b_1 = -1$, $w_2 = 3$, $b_2 = 0$ and $y = 2$, compute the gradient with respect to all four parameters.

Solution on p. 262 →

Exercise 11.24. A network has layer sizes $3 \to 4 \to 2$. Give the shapes of $\boldsymbol{\delta}^{(1)}$, $\boldsymbol{\delta}^{(2)}$, $\frac{\partial L}{\partial \boldsymbol{W}^{(1)}}$ and $\frac{\partial L}{\partial \boldsymbol{W}^{(2)}}$.

Solution on p. 263 →

11.7 Vanishing and exploding gradients

The backpropagation equations multiply the error signal by a matrix and by the activation derivatives once per layer. In a deep network this means long products, and long products of numbers tend to vanish or explode: in Section 2.5 we saw that $0.9^{100} \approx 0.000027$ and $1.1^{100} \approx 13\,781$.

Picture this (A chain of one-neuron layers)**.** Imagine a network with a single neuron per layer, $z^{(l)} = w_l a^{(l-1)}$ and $a^{(l)} = \varphi(z^{(l)})$. By the chain rule,

$$\frac{\partial L}{\partial a^{(0)}} = \frac{\partial L}{\partial a^{(L)}} \prod_{l=1}^{L} w_l \, \varphi'(z^{(l)}).$$

If every factor $w_l \varphi'(z^{(l)})$ is about c, the gradient reaching the first layer is about c^L times the gradient at the last (Figure 11.5). With $c = 0.75$ and 50 layers, it shrinks by a factor of about 10^{-6}; with $c = 1.25$, it grows by a factor of 70 000.

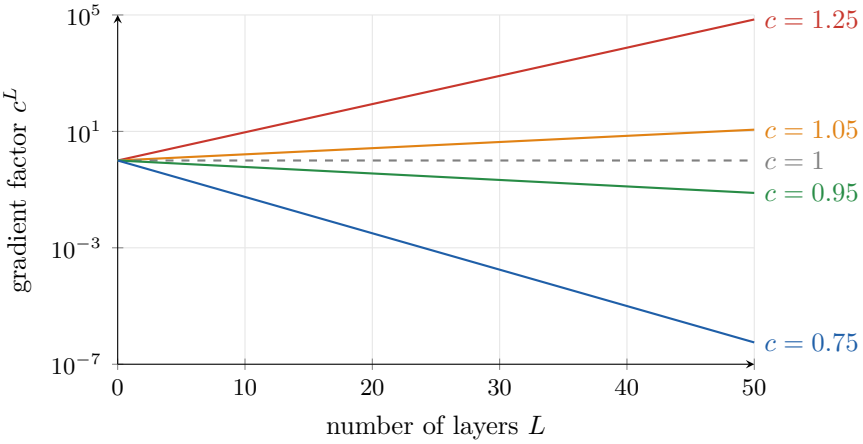


Figure 11.5: If each layer multiplies the gradient by a factor c, the gradient reaching the first of L layers is multiplied by c^L (logarithmic scale). Only factors very close to 1 keep it at a usable size.

For full layers the factors are matrices, and their effect on the length of $\boldsymbol{\delta}$ is governed by their singular values (Section 5.8), but the conclusion is the same: gradients either **vanish**, so that the early layers learn extremely slowly, or **explode**, so that training becomes unstable.

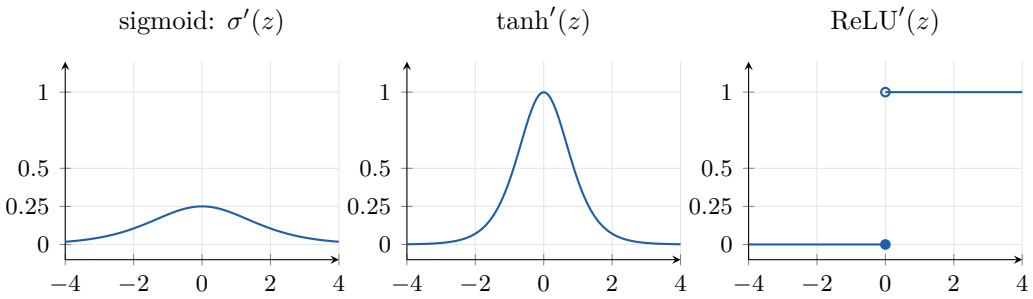


Figure 11.6: Derivatives of three activation functions. The sigmoid's derivative never exceeds $\frac{1}{4}$; $\tanh'$ reaches 1 at 0; the ReLU passes gradients through unchanged wherever the neuron is active.

Example 11.22 (Why deep sigmoid networks were hard to train)**.** The derivative of the sigmoid is $\sigma(1-\sigma) \le \frac{1}{4}$, with the maximum at $z = 0$ (Figure 11.6). With weights of size about 1, every sigmoid layer shrinks the gradient by a factor of at least 4; after 10 layers the factor is at most $\left(\frac{1}{4}\right)^{10} \approx 10^{-6}$. The early layers of such a network barely learn, which is a main reason why deep networks were long considered untrainable.

11.7.1 Remedies

Modern deep learning combines several defences, each built on mathematics from this book.

- **ReLU-type activations.** $\mathrm{ReLU}'(z) = 1$ for active neurons, so the activations do not shrink the gradient.
- **Careful initialisation.** Choosing the variance of the initial weights as $\frac{1}{n}$ or $\frac{2}{n}$ (Xavier or He initialisation, Section 9.8) keeps the variances of activations and gradients roughly constant from layer to layer, so that the factors start out close to 1.
- **Normalisation layers.** Batch and layer normalisation standardise the activations inside the network (Section 9.8).
- **Residual connections.** A residual block computes $\boldsymbol{a}^{(l+1)} = \boldsymbol{a}^{(l)} + F(\boldsymbol{a}^{(l)})$, whose Jacobian is $\boldsymbol{I} + \boldsymbol{J}_F$. The identity term gives the gradient a direct path backwards that does not shrink. Residual connections made networks with hundreds of layers trainable and are part of every transformer.
- **Gradient clipping.** Against exploding gradients: if $\|\boldsymbol{g}\| > \tau$ for a threshold τ, replace $\boldsymbol{g}$ by $\tau\,\boldsymbol{g}/\|\boldsymbol{g}\|$, which keeps the direction but limits the length (Section 4.3).

Exercises for Section 11.7

Exercise 11.25. Bound the factor by which 20 sigmoid layers with weights of size 1 can shrink a gradient.

Solution on p. 263 →

Exercise 11.26. Each of 20 blocks applies a function with derivative $f'(x) = -0.05$ (take it to be constant). Compare the derivative of the composition of the 20 blocks with and without residual connections, $x \mapsto x + f(x)$.

Solution on p. 263 →

Exercise 11.27. Clip the gradient $\boldsymbol{g} = (6, -8)$ to length at most $\tau = 5$.

Solution on p. 263 →

Exercise 11.28. Why do tanh networks suffer less from vanishing gradients than sigmoid networks?

Solution on p. 263 →

11.8 Automatic differentiation

Nobody derives backpropagation formulas by hand for real networks. Libraries such as PyTorch, JAX and TensorFlow compute gradients by **automatic differentiation** (autodiff), which is neither symbolic differentiation (manipulating formulas, which grow out of control) nor numerical differentiation (which is inexact and slow, Section 6.8).

The idea is simple. Every program, however complicated, is built from elementary operations ($+$, $\times$, exp, ReLU, matrix products, ...), and the derivative of each elementary operation is known. During the forward pass the library records every operation in a computation graph (Figure 7.5), together with its local derivatives. The backward pass then visits the graph in reverse order and applies the chain rule, adding up the contributions of all paths, exactly as we did by hand in Example 7.14.

The following complete, working implementation of this idea takes about forty lines of Python. Each `Value` remembers the values it was computed from and the local derivative with respect to each of them.

```
import math

class Value:
    """A number that remembers how it was computed."""

    def __init__(self, data, parents=(), local_grads=()):
        self.data = data                # value from the forward pass
        self.grad = 0.0                 # d(output)/d(self)
        self.parents = parents          # Values it was computed from
        self.local_grads = local_grads  # d(self)/d(parent) for each

    def __add__(self, other):
        other = other if isinstance(other, Value) else Value(other)
        return Value(self.data + other.data, (self, other), (1.0, 1.0))

    def __mul__(self, other):
        other = other if isinstance(other, Value) else Value(other)
        return Value(self.data * other.data, (self, other),
                     (other.data, self.data))

    def relu(self):
        return Value(max(self.data, 0.0), (self,),
                     (1.0 if self.data > 0 else 0.0,))

    def backward(self):
        # List the nodes so that each comes after its parents.
        order, seen = [], set()
        def visit(v):
            if id(v) not in seen:
                seen.add(id(v))
                for p in v.parents:
                    visit(p)
                order.append(v)
        visit(self)
        # Chain rule, from the output back towards the inputs.
        self.grad = 1.0
        for v in reversed(order):
            for p, g in zip(v.parents, v.local_grads):
                p.grad += v.grad * g

x, y = Value(2.0), Value(3.0)
f = (x + y) * (y + 1)
f.backward()
print(f.data, x.grad, y.grad)    # 20.0 4.0 9.0
```

The line `p.grad += v.grad * g` is the chain rule: the derivative of the output with respect

to a parent receives, from each child, the child's gradient times the local derivative. The `+=` adds the contributions of different paths, which is why y, used twice, receives $4 + 5 = 9$. The output reproduces Example 7.14. Real libraries work the same way, with vectors and matrices instead of single numbers:

```
import torch

x = torch.tensor(2.0, requires_grad=True)
y = torch.tensor(3.0, requires_grad=True)
f = (x + y) * (y + 1)
f.backward()
print(x.grad, y.grad)            # tensor(4.) tensor(9.)
```

Remark (Reverse mode and forward mode)**.** What we have described is *reverse-mode* automatic differentiation: one backward pass gives the derivatives of one output with respect to all inputs. That is exactly the situation in training, with one scalar loss and millions of parameters. *Forward-mode* autodiff instead carries derivatives along with the values in the forward pass, giving the derivatives of all outputs with respect to one input per pass; it is efficient when there are few inputs and many outputs.

Why this matters for ML (Gradient checking)**.** When you implement a new operation or a backward pass by hand, compare its gradients with the central differences

$$\frac{L(\theta + h) - L(\theta - h)}{2h}$$

of Section 6.8, on a few random inputs. Relative differences of about 10^{-7} indicate a correct implementation; differences of 10^{-2} indicate a bug.

Exercises for Section 11.8

Exercise 11.29. Using the `Value` class, the program `x = Value(2.0); f = x * x * x; f.backward()` computes $f = x^3$. Trace the backward pass and confirm that `x.grad` ends up as 12.

Solution on p. 263 →

Exercise 11.30. Describe how to add a tanh operation to the `Value` class.

Solution on p. 263 →

Exercise 11.31. A model has 10^9 parameters and one scalar loss. Roughly how many evaluations of the loss would numerical differentiation need to obtain the full gradient, and how many passes does reverse-mode autodiff need?

Solution on p. 263 →

11.9 A glimpse of attention

The transformer architecture behind today's language models is built around one operation, *attention*. A text is split into tokens, each represented by a vector; attention lets each token collect information from the others, weighting them by relevance. Mathematically it uses nothing beyond this book.

Definition 11.23 (Scaled dot-product attention). Given a *query* vector $\boldsymbol{q} \in \mathbb{R}^d$, *key* vectors $\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n \in \mathbb{R}^d$ and *value* vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$, attention computes

$$s_j = \frac{\boldsymbol{q}^\top \boldsymbol{k}_j}{\sqrt{d}}, \qquad (\alpha_1, \ldots, \alpha_n) = \operatorname{softmax}(s_1, \ldots, s_n), \qquad \text{output} = \sum_{j=1}^{n} \alpha_j \boldsymbol{v}_j.$$

For many queries at once, stored as the rows of matrices $\boldsymbol{Q}$, $\boldsymbol{K}$ and $\boldsymbol{V}$,

$$\operatorname{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \operatorname{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d}}\right)\boldsymbol{V}$$

with the softmax applied to each row.

Read it piece by piece: the dot products $\boldsymbol{q}^\top \boldsymbol{k}_j$ measure how well each key matches the query (Section 4.4); softmax turns the scores into weights that are positive and add up to 1; and the output is the weighted average of the values, that is, the expected value under the distribution α (Section 9.7). In a transformer, the queries, keys and values are computed from the token vectors by learned matrices, $\boldsymbol{Q} = \boldsymbol{X}\boldsymbol{W}_Q$, $\boldsymbol{K} = \boldsymbol{X}\boldsymbol{W}_K$ and $\boldsymbol{V} = \boldsymbol{X}\boldsymbol{W}_V$.

Example 11.24 (Attention by hand). Let $d = 2$, keys $\boldsymbol{k}_1 = (1, 0)$, $\boldsymbol{k}_2 = (0, 1)$, $\boldsymbol{k}_3 = (1, 1)$, values equal to the keys, and the query $\boldsymbol{q} = (1, 0)$. The scores are $\frac{1}{\sqrt{2}}(1, 0, 1) \approx (0.707, 0, 0.707)$, the weights are $\operatorname{softmax}(0.707, 0, 0.707) \approx (0.401, 0.198, 0.401)$, and the output is

$$0.401\,(1, 0) + 0.198\,(0, 1) + 0.401\,(1, 1) \approx (0.802, 0.599).$$

The query attends mostly to the two keys that point in its direction.

Intuition (Why divide by $\sqrt{d}$?). If the components of $\boldsymbol{q}$ and $\boldsymbol{k}$ are independent with mean 0 and variance 1, then $\boldsymbol{q}^\top \boldsymbol{k} = \sum_{i=1}^{d} q_i k_i$ is a sum of d independent terms of variance 1, so it has variance d (Section 9.8). For $d = 64$ the scores would have standard deviation 8, and softmax of such large numbers is nearly one-hot, with almost zero gradients (Section 11.5). Dividing by $\sqrt{d}$ brings the variance back to 1.

Exercises for Section 11.9

Exercise 11.32. Repeat Example 11.24 with the query $\boldsymbol{q} = (0, 1)$.

Solution on p. 263 →

Exercise 11.33. What is the output of attention if all the scores are equal? If one score is much larger than all the others?

Solution on p. 263 →

Exercise 11.34. Queries and keys have $d = 256$ components, each with mean 0 and variance 1, independently. What is the standard deviation of $\boldsymbol{q}^\top \boldsymbol{k}$, and of the scaled score?

Solution on p. 263 →

11.10 Putting it all together: training a network

We can now describe, and justify step by step, the complete procedure by which neural networks are trained.

Algorithm 11.25 (Training a neural network)**.** To train a classifier:

Step 1. Prepare the data. Split it into training, validation and test sets, and standardise the features using statistics of the training set (Sections 9.8 and 10.9).

Step 2. Initialise. Draw the weights at random with variance $\frac{2}{n}$ or $\frac{1}{n}$; set the biases to 0 (Section 9.8).

Step 3. Loop over epochs and mini-batches. For each mini-batch: run the forward pass and compute the average cross-entropy loss from the logits (Section 11.5); run the backward pass to get the gradients (Section 11.6); update the parameters with SGD, momentum or Adam, possibly with weight decay and a learning-rate schedule (Chapter 8 and Section 10.7).

Step 4. Monitor. Track the training and validation losses; choose hyperparameters and the stopping time on the validation set (Section 10.9).

Step 5. Evaluate. Measure the final model once on the test set, with error bars (Section 10.4).

The following program carries this out for a small problem: points inside the unit circle belong to class 1 and points in a surrounding ring to class 0 (compare Figure 3.6). No straight line separates the classes, but a network with one hidden layer of 16 ReLU neurons learns to. For brevity the program uses all 400 training points in every step (full-batch gradient descent) and no validation set; every line corresponds to an equation of this chapter.

```
import numpy as np

rng = np.random.default_rng(0)

# Data: class 1 inside the unit circle, class 0 in a ring around it.
n = 400
r = np.concatenate([rng.uniform(0, 1, n // 2),
                    rng.uniform(1.5, 2.5, n // 2)])
angle = rng.uniform(0, 2 * np.pi, n)
X = np.column_stack([r * np.cos(angle), r * np.sin(angle)])
y = np.concatenate([np.ones(n // 2), np.zeros(n // 2)])

# One hidden layer of 16 ReLU units and one output logit.
H = 16
W1 = rng.normal(0, np.sqrt(2 / 2), (H, 2))   # He initialisation
b1 = np.zeros(H)
w2 = rng.normal(0, np.sqrt(1 / H), H)
b2 = 0.0

lr = 0.5
for step in range(1, 501):
    # Forward pass for all examples at once (one row each).
    Z1 = X @ W1.T + b1              # pre-activations, n x H
    A1 = np.maximum(Z1, 0)          # ReLU
    z2 = A1 @ w2 + b2               # output logits, length n
```

```
    # Binary cross-entropy, computed stably from the logits.
    loss = np.mean(np.logaddexp(0, z2) - y * z2)

    # Backward pass.
    q = 1 / (1 + np.exp(-z2))        # predicted P(y = 1)
    d2 = (q - y) / n                 # dL/dz2
    grad_w2 = A1.T @ d2
    grad_b2 = d2.sum()
    D1 = np.outer(d2, w2) * (Z1 > 0)     # dL/dZ1
    grad_W1 = D1.T @ X
    grad_b1 = D1.sum(axis=0)

    # Gradient descent step.
    W1 -= lr * grad_W1
    b1 -= lr * grad_b1
    w2 -= lr * grad_w2
    b2 -= lr * grad_b2

    if step in (1, 10, 50, 100, 500):
        accuracy = np.mean((z2 > 0) == y)
        print(f"step {step:3d}  loss {loss:.3f}  accuracy {accuracy:.3f}")
```

It prints:

```
step   1  loss 0.815  accuracy 0.552
step  10  loss 0.445  accuracy 0.807
step  50  loss 0.165  accuracy 1.000
step 100  loss 0.073  accuracy 1.000
step 500  loss 0.010  accuracy 1.000
```

The loss falls steadily, and after 50 steps every training point is classified correctly. On 1000 new points from the same distribution the network is also right every time: it generalises, because the gap between the two classes leaves room for a decision boundary that is not tuned to individual training points (Figure 11.7).

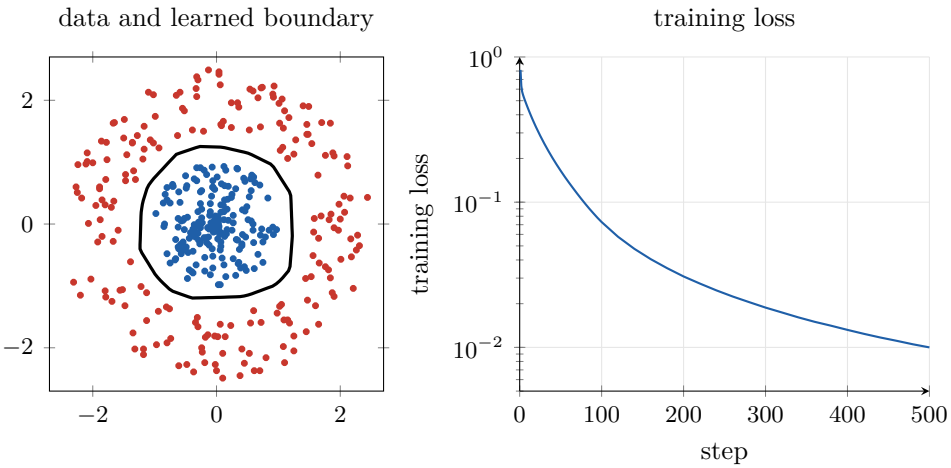


Figure 11.7: Left: the training data (blue: class 1, red: class 0) and the decision boundary learned by the network, where its output probability is $\frac{1}{2}$. As a ReLU network computes a piecewise linear function, the boundary is a polygon. Right: the training loss, on a logarithmic scale.

Where each idea came from

Ingredient	Mathematics
layers $\boldsymbol{W}\boldsymbol{a}+\boldsymbol{b}$, shapes, outer products	linear algebra (Chapters 4 and 5)
activations, e^x, ln, sigmoid	functions (Chapter 2)
backpropagation, autodiff	chain rule, gradients (Chapters 6 and 7)
gradient descent, SGD, Adam	optimisation (Chapter 8)
initialisation, normalisation, attention scaling	variance (Chapter 9)
softmax, cross-entropy, KL divergence	probability, likelihood (Chapters 9 and 10, this chapter)
validation, regularisation, error bars	statistics (Chapter 10)

Exercises for Section 11.10

Exercise 11.35. How many parameters does the network in the program have?

Solution on p. 263 →

Exercise 11.36. In the program, why is `d2` divided by n?

Solution on p. 263 →

Exercise 11.37. What would happen in the program if all the weights and biases were initialised to 0?

Solution on p. 263 →

Exercise 11.38. What would you expect to happen if the learning rate were made much larger, say 100?

Solution on p. 263 →

11.11 Chapter summary

Key takeaways

- A feed-forward network computes $\boldsymbol{z}^{(l)}=\boldsymbol{W}^{(l)}\boldsymbol{a}^{(l-1)}+\boldsymbol{b}^{(l)}$, $\boldsymbol{a}^{(l)}=\varphi(\boldsymbol{z}^{(l)})$; without non-linear activations it collapses to one linear map.
- $\operatorname{softmax}(\boldsymbol{z})_k=e^{z_k}/\sum_j e^{z_j}$ turns logits into probabilities; it is shift invariant, so subtract $\max_j z_j$ before exponentiating.
- Entropy: $H(\boldsymbol{p})=-\sum_k p_k\log p_k$. Cross-entropy: $H(\boldsymbol{p},\boldsymbol{q})=-\sum_k p_k\log q_k$. KL divergence: $D_{\mathrm{KL}}(\boldsymbol{p}\,\|\,\boldsymbol{q})=H(\boldsymbol{p},\boldsymbol{q})-H(\boldsymbol{p})\geq 0$. The cross-entropy loss is the negative log-likelihood.
- Matrix calculus: $\frac{\partial L}{\partial \boldsymbol{W}}=\boldsymbol{\delta}\boldsymbol{x}^{\top}$, $\frac{\partial L}{\partial \boldsymbol{x}}=\boldsymbol{W}^{\top}\boldsymbol{\delta}$ for $\boldsymbol{z}=\boldsymbol{W}\boldsymbol{x}+\boldsymbol{b}$; activations multiply by $\varphi'(\boldsymbol{z})$ componentwise. Gradients have the shapes of the variables.
- Softmax with cross-entropy has gradient $\boldsymbol{q}-\boldsymbol{y}$: prediction minus truth.
- Backpropagation computes all gradients with one backward pass, using the recursion $\boldsymbol{\delta}^{(l)}=(\boldsymbol{W}^{(l+1)\top}\boldsymbol{\delta}^{(l+1)})\odot\varphi'(\boldsymbol{z}^{(l)})$.
- Products over many layers make gradients vanish or explode; ReLU, good

initialisation, normalisation, residual connections and clipping help.
- Automatic differentiation records the computation and applies the chain rule in reverse.
- Attention: $\text{softmax}(\boldsymbol{Q}\boldsymbol{K}^\top/\sqrt{d})\boldsymbol{V}$, a weighted average of values.

You have now seen the mathematical core of modern machine learning, from counting and fractions to backpropagation and attention. The last chapter suggests where to go from here.

Chapter 12

Where to Go Next

Motivation. You began this book with counting, fractions and the order of operations. You have finished it with the mathematics of backpropagation, cross-entropy and attention. That is a long road, and you now have the prerequisites that textbooks and courses on machine learning take for granted. This short chapter looks back at the big ideas, suggests projects that turn knowledge into skill, sketches the mathematics that lies beyond these notes, and recommends books, courses and videos for the next stage of the journey.

12.1 The big ideas

Beneath the many definitions and formulas, a handful of ideas carry most of the weight. If you remember nothing else, remember these.

Seven big ideas

1. **Data are vectors, models are functions, layers are matrices.** Anything a computer learns from (an image, a sentence, a customer) becomes a list of numbers, and a model is a function of those numbers with adjustable parameters (Chapters 2 and 4).
2. **Geometry reveals structure.** Lengths, angles and projections measure similarity and error; eigenvectors and the SVD find the most important directions in data (Chapters 3 and 5).
3. **Learning is optimisation.** Training means choosing the parameters that minimise a loss, usually by following the negative gradient downhill (Chapter 8).
4. **Derivatives measure sensitivity, and the chain rule composes them.** Backpropagation is the chain rule, organised so that all the derivatives of a huge network cost little more than evaluating it (Chapters 6, 7 and 11).
5. **Uncertainty is described by probability.** Models output distributions, and the standard losses are negative log-likelihoods (Chapters 9 and 10).
6. **What matters is new data.** Fitting the training set is easy; the goal is generalisation, governed by the trade-off between bias and variance and measured honestly on held-out data (Chapter 10).
7. **Numbers on a computer need care.** Work with log-probabilities, subtract the maximum before exponentiating, scale initial weights to keep variances stable, and check gradients numerically (Chapters 2, 6 and 11).

12.2 Consolidate by building

Mathematics becomes yours when you use it. The best next step is to implement the methods of this book yourself, in Python with NumPy, checking every result against the theory. The following projects are arranged roughly in order of difficulty; each lists the chapters it relies on.

Project 1. Linear regression three ways. Fit a line to a real data set (for example, house prices against size) with the normal equations, with gradient descent and with stochastic gradient descent. Check that all three agree, and plot the loss against the number of steps for several learning rates. (Chapters 5 and 8.)

Project 2. PCA on images. Load a small data set of images of handwritten digits (one is included in the scikit-learn library), compute the principal components with the SVD, plot the first few as images, and reconstruct digits from 5, 20 and 50 components. (Chapter 5.)

Project 3. Logistic regression with gradient checking. Implement binary logistic regression with the gradient $(q-y)\boldsymbol{x}$, verify the gradient with central differences, add L^2 regularisation, and report the test accuracy with a 95% confidence interval. (Chapters 6, 10 and 11.)

Project 4. Simulate the limit theorems. Check the law of large numbers and the central limit theorem by simulation, for dice, for a skewed distribution such as the exponential, and for a distribution whose mean and variance are infinite (try $1/U$ for U uniform on $[0,1]$, and see what goes wrong). (Chapters 9 and 10.)

Project 5. Grow the autodiff engine. Extend the `Value` class of Section 11.8 with subtraction, division, powers, exp, ln and tanh; build a small network from it and train it on the circle data of Section 11.10. (Chapter 11.)

Project 6. A digit classifier from scratch. Train a network with one or two hidden layers on the MNIST digits, with softmax and cross-entropy, mini-batches and Adam, all in NumPy. Plot training and validation curves, experiment with initialisation and learning rates, and look at the digits it gets wrong. (Chapters 8, 10 and 11.)

Project 7. Attention by hand. Implement scaled dot-product attention for a short sequence of random vectors, verify that each row of weights sums to 1, and observe what happens to the weights without the factor $1/\sqrt{d}$ as d grows. (Chapters 9 and 11.)

> **Remark** (How to read mathematics in papers and textbooks)**.** When you meet an unfamiliar formula, do what this book has done throughout: write down the shape of every vector and matrix; try the formula on a tiny example with numbers (2×2 matrices, three data points); draw a picture; and ask what happens in extreme cases (a parameter equal to 0, or very large). Appendix A lists the notation used here, which is standard in the machine learning literature.

12.3 Mathematics beyond these notes

These notes cover the prerequisites. The following topics build directly on them and appear in more advanced work.

More linear algebra
Further matrix factorisations (LU, QR, Cholesky), numerical stability and the condition number, sparse matrices, and tensors (arrays with more than two indices), which are the basic objects of deep learning libraries. Builds on Chapters 4 and 5.

Fourier analysis and convolution
Representing signals as sums of sines and cosines; convolution, the operation behind

convolutional neural networks for images and audio. Builds on Chapters 3 and 6.

Real analysis
Rigorous treatment of limits, sequences, series and continuity, needed to read proofs that algorithms converge. Builds on Chapter 6.

Convex optimisation and duality
The Karush–Kuhn–Tucker conditions, which extend Lagrange multipliers to inequality constraints, duality, and convergence rates; support vector machines are the classic application. Builds on Chapter 8.

Probability theory and stochastic processes
Measure-theoretic foundations; Markov chains and Markov chain Monte Carlo sampling; stochastic differential equations, which describe the noising process of diffusion models. Builds on Chapter 9.

Bayesian machine learning
Conjugate priors, Gaussian processes, variational inference and probabilistic graphical models. Builds on Chapters 9 and 10.

Statistical learning theory
Why and when learning from a finite sample works: VC dimension, PAC learning and generalisation bounds. Builds on Chapter 10.

Information theory
Mutual information, coding theorems and rate–distortion theory, which appear in representation learning and compression. Builds on Section 11.3.

Reinforcement learning
Markov decision processes, Bellman equations and policy gradients, for agents that learn from rewards. Builds on Chapters 8 and 9.

Graphs
Graph theory and the spectral analysis of graphs, the basis of graph neural networks. Builds on Chapter 5.

12.4 Recommended resources

There are excellent books and courses at every level, many of them free. The following selection is organised by purpose; Figure 12.1 suggests one path through it.

12.4.1 Mathematics

- Marc Peter Deisenroth, A. Aldo Faisal and Cheng Soon Ong, *Mathematics for Machine Learning* (Cambridge University Press, 2020; free PDF online). The natural next book: the same topics as Chapters 4, 5 and 7 to 9 at a faster pace and greater depth, followed by applications to regression, PCA, Gaussian mixture models and support vector machines.
- Gilbert Strang, *Introduction to Linear Algebra* (Wellesley-Cambridge Press), with his celebrated MIT 18.06 video lectures (free on MIT OpenCourseWare). Intuitive and applied. His *Linear Algebra and Learning from Data* connects linear algebra to machine learning directly.
- Stephen Boyd and Lieven Vandenberghe, *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares* (Cambridge University Press, 2018; free online). Gentle and full of applications.
- Sheldon Axler, *Linear Algebra Done Right* (Springer; the fourth edition is open access). For readers who want a proof-based treatment.

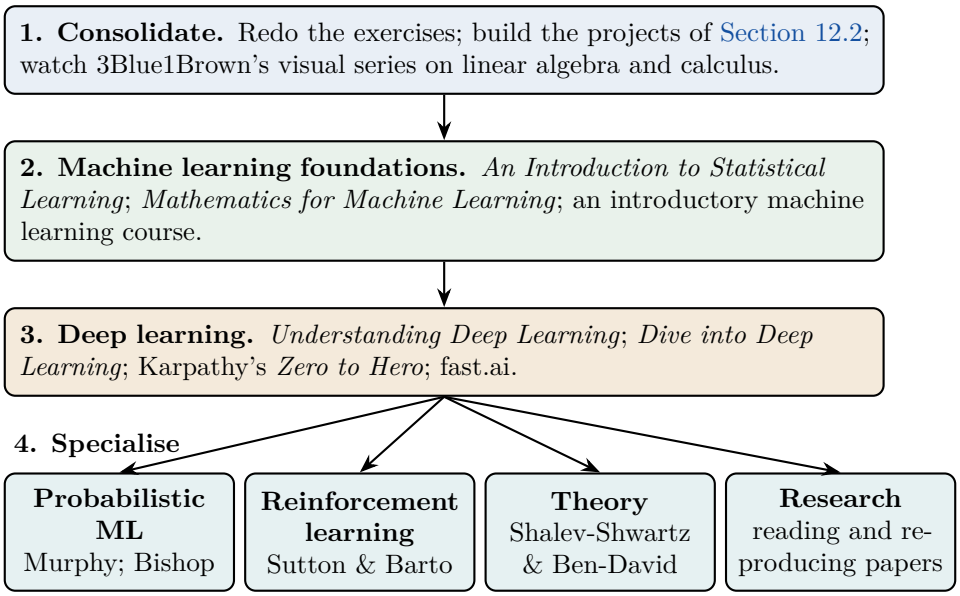


Figure 12.1: A suggested path for further study.

- 3Blue1Brown (Grant Sanderson), *Essence of Linear Algebra* and *Essence of Calculus* (free video series). Unrivalled geometric intuition; ideal alongside Chapters 4 and 6.
- Joseph K. Blitzstein and Jessica Hwang, *Introduction to Probability* (CRC Press, 2nd edition 2019), with the free Harvard Stat 110 lectures. A superb, example-driven course in probability.
- Stephen Boyd and Lieven Vandenberghe, *Convex Optimization* (Cambridge University Press, 2004; free PDF). The standard reference on convexity, duality and optimisation algorithms.
- David J. C. MacKay, *Information Theory, Inference, and Learning Algorithms* (Cambridge University Press, 2003; free PDF). A delightful book linking information theory, Bayesian inference and neural networks.

12.4.2 Machine learning

- Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, *An Introduction to Statistical Learning* (Springer; editions with labs in R and in Python; free PDF). The friendliest introduction to the classical methods. Its big sibling, *The Elements of Statistical Learning* by Hastie, Tibshirani and Friedman, is more mathematical.
- Christopher M. Bishop, *Pattern Recognition and Machine Learning* (Springer, 2006). A classic, with a consistently probabilistic viewpoint.
- Kevin P. Murphy, *Probabilistic Machine Learning: An Introduction* (MIT Press, 2022) and *Advanced Topics* (2023); free drafts online. Comprehensive and up to date.
- Shai Shalev-Shwartz and Shai Ben-David, *Understanding Machine Learning: From Theory to Algorithms* (Cambridge University Press, 2014; free online). The theory of generalisation.
- Richard S. Sutton and Andrew G. Barto, *Reinforcement Learning: An Introduction* (MIT Press, 2nd edition 2018; free online).

12.4.3 Deep learning

- Simon J. D. Prince, *Understanding Deep Learning* (MIT Press, 2023; free PDF). Clear explanations and outstanding figures; the best next step after Chapter 11.

- Christopher M. Bishop and Hugh Bishop, *Deep Learning: Foundations and Concepts* (Springer, 2024). A thorough modern textbook.
- Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning* (MIT Press, 2016; free online). Its first part reviews linear algebra, probability, information theory and numerical computation.
- Aston Zhang, Zachary C. Lipton, Mu Li and Alexander J. Smola, *Dive into Deep Learning* (free online at `d2l.ai`). Every idea is accompanied by runnable code.
- Andrej Karpathy, *Neural Networks: Zero to Hero* (free video series). Starts by building an autodiff engine much like the one in Section 11.8 and ends with a small GPT-style language model.
- fast.ai, *Practical Deep Learning for Coders* (free course). A top-down, hands-on complement to the bottom-up approach of these notes.

12.5 A final word

Nobody reads mathematics once and understands it fully. Expect to return to these chapters: when a paper writes $\nabla_{\boldsymbol{\theta}} \mathbb{E}_{\boldsymbol{x}\sim p}[\ell(\boldsymbol{x};\boldsymbol{\theta})]$ and you need to recall why the expectation and the gradient can be swapped for a mini-batch estimate; when a training run diverges and you suspect the learning rate or the conditioning; when a model is confidently wrong and you want to reason about its probabilities. Each time, the ideas will feel more familiar and the formulas more like a language you speak.

Mistakes and confusion are part of the process, not signs of failure. Work small examples, draw pictures, check shapes, and write code. Machine learning rewards exactly the habits this book has tried to build: careful definitions, concrete numbers before abstraction, and the patience to ask *why*.

Good luck, and enjoy the journey.

Appendix A

Notation and the Greek Alphabet

This appendix collects the notation used in these notes, which follows the conventions of most machine learning textbooks and research papers. The last column says where each symbol is introduced.

Conventions

- Ordinary numbers (*scalars*) are written in italic: x, a, λ.
- Vectors are bold lower-case letters, $\boldsymbol{x}$, $\boldsymbol{w}$, $\boldsymbol{\theta}$, and are column vectors; their components are $x_1, \dots, x_n$.
- Matrices are bold capital letters, $\boldsymbol{A}$, $\boldsymbol{W}$, $\boldsymbol{\Sigma}$, with entries A_{ij} (row i, column j).
- Random variables are italic capitals, X, Y; their values are lower-case, x, y. Random vectors are bold capitals, $\boldsymbol{X}$.
- A hat marks an estimate or a prediction: $\hat{\theta}$, $\hat{y}$.
- ln is the natural logarithm. In information theory, log means $\log_2$ when measuring in bits and ln when measuring in nats.
- Superscripts in parentheses number the layers of a network: $\boldsymbol{W}^{(l)}$ is the weight matrix of layer l, not a power.

Symbols

Symbol	Meaning	Introduced in
Numbers and sets		
$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$	natural numbers, integers, rational and real numbers	Section 1.1
$x \in A$	x is an element of the set A	Section 1.1
$A \subset B$	A is a subset of B	Section 1.1
$\{x \in \mathbb{R} : x > 0\}$	the set of all x with the stated property	Section 1.1
$[a, b]$, (a, b)	closed and open intervals	Section 1.1
$A \cup B$, $A \cap B$, A^c	union, intersection, complement	Section 9.1
$\emptyset$	the empty set	Section 9.1
$\lvert A \rvert$	number of elements of a finite set A	Section 9.2
$\mathbb{R}^n$	the set of vectors with n real components	Section 4.1
Arithmetic and algebra		
a^n, $\sqrt{a}$, $a^{1/n}$	powers and roots	Section 1.4
$\sum_{i=1}^{n} x_i$	the sum $x_1 + x_2 + \cdots + x_n$	Section 1.7
$\prod_{i=1}^{n} x_i$	the product $x_1 x_2 \cdots x_n$	Section 1.7

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Symbol	Meaning	Introduced in
$n!$	factorial, $1 \times 2 \times \cdots \times n$	Section 1.7
$\binom{n}{k}$	binomial coefficient, "n choose k"	Section 9.2
$\lvert x \rvert$	absolute value	Section 1.10
$\approx$	is approximately equal to	throughout
$\propto$	is proportional to	Section 10.7
Functions		
$f\colon A \to B$	a function from A to B	Section 2.1
$f \circ g$	composition, $(f \circ g)(x) = f(g(x))$	Section 2.7
f^{-1}	inverse function	Section 2.8
e, $e^x = \exp(x)$	Euler's number 2.718…, the exponential function	Section 2.5
$\ln x$, $\log_b x$	natural logarithm, logarithm to base b	Section 2.6
sin, cos, tan	trigonometric functions	Section 3.5
$\sigma(x)$	sigmoid function $\frac{1}{1+e^{-x}}$ (but see also σ below)	Section 2.8
$\mathrm{ReLU}(x)$, $\tanh(x)$	activation functions	Section 2.10
$\mathrm{softmax}(\boldsymbol{z})$	softmax function	Section 11.2
Linear algebra		
$\boldsymbol{x}$, x_i	a vector and its i-th component	Section 4.1
$\boldsymbol{A}$, A_{ij}	a matrix and its entry in row i, column j	Section 4.5
$\mathbf{0}$, $\mathbf{1}$	the vector of zeros, the vector of ones	Section 4.1
$\boldsymbol{I}$	identity matrix	Section 4.5
$\boldsymbol{x} \cdot \boldsymbol{y} = \boldsymbol{x}^\top \boldsymbol{y}$	dot product	Section 4.4
$\lVert \boldsymbol{x} \rVert$	length (Euclidean or L^2 norm)	Section 4.3
$\lVert \boldsymbol{x} \rVert_1$, $\lVert \boldsymbol{x} \rVert_p$, $\lVert \boldsymbol{x} \rVert_\infty$	L^1, L^p and L^∞ norms	Section 5.4
$\boldsymbol{A}^\top$	transpose	Section 4.9
$\boldsymbol{x}\boldsymbol{y}^\top$	outer product	Section 4.9
$\boldsymbol{A}^{-1}$	inverse matrix	Section 4.10
$\det \boldsymbol{A}$	determinant	Section 4.10
$\mathrm{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$	span: all linear combinations	Section 5.1
$\mathrm{rank}\, \boldsymbol{A}$	rank	Section 5.3
$\mathrm{proj}_{\boldsymbol{a}}(\boldsymbol{b})$	projection of $\boldsymbol{b}$ onto the line through $\boldsymbol{a}$	Section 5.6
$\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}$	eigenvalue λ and eigenvector $\boldsymbol{v}$	Section 5.7
$\mathrm{tr}\, \boldsymbol{A}$	trace, the sum of the diagonal entries	Section 5.7
$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$	singular value decomposition	Section 5.8
$\mathrm{diag}(d_1, \ldots, d_n)$	diagonal matrix with the given diagonal	Section 7.6
$\boldsymbol{u} \odot \boldsymbol{v}$	componentwise (Hadamard) product	Section 11.4

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Symbol	Meaning	Introduced in
Calculus		
$\lim_{x \to a} f(x)$	limit	Section 6.2
$f'(x)$, $\frac{\mathrm{d}f}{\mathrm{d}x}$	derivative	Section 6.3
$f''(x)$	second derivative	Section 6.5
$\int_a^b f(x)\,\mathrm{d}x$	definite integral	Section 6.7
$\frac{\partial f}{\partial x}$	partial derivative	Section 7.2
∇f, $\nabla_{\boldsymbol{x}} f$	gradient (vector of partial derivatives)	Section 7.3
$D_{\boldsymbol{u}} f$	directional derivative	Section 7.4
$\boldsymbol{J}$	Jacobian matrix	Section 7.6
$\boldsymbol{H}$	Hessian matrix	Section 7.7
$\frac{\partial L}{\partial \boldsymbol{W}}$	matrix of the derivatives $\frac{\partial L}{\partial W_{ij}}$	Section 11.4
Optimisation		
$\boldsymbol{\theta}$	the parameters of a model	Section 8.1
$L(\boldsymbol{\theta})$, ℓ	loss function, loss of one example	Section 8.1
$\arg\min_x f(x)$, $\arg\max_x f(x)$	the point where f is smallest, largest	Section 8.1
η	learning rate	Section 8.3
λ	regularisation strength; Lagrange multiplier	Sections 5.4 and 8.9
Probability		
Ω	sample space	Section 9.1
$P(A)$, $P(A \mid B)$	probability, conditional probability	Sections 9.2 and 9.3
X, $p(x)$, $F(x)$	random variable, its PMF or PDF, its CDF	Section 9.6
$\mathbb{E}[X]$, μ	expected value (mean)	Section 9.7
$\mathrm{Var}(X)$, σ^2, σ	variance, standard deviation	Section 9.8
$\mathrm{Cov}(X, Y)$, $\mathrm{Corr}(X, Y)$, ρ	covariance, correlation	Section 9.11
$\boldsymbol{\Sigma}$	covariance matrix	Section 9.11
$X \sim \mathcal{N}(\mu, \sigma^2)$	X has the normal distribution with mean μ, variance σ^2	Section 9.9
$\mathrm{Bin}(n, p)$	binomial distribution	Section 9.9
i.i.d.	independent and identically distributed	Section 9.4
Statistics		
$\bar{x}$, s^2	sample mean, sample variance	Section 10.1
$\hat{\theta}$	estimate of the parameter θ	Section 10.2
Bias, MSE, SE	bias, mean squared error, standard error	Section 10.2
$\mathcal{L}(\theta)$	likelihood	Section 10.5

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Symbol	Meaning	Introduced in
$\mathcal{D}$	a data set	Section 10.7
H_0, H_1	null and alternative hypotheses	Section 10.8
Information theory and deep learning		
$H(\boldsymbol{p})$	entropy	Section 11.3
$H(\boldsymbol{p}, \boldsymbol{q})$	cross-entropy	Section 11.3
$D_{\mathrm{KL}}(\boldsymbol{p} \,\|\, \boldsymbol{q})$	Kullback–Leibler divergence	Section 11.3
$\boldsymbol{W}^{(l)}$, $\boldsymbol{b}^{(l)}$	weights and biases of layer l	Section 11.1
$\boldsymbol{z}^{(l)}$, $\boldsymbol{a}^{(l)}$	pre-activations and activations of layer l	Section 11.1
φ	activation function	Section 11.1
$\boldsymbol{q}$, $\boldsymbol{y}$	predicted probabilities, one-hot label	Section 11.5
$\boldsymbol{\delta}^{(l)}$	error signal $\frac{\partial L}{\partial \boldsymbol{z}^{(l)}}$	Section 11.6
$\boldsymbol{Q}$, $\boldsymbol{K}$, $\boldsymbol{V}$	queries, keys and values in attention	Section 11.9

The Greek alphabet

Mathematics uses Greek letters as additional symbols. The table lists all of them, with their typical meanings in machine learning. Letters that look exactly like Latin ones (such as capital alpha, A) are not used as separate symbols.

Lower	Upper	Name	Typical uses
α	A	alpha	significance level; attention weights; step sizes
β	B	beta	momentum coefficients β_1, β_2; regression coefficients
γ	Γ	gamma	discount factor in reinforcement learning; the gamma function
δ	Δ	delta	a small change (Δx); error signals $\boldsymbol{\delta}$
ϵ, ε	E	epsilon	a small positive number; noise
ζ	Z	zeta	occasionally a variable
η	H	eta	learning rate
θ	Θ	theta	parameters of a model; angles
ι	I	iota	rarely used
κ	K	kappa	condition number
λ	Λ	lambda	eigenvalues ($\boldsymbol{\Lambda}$: diagonal matrix of them); regularisation strength; rates
μ	M	mu	mean
ν	N	nu	degrees of freedom
ξ	Ξ	xi	random variables; slack variables
o	O	omicron	not used (looks like the letter o)
π	Π	pi	the number 3.14159…; a policy; Π: product
ρ	P	rho	correlation
σ	Σ	sigma	standard deviation; sigmoid; Σ: sum, covariance matrix
τ	T	tau	temperature; thresholds; time
υ	Υ	upsilon	rarely used
ϕ, φ	Φ	phi	activation functions; feature maps; Φ: normal CDF
χ	X	chi	the χ^2 (chi-squared) distribution
ψ	Ψ	psi	feature maps; parameters
ω	Ω	omega	frequencies; Ω: sample space

Appendix B

Formula Sheet

The most important formulas of each chapter, collected for quick reference. Each heading names the chapter where the formulas are explained, with examples and conditions of validity. Notation is listed in Appendix A.

Numbers and algebra

Chapter 1

Powers, products and sums:

$$a^m a^n = a^{m+n}, \qquad (a^m)^n = a^{mn},$$
$$a^{-n} = \frac{1}{a^n}, \qquad a^{1/n} = \sqrt[n]{a},$$
$$(a+b)^2 = a^2 + 2ab + b^2,$$
$$(a+b)(a-b) = a^2 - b^2,$$
$$\sum_{i=1}^{n}(ax_i + b) = a\sum_{i=1}^{n} x_i + nb.$$

$|x-a| < r$ means $a-r < x < a+r$. Multiplying an inequality by a negative number reverses it.

Functions

Chapter 2

Slope of the line through (x_1, y_1) and (x_2, y_2): $m = \frac{y_2-y_1}{x_2-x_1}$. Roots of ax^2+bx+c and the vertex:

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}, \qquad x_{\text{vertex}} = -\frac{b}{2a}.$$

Exponentials and logarithms:

$$e^a e^b = e^{a+b}, \qquad \ln(ab) = \ln a + \ln b,$$
$$\ln(a^r) = r \ln a, \qquad \log_b x = \frac{\ln x}{\ln b}.$$

Sigmoid and its inverse, the logit:

$$\sigma(x) = \frac{1}{1+e^{-x}}, \qquad 1 - \sigma(x) = \sigma(-x),$$
$$\sigma^{-1}(p) = \ln \frac{p}{1-p}.$$

Geometry and trigonometry

Chapter 3

Distance: $d(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{\sum_i (x_i - y_i)^2}$. Angles: π radians $= 180°$.

$$\sin^2\theta + \cos^2\theta = 1, \qquad \tan\theta = \frac{\sin\theta}{\cos\theta},$$
$$c^2 = a^2 + b^2 - 2ab\cos C \quad \text{(law of cosines)},$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta,$$
$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta.$$

Rotation of (x, y) by the angle α:

$$x' = x\cos\alpha - y\sin\alpha, \qquad y' = x\sin\alpha + y\cos\alpha.$$

Vectors and matrices

Chapter 4

Dot products, norms and matrix algebra:

$$\boldsymbol{x}\cdot\boldsymbol{y} = \textstyle\sum_i x_i y_i = \|\boldsymbol{x}\|\|\boldsymbol{y}\|\cos\theta,$$
$$\|\boldsymbol{x}\| = \sqrt{\boldsymbol{x}\cdot\boldsymbol{x}}, \qquad |\boldsymbol{x}\cdot\boldsymbol{y}| \le \|\boldsymbol{x}\|\|\boldsymbol{y}\|,$$
$$\|\boldsymbol{x}+\boldsymbol{y}\| \le \|\boldsymbol{x}\| + \|\boldsymbol{y}\|,$$
$$(\boldsymbol{AB})_{ij} = \textstyle\sum_k A_{ik}B_{kj},$$
$$(\boldsymbol{AB})^\top = \boldsymbol{B}^\top\boldsymbol{A}^\top, \qquad (\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1},$$
$$\det(\boldsymbol{AB}) = \det\boldsymbol{A}\,\det\boldsymbol{B},$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Core linear algebra

Chapter 5

Norms, projections and decompositions:

$$\|\boldsymbol{x}\|_p = \Big(\sum_i |x_i|^p\Big)^{1/p}, \qquad \|\boldsymbol{x}\|_\infty = \max_i |x_i|,$$
$$\operatorname{proj}_{\boldsymbol{a}}(\boldsymbol{b}) = \frac{\boldsymbol{a}\cdot\boldsymbol{b}}{\boldsymbol{a}\cdot\boldsymbol{a}}\,\boldsymbol{a},$$
$$\boldsymbol{A}^\top\boldsymbol{A}\hat{\boldsymbol{x}} = \boldsymbol{A}^\top\boldsymbol{b} \quad \text{(least squares)},$$
$$\boldsymbol{Av} = \lambda\boldsymbol{v}, \qquad \det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0,$$
$$\textstyle\sum_i \lambda_i = \operatorname{tr}\boldsymbol{A}, \qquad \prod_i \lambda_i = \det\boldsymbol{A},$$
$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top \quad \text{(symmetric } \boldsymbol{A}\text{)},$$
$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top \quad \text{(SVD, any } \boldsymbol{A}\text{)}.$$

$\boldsymbol{A}$ is positive definite if $\boldsymbol{x}^\top\boldsymbol{Ax} > 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$; for symmetric $\boldsymbol{A}$ this means all eigenvalues are positive.

Calculus of one variable

Chapter 6

The derivative and its numerical estimate:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \approx \frac{f(x+h)-f(x-h)}{2h} \quad \text{(small } h\text{)}.$$

Rules, Taylor approximation and integrals:

$$(x^n)' = nx^{n-1}, \qquad (e^x)' = e^x,$$
$$(\ln x)' = \frac{1}{x}, \qquad (\sin x)' = \cos x,$$
$$(\cos x)' = -\sin x,$$
$$(fg)' = f'g + fg',$$
$$\Big(\frac{f}{g}\Big)' = \frac{f'g - fg'}{g^2},$$
$$\big(f(g(x))\big)' = f'(g(x))\,g'(x),$$
$$\sigma' = \sigma(1-\sigma),$$
$$f(a+h) \approx f(a) + f'(a)\,h + \tfrac{1}{2}f''(a)\,h^2,$$
$$\int_a^b f'(x)\,\mathrm{d}x = f(b) - f(a),$$
$$\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}.$$

Multivariable calculus

Chapter 7

Gradients, chain rule and approximation:

$$\nabla f = \Big(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\Big),$$
$$D_{\boldsymbol{u}} f = \nabla f \cdot \boldsymbol{u},$$
$$\frac{\mathrm{d}}{\mathrm{d}t} f(\boldsymbol{x}(t)) = \textstyle\sum_i \frac{\partial f}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}t},$$
$$\boldsymbol{J}_{\boldsymbol{f}\circ\boldsymbol{g}} = \boldsymbol{J}_{\boldsymbol{f}}\,\boldsymbol{J}_{\boldsymbol{g}},$$
$$f(\boldsymbol{x}+\boldsymbol{h}) \approx f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top \boldsymbol{h} + \tfrac{1}{2}\boldsymbol{h}^\top \boldsymbol{H}\boldsymbol{h},$$
$$\nabla(\boldsymbol{a}^\top \boldsymbol{x}) = \boldsymbol{a},$$
$$\nabla(\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}) = (\boldsymbol{A} + \boldsymbol{A}^\top)\boldsymbol{x},$$
$$\nabla\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|^2 = 2\boldsymbol{A}^\top(\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}).$$

At a critical point ($\nabla f = \boldsymbol{0}$): positive definite Hessian means a minimum, negative definite a maximum, mixed signs a saddle.

Optimisation

Chapter 8

Gradient descent: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla L(\boldsymbol{\theta})$.
SGD: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \frac{1}{b}\sum_{i\in B} \nabla \ell_i(\boldsymbol{\theta})$.
Momentum: $\boldsymbol{v} \leftarrow \beta \boldsymbol{v} + \boldsymbol{g}$, then $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \boldsymbol{v}$.
Newton: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \boldsymbol{H}^{-1}\nabla L(\boldsymbol{\theta})$.

Adam, with gradient $\boldsymbol{g}_t$ at step t (componentwise):

$$\boldsymbol{m}_t = \beta_1 \boldsymbol{m}_{t-1} + (1-\beta_1)\boldsymbol{g}_t,$$
$$\boldsymbol{s}_t = \beta_2 \boldsymbol{s}_{t-1} + (1-\beta_2)\boldsymbol{g}_t^2,$$
$$\hat{\boldsymbol{m}}_t = \frac{\boldsymbol{m}_t}{1-\beta_1^t}, \qquad \hat{\boldsymbol{s}}_t = \frac{\boldsymbol{s}_t}{1-\beta_2^t},$$
$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} - \eta\,\frac{\hat{\boldsymbol{m}}_t}{\sqrt{\hat{\boldsymbol{s}}_t} + \epsilon}.$$

Ridge regression: $(\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I})\boldsymbol{w} = \boldsymbol{X}^\top \boldsymbol{y}$.
Lagrange multipliers: $\nabla f = \lambda \nabla g$ and $g = 0$.
Convexity: for all $\lambda \in [0,1]$,

$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \le \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).$$

Probability

Chapter 9

Rules of probability, expectation and variance:

$$P(A^c) = 1 - P(A),$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)},$$
$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)} \quad \text{(Bayes)},$$
$$P(A) = \textstyle\sum_i P(A \mid B_i)\,P(B_i),$$
$$\mathbb{E}[X] = \textstyle\sum_x x\,p(x),$$
$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b,$$
$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$
$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X),$$
$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$
$$\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y},$$
$$\mathrm{Var}(X+Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y + 2\,\mathrm{Cov}(X,Y).$$

Binomial: $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$, mean np, variance $np(1-p)$. Normal: $p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$; 68%, 95% and 99.7% lie within 1, 2 and 3 standard deviations of the mean. Multivariate normal:

$$p(\boldsymbol{x}) = \frac{\exp\big(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\big)}{\sqrt{(2\pi)^d \det \boldsymbol{\Sigma}}}.$$

Statistics

Chapter 10

Estimation and inference:

$$\bar{x} = \tfrac{1}{n}\textstyle\sum_i x_i, \qquad s^2 = \tfrac{1}{n-1}\textstyle\sum_i (x_i - \bar{x})^2,$$

$$\mathrm{SE}(\bar{X}) = \sigma/\sqrt{n}, \qquad 95\%\ \text{CI: } \bar{x} \pm 1.96\, s/\sqrt{n},$$

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}),$$

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_\theta \textstyle\sum_i \ln p(x_i \mid \theta),$$

$$\hat{\theta}_{\mathrm{MAP}} = \arg\min_\theta \big(-\ln p(\mathcal{D} \mid \theta) - \ln p(\theta)\big).$$

Coin: $\hat{\theta} = h/n$. Normal: $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$. Gaussian noise gives squared error; Bernoulli labels give binary cross-entropy $-\sum_i [y_i \ln q_i + (1 - y_i)\ln(1 - q_i)]$. Expected test error = bias^2 + variance + noise.

Deep learning

Chapter 11

Layers, softmax and information theory:

$$\boldsymbol{z}^{(l)} = \boldsymbol{W}^{(l)}\boldsymbol{a}^{(l-1)} + \boldsymbol{b}^{(l)},$$

$$\boldsymbol{a}^{(l)} = \varphi(\boldsymbol{z}^{(l)}),$$

$$\mathrm{softmax}(\boldsymbol{z})_k = \frac{e^{z_k}}{\sum_j e^{z_j}},$$

$$L = -\ln q_c \implies \frac{\partial L}{\partial \boldsymbol{z}} = \boldsymbol{q} - \boldsymbol{y},$$

$$H(\boldsymbol{p}) = -\textstyle\sum_k p_k \log p_k,$$

$$H(\boldsymbol{p}, \boldsymbol{q}) = -\textstyle\sum_k p_k \log q_k,$$

$$D_{\mathrm{KL}}(\boldsymbol{p} \,\|\, \boldsymbol{q}) = \textstyle\sum_k p_k \log \frac{p_k}{q_k}$$

$$= H(\boldsymbol{p}, \boldsymbol{q}) - H(\boldsymbol{p}) \geq 0.$$

For $\boldsymbol{z} = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{b}$ with $\boldsymbol{\delta} = \frac{\partial L}{\partial \boldsymbol{z}}$: $\frac{\partial L}{\partial \boldsymbol{W}} = \boldsymbol{\delta}\boldsymbol{x}^\top$, $\frac{\partial L}{\partial \boldsymbol{x}} = \boldsymbol{W}^\top\boldsymbol{\delta}$, $\frac{\partial L}{\partial \boldsymbol{b}} = \boldsymbol{\delta}$. Backpropagation:

$$\boldsymbol{\delta}^{(l)} = \big(\boldsymbol{W}^{(l+1)\top}\boldsymbol{\delta}^{(l+1)}\big) \odot \varphi'(\boldsymbol{z}^{(l)}).$$

Attention: $\mathrm{softmax}\big(\boldsymbol{Q}\boldsymbol{K}^\top/\sqrt{d}\big)\boldsymbol{V}$.

Appendix C

Solutions to the Exercises

This appendix contains a solution or solution sketch for every exercise in the notes. Click on an exercise number to jump back to the exercise.

Try each exercise seriously before you look here. If your answer differs, do not just read the solution: find the step where your reasoning went a different way, and make sure you understand why. Many exercises can be solved in more than one way, so a correct answer reached by a different route is just as good.

Chapter 1: Numbers, Arithmetic, and Algebraic Thinking

Exercise 1.1. 7: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$. -3: $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$. 0: $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ (not $\mathbb{N}$ with our convention). $0.25 = \frac{1}{4}$: $\mathbb{Q}, \mathbb{R}$. $-\frac{2}{3}$: $\mathbb{Q}, \mathbb{R}$. $\sqrt{9} = 3$: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$. $\sqrt{5}$: only $\mathbb{R}$ (it is irrational). π: only $\mathbb{R}$.

Exercise 1.2. $[-2, 5)$ (filled dot at -2, open dot at 5); $(3, \infty)$ (open dot at 3, arrow to the right); $(-\infty, 0]$ (arrow to the left, filled dot at 0); $[0, 1]$ (filled dots at both ends). The last one is the set of all possible probabilities.

Exercise 1.3. $0.75 = \frac{75}{100} = \frac{3}{4}$. For the second, let $x = 0.272727\ldots$. Then $100x = 27.272727\ldots$, and subtracting, $100x - x = 27$, so $99x = 27$ and $x = \frac{27}{99} = \frac{3}{11}$. (Check: $3 \div 11 = 0.2727\ldots$.)

Exercise 1.4. There are 256 values (0 through 255; do not forget 0). The rescaled values lie in $[0, 1]$, since $0/255 = 0$ and $255/255 = 1$. A pixel value of 51 becomes $51/255 = 0.2$.

Exercise 1.5. (a) 5; (b) -5; (c) $-3 + 8 = 5$; (d) -24; (e) 35; (f) 9; (g) 0.

Exercise 1.6. $7 \times 98 = 7 \times (100 - 2) = 700 - 14 = 686$. $25 \times 44 = 25 \times (40 + 4) = 1000 + 100 = 1100$.

Exercise 1.7. Take $a = 16$, $b = 4$, $c = 2$: $(16 \div 4) \div 2 = 4 \div 2 = 2$, while $16 \div (4 \div 2) = 16 \div 2 = 8$.

Exercise 1.8. "$\frac{x}{x} = 1$" holds only for $x \neq 0$. If $\frac{0}{0}$ were a number q, we would need $0 \times q = 0$; but every number q satisfies this, so there is no single answer. For this reason $\frac{0}{0}$ is left undefined (it is called *indeterminate*).

Exercise 1.9. (a) $\frac{3}{4}$ (divide by 6); (b) $\frac{5}{7}$ (divide by 7); (c) $\frac{10}{3}$ (divide by 12).

Exercise 1.10. (a) $\frac{6}{15} + \frac{5}{15} = \frac{11}{15}$. (b) $\frac{21}{24} - \frac{4}{24} = \frac{17}{24}$. (c) $\frac{24}{36} = \frac{2}{3}$. (d) $\frac{5}{6} \times \frac{3}{10} = \frac{15}{60} = \frac{1}{4}$.

Exercise 1.11. (a) $0.15 \times 80 = 12$. (b) $\frac{45}{180} = 0.25 = 25\%$. (c) $\frac{50-40}{40} = 25\%$. (d) $\frac{40-50}{50} = -20\%$.

Exercise 1.12. (a) 4 percentage points. (b) $\frac{84-80}{80} = 5\%$. (c) The error rate falls from 20% to 16%, a relative reduction of $\frac{20-16}{20} = 20\%$.

Exercise 1.13. Spam is 1 part out of $3 + 1 = 4$, so there are $\frac{1}{4} \times 1000 = 250$ spam emails. The training set has $0.8 \times 1000 = 800$ emails, of which $\frac{1}{4} \times 800 = 200$ are spam.

Exercise 1.14. (a) $2^7 = 128$; (b) $5^2 = 25$; (c) $3^6 = 729$; (d) 1; (e) $\frac{1}{16}$; (f) 3; (g) $(\sqrt[3]{8})^2 = 4$; (h) $\frac{1}{(\sqrt[4]{16})^3} = \frac{1}{2^3} = \frac{1}{8}$.

Exercise 1.15. (a) $x^{3+5-2} = x^6$. (b) $2^3(x^2)^3 = 8x^6$. (c) x^2 (which is never negative, so no absolute value is needed). (d) $x^{-2}/x^{-3} = x^{-2-(-3)} = x^1 = x$.

Exercise 1.16. (a) False: $a = b = 1$ gives 4 versus 2. (b) True. (c) False: $a = 9$, $b = 16$ gives 5 versus 7. (d) True. (e) False: $a = b = 1$ gives $2^2 = 4$ versus $2 + 2 = 4$, which happens to agree, but $a = 1$, $b = 2$ gives $2^3 = 8$ versus $2 + 4 = 6$. (The correct rule is $2^{a+b} = 2^a \cdot 2^b$.)

Exercise 1.17. $0.00005 = 5 \times 10^{-5}$; $3\,400\,000 = 3.4 \times 10^6$; $(4 \times 2.5) \times 10^{3-5} = 10 \times 10^{-2} = 10^{-1} = 0.1$.

Exercise 1.18. $7 \times 10^9 \times 2 = 1.4 \times 10^{10}$ bytes, which is $\frac{1.4 \times 10^{10}}{10^9} = 14$ gigabytes.

Exercise 1.19. (a) $6 + 12 - 2 = 16$. (b) $10 \times 1 = 10$. (c) $8 \div 2 = 4$. (d) $50 - 9 = 41$. (e) $-9 + 9 = 0$. (f) $(8 - 6)^3 = 2^3 = 8$.

Exercise 1.20. (a) $(2+3) \times 4 - 1 = 19$. (b) $2 + 3 \times (4 - 1) = 11$. (c) $(2+3) \times (4-1) = 15$.

Exercise 1.21. Numerator $8 + 4 = 12$, denominator $6 - 6 + 3 = 3$, so the value is 4. And $\sqrt{9 + 16} = \sqrt{25} = 5$.

Exercise 1.22. At $x = 3$: $18 - 9 + 1 = 10$. At $x = -1$: $2 + 3 + 1 = 6$. At $x = \frac{1}{2}$: $\frac{1}{2} - \frac{3}{2} + 1 = 0$.

Exercise 1.23. (a) $3a + 8b$. (b) $2x - 6 - 3x - 3 = -x - 9$. (c) $x^2 + 2x - x^2 + 4x = 6x$.

Exercise 1.24. (a) $x^2 - 2x + 4x - 8 = x^2 + 2x - 8$. (b) $4x^2 - 12x + 9$. (c) $x^2 - 25$. (d) $a^3 + a^2b + ab^2 - a^2b - ab^2 - b^3 = a^3 - b^3$.

Exercise 1.25. (a) $6(2x + 3)$. (b) $(x + 3)(x + 4)$, since $3 + 4 = 7$ and $3 \times 4 = 12$. (c) $(x - 3)(x + 3)$. (d) $(x - 3)(x + 2)$, since $-3 + 2 = -1$ and $(-3) \times 2 = -6$.

Exercise 1.26. $3000 \times 80 + 50\,000 = 240\,000 + 50\,000 = 290\,000$. The error is $300\,000 - 290\,000 = 10\,000$: the model under-predicts by 10 000.

Exercise 1.27. (a) $1 + 3 + 5 + 7 + 9 = 25$. (b) $1 + 4 + 9 + 16 = 30$. (c) There are four terms ($j = 0, 1, 2, 3$), each equal to 3, so the sum is 12. (d) $5! = 120$.

Exercise 1.28. (a) $\sum_{i=1}^{100} i$. (b) $\sum_{i=1}^{n} x_i^2$. (c) $\sum_{i=1}^{3} w_i x_i$. (This last kind of sum, a *weighted sum*, is what a single artificial neuron computes.)

Exercise 1.29. $\sum_i x_i = 12$, $\sum_i y_i = 6$, $\sum_i x_i y_i = 3+0+2+10 = 15$, and $(\sum_i x_i)(\sum_i y_i) = 72$. The last two differ, confirming that a sum of products is not a product of sums.

Exercise 1.30. Split the sum and use $\sum_{i=1}^{n} c = nc$ with the constant $c = \bar{x}$:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n} x_i - n\bar{x} = n\bar{x} - n\bar{x} = 0,$$

because $\sum_{i=1}^{n} x_i = n\bar{x}$ by the definition of the mean.

Exercise 1.31. Errors $y_i - \hat{y}_i$: $-1, 0, 2$. Squares: $1, 0, 4$, so MSE $= \frac{5}{3} \approx 1.67$. Absolute errors: $1, 0, 2$, so the mean absolute error is $\frac{3}{3} = 1$.

Exercise 1.32. (a) $4x = 20$, $x = 5$. (b) $2x + 6 = 5x - 3$, so $9 = 3x$ and $x = 3$; check: $12 = 12$. (c) $\frac{x}{3} = 6$, $x = 18$. (d) Multiply by 15: $3(2x - 1) = 5(x + 2)$, so $6x - 3 = 5x + 10$ and $x = 13$; check: $\frac{25}{5} = 5 = \frac{15}{3}$.

Exercise 1.33. (a) $t = \frac{v-u}{a}$ (assuming $a \neq 0$). (b) $h = \frac{2A}{b}$ (assuming $b \neq 0$). (c) Multiply by σ: $z\sigma = x - \mu$, so $x = \mu + \sigma z$.

Exercise 1.34. (a) Adding: $2x = 10$, so $x = 5$, $y = 2$. (b) Adding: $4x = 16$, so $x = 4$ and then $y = 2$. (c) The second equation is twice the first, so they carry the same information: every pair with $y = 5 - 2x$ is a solution (infinitely many solutions). (d) The two equations contradict each other ($1 \neq 3$), so there is no solution.

Exercise 1.35. $2w + b = 5$ and $4w + b = 9$. Subtracting: $2w = 4$, so $w = 2$ and $b = 5 - 4 = 1$. The line is $y = 2x + 1$.

Exercise 1.36. Let a and c be the numbers of adult and child tickets: $a + c = 20$ and $12a + 7c = 190$. Substituting $c = 20 - a$: $12a + 140 - 7a = 190$, so $5a = 50$ and $a = 10$ (and $c = 10$).

Exercise 1.37. (a) $2x < 6$, so $x < 3$: $(-\infty, 3)$. (b) $-3x \leq 6$; dividing by -3 flips the sign: $x \geq -2$: $[-2, \infty)$. (c) Add 1: $-2 \leq 2x < 8$; divide by 2: $-1 \leq x < 4$: $[-1, 4)$.

Exercise 1.38. (a) 5; (b) -2; (c) 0; (d) 2.1; (e) 4.

Exercise 1.39. $\text{ReLU}(x) = x$ exactly when $x \geq 0$; $\text{ReLU}(x) = 0$ exactly when $x \leq 0$ (both hold at $x = 0$).

Exercise 1.40. $\text{clip}(-3) = \min(-1, 1) = -1$; $\text{clip}(0.5) = \min(0.5, 1) = 0.5$; $\text{clip}(2) = \min(2, 1) = 1$. It leaves numbers in $[-1, 1]$ unchanged and replaces anything outside by the nearest endpoint.

Exercise 1.41. $-6 < -16$ is false: -6 lies to the right of -16 on the number line. Multiplying by a negative number reverses the inequality, so the correct statement is $-6 > -16$.

Exercise 1.42. 8; $|-7| = 7$; $2 \cdot 5 = 10$; $|5| = 5$; -6.

Exercise 1.43. (a) $x + 2 = 5$ or $x + 2 = -5$, so $x = 3$ or $x = -7$. (b) $-3 \leq x - 1 \leq 3$, so $-2 \leq x \leq 4$: $[-2, 4]$. (c) $2x - 4 > 6$ or $2x - 4 < -6$, so $x > 5$ or $x < -1$.

Exercise 1.44. $|-3.5 - 2| = |-5.5| = 5.5$.

Exercise 1.45. $|4 + (-7)| = 3 \leq |4| + |-7| = 11$. Equality holds exactly when a and b do not have opposite signs, i.e. when both are ≥ 0 or both are ≤ 0.

Exercise 1.46. Errors: $-1, 3, 0$. MAE $= \frac{1+3+0}{3} = \frac{4}{3} \approx 1.33$. MSE $= \frac{1+9+0}{3} = \frac{10}{3} \approx 3.33$. The second data point contributes 9 of the 10 units of squared error: squaring makes the largest error dominate.

Chapter 2: Functions and Graphs

Exercise 2.1. (a) Yes. (b) No: for $x > 0$ there are two such y, and for $x < 0$ there is none. (c) Not on all of $\mathbb{R}$, because $x = 0$ has no output; it is a function on $\{x \in \mathbb{R} : x \neq 0\}$. (d) Yes.

Exercise 2.2. $f(0) = 0$, $f(2) = 4 - 6 = -2$, $f(-1) = 1 + 3 = 4$, $f(a) = a^2 - 3a$, and $f(a+1) = (a+1)^2 - 3(a+1) = a^2 + 2a + 1 - 3a - 3 = a^2 - a - 2$.

Exercise 2.3. (a) Need $x - 2 \geq 0$: $[2, \infty)$. (b) Need $x \neq -3$. (c) Need $4 - x^2 \geq 0$, i.e. $x^2 \leq 4$, i.e. $|x| \leq 2$: $[-2, 2]$.

Exercise 2.4. $x^2 \geq 0$, so $f(x) \geq 1$, and every value ≥ 1 occurs: the range is $[1, \infty)$. Similarly $-|x| \leq 0$ and every value ≤ 0 occurs: the range of g is $(-\infty, 0]$.

Exercise 2.5. The domain is the set of all possible emails; the codomain is the two-element set $\{\text{spam}, \text{not spam}\}$. If it were not a function, the same email could receive different verdicts, so its behaviour could not be predicted, tested or relied upon.

Exercise 2.6. Values: $f(\pm 3) = -5$, $f(\pm 2) = 0$, $f(\pm 1) = 3$, $f(0) = 4$. The graph is an upside-down U. Roots at $x = -2$ and $x = 2$; y-intercept 4; maximum value 4 at $x = 0$.

Exercise 2.7. The graph is above the x-axis for $x < -1$ and for $x > 3$. It is increasing on $[1, \infty)$ (and decreasing on $(-\infty, 1]$).

Exercise 2.8. No: the vertical line $x = 0$ meets it twice, at $(0, 1)$ and $(0, -1)$. The upper half is the graph of $f(x) = \sqrt{1 - x^2}$ and the lower half is the graph of $g(x) = -\sqrt{1 - x^2}$, both with domain $[-1, 1]$.

Exercise 2.9. $m = \frac{-4-2}{3-1} = -3$. Then $2 = -3 \cdot 1 + b$ gives $b = 5$; the line is $y = -3x + 5$.

Exercise 2.10. $y - 1 = -2(x - 3)$, so $y = -2x + 7$.

Exercise 2.11. (a) By $1.8 \times 10 = 18$ degrees Fahrenheit. (b) Solve $C = 1.8C + 32$: $-0.8C = 32$, so $C = -40$. Indeed $-40\,^\circ\text{C} = -40\,^\circ\text{F}$.

Exercise 2.12. Each extra square metre adds 2500 to the predicted price (the slope); the intercept 40 000 is the prediction for a house of size 0. A size increase of 12 raises the prediction by $12 \times 2500 = 30\,000$. The intercept is not meaningful on its own (there are no houses of size 0); it just positions the line so that it fits the data in the realistic range.

Exercise 2.13. Yes, both have slope 2 (and different intercepts, so they never meet). For the second pair, $2x + 1 = -x + 4$ gives $x = 1$ and $y = 3$: they meet at $(1, 3)$.

Exercise 2.14. (a) Degree 3, leading coefficient 4. (b) Expanding gives $x^2 - x - 2$: degree 2, leading coefficient 1. (c) Degree 0, leading coefficient 5.

Exercise 2.15. (a) $(x-2)(x-3) = 0$: $x = 2$ or 3. (b) $(x-4)(x+4) = 0$: $x = \pm 4$. (c) $x(x+3) = 0$: $x = 0$ or -3.

Exercise 2.16. (a) $D = 1 - 4 = -3 < 0$: none. (b) $D = 36 - 36 = 0$: one, $x = 3$. (c) $D = 9 + 16 = 25$: two, $x = \frac{3 \pm 5}{4}$, i.e. $x = 2$ or $x = -\frac{1}{2}$. (d) $D = 16 - 4 = 12$: two, $x = \frac{-4 \pm \sqrt{12}}{2} = -2 \pm \sqrt{3}$, i.e. about -0.268 and -3.732.

Exercise 2.17. (a) $x^2+4x+7 = (x+2)^2-4+7 = (x+2)^2+3$: vertex $(-2,3)$, minimum value 3. (b) $-x^2+6x-5 = -(x^2-6x)-5 = -[(x-3)^2-9]-5 = -(x-3)^2+4$: vertex $(3,4)$, maximum value 4.

Exercise 2.18. $L(c) = (1-c)^2+(3-c)^2+(8-c)^2 = 3c^2-24c+74$. The vertex is at $c = \frac{24}{6} = 4$, which is the mean $\frac{1+3+8}{3} = 4$. The minimum value is $L(4) = 9+1+16 = 26$.

Exercise 2.19. 32, $\frac{1}{8}$, 1, $3^2 = 9$, 2, 1.

Exercise 2.20. $N(t) = 500 \cdot 3^t$, and $N(4) = 500 \cdot 81 = 40\,500$.

Exercise 2.21. $A(t) = 160 \cdot \left(\frac{1}{2}\right)^{t/8}$. After 24 days (three half-lives), $A = 160 \cdot \frac{1}{8} = 20$ grams.

Exercise 2.22. At $x = 10$: $x^3 = 1000$ and $2^x = 1024$, nearly equal. At $x = 20$: $x^3 = 8000$ but $2^x = 1\,048\,576$, more than a hundred times larger.

Exercise 2.23. $0.5^{50} \approx 8.9 \times 10^{-16}$: the signal has vanished. And $1.5^{50} \approx 6.4 \times 10^{8}$: the signal has exploded. Factors only slightly different from 1 have enormous effects over many layers.

Exercise 2.24. 5, -3, 4, 5, 0, -3.

Exercise 2.25. (a) $2\ln x + \ln y - \ln z$. (b) $\ln \frac{8}{2} = \ln 4 = \ln 2^2 = 2\ln 2$.

Exercise 2.26. $x = \frac{\ln 50}{\ln 3} \approx \frac{3.912}{1.099} \approx 3.561$. And $e^x = 8$, so $x = \ln 8 \approx 2.079$.

Exercise 2.27. We need $0.9^t < 0.1$. Taking logarithms, $t \ln 0.9 < \ln 0.1$; since $\ln 0.9 < 0$, dividing by it flips the inequality: $t > \frac{\ln 0.1}{\ln 0.9} \approx 21.85$. So the first such epoch is $t = 22$. (Check: $0.1 \times 0.9^{21} \approx 0.0109$ and $0.1 \times 0.9^{22} \approx 0.0098$.)

Exercise 2.28. $\ln(0.05^{1000}) = 1000 \ln 0.05 \approx 1000 \times (-2.996) = -2996$. Converting to base 10: $\log_{10}(0.05^{1000}) = \frac{-2996}{\ln 10} \approx -1301$, so the product is about 10^{-1301}, which underflows to 0 on a computer. The log-probability -2996 is harmless.

Exercise 2.29. $f(g(2)) = f(4) = 7$ and $g(f(2)) = g(5) = 25$. In general $f(g(x)) = x^2+3$ and $g(f(x)) = (x+3)^2$.

Exercise 2.30. (a) $h = f \circ g$ with $g(x) = 3x+1$ and $f(u) = \sqrt{u}$. (b) One possible chain: $x \mapsto -x \mapsto e^{-x} \mapsto 1+e^{-x} \mapsto \frac{1}{1+e^{-x}}$, i.e. $k = f_4 \circ f_3 \circ f_2 \circ f_1$ with $f_1(x) = -x$, $f_2(u) = e^u$, $f_3(v) = 1+v$, $f_4(w) = 1/w$. (This is the sigmoid function of Section 2.10.)

Exercise 2.31. (a) $y = -3(2x-1)+4 = -6x+7$, a linear function. (b) $x = 0$: $h = \mathrm{ReLU}(-1) = 0$, $y = 4$. $x = 1$: $h = 1$, $y = 1$. $x = 2$: $h = 3$, $y = -5$. The output changes by -3 from $x = 0$ to $x = 1$ but by -6 from $x = 1$ to $x = 2$; a linear function changes by the same amount for equal steps, so this is not linear. The ReLU has made the network non-linear.

Exercise 2.32. (a) $y = 5x-2 \Rightarrow x = \frac{y+2}{5}$, so $f^{-1}(y) = \frac{y+2}{5}$. Check: $f^{-1}(f(x)) = \frac{5x-2+2}{5} = x$. (b) $g^{-1}(y) = 4y+3$. (c) $y = x^3+1 \Rightarrow x = \sqrt[3]{y-1}$, so $h^{-1}(y) = \sqrt[3]{y-1}$.

Exercise 2.33. Not on $\mathbb{R}$, since $f(1) = f(-1) = 2$. On $[0,\infty)$ it is one-to-one: from $y = x^2+1$ with $x \geq 0$ we get $x = \sqrt{y-1}$. So $f^{-1}(y) = \sqrt{y-1}$, defined for $y \geq 1$ (the range of f).

Exercise 2.34. $\sigma(0) = \frac{1}{1+1} = 0.5$; $\text{logit}(0.5) = \ln 1 = 0$; $\text{logit}(0.9) = \ln \frac{0.9}{0.1} = \ln 9 \approx 2.197$. Then $\sigma(\ln 9) = \frac{1}{1+e^{-\ln 9}} = \frac{1}{1+1/9} = \frac{9}{10} = 0.9$.

Exercise 2.35. Solving for x: $x = \mu + sz = 50 + 10z$. For $z = 1.3$, $x = 63$.

Exercise 2.36. Shift x^2 one unit to the left and four units down. The vertex is at $(-1, -4)$. Roots: $(x+1)^2 = 4$, so $x + 1 = \pm 2$, i.e. $x = 1$ or $x = -3$.

Exercise 2.37. $g(x) = 2|x - 3|$. Its graph is a V with its tip at $(3, 0)$ and slopes ± 2.

Exercise 2.38. The output is 0.5 when $4x - 2 = 0$, i.e. $x = 0.5$. It is four times as steep as $\sigma(x)$. With weight -4, the function $\sigma(-4x - 2)$ decreases instead of increasing, and its midpoint moves to $x = -0.5$.

Exercise 2.39. e^{x-2}: shifted 2 to the right. e^{-x}: reflected in the y-axis (a decaying exponential). $-e^x$: reflected in the x-axis (always negative). $e^x - 1$: shifted down by 1, so it passes through the origin and approaches -1 for very negative x.

Exercise 2.40. $\sigma(0) = \frac{1}{2}$; $\tanh(0) = \frac{1-1}{1+1} = 0$; $\text{ReLU}(0) = 0$; $\text{softplus}(0) = \ln(1 + 1) = \ln 2 \approx 0.693$.

Exercise 2.41. $1 - \sigma(x) = 1 - \frac{1}{1+e^{-x}} = \frac{e^{-x}}{1+e^{-x}}$. Multiplying top and bottom by e^x gives $\frac{1}{e^x+1} = \sigma(-x)$. Interpretation: if the logit z gives probability $\sigma(z)$ for "yes", then the probability of "no", $1 - \sigma(z)$, is obtained by flipping the sign of the logit.

Exercise 2.42. $2\sigma(2x) - 1 = \frac{2}{1+e^{-2x}} - 1 = \frac{1-e^{-2x}}{1+e^{-2x}}$. Multiplying numerator and denominator by e^x gives $\frac{e^x - e^{-x}}{e^x + e^{-x}} = \tanh(x)$.

Exercise 2.43. Leaky ReLU: $\max(-0.05, -5) = -0.05$, 0, and 3. $\text{softplus}(5) = \ln(1 + e^5) \approx 5.0067$ and $\text{softplus}(-5) = \ln(1 + e^{-5}) \approx 0.0067$, very close to $\text{ReLU}(5) = 5$ and $\text{ReLU}(-5) = 0$. Softplus differs noticeably from ReLU only near 0.

Chapter 3: Geometry, Coordinates and Trigonometry

Exercise 3.1. $(4, -1)$: quadrant IV. $(-3, -5)$: quadrant III. $(0, 2)$: on the y-axis. $(-1, 6)$: quadrant II.

Exercise 3.2. Each grayscale image has $32 \times 32 = 1024$ numbers, so it is a point in $\mathbb{R}^{1024}$. A colour image has $32 \times 32 \times 3 = 3072$ numbers, so it lives in $\mathbb{R}^{3072}$. (The number of images, 1000, is the number of points, not the dimension.)

Exercise 3.3. (a) A vertical line crossing the x-axis at 2. (b) The upper half of the plane, not including the x-axis. (c) The horizontal plane containing the x- and y-axes (the "floor").

Exercise 3.4. $d = \sqrt{(4+2)^2 + (-5-3)^2} = \sqrt{36 + 64} = 10$. Midpoint: $\left(\frac{-2+4}{2}, \frac{3-5}{2}\right) = (1, -1)$.

Exercise 3.5. $\sqrt{2^2 + 4^2 + (-4)^2} = \sqrt{4 + 16 + 16} = \sqrt{36} = 6$.

Exercise 3.6. The differences are $(-1, 2, 0, -3)$. Euclidean: $\sqrt{1 + 4 + 0 + 9} = \sqrt{14} \approx 3.74$. Manhattan: $1 + 2 + 0 + 3 = 6$.

Exercise 3.7. Squared side lengths: between $(1,1)$ and $(4,2)$: $9+1=10$; between $(4,2)$ and $(3,5)$: $1+9=10$; between $(1,1)$ and $(3,5)$: $4+16=20$. Since $10+10=20$, it is a right triangle, with the right angle at $(4,2)$ (opposite the longest side). It is also isosceles.

Exercise 3.8. Distances from $(3.5,3)$: to P_1: $\sqrt{6.25+4}\approx 3.20$; to P_2: $\sqrt{2.25+4}=2.5$; to P_3: $\sqrt{0.25+0}=0.5$; to P_4: $\sqrt{2.25+1}\approx 1.80$; to P_5: $\sqrt{0.25+2.25}\approx 1.58$. With $k=1$ the nearest point is P_3, so the answer is B. With $k=3$ the nearest are P_3, P_5, P_4, all labelled B, so again B.

Exercise 3.9. $3y=-2x+6$, so $y=-\frac{2}{3}x+2$. It crosses the y-axis at $(0,2)$ and the x-axis where $y=0$, i.e. at $(3,0)$.

Exercise 3.10. $(1,1)$: $2-1+1=2>0$, positive side. $(0,3)$: $0-3+1=-2<0$, negative side. $(1,3)$: $2-3+1=0$, on the line.

Exercise 3.11. $(1,0)$: $2-0-1=1>0$, class 1. $(0,1)$: $0-1-1=-2<0$, class 0. $(2,3)$: $4-3-1=0$: exactly on the boundary (undecided). The boundary $2x_1-x_2-1=0$ is $x_2=2x_1-1$.

Exercise 3.12. $(x+1)^2-1+(y-4)^2-16+8=0$, so $(x+1)^2+(y-4)^2=9$: centre $(-1,4)$, radius 3. For $(1,2)$: $(2)^2+(-2)^2=8<9$, so it is inside.

Exercise 3.13. s is the squared distance from the origin. All blue points lie within distance 1.9 of the origin, so they have $s<1.9^2=3.61$, while all red points lie further out, so they have $s>3.61$. Hence the rule "A if $s<3.61$, B otherwise" classifies every point correctly. The non-linear feature s has made the problem linearly separable.

Exercise 3.14. $\frac{\pi}{4}$; $\frac{2\pi}{3}$; $\frac{3\pi}{2}$; $\frac{\pi}{180}\approx 0.01745$.

Exercise 3.15. $30°$; $225°$; $2\times\frac{180°}{\pi}\approx 114.6°$.

Exercise 3.16. Twenty minutes is a third of a full turn: $\frac{2\pi}{3}$ radians. Clock hands turn clockwise, so with the mathematical convention the angle is $-\frac{2\pi}{3}$.

Exercise 3.17. $s=r\theta=5\times 0.8=4$. And $\theta=\frac{s}{r}=\frac{\pi}{2}$ (a right angle).

Exercise 3.18. Opposite: $10\sin 30°=10\times\frac{1}{2}=5$. Adjacent: $10\cos 30°=10\times\frac{\sqrt{3}}{2}=5\sqrt{3}\approx 8.66$.

Exercise 3.19. The tree (opposite) and the shadow (adjacent) form a right triangle with angle $35°$ at the tip of the shadow, so height $=12\tan 35°\approx 12\times 0.700=8.4\,\text{m}$.

Exercise 3.20. $\sin 150°=\frac{1}{2}$; $\cos 150°=-\frac{\sqrt{3}}{2}$; $\sin 225°=-\frac{\sqrt{2}}{2}$; $\cos 300°=\frac{1}{2}$; $\tan 45°=1$; $\sin(-30°)=-\frac{1}{2}$.

Exercise 3.21. By the Pythagorean identity, $\cos^2\theta=1-0.36=0.64$, and cosine is positive in the first quadrant, so $\cos\theta=0.8$. Then $\tan\theta=\frac{0.6}{0.8}=0.75$. (This is the 3–4–5 triangle scaled down.)

Exercise 3.22. $c^2=25+49-2\cdot 5\cdot 7\cos 60°=74-70\times\frac{1}{2}=39$, so $c=\sqrt{39}\approx 6.24$.

Exercise 3.23. $y=3\sin(2x)$: amplitude 3, period $\frac{2\pi}{2}=\pi$. $y=\cos(x/2)$: amplitude 1, period $\frac{2\pi}{1/2}=4\pi$.

Exercise 3.24. $(1,1)$: $r=\sqrt{2}$, $\theta=\frac{\pi}{4}$. $(0,-3)$: $r=3$, $\theta=\frac{3\pi}{2}$ (equivalently $-\frac{\pi}{2}$). $(-2,0)$: $r=2$, $\theta=\pi$.

Exercise 3.25. $x=4\cos\frac{3\pi}{4}=-2\sqrt{2}\approx-2.83$, $y=4\sin\frac{3\pi}{4}=2\sqrt{2}\approx 2.83$. And $(\cos\pi,\sin\pi)=(-1,0)$.

Exercise 3.26. $x'=3\cos 30^\circ-4\sin 30^\circ\approx 2.598-2=0.598$; $y'=3\sin 30^\circ+4\cos 30^\circ\approx 1.5+3.464=4.964$. Then $\sqrt{0.598^2+4.964^2}\approx\sqrt{25.0}=5=\sqrt{3^2+4^2}$.

Exercise 3.27. In polar coordinates a rotation by α adds α to the angle and leaves r unchanged. Doing it twice adds α and then β, i.e. $\alpha+\beta$ in total. Rotating by α and then $-\alpha$ adds 0: every point returns to where it started, so the second rotation is the inverse of the first.

Exercise 3.28. The two points are

$$(\cos 10^\circ,\sin 10^\circ)\approx(0.985,0.174)\quad\text{and}\quad(\cos 30^\circ,\sin 30^\circ)\approx(0.866,0.5).$$

The angle between them is $30^\circ-10^\circ=20^\circ$. For positions 11 and 13 the angles are 110° and 130°, and the angle between them is again 20°: it depends only on the difference of the positions.

Chapter 4: Linear Algebra I: Vectors and Matrices

Exercise 4.1. $\boldsymbol{x}=(120,3,15,2.5)$, written as a column; it lives in $\mathbb{R}^4$.

Exercise 4.2. Red $=(1,0,0)$, green $=(0,1,0)$, blue $=(0,0,1)$. The entries of a one-hot vector always add up to 1.

Exercise 4.3. The displacement is $Q-P=(5-1,1-4)=(4,-3)$. Its length is $\sqrt{16+9}=5$.

Exercise 4.4. $\boldsymbol{u}+\boldsymbol{v}=(1,3,3)$; $\boldsymbol{u}-\boldsymbol{v}=(3,-5,3)$; $3\boldsymbol{u}=(6,-3,9)$; $2\boldsymbol{u}-3\boldsymbol{v}=(4,-2,6)-(-3,12,0)=(7,-14,6)$.

Exercise 4.5. Component by component: $a+b=5$ and $a-b=1$. Adding gives $2a=6$, so $a=3$ and $b=2$. Check: $3(1,1)+2(1,-1)=(5,1)$.

Exercise 4.6. $\frac{1}{3}((1,2)+(3,6)+(5,1))=\frac{1}{3}(9,9)=(3,3)$.

Exercise 4.7. $\eta\boldsymbol{g}=(0.04,-0.06,0.1)$, so the new weights are $(0.5-0.04,-1.2+0.06,2.0-0.1)=(0.46,-1.14,1.9)$.

Exercise 4.8. 5; 3; $\sqrt{4}=2$.

Exercise 4.9. $\|(6,-8)\|=\sqrt{36+64}=10$, so the unit vector is $(0.6,-0.8)$.

Exercise 4.10. $\|(1,2,3)-(4,6,3)\|=\|(-3,-4,0)\|=5$.

Exercise 4.11. $\|\boldsymbol{u}+\boldsymbol{v}\|=\|(1,1)\|=\sqrt{2}\approx 1.41\le 1+1=2$. Equality holds for vectors pointing the same way, e.g. $\boldsymbol{u}=(1,0)$, $\boldsymbol{v}=(2,0)$: $\|(3,0)\|=3=1+2$.

Exercise 4.12. $8+0-6=2$ (not orthogonal) and $1-1+1-1=0$ (orthogonal).

Exercise 4.13. (a) $\cos\theta = \frac{-1}{\sqrt{2}\cdot 1} = -\frac{1}{\sqrt{2}}$, so $\theta = 135°$. (b) The dot product is $2-4+2=0$, so $\theta = 90°$.

Exercise 4.14. $3k - 12 = 0$, so $k = 4$.

Exercise 4.15. $\text{dog}\cdot\text{car} = 0.08 + 0.18 + 0.18 = 0.44$, and $\|\text{dog}\| \approx 1.221$, $\|\text{car}\| \approx 0.927$, so the similarity is $\frac{0.44}{1.132} \approx 0.39$. The least similar pair is cat and car (≈ 0.30), and the most similar is cat and dog (≈ 0.99).

Exercise 4.16. $\boldsymbol{w} = (3,4)$ is perpendicular to the line (Proposition 4.12). Substituting $(3t, 4t)$ into the equation: $9t + 16t = 12$, so $t = 0.48$ and the closest point is $(1.44, 1.92)$. Its distance from the origin is $0.48 \times \|(3,4)\| = 0.48 \times 5 = 2.4$. (In general the distance from the origin to $\boldsymbol{w}\cdot\boldsymbol{x} + b = 0$ is $|b|/\|\boldsymbol{w}\|$.)

Exercise 4.17. $z = \boldsymbol{w}\cdot\boldsymbol{x} + b = 1 - 1 + 1 + 0.1 = 1.1$, and the output is $\sigma(1.1) = \frac{1}{1+e^{-1.1}} \approx \frac{1}{1.333} \approx 0.750$.

Exercise 4.18. 2×3; $a_{21} = 2$; $a_{13} = -1$; first row $(3, 0, -1)$; second column $(0, 5)$.

Exercise 4.19. The first and the third (every diagonal matrix is symmetric). The second is not: $a_{12} = 1$ but $a_{21} = -1$.

Exercise 4.20. $\left(\begin{smallmatrix} 2 & 4 \\ 6 & 8 \end{smallmatrix}\right) - \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2 & 3 \\ 5 & 8 \end{smallmatrix}\right)$.

Exercise 4.21. Each image has $28 \times 28 = 784$ pixels, so the data matrix is 500×784.

Exercise 4.22. $(1 + 4 + 0,\ 0 + 2 - 3) = (5, -1)$.

Exercise 4.23. Rows: $(2 - 1,\ 1 - 3) = (1, -2)$. Columns: $1\cdot(2,1) + (-1)\cdot(1,3) = (2-1, 1-3) = (1, -2)$.

Exercise 4.24. $\boldsymbol{Ay}$ is defined (the number of columns of $\boldsymbol{A}$, 2, matches the length of $\boldsymbol{y}$) and lies in $\mathbb{R}^3$. $\boldsymbol{Ax}$ is not defined.

Exercise 4.25. $\boldsymbol{Xw} = (2-2,\ 6-0,\ -2-1) = (0, 6, -3)$; adding b gives $\hat{\boldsymbol{y}} = (0.5, 6.5, -2.5)$.

Exercise 4.26. (a) $\boldsymbol{e}_1 \mapsto (0,1)$ and $\boldsymbol{e}_2 \mapsto (1,0)$, so $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, which swaps the two coordinates. (b) $2\boldsymbol{I} = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right)$. (c) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$, i.e. $-\boldsymbol{I}$; this also equals $\boldsymbol{R}_{180°}$.

Exercise 4.27. T is not linear, since $T(\mathbf{0}) = (1,0) \neq \mathbf{0}$ (it is a translation). S is linear, with matrix $\left(\begin{smallmatrix} 0 & 1 \\ 3 & 0 \end{smallmatrix}\right)$ (columns $S(\boldsymbol{e}_1) = (0,3)$ and $S(\boldsymbol{e}_2) = (1,0)$).

Exercise 4.28. $\boldsymbol{A} = \left(\begin{smallmatrix} 2 & -1 \\ 1 & 3 \end{smallmatrix}\right)$ and $\boldsymbol{A}(4,2) = 4(2,1) + 2(-1,3) = (6, 10)$.

Exercise 4.29. $\boldsymbol{AB} = \left(\begin{smallmatrix} 3 & 1 \\ 6 & 4 \end{smallmatrix}\right)$ and $\boldsymbol{BA} = \left(\begin{smallmatrix} 5 & 1 \\ 4 & 2 \end{smallmatrix}\right)$.

Exercise 4.30. $\boldsymbol{AB}$: 2×4. $\boldsymbol{BC}$: 3×2. $\boldsymbol{CA}$: 4×3. $\boldsymbol{AC}$: not defined ($3 \neq 4$). $\boldsymbol{ABC}$: 2×2.

Exercise 4.31. $\boldsymbol{W}_2\boldsymbol{W}_1 = \begin{pmatrix} 1\cdot 1 + (-1)\cdot 0 & 1\cdot 2 + (-1)\cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \end{pmatrix}$. Step by step: $\boldsymbol{W}_1\boldsymbol{x} = (3,1)$ and $\boldsymbol{W}_2(3,1) = 3 - 1 = 2$. In one go: $(1,1)\cdot(1,1) = 2$. They agree.

Exercise 4.32.

$$\boldsymbol{R}_\alpha \boldsymbol{R}_\beta = \begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & -\cos\alpha\sin\beta - \sin\alpha\cos\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta & -\sin\alpha\sin\beta + \cos\alpha\cos\beta \end{pmatrix},$$

while $\boldsymbol{R}_{\alpha+\beta}$ has first column $(\cos(\alpha+\beta), \sin(\alpha+\beta))$. Comparing the first columns gives $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$ and $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$.

Exercise 4.33. Row by column: $\left(\begin{smallmatrix} 1\cdot1+1\cdot0 & 1\cdot1+1\cdot1 \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$. Geometrically, the matrix is a horizontal shear that moves each point sideways by its height; doing it twice moves each point sideways by twice its height.

Exercise 4.34. $\boldsymbol{A}^\top = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \\ -1 & 5 \end{smallmatrix}\right)$ and $\boldsymbol{A}\boldsymbol{A}^\top = \left(\begin{smallmatrix} 4+0+1 & 2+0-5 \\ 2+0-5 & 1+9+25 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 5 & -3 \\ -3 & 35 \end{smallmatrix}\right)$, which is symmetric, as it should be.

Exercise 4.35. $(\boldsymbol{A}\boldsymbol{B})^\top = \left(\begin{smallmatrix} 4 & 10 \\ 6 & 12 \end{smallmatrix}\right)$. And $\boldsymbol{B}^\top\boldsymbol{A}^\top = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 4 & 10 \\ 6 & 12 \end{smallmatrix}\right)$. They agree.

Exercise 4.36. $\boldsymbol{u}^\top\boldsymbol{v} = 3 - 2 = 1$. $\boldsymbol{u}\boldsymbol{v}^\top = \left(\begin{smallmatrix} 3 & -1 \\ 6 & -2 \end{smallmatrix}\right)$ and $\boldsymbol{v}\boldsymbol{u}^\top = \left(\begin{smallmatrix} 3 & 6 \\ -1 & -2 \end{smallmatrix}\right) = (\boldsymbol{u}\boldsymbol{v}^\top)^\top$.

Exercise 4.37. $\boldsymbol{X}^\top\boldsymbol{X}$ is $d \times d$ and $\boldsymbol{X}\boldsymbol{X}^\top$ is $n \times n$. With $n = 10^6$ and $d = 100$, $\boldsymbol{X}^\top\boldsymbol{X}$ has 10^4 entries while $\boldsymbol{X}\boldsymbol{X}^\top$ has 10^{12}, so $\boldsymbol{X}^\top\boldsymbol{X}$ is vastly cheaper.

Exercise 4.38. $\det = 24 - 14 = 10$, so the inverse is $\frac{1}{10}\left(\begin{smallmatrix} 6 & -7 \\ -2 & 4 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{smallmatrix}\right)$. Check: $\left(\begin{smallmatrix} 4 & 7 \\ 2 & 6 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2.4-1.4 & -2.8+2.8 \\ 1.2-1.2 & -1.4+2.4 \end{smallmatrix}\right) = \boldsymbol{I}$.

Exercise 4.39. $\det = 6k - 6 = 0$ when $k = 1$. Then both columns, $(1, 3)$ and $(2, 6)$, lie on the line through $(1, 3)$, and the matrix squashes the whole plane onto that line.

Exercise 4.40. $1 \cdot (24 - 0) - 2 \cdot (0 - 0) + 3 \cdot (0 - 0) = 24$, which equals $1 \cdot 4 \cdot 6$, as expected for a triangular matrix.

Exercise 4.41. $\det(\boldsymbol{A}\boldsymbol{B}) = 3 \times (-2) = -6$. $\det(\boldsymbol{A}^{-1}) = \frac{1}{3}$. For $2\boldsymbol{A}$: the extra factor 2 doubles lengths in both directions, which multiplies areas by 4, so $\det(2\boldsymbol{A}) = 4 \times 3 = 12$. (In general, for an $n \times n$ matrix, $\det(c\boldsymbol{A}) = c^n \det \boldsymbol{A}$.)

Exercise 4.42. $\boldsymbol{A} = \left(\begin{smallmatrix} 2 & -1 \\ 1 & 3 \end{smallmatrix}\right)$, $\boldsymbol{b} = (1, 11)$. $\det \boldsymbol{A} = 6 + 1 = 7$, so $\boldsymbol{A}^{-1} = \frac{1}{7}\left(\begin{smallmatrix} 3 & 1 \\ -1 & 2 \end{smallmatrix}\right)$ and $\boldsymbol{x} = \frac{1}{7}(3 + 11, -1 + 22) = \frac{1}{7}(14, 21) = (2, 3)$. Check: $4 - 3 = 1$ and $2 + 9 = 11$.

Exercise 4.43. $R_2 - 2R_1$ gives $(0, -1, -3 \mid -11)$; $R_3 - R_1$ gives $(0, -2, 1 \mid -1)$. Then $R_3 - 2R_2$ gives $(0, 0, 7 \mid 21)$, so $x_3 = 3$. From $-x_2 - 3x_3 = -11$: $x_2 = 2$. From the first equation: $x_1 = 6 - 2 - 3 = 1$. Solution $(1, 2, 3)$.

Exercise 4.44. (a) Infinitely many: the second equation is twice the first (the same line). (b) None: doubling the first gives $2x + 4y = 8 \neq 5$ (parallel lines). (c) Exactly one: subtracting gives $3y = 3$, so $y = 1$ and $x = 2$.

Exercise 4.45. $\left(\begin{smallmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} w_1 \\ w_0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2 \\ 3 \\ 5 \end{smallmatrix}\right)$. The first two equations, $w_1 + w_0 = 2$ and $2w_1 + w_0 = 3$, force $w_1 = 1$ and $w_0 = 1$; but then the third gives $3 + 1 = 4 \neq 5$. The points do not lie on a line, so the system has no solution; Chapter 5 finds the best approximate one.

Chapter 5: Linear Algebra II: Core Ideas for Machine Learning

Exercise 5.1. (a) The line through the origin with slope 2 (the line $x_2 = 2x_1$). (b) The same line: $(2, 4) = 2(1, 2)$ adds no new direction. (c) The whole plane $\mathbb{R}^2$, since the two vectors point in different directions.

Exercise 5.2. We need $a + b$, $a + 2b$, b to equal the three components. For $(3, 5, 1)$: $b = 1$, then $a = 2$ from the first component, but the second gives $2 + 2 = 4 \neq 5$: not in the span. For $(3, 4, 1)$: $b = 1$, $a = 2$, and $a + 2b = 4$ ✓: it is in the span, equal to $2(1, 1, 0) + (1, 2, 1)$.

Exercise 5.3. The predictions $\boldsymbol{Xw}$ lie in the column space of the 3×2 data matrix $\boldsymbol{X}$, which is at most a plane through the origin in $\mathbb{R}^3$. A target vector $\boldsymbol{y}$ off that plane cannot be fitted exactly, so in general the answer is no.

Exercise 5.4. (a) Dependent: $(2, 6) = 2(1, 3)$. (b) The matrix with these columns has determinant $1(0 - 1) - 0 + 1(0 - 1) = -2 \neq 0$, so they are independent. (c) Dependent: three vectors in $\mathbb{R}^2$ are always dependent.

Exercise 5.5. They are dependent when $\det \left(\begin{smallmatrix} 1 & k \\ k & 4 \end{smallmatrix}\right) = 4 - k^2 = 0$, i.e. $k = 2$ or $k = -2$. (For $k = 2$: $(2, 4) = 2(1, 2)$.)

Exercise 5.6. In every row exactly one of the three one-hot entries is 1, so the three one-hot columns add up to the column of ones: $\mathbf{1} - \boldsymbol{c}_{\text{red}} - \boldsymbol{c}_{\text{green}} - \boldsymbol{c}_{\text{blue}} = \mathbf{0}$, a non-trivial combination giving zero.

Exercise 5.7. Solve $a(1, 1) + b(-1, 1) = (1, 5)$: $a - b = 1$ and $a + b = 5$, so $a = 3$ and $b = 2$. The coordinates are $(3, 2)$.

Exercise 5.8. (a) 1: the second row is three times the first. (b) 2: the two rows are independent (and the rank cannot exceed 2). (c) 2: elimination gives $R_2 - 4R_1 = (0, -3, -6)$ and $R_3 - 7R_1 = (0, -6, -12)$, which is twice the new second row, so one row becomes zero. (Equivalently, the third column minus twice the second plus the first is zero.)

Exercise 5.9. $\boldsymbol{Ax} = \mathbf{0}$ means $x_1 + 2x_2 = 0$ (the second equation is a multiple of the first), so $\boldsymbol{x} = t(-2, 1)$. The null space has dimension 1, the rank is 1, and $1 + 1 = 2 =$ number of columns.

Exercise 5.10. 10 outer products of two vectors with 1000 entries each need $10 \times 2000 = 20\,000$ numbers, versus 10^6 entries: 2% of the storage.

Exercise 5.11. $(1, -2, 2)$: 5, 3, 2. $(-3, 0, 4, 0)$: 7, 5, 4.

Exercise 5.12. $(0.6, 0.6)$: L^1 norm 1.2 (outside), L^2 norm ≈ 0.85 (inside), L^∞ norm 0.6 (inside). $(0.9, 0.2)$: 1.1 (outside), ≈ 0.92 (inside), 0.9 (inside). $(0.5, -0.5)$: 1 (on the boundary of the diamond), ≈ 0.71 (inside), 0.5 (inside).

Exercise 5.13. $\sqrt{1 + 4 + 4 + 16} = \sqrt{25} = 5$.

Exercise 5.14. $\|(3, 4)\| = 5 > 1$, so it becomes $(0.6, 0.8)$: same direction, length 1. $\|(0.3, 0.4)\| = 0.5 \leq 1$, so it is left unchanged.

Exercise 5.15. Ridge: $0.1 \times (4 + 1 + 0.25) = 0.525$. Lasso: $0.1 \times (2 + 1 + 0.5) = 0.35$.

Exercise 5.16. Dot products: $\boldsymbol{q}_1 \cdot \boldsymbol{q}_2 = \frac{1}{9}(2+2-4) = 0$, $\boldsymbol{q}_1 \cdot \boldsymbol{q}_3 = \frac{1}{9}(2-4+2) = 0$, $\boldsymbol{q}_2 \cdot \boldsymbol{q}_3 = \frac{1}{9}(4-2-2) = 0$; each has squared length $\frac{1}{9}(1+4+4) = 1$. Three orthonormal vectors in $\mathbb{R}^3$ are independent and hence a basis. Coordinates: $\boldsymbol{v} \cdot \boldsymbol{q}_1 = \frac{1}{3}(3+0+6) = 3$, $\boldsymbol{v} \cdot \boldsymbol{q}_2 = \frac{1}{3}(6+0-6) = 0$, $\boldsymbol{v} \cdot \boldsymbol{q}_3 = \frac{1}{3}(6+0+3) = 3$. Check: $3\boldsymbol{q}_1 + 3\boldsymbol{q}_3 = (1,2,2) + (2,-2,1) = (3,0,3)$.

Exercise 5.17. $\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}(1,1,0)$. $\boldsymbol{v}_2 \cdot \boldsymbol{q}_1 = \frac{1}{\sqrt{2}}$, so $\boldsymbol{w}_2 = (1,0,1) - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1,1,0) = (0.5,-0.5,1)$, with length $\sqrt{1.5}$. So $\boldsymbol{q}_2 = \frac{1}{\sqrt{6}}(1,-1,2)$. Check: $\boldsymbol{q}_1 \cdot \boldsymbol{q}_2 = \frac{1}{\sqrt{12}}(1-1+0) = 0$.

Exercise 5.18. $\boldsymbol{R}_\alpha^\top \boldsymbol{R}_\alpha = \left(\begin{smallmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{smallmatrix}\right)\left(\begin{smallmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{smallmatrix}\right) = \left(\begin{smallmatrix} \cos^2\alpha+\sin^2\alpha & 0 \\ 0 & \sin^2\alpha+\cos^2\alpha \end{smallmatrix}\right) = \boldsymbol{I}$. Hence $\boldsymbol{R}_\alpha^{-1} = \boldsymbol{R}_\alpha^\top = \boldsymbol{R}_{-\alpha}$, the rotation by $-\alpha$.

Exercise 5.19. $\frac{\boldsymbol{a}\cdot\boldsymbol{b}}{\boldsymbol{a}\cdot\boldsymbol{a}} = \frac{4}{2} = 2$, so $\boldsymbol{p} = (2,2)$. Error $\boldsymbol{e} = (1,-1)$, and $\boldsymbol{e} \cdot \boldsymbol{a} = 1 - 1 = 0$.

Exercise 5.20. $\boldsymbol{a} \cdot \boldsymbol{b} = 4$, $\boldsymbol{a} \cdot \boldsymbol{a} = 2$, so $\boldsymbol{p} = 2\boldsymbol{a} = (2,0,2)$. The error is $(-1,2,1)$ (check: $-1+0+1=0$), and the distance is $\sqrt{1+4+1} = \sqrt{6} \approx 2.45$.

Exercise 5.21. $\boldsymbol{X} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{smallmatrix}\right)$, $\boldsymbol{y} = (1,3,4)$. $\boldsymbol{X}^\top\boldsymbol{X} = \left(\begin{smallmatrix} 5 & 3 \\ 3 & 3 \end{smallmatrix}\right)$ and $\boldsymbol{X}^\top\boldsymbol{y} = (11,8)$. With $\det = 6$: $\hat{\boldsymbol{w}} = \frac{1}{6}\left(\begin{smallmatrix} 3 & -3 \\ -3 & 5 \end{smallmatrix}\right)(11,8) = \frac{1}{6}(9,7) = (1.5, 1.1\overline{6})$. The line is $y = 1.5x + \frac{7}{6}$, with residuals $(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6})$, which sum to zero.

Exercise 5.22. $\boldsymbol{X}^\top\boldsymbol{X} = 3$ (the number of data points) and $\boldsymbol{X}^\top\boldsymbol{y} = 2+4+9 = 15$, so $c = 5$: the mean, as found with the vertex formula in Section 2.4.

Exercise 5.23. $\boldsymbol{P}^2 = \frac{\boldsymbol{a}(\boldsymbol{a}^\top\boldsymbol{a})\boldsymbol{a}^\top}{(\boldsymbol{a}^\top\boldsymbol{a})^2} = \frac{\boldsymbol{a}\boldsymbol{a}^\top}{\boldsymbol{a}^\top\boldsymbol{a}} = \boldsymbol{P}$, using associativity and the fact that $\boldsymbol{a}^\top\boldsymbol{a}$ is a number. And $(\boldsymbol{a}\boldsymbol{a}^\top)^\top = \boldsymbol{a}\boldsymbol{a}^\top$. Geometrically: once a vector has been projected onto the line, it is on the line, so projecting it again leaves it where it is.

Exercise 5.24. (a) 3 with eigenvector $\boldsymbol{e}_1$, and -2 with eigenvector $\boldsymbol{e}_2$. (b) $\mathrm{tr} = 7$, $\det = 6$: $\lambda^2 - 7\lambda + 6 = 0$, so $\lambda = 6$ or $\lambda = 1$. For 6: $\left(\begin{smallmatrix} -1 & 2 \\ 2 & -4 \end{smallmatrix}\right)\boldsymbol{v} = \boldsymbol{0}$ gives $\boldsymbol{v} = (2,1)$. For 1: $\left(\begin{smallmatrix} 4 & 2 \\ 2 & 1 \end{smallmatrix}\right)\boldsymbol{v} = \boldsymbol{0}$ gives $\boldsymbol{v} = (1,-2)$. Indeed $(2,1)\cdot(1,-2) = 0$, as the spectral theorem predicts for a symmetric matrix.

Exercise 5.25. $\left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)(1,-1) = (2,-2) = 2(1,-1)$: eigenvalue 2. The eigenvalues add up to the trace 6, so the other one is 4 (with eigenvector $(1,1)$).

Exercise 5.26. The characteristic equation is $\lambda^2 - 0 \cdot \lambda + 1 = 0$, i.e. $\lambda^2 = -1$, which has no real solution. (It has the complex solutions $\pm i$, which is how rotations are described in the complex numbers.)

Exercise 5.27. $\lambda^2 - 4\lambda - 5 = 0$, i.e. $(\lambda - 5)(\lambda + 1) = 0$: $\lambda = 5$ and $\lambda = -1$. Sum $4 =$ trace; product $-5 =$ determinant $3 - 8$.

Exercise 5.28. $\boldsymbol{W}^{10}\boldsymbol{h} = (0.5^{10}, 1.2^{10}) \approx (0.00098, 6.19)$. The component along the eigenvector with eigenvalue 0.5 has almost vanished, while the one with eigenvalue 1.2 has grown more than sixfold; over 100 steps it would grow by a factor of about 8×10^7. This is the vanishing/exploding behaviour of recurrent networks.

Exercise 5.29. $\boldsymbol{A}^\top\boldsymbol{A} = \left(\begin{smallmatrix} 9 & 0 \\ 0 & 4 \end{smallmatrix}\right)$, so the singular values are 3 and 2. Singular values are lengths of the ellipse's axes and are never negative; the sign of -2 is absorbed into one of the orthogonal matrices (a reflection).

Exercise 5.30. $\boldsymbol{A}^\top\boldsymbol{A} = \left(\begin{smallmatrix} 2 & 2 \\ 2 & 2 \end{smallmatrix}\right)$ has eigenvalues 4 and 0, so $\sigma_1 = 2$, $\sigma_2 = 0$ (rank 1). With $\boldsymbol{u}_1 = \boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)$: $2 \cdot \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right) = \boldsymbol{A}$.

Exercise 5.31. $40 \times (800 + 1000 + 1) = 72\,040$ numbers, versus 800 000: about 9%.

Exercise 5.32. Error $\sqrt{1^2 + 0.1^2} = \sqrt{1.01} \approx 1.005$. Since $\|\boldsymbol{A}\|_F^2 = \sum \sigma_i^2 = 126.01$, the rank-2 approximation captures $\frac{100+25}{126.01} \approx 99.2\%$ of it.

Exercise 5.33. $3x_1^2 - 2x_1x_2 + 2x_2^2$. Since $a = 3 > 0$ and $ac - b^2 = 6 - 1 = 5 > 0$, it is positive definite.

Exercise 5.34. (a) $\det = 1 - 9 = -8 < 0$, so the eigenvalues have opposite signs: indefinite. (b) $\det = 0$ and trace $5 > 0$, so the eigenvalues are 5 and 0: positive semidefinite (the form is $(2x_1 + x_2)^2$). (c) Eigenvalues -2 and -1: negative definite.

Exercise 5.35. Need $1 > 0$ (true) and $4 - c^2 > 0$, i.e. $-2 < c < 2$.

Exercise 5.36. Mean $(3, 3)$; centred points $(-1,-3)$, $(-3,-1)$, $(1,3)$, $(3,1)$. $\boldsymbol{C} = \frac{1}{4}\left(\begin{smallmatrix} 1+9+1+9 & 3+3+3+3 \\ 12 & 9+1+9+1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 5 & 3 \\ 3 & 5 \end{smallmatrix}\right)$. Eigenvalues 8 and 2 (trace 10, determinant 16), with eigenvectors $\frac{1}{\sqrt{2}}(1,1)$ and $\frac{1}{\sqrt{2}}(1,-1)$. The first component explains $\frac{8}{10} = 80\%$ of the spread. The projections onto it are $\pm\frac{4}{\sqrt{2}} \approx \pm 2.83$.

Exercise 5.37. PCA looks for directions of largest spread *around the mean*. Without centring, $\frac{1}{n}\boldsymbol{X}^\top\boldsymbol{X}$ measures spread around the origin, so if the data is far from the origin the first "component" mostly points from the origin towards the centre of the data cloud, describing where the data is rather than how it varies.

Exercise 5.38. The total is 10. One component keeps 60%, two keep 90%. So two components suffice.

Chapter 6: Calculus of One Variable

Exercise 6.1. On $[1, 3]$: $\frac{10-2}{2} = 4$. On $[1, 1.1]$: $\frac{2.21-2}{0.1} = 2.1$. (As the interval shrinks, the rate approaches 2, the derivative at 1.)

Exercise 6.2. $\frac{0.5-2.3}{1000} = -0.0018$ per step. The negative sign means the loss is decreasing on average, i.e. the model is improving.

Exercise 6.3. $P(10) = 50 \cdot 1.02^{10} \approx 60.95$, so the average rate is $\frac{60.95-50}{10} \approx 1.095$ thousand per year.

Exercise 6.4. (a) $\frac{(x-2)(x+2)}{x-2} = x + 2 \to 4$. (b) 7 (the function is continuous, so just substitute). (c) $\frac{x(x+3)}{x} = x + 3 \to 3$.

Exercise 6.5. (a) $\frac{3+1/x}{1-2/x} \to \frac{3}{1} = 3$. (b) 0. (c) 0, because $e^{-x} \to \infty$ as $x \to -\infty$, so $\frac{1}{1+e^{-x}} \to 0$.

Exercise 6.6. ReLU: the left-hand limit is 0, the right-hand limit is 0, and $\mathrm{ReLU}(0) = 0$, so it is continuous. The step function has left limit 0 and right limit 1, so the limit does not exist and it is not continuous at 0.

Exercise 6.7. (a) $\frac{3(x+h)+2-3x-2}{h} = \frac{3h}{h} = 3$, so $f'(x) = 3$. (b) $\frac{(x+h)^2+(x+h)-x^2-x}{h} = \frac{2xh+h^2+h}{h} = 2x + h + 1 \to 2x + 1$.

Exercise 6.8. $\frac{1}{h}\left(\frac{1}{x+h} - \frac{1}{x}\right) = \frac{1}{h} \cdot \frac{x-(x+h)}{x(x+h)} = \frac{-1}{x(x+h)} \to -\frac{1}{x^2}$ as $h \to 0$.

Exercise 6.9. The point is $(3, 9)$ and the slope is $2 \cdot 3 = 6$. Point–slope form: $y-9 = 6(x-3)$, i.e. $y = 6x - 9$.

Exercise 6.10. $|x - 2|$ has its corner at $x = 2$; $\mathrm{ReLU}(x - 1)$ has its corner at $x = 1$. Both are differentiable everywhere else.

Exercise 6.11. (a) $12x^2 - 2$. (b) $\frac{1}{2\sqrt{x}} - \frac{1}{x^2}$. (c) $5e^x - \frac{3}{x}$. (d) $\frac{3}{2}x^{1/2}$.

Exercise 6.12. (a) $(x^2 + 2x)e^x$. (b) $1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1$. (c) $\frac{2x(x+1)-x^2}{(x+1)^2} = \frac{x^2+2x}{(x+1)^2}$.

Exercise 6.13. (a) $4(2x + 1)^3 \cdot 2 = 8(2x + 1)^3$. (b) $3e^{3x}$. (c) $\frac{2x}{x^2+1}$. (d) $2x\cos(x^2)$. (e) $-x\,e^{-x^2/2}$ (this function is the shape of the normal distribution of Chapter 9).

Exercise 6.14. By the chain rule, $\frac{\mathrm{d}}{\mathrm{d}x}\ln(1 + e^x) = \frac{1}{1+e^x} \cdot e^x = \frac{e^x}{1+e^x}$. Dividing top and bottom by e^x gives $\frac{1}{e^{-x}+1} = \sigma(x)$.

Exercise 6.15. By the chain rule, $L'(w) = 2(y - wx) \cdot (-x) = -2x(y - wx) = -6(5 - 3w)$. At $w = 1$: $L'(1) = -12 < 0$, so the loss decreases when w increases: increase w. (The loss is zero at $w = \frac{5}{3}$.)

Exercise 6.16. With $f = e^x - e^{-x}$ and $g = e^x + e^{-x}$ we have $f' = g$ and $g' = f$. So $\tanh' = \frac{f'g-fg'}{g^2} = \frac{g^2-f^2}{g^2} = 1 - \left(\frac{f}{g}\right)^2 = 1 - \tanh^2 x$.

Exercise 6.17. $f'(x) = 3x^2-12x+9 = 3(x-1)(x-3)$, so $x = 1$ and $x = 3$. $f''(x) = 6x-12$: $f''(1) = -6 < 0$ (local maximum, $f(1) = 5$) and $f''(3) = 6 > 0$ (local minimum, $f(3) = 1$).

Exercise 6.18. $f'(x) = 2x - 4 = 0$ at $x = 2$; $f''(x) = 2 > 0$, so it is a minimum, with value $f(2) = 3$. Completing the square: $f(x) = (x - 2)^2 + 3$, the same answer.

Exercise 6.19. $L'(c) = 4(c - 1) + 2(c - 4) = 6c - 12 = 0$, so $c = 2$; $L'' = 6 > 0$. This is the weighted mean $\frac{2\cdot1+1\cdot4}{2+1} = 2$: the data point 1 counts twice.

Exercise 6.20. $f'(x) = e^{-x} - xe^{-x} = (1 - x)e^{-x}$, which is zero at $x = 1$, positive before and negative after. So $x = 1$ is a maximum, with value $f(1) = e^{-1} \approx 0.368$.

Exercise 6.21. $f'(x) = 4x^3$ and $f''(x) = 12x^2$, so $f'(0) = f''(0) = 0$: inconclusive. But $x^4 \geq 0 = f(0)$ for all x, so 0 is a (global) minimum. (By contrast, x^3 also has $f'(0) = f''(0) = 0$ and no extremum at all.)

Exercise 6.22. $f(0) = 0$ and $f'(x) = \frac{1}{1+x}$, so $f'(0) = 1$ and $\ln(1 + x) \approx x$. Hence $\ln 1.05 \approx 0.05$ (true value 0.04879).

Exercise 6.23. $\cos 0 = 1$, $(\cos)'(0) = -\sin 0 = 0$, $(\cos)''(0) = -\cos 0 = -1$. So $\cos x \approx 1 - \frac{x^2}{2}$ and $\cos(0.2) \approx 1 - 0.02 = 0.98$ (true value 0.98007).

Exercise 6.24. $f(1) = 1$, $f'(x) = -x^{-2}$ so $f'(1) = -1$, and $f''(x) = 2x^{-3}$ so $f''(1) = 2$. Thus $f(1 + h) \approx 1 - h + h^2$, and $\frac{1}{1.1} \approx 1 - 0.1 + 0.01 = 0.91$ (true value 0.90909).

Exercise 6.25. Since $f'(a) = 0$: $f(a \pm 0.1) \approx 2 + \frac{1}{2} \cdot 8 \cdot 0.01 = 2.04$ in both directions.

Exercise 6.26. (a) $\left[x^3\right]_0^2 = 8$. (b) $\left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{2}{3}(8-1) = \frac{14}{3} \approx 4.67$. (c) $\left[e^x + x\right]_0^1 = (e+1) - 1 = e \approx 2.718$. (d) $\left[\ln x\right]_1^2 = \ln 2 \approx 0.693$.

Exercise 6.27. $\frac{x^6}{6} + C$; $-\frac{1}{x} + C$ (since x^{-2} integrates to $\frac{x^{-1}}{-1}$); $\frac{1}{2}e^{2x} + C$; $\sin x + C$.

Exercise 6.28. Width 0.5; left ends $0, 0.5, 1, 1.5$; sum $0.5(0+0.5+1+1.5) = 1.5$. The exact value is $\left[\frac{x^2}{2}\right]_0^2 = 2$ (the area of a triangle with base 2 and height 2). Left-end rectangles underestimate an increasing function.

Exercise 6.29. $\int_0^1 2x\,\mathrm{d}x = \left[x^2\right]_0^1 = 1$. And $\int_0^1 x \cdot 2x\,\mathrm{d}x = \left[\frac{2x^3}{3}\right]_0^1 = \frac{2}{3}$: values near 1 are more likely than values near 0, so the average is above $\frac{1}{2}$.

Exercise 6.30. $\lim_{b\to\infty}\left[-\frac{1}{2}e^{-2x}\right]_0^b = \lim_{b\to\infty}\frac{1}{2}\left(1 - e^{-2b}\right) = \frac{1}{2}$.

Exercise 6.31. Forward: $\frac{e^{1.01}-e}{0.01} \approx 2.7319$. Central: $\frac{e^{1.01}-e^{0.99}}{0.02} \approx 2.71833$. Exact: $e \approx 2.71828$. The central difference is accurate to about four decimal places, the forward difference to about one.

Exercise 6.32. Evaluate the loss at $w = 0.5 + h$ and $w = 0.5 - h$ for, say, $h = 10^{-5}$, and compute $\frac{L(0.5+h)-L(0.5-h)}{2h}$. It should agree with the formula's value -0.2115 to many decimal places (it gives -0.21151). A significant disagreement would reveal a mistake in the formula or in the code.

Chapter 7: Multivariable Calculus

Exercise 7.1. $f(2,1) = 4 - 1 + 3 = 6$. The level curve $x^2 - y + 3 = 3$ is $y = x^2$, a parabola.

Exercise 7.2. $x + 2y = c$ are parallel straight lines with slope $-\frac{1}{2}$ (a tilted plane has straight, evenly spaced contours). $x^2 + y^2 = c$ are circles centred at the origin with radius $\sqrt{c}$ (a round bowl).

Exercise 7.3. $L(2,0) = \frac{1}{2}[0+0] = 0$: a perfect fit ($y = 2x$). $L(1,1) = \frac{1}{2}[0 + (4-3)^2] = 0.5$. The parameters $(2,0)$ fit better.

Exercise 7.4. $\frac{\partial f}{\partial x} = 6xy - 2y^3$ and $\frac{\partial f}{\partial y} = 3x^2 - 6xy^2$.

Exercise 7.5. $\frac{\partial f}{\partial x} = e^y + \frac{1}{x}$ and $\frac{\partial f}{\partial y} = x\,e^y$.

Exercise 7.6. $f_x = y\cos(xy)$ and $f_y = x\cos(xy)$. Then $\frac{\partial}{\partial y}(y\cos(xy)) = \cos(xy) - xy\sin(xy)$ (product rule), and $\frac{\partial}{\partial x}(x\cos(xy)) = \cos(xy) - xy\sin(xy)$. They agree.

Exercise 7.7. The error is $y - wx - b = 7 - 2 - 1 = 4$. So $\frac{\partial L}{\partial w} = -2 \cdot 2 \cdot 4 = -16$ and $\frac{\partial L}{\partial b} = -2 \cdot 4 = -8$. Both are negative: increasing either parameter reduces the loss, and w has the larger effect.

Exercise 7.8. $\nabla f = (2xy, x^2 + 3y^2)$, so $\nabla f(1,2) = (4, 1 + 12) = (4, 13)$.

Exercise 7.9. $\nabla f = (2x - 2, 2y + 4) = \mathbf{0}$ gives $(x, y) = (1, -2)$. (Completing the square, $f = (x-1)^2 + (y+2)^2 - 5$, so this is the minimum, with value -5.)

Exercise 7.10. $f(1,1) = 2$ and $\nabla f(1,1) = (2,2)$, so $f(1.1, 0.9) \approx 2 + (2,2) \cdot (0.1, -0.1) = 2$. The exact value is $1.21 + 0.81 = 2.02$; the small error comes from the curvature.

Exercise 7.11. Residuals $r_i = y_i - wx_i - b$: $1 - 1 = 0$ and $3 - 2 = 1$. With $n = 2$: $\frac{\partial L}{\partial w} = -\frac{2}{2}(1 \cdot 0 + 2 \cdot 1) = -2$ and $\frac{\partial L}{\partial b} = -\frac{2}{2}(0 + 1) = -1$. So $\nabla L = (-2, -1)$.

Exercise 7.12. $\boldsymbol{u} = \frac{1}{\sqrt{2}}(1,1)$ and $\nabla f(1,1) = (2,6)$, so $D_{\boldsymbol{u}} f = \frac{2+6}{\sqrt{2}} = 4\sqrt{2} \approx 5.66$.

Exercise 7.13. $\nabla f = (y, x) = (3, 2)$. Fastest increase in the direction $\frac{1}{\sqrt{13}}(3,2)$, at rate $\sqrt{13} \approx 3.61$. No change in the directions perpendicular to $(3,2)$, i.e. $\pm\frac{1}{\sqrt{13}}(2,-3)$.

Exercise 7.14. At an interior minimum of a differentiable function, $\nabla f = \mathbf{0}$, so $D_{\boldsymbol{u}} f = \mathbf{0} \cdot \boldsymbol{u} = 0$ in every direction.

Exercise 7.15. $\frac{\mathrm{d}z}{\mathrm{d}t} = 2x \cdot 2 + 2y \cdot 2t = 8t + 4t^3$ (using $x = 2t$, $y = t^2$). Check: $z = 4t^2 + t^4$, so $\frac{\mathrm{d}z}{\mathrm{d}t} = 8t + 4t^3$.

Exercise 7.16. At $t = 1$: $x = 4$, $y = 1$. $\frac{\mathrm{d}f}{\mathrm{d}t} = y \cdot 3 + x \cdot 2t = 3 + 8 = 11$. (Check: $f = 3t^3 + t^2$, so $f' = 9t^2 + 2t = 11$ at $t = 1$.)

Exercise 7.17. Forward: $q = -10$, $f = q + z = -14$. Backward: $\frac{\partial f}{\partial q} = 1$ and $\frac{\partial f}{\partial z} = 1$; then $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \cdot y = 5$ and $\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \cdot x = -2$.

Exercise 7.18. With $z = w_1 x_1 + w_2 x_2 + b$: $\frac{\partial L}{\partial w_i} = 2(\hat{y} - y)\,\sigma'(z)\,x_i$ and $\frac{\partial L}{\partial b} = 2(\hat{y} - y)\,\sigma'(z)$. Here $z = 0.5 - 1 = -0.5$, $\hat{y} = \sigma(-0.5) \approx 0.378$ and $\sigma'(z) = \hat{y}(1 - \hat{y}) \approx 0.235$. So $2(\hat{y} - y)\sigma'(z) \approx 2(-0.622)(0.235) \approx -0.293$, giving $\frac{\partial L}{\partial w_1} \approx -0.293$, $\frac{\partial L}{\partial w_2} \approx -0.585$ and $\frac{\partial L}{\partial b} \approx -0.293$.

Exercise 7.19. $\begin{pmatrix} y & x \\ 1 & 1 \\ 2x & 0 \end{pmatrix}$, a 3×2 matrix (three outputs, two inputs).

Exercise 7.20. It is $\boldsymbol{A}$ itself, the same at every point, and it does not depend on $\boldsymbol{b}$ (a shift does not affect rates of change).

Exercise 7.21. Diagonal, with entries $1 - \tanh^2 z_i$: at $(0,1)$ this is $\mathrm{diag}(1, 1 - \tanh^2 1) \approx \mathrm{diag}(1, 0.42)$.

Exercise 7.22. $f_x = 3x^2 + 2y$, $f_y = 2x + 2y$; so $f_{xx} = 6x$, $f_{xy} = f_{yx} = 2$, $f_{yy} = 2$, and $\boldsymbol{H} = \begin{pmatrix} 6x & 2 \\ 2 & 2 \end{pmatrix}$.

Exercise 7.23. $\nabla f = (2x - 4, 2y + 2) = \mathbf{0}$ at $(2, -1)$. $\boldsymbol{H} = 2\boldsymbol{I}$ is positive definite, so it is a minimum, with value $4 + 1 - 8 - 2 = -5$.

Exercise 7.24. $\nabla f = (y, x)$ vanishes at the origin. $\boldsymbol{H} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has eigenvalues 1 and -1 (with eigenvectors $(1,1)$ and $(1,-1)$), so it is indefinite: f increases along the line $y = x$ (where $f = x^2$) and decreases along $y = -x$ (where $f = -x^2$).

Exercise 7.25. $\nabla f = (3x^2 - 12, 3y^2 - 3) = \mathbf{0}$ gives $x = \pm 2$, $y = \pm 1$: four critical points. $\boldsymbol{H} = \mathrm{diag}(6x, 6y)$. At $(2,1)$: both positive, a minimum ($f = 8 - 24 + 1 - 3 = -18$). At $(-2,-1)$: both negative, a maximum ($f = 18$). At $(2,-1)$ and $(-2,1)$: opposite signs, saddle points.

Exercise 7.26. $\nabla f = (4x^3, 4y^3)$ and $\boldsymbol{H} = \operatorname{diag}(12x^2, 12y^2)$, which is the zero matrix at the origin: inconclusive. But $f \geq 0 = f(0,0)$ everywhere, so the origin is a (global) minimum.

Exercise 7.27. $f(0,0) = 1$, $\nabla f = (1,1)$, $\boldsymbol{H} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, so $f(\boldsymbol{h}) \approx 1 + h_1 + h_2 + \frac{1}{2}(h_1 + h_2)^2$. At $(0.1, 0.05)$: $1 + 0.15 + 0.01125 = 1.16125$ (exact: $e^{0.15} \approx 1.16183$).

Exercise 7.28. $\nabla f = \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}$ and $\boldsymbol{H} = \boldsymbol{A}$ (positive definite). Setting the gradient to zero: $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b} = (1,1)$, the minimiser.

Exercise 7.29. $\nabla f(1,1) = (2,6)$ and $f(1,1) = 4$. With $\eta = 0.1$: new point $(0.8, 0.4)$, $f = 0.64 + 0.48 = 1.12$, a big improvement. With $\eta = 0.5$: new point $(0, -2)$, $f = 12$, *worse* than before. The step overshot in the steep y-direction (curvature 6): $\eta = 0.5 > 2/6$ is too large for that direction.

Exercise 7.30. Inner: $\int_0^1 (x^2 + y)\,\mathrm{d}x = \frac{1}{3} + y$. Outer: $\int_0^2 \left(\frac{1}{3} + y\right)\mathrm{d}y = \frac{2}{3} + 2 = \frac{8}{3}$.

Exercise 7.31. $\int_0^1 \int_0^1 4xy\,\mathrm{d}x\,\mathrm{d}y = 4\left(\int_0^1 x\,\mathrm{d}x\right)\left(\int_0^1 y\,\mathrm{d}y\right) = 4 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1$. It is also non-negative, so it is a valid density.

Chapter 8: Optimization

Exercise 8.1. $\min f = 1$ and $\arg\min f = -2$. Maximising g is minimising $-g(x) = (x-1)^2 - 4$, whose minimiser is $x = 1$; so the maximum of g is $g(1) = 4$.

Exercise 8.2. $L(w) = \frac{1}{3}\left(|1 - w| + |2 - 2w| + |2 - 3w|\right)$. $L(0.5) = \frac{1}{3}(0.5 + 1 + 0.5) \approx 0.667$ and $L(1) = \frac{1}{3}(0 + 0 + 1) \approx 0.333$, so $w = 1$ is better.

Exercise 8.3. $\nabla f = (2(x-1), 4(y+3)) = \boldsymbol{0}$ at $(1, -3)$; the Hessian $\operatorname{diag}(2,4)$ is positive definite. The minimum value is 5.

Exercise 8.4. $L'(w) = \sum_i 2(y_i - wx_i)(-x_i) = -2\left(\sum_i x_i y_i - w\sum_i x_i^2\right)$. Setting it to zero gives the formula, and $L''(w) = 2\sum_i x_i^2 > 0$ (unless all $x_i = 0$), so it is a minimum.

Exercise 8.5. The derivative is $-2\sum_i x_i(y_i - wx_i) + 2\lambda w$; setting it to zero gives $w\left(\sum_i x_i^2 + \lambda\right) = \sum_i x_i y_i$. As $\lambda \to \infty$ the denominator grows without bound and $w \to 0$: a very strong penalty forces the weight to zero.

Exercise 8.6. $f'(x) = 2x - 4$. $x_1 = 0 - 0.25(-4) = 1$; $x_2 = 1 - 0.25(-2) = 1.5$; $x_3 = 1.5 - 0.25(-1) = 1.75$. It approaches the minimiser $x = 2$, halving the distance at each step.

Exercise 8.7. $\nabla f = (2x, 4y)$. Step 1: $\nabla f(2,1) = (4,4)$, new point $(1.6, 0.6)$, $f = 2.56 + 0.72 = 3.28$ (down from 6). Step 2: $\nabla f = (3.2, 2.4)$, new point $(1.28, 0.36)$, $f = 1.6384 + 0.2592 \approx 1.90$.

Exercise 8.8. We need $3 \cdot 0.8^k < 0.01$, i.e. $0.8^k < \frac{1}{300}$, i.e. $k > \frac{\ln 300}{\ln 1.25} \approx 25.6$. So 26 steps.

Exercise 8.9. $f'' = 10$, so we need $0 < \eta < 0.2$. The update is $x_{k+1} = (1 - 10\eta)x_k$; with $\eta = 0.1$ the factor is 0, so one step lands exactly on the minimiser 0.

Exercise 8.10. The Hessian is $\operatorname{diag}(2, 100)$, so we need $\eta < \frac{2}{100} = 0.02$. With $\eta = 0.019$ the x-coordinate is multiplied by $1 - 0.019 \cdot 2 = 0.962$ per step, so it takes about 60 steps just to halve it. The condition number is 50: the steep direction forces a small learning rate, which makes progress along the flat direction very slow.

Exercise 8.11. Epoch 75 is after two halvings: $0.1 \times 0.25 = 0.025$. Cosine: $\eta_{50} = 0.05(1 + \cos\frac{\pi}{2}) = 0.05$ and $\eta_{100} = 0.05(1 + \cos\pi) = 0$.

Exercise 8.12. x^4: yes ($f'' = 12x^2 \geq 0$). $-x^2$: no (it is concave). e^{-x}: yes ($f'' = e^{-x} > 0$). x^3: no ($f'' = 6x < 0$ for $x < 0$). $|x - 2|$: yes (a shifted norm). $\max(0, x)$: yes (a maximum of two linear functions).

Exercise 8.13. Its Hessian is $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, with eigenvalues 3 and 1, so it is positive definite everywhere; by Theorem 8.8 f is (strictly) convex.

Exercise 8.14. The first is not: it contains -1 and 1 but not 0, which lies between them. The second, the interval $[-1, 1]$, is convex.

Exercise 8.15. $50\,000/100 = 500$ updates per epoch, so $10\,000$ updates in 20 epochs.

Exercise 8.16. The single-example gradient is $2(c - y)$, so $c \leftarrow c - 0.5(c - y)$: $c_1 = 0 - 0.5(0 - 6) = 3$; $c_2 = 3 - 0.5(3 - 2) = 2.5$; $c_3 = 2.5 - 0.5(2.5 - 8) = 5.25$. The target is the mean 5.

Exercise 8.17. With $m = n/b$ batches $B_1, \ldots, B_m$,

$$\frac{1}{m}\sum_{j=1}^{m}\frac{1}{b}\sum_{i\in B_j}\nabla\ell_i = \frac{1}{mb}\sum_{i=1}^{n}\nabla\ell_i = \frac{1}{n}\sum_{i=1}^{n}\nabla\ell_i = \nabla L,$$

since the batches together contain every example exactly once.

Exercise 8.18. $v_1 = 1$, $v_2 = 1.9$, $v_3 = 2.71$, so the steps are 0.1, 0.19, 0.271. The velocity approaches $\frac{1}{1-0.9} = 10$, so the steps approach 1: ten times the step of plain gradient descent.

Exercise 8.19. $\hat{\boldsymbol{m}}_1 = \boldsymbol{g}_1$ and $\hat{\boldsymbol{s}}_1 = \boldsymbol{g}_1^2$, so the step is $-\eta\,\boldsymbol{g}_1/|\boldsymbol{g}_1| = -0.001 \cdot (1, -1) = (-0.001, +0.001)$. Both parameters move by 0.001, although their gradients differ by a factor of 15 000.

Exercise 8.20. $f'(x) = 2x - 6$ and $f'' = 2$, so $x_1 = x_0 - \frac{2x_0 - 6}{2} = 3$, the exact minimiser, whatever x_0 is. For a quadratic, the quadratic approximation is the function itself.

Exercise 8.21. Newton: $x_1 = 1 - \frac{e-2}{e} \approx 0.736$. Gradient descent: $x_1 = 1 - 0.1(e - 2) \approx 0.928$. Newton is much closer.

Exercise 8.22. $(2x, 2y) = \lambda(1, 1)$ gives $x = y$, so $x = y = 2$, with $\lambda = 4$ and minimum value 8.

Exercise 8.23. $(1, 1) = \lambda(2x, 2y)$ gives $x = y = \frac{1}{2\lambda}$. On the circle, $2x^2 = 2$, so $x = y = \pm 1$. The maximum is $f(1, 1) = 2$ and the minimum is $f(-1, -1) = -2$.

Exercise 8.24. $N = 20$, so $\boldsymbol{p} = (0.6, 0.3, 0.1)$.

Exercise 8.25. It projects onto the allowed set by replacing each negative entry by the nearest allowed value 0: the new weights are $(0.4, 0, 0, 0)$. (This is the ReLU applied to the weight vector.)

Chapter 9: Probability

Exercise 9.1. $\Omega = \{HH, HT, TH, TT\}$. "At least one head" is $\{HH, HT, TH\}$.

Exercise 9.2. $A \cup B = \{2, 4, 5, 6\}$, $A \cap B = \{4, 6\}$, $A^c = \{1, 3, 5\}$. They are not disjoint, since $A \cap B \neq \emptyset$.

Exercise 9.3. $P(A \cup B) = 0.3 + 0.5 - 0.1 = 0.7$; $P(A^c) = 0.7$; neither: $1 - P(A \cup B) = 0.3$.

Exercise 9.4. (a) $(2,6), (3,5), (4,4), (5,3), (6,2)$: $\frac{5}{36}$. (b) 6 doubles: $\frac{6}{36} = \frac{1}{6}$. (c) Sums $10, 11, 12$ occur in $3 + 2 + 1 = 6$ ways: $\frac{6}{36} = \frac{1}{6}$.

Exercise 9.5. The complement, "no heads", has probability $\left(\frac{1}{2}\right)^5 = \frac{1}{32}$, so the answer is $\frac{31}{32} \approx 0.969$.

Exercise 9.6. $\binom{8}{3} = 56$ committees. Those containing Alice are determined by choosing the other 2 members from the remaining 7: $\binom{7}{2} = 21$. So the probability is $\frac{21}{56} = \frac{3}{8}$. (Sanity check: each of the 8 people has the same chance, and 3 of the 8 seats are filled.)

Exercise 9.7. $P(\text{free} \mid \text{not spam}) = \frac{5}{60} \approx 0.083$ and $P(\text{not spam} \mid \text{no free}) = \frac{55}{65} \approx 0.846$.

Exercise 9.8. $0.7 \times 0.01 + 0.3 \times 0.04 = 0.007 + 0.012 = 0.019$, i.e. 1.9%.

Exercise 9.9. By the multiplication rule, $P = \frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221} \approx 0.0045$ (after one ace is drawn, 3 aces remain among 51 cards).

Exercise 9.10. $0.05 \times 0.001 \times 0.1 = 5 \times 10^{-6}$, and $\ln(5 \times 10^{-6}) \approx -12.2$. (Language models work with such log-probabilities, which add up over the words instead of multiplying.)

Exercise 9.11. $P(A \cap B) = P(\{2\}) = \frac{1}{6}$ and $P(A)P(B) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$: independent. A and C are disjoint with positive probabilities, so $P(A \cap C) = 0 \neq \frac{1}{4}$: dependent.

Exercise 9.12. $0.9^{10} \approx 0.349$. Even a fairly accurate model usually makes at least one mistake in ten tries.

Exercise 9.13. $P(D \mid +) = \frac{0.9 \times 0.02}{0.9 \times 0.02 + 0.1 \times 0.98} = \frac{0.018}{0.018 + 0.098} \approx 0.155$.

Exercise 9.14. $\frac{0.9 \times 0.155}{0.9 \times 0.155 + 0.1 \times 0.845} \approx \frac{0.1397}{0.1397 + 0.0845} \approx 0.62$. A second positive test raises the probability from about 16% to about 62%.

Exercise 9.15. $P(\text{red}) = \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{2}$, so $P(\text{urn 1} \mid \text{red}) = \frac{(3/4)(1/2)}{1/2} = \frac{3}{4}$.

Exercise 9.16. Spam: $0.4 \times 0.05 = 0.02$; ham: $0.6 \times 0.3 = 0.18$. So $P(\text{spam} \mid \text{meeting}) = \frac{0.02}{0.02 + 0.18} = 0.1$.

Exercise 9.17. $p(0) = \frac{1}{8}$, $p(1) = \frac{3}{8}$, $p(2) = \frac{3}{8}$, $p(3) = \frac{1}{8}$ (there are $\binom{3}{k}$ sequences with k heads among 8). $F(1) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$.

Exercise 9.18. $p(x) = \frac{1}{4}$ on $[0, 4]$. $P(1 \leq X \leq 2.5) = 1.5 \times \frac{1}{4} = 0.375$; $P(X = 2) = 0$.

Exercise 9.19. $p \geq 0$ and $\int_0^1 3x^2 \, \mathrm{d}x = \left[x^3\right]_0^1 = 1$. $P(X \leq 0.5) = 0.5^3 = 0.125$. No problem: densities may exceed 1; only areas must be at most 1.

Exercise 9.20. $P(T > 2) = e^{-0.5\times 2} = e^{-1} \approx 0.368$.

Exercise 9.21. $\mathbb{E}[X] = 0 + 0.5 + 0.6 = 1.1$ and $\mathbb{E}[X^2] = 0 + 0.5 + 1.2 = 1.7$.

Exercise 9.22. $\mathbb{E} = 10 \times 0.1 - 2 \times 0.9 = 1 - 1.8 = -0.8$. On average you lose 0.8 per game, so playing many times is a losing strategy.

Exercise 9.23. By linearity, $30 \times \frac{1}{6} = 5$ sixes and $200 \times 0.8 = 160$ correct predictions.

Exercise 9.24. $\mathbb{E}[X] = \int_0^1 x \cdot 3x^2 \,\mathrm{d}x = \left[\frac{3x^4}{4}\right]_0^1 = 0.75$.

Exercise 9.25. $\mathrm{Var}(X) = 1.7 - 1.1^2 = 0.49$ and $\sigma = 0.7$.

Exercise 9.26. $\mathbb{E}[X] = p$ and $\mathbb{E}[X^2] = p$ (since $X^2 = X$), so $\mathrm{Var}(X) = p - p^2 = p(1-p)$. This parabola is largest at $p = \frac{1}{2}$, where the variance is $\frac{1}{4}$.

Exercise 9.27. $\mathrm{Var}(3X - 5) = 9 \times 4 = 36$, so the standard deviation is 6.

Exercise 9.28. $z = \frac{82-70}{8} = 1.5$. A z-score of -0.5 corresponds to $70 - 0.5 \times 8 = 66$.

Exercise 9.29. Each term $w_i x_i$ has variance $\mathbb{E}[w_i^2]\mathbb{E}[x_i^2] = v$, and the 100 terms are independent, so the sum has variance $100v$. We need $v = 0.01 = \frac{1}{100}$, as in Xavier initialisation.

Exercise 9.30. $P(X = 2) = \binom{5}{2}0.5^5 = \frac{10}{32} = 0.3125$. Mean 2.5, variance $5 \times 0.5 \times 0.5 = 1.25$.

Exercise 9.31. 70 is 2σ above the mean, so $P(X > 70) \approx \frac{5\%}{2} = 2.5\%$ (more precisely 2.3%). 40 to 60 is $\mu \pm \sigma$: about 68%.

Exercise 9.32. $3Z + 2 \sim \mathcal{N}(2, 9)$. Uniform on $[2, 8]$: mean 5, variance $\frac{36}{12} = 3$.

Exercise 9.33. The number correct is $\mathrm{Bin}(1000, 0.9)$: mean 900, standard deviation $\sqrt{1000 \times 0.9 \times 0.1} = \sqrt{90} \approx 9.5$. Dividing by 1000, the measured accuracy has mean 0.9 and standard deviation about 0.0095, roughly one percentage point. Differences between models smaller than this may be due to chance.

Exercise 9.34. $P(X = 0) = \frac{3^0 e^{-3}}{0!} = e^{-3} \approx 0.050$.

Exercise 9.35. $P(\text{sun} \mid \text{no}) = \frac{0.50}{0.55} \approx 0.91$ and $P(\text{yes} \mid \text{rain}) = \frac{0.35}{0.40} = 0.875$.

Exercise 9.36. Each of the four combinations HH, HT, TH, TT has probability $\frac{1}{4}$, and each marginal is $\left(\frac{1}{2}, \frac{1}{2}\right)$; indeed $\frac{1}{4} = \frac{1}{2} \times \frac{1}{2}$ in every cell.

Exercise 9.37. $\int_0^1 \int_0^1 (x + y) \,\mathrm{d}y \,\mathrm{d}x = \int_0^1 \left(x + \frac{1}{2}\right) \mathrm{d}x = \frac{1}{2} + \frac{1}{2} = 1$. The marginal is $p(x) = \int_0^1 (x + y) \,\mathrm{d}y = x + \frac{1}{2}$ for $0 \le x \le 1$.

Exercise 9.38. Means 2.5 and 5. Deviations: $(-1.5, -0.5, 0.5, 1.5)$ and $(-3, -1, 0, 4)$. Products: $4.5, 0.5, 0, 6$, with mean $\frac{11}{4} = 2.75$. Variances: $\frac{2.25+0.25+0.25+2.25}{4} = 1.25$ and $\frac{9+1+0+16}{4} = 6.5$. Correlation: $\frac{2.75}{\sqrt{1.25 \times 6.5}} \approx 0.965$.

Exercise 9.39. $\mathrm{Var}(X + Y) = 4 + 9 - 4 = 9$ and $\rho = \frac{-2}{2\times 3} = -\frac{1}{3}$.

Exercise 9.40. $\mathbb{E}[X] = 0$ and $\mathbb{E}[XY] = \mathbb{E}[X^3] = \frac{1}{3}(-1+0+1) = 0$, so $\operatorname{Cov}(X,Y) = 0 - 0 \cdot \mathbb{E}[Y] = 0$. But $P(X=0, Y=0) = \frac{1}{3}$ while $P(X=0)P(Y=0) = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$, so they are dependent.

Exercise 9.41. The first has determinant $8 - 9 = -1 < 0$, so it has a negative eigenvalue and is not positive semidefinite: it cannot be a covariance matrix (equivalently, the implied correlation $\frac{3}{\sqrt{8}} \approx 1.06$ would exceed 1). The second has positive diagonal and determinant $4 > 0$, so it is valid, with correlation $\frac{2}{\sqrt{4 \times 2}} \approx 0.71$.

Chapter 10: Statistics for Machine Learning

Exercise 10.1. Mean $\frac{36}{6} = 6$. Sorted: $3, 4, 5, 6, 8, 10$, so the median is $\frac{5+6}{2} = 5.5$. Deviations $-2, 2, 0, -1, -3, 4$; squares add up to 34, so $s^2 = \frac{34}{5} = 6.8$ and $s \approx 2.61$.

Exercise 10.2. Mean $\frac{126}{6} = 21$; the median is unchanged at 5.5.

Exercise 10.3. $\bar{y} = 2\bar{x} + 3$ and the median becomes $2m + 3$ (the order of the values is unchanged). The deviations from the mean are doubled and the shift cancels, so $s_y = 2s_x$.

Exercise 10.4. Each term $|x_i - c|$ has slope $+1$ in c if $x_i < c$ (moving c right moves it away from x_i) and -1 if $x_i > c$. Adding up gives the stated slope. It is negative while fewer than half of the values lie below c and positive once more than half do, so g decreases up to the median and increases after it.

Exercise 10.5. $\frac{10}{\sqrt{n}} = 1$ gives $n = 100$; $\frac{10}{\sqrt{n}} = 0.5$ gives $n = 400$.

Exercise 10.6. $\mathbb{E}[X_1] = \mu$, so it is unbiased, but its variance is σ^2, versus $\frac{\sigma^2}{25}$ for $\bar{X}$. Being unbiased is not enough; $\bar{X}$ uses all the data and is much more precise.

Exercise 10.7. By Theorem 10.10 its expectation is $\frac{n-1}{n}\sigma^2 = 0.8\sigma^2$, so its bias is $-0.2\sigma^2$.

Exercise 10.8. $\operatorname{MSE}(A) = 0 + 4 = 4$ and $\operatorname{MSE}(B) = 1 + 1 = 2$, so the biased estimator B is better on average.

Exercise 10.9. We need $\frac{35/12}{n \times 0.1^2} \le 0.05$, i.e. $n \ge \frac{2.9167}{0.0005} \approx 5833.3$, so $n = 5834$ rolls suffice. (Chebyshev is pessimistic; the central limit theorem shows that about 1100 are enough.)

Exercise 10.10. $\frac{1}{5}(0.01 + 0.16 + 0.25 + 0.64 + 0.81) = \frac{1.87}{5} = 0.374$. The exact value is $\int_0^1 x^2 \, \mathrm{d}x = \frac{1}{3} \approx 0.333$.

Exercise 10.11. The standard deviation scales like $1/\sqrt{B}$. Dividing it by 4 requires multiplying B by 16: $B = 512$.

Exercise 10.12. The next toss is independent of the past: $\frac{1}{2}$. Eleven heads in a row had probability $\left(\frac{1}{2}\right)^{11} = \frac{1}{2048}$ before the first toss; the rarity of the whole sequence does not change the next toss.

Exercise 10.13. Approximately $\mathcal{N}\left(3.5, \frac{35/12}{100}\right)$, with standard deviation $\sqrt{0.02917} \approx 0.171$. Since 3.84 is $\frac{0.34}{0.171} \approx 2$ standard deviations above the mean, the probability is about 2.5% (more precisely 0.023).

Exercise 10.14. $\hat{p} = 0.9$ and $\text{SE} = \sqrt{\frac{0.9\times0.1}{400}} = 0.015$, so the interval is $0.9 \pm 0.029 = [0.871, 0.929]$.

Exercise 10.15. We need $1.96\sqrt{\frac{0.9\times0.1}{n}} \le 0.01$, i.e. $n \ge \frac{1.96^2\times0.09}{0.0001} \approx 3457$: about 3500 examples.

Exercise 10.16. $\mathbb{E}[S] = 12 \times \frac{1}{2} - 6 = 0$ and $\text{Var}(S) = 12 \times \frac{1}{12} = 1$. As a sum of many independent pieces, S is approximately standard normal; this was once a popular way of generating normal random numbers.

Exercise 10.17. $\hat{\theta} = \frac{12}{40} = 0.3$.

Exercise 10.18. $\ln \mathcal{L}(\lambda) = \sum_i (x_i \ln \lambda - \lambda - \ln x_i!)$. Its derivative $\frac{1}{\lambda}\sum_i x_i - n$ vanishes at $\hat{\lambda} = \frac{1}{n}\sum_i x_i = \bar{x}$ (and the second derivative $-\sum_i x_i/\lambda^2$ is negative). For the data, $\hat{\lambda} = \frac{10}{5} = 2$.

Exercise 10.19. $\ln \mathcal{L}(\lambda) = n \ln \lambda - \lambda \sum_i t_i$, with derivative $\frac{n}{\lambda} - \sum_i t_i = 0$, so $\hat{\lambda} = \frac{n}{\sum_i t_i} = \frac{1}{\bar{t}} = \frac{1}{1.25} = 0.8$.

Exercise 10.20. $\hat{\mu} = \bar{x} = 5$ and $\hat{\sigma}^2 = \frac{32}{8} = 4$, slightly smaller than $s^2 = \frac{32}{7} \approx 4.57$.

Exercise 10.21. Model B, whose log-likelihood is larger. The ratio of the likelihoods is $e^{-118.2}/e^{-120.5} = e^{2.3} \approx 10$.

Exercise 10.22. The losses are $-\ln 0.9 \approx 0.105$, $-\ln 0.7 \approx 0.357$, $-\ln 0.4 \approx 0.916$ and $-\ln 0.7 \approx 0.357$, with total 1.735 and average 0.434. The third example, a positive example given only 0.4, contributes most.

Exercise 10.23. $-\ln \mathcal{L} = \sum_i \left(\ln(2b) + \frac{1}{b}|y_i - f(\boldsymbol{x}_i)|\right) = n \ln(2b) + \frac{1}{b}\sum_i |y_i - f(\boldsymbol{x}_i)|$. The first term and the factor $\frac{1}{b}$ do not depend on the model, so minimising the NLL means minimising $\sum_i |y_i - f(\boldsymbol{x}_i)|$.

Exercise 10.24. The loss is $-h \ln q - (n-h)\ln(1-q)$, the negative log-likelihood of the coin in Example 10.20, so it is minimised by $q = \frac{h}{n}$, the fraction of positive labels.

Exercise 10.25. $-\ln 0.9 \approx 0.105$, $-\ln 0.5 \approx 0.693$, $-\ln 0.1 \approx 2.303$ and $-\ln 0.001 \approx 6.908$. The loss grows slowly at first and then very fast as the prediction becomes confidently wrong.

Exercise 10.26. $\hat{\theta}_{\text{MAP}} = \frac{7+1}{10+2} = \frac{2}{3} \approx 0.667$, pulled slightly from the MLE 0.7 towards $\frac{1}{2}$.

Exercise 10.27. $\lambda = \frac{\sigma^2}{\tau^2} = 10$. As $\tau^2 \to \infty$ (a vaguer and vaguer prior), $\lambda \to 0$ and we recover least squares.

Exercise 10.28. The posterior is proportional to $\mathcal{L}(\theta) \cdot 1$, so it has the same maximiser as the likelihood. ($-\ln p(\theta) = 0$: no penalty.)

Exercise 10.29. MLE: $\frac{0}{50} = 0$. Laplace: $\frac{0+1}{50+2} = \frac{1}{52} \approx 0.019$. With the MLE, every email containing the word gets a spam likelihood of exactly 0, so it can never be classified as spam, no matter how "spammy" its other words are.

Exercise 10.30. Under H_0 the number of heads has mean 50 and standard deviation $\sqrt{100 \times 0.25} = 5$, so $z = \frac{61-50}{5} = 2.2$. The two-sided p-value is $P(|Z| \geq 2.2) \approx 0.028 < 0.05$: we reject fairness.

Exercise 10.31. Only (b) and (d). Statement (a) confuses $P(\text{data} \mid H_0)$ with $P(H_0 \mid \text{data})$, and (c) confuses significance with size: a tiny effect can be significant if there is enough data.

Exercise 10.32. $1 - 0.95^{20} \approx 0.64$. Testing each at $\alpha = \frac{0.05}{20} = 0.0025$ (the *Bonferroni correction*) gives $1 - 0.9975^{20} \approx 0.049$.

Exercise 10.33. The 95% confidence interval $[0.871, 0.929]$ contains 0.88, so the difference is not significant at the 5% level. (Directly: $z = \frac{0.9-0.88}{\sqrt{0.88 \times 0.12/400}} \approx 1.23$, with two-sided p-value about 0.22.)

Exercise 10.34. (a) Overfitting (high variance): get more data, regularise, augment the data or use a simpler model. (b) Underfitting (high bias): use a bigger model or better features, train longer, or reduce regularisation.

Exercise 10.35. It is trained 5 times, each time on 800 examples, and validated on the remaining 200.

Exercise 10.36. $0.04 + 0.10 + 0.25 = 0.39$. Bias and variance can be reduced (by a better model class, more data, regularisation); the noise 0.25 cannot.

Exercise 10.37. The test errors contain random fluctuations, and choosing the minimum selects the model whose fluctuation happened to be most favourable, the multiple-comparisons effect. An honest estimate requires choosing the model on a validation set and then evaluating it once on an untouched test set.

Exercise 10.38. More data reduces the variance of the learned function (it depends less on the particular sample), which is the main problem of an overfitting model. The bias of an underfitting model comes from its limited model class and does not go away with more data.

Chapter 11: Mathematics for Deep Learning

Exercise 11.1. $2 \cdot 3 + 3 + 3 \cdot 1 + 1 = 13$. For 10–20–20–5: $(200+20)+(400+20)+(100+5) = 745$.

Exercise 11.2. $\boldsymbol{W}^{(2)}\boldsymbol{W}^{(1)} = (1 \;\; 1)$ and $\boldsymbol{W}^{(2)}\boldsymbol{b}^{(1)} + b^{(2)} = 1 + 2 = 3$, so the network computes $x_1 + x_2 + 3$.

Exercise 11.3. $\boldsymbol{W}^{(1)}\boldsymbol{x} + \boldsymbol{b}^{(1)} = (-1 + 0.5, \; -2 - 1) = (-0.5, -3)$, so $\boldsymbol{a}^{(1)} = (0, 0)$ and the output is $z^{(2)} = 0 + 0 - 2 = -2$.

Exercise 11.4. The exponentials are 1, 2 and 3, with sum 6, so the result is $\left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right)$.

Exercise 11.5. It equals $\text{softmax}(0, 1, 2)$. The exponentials are 1, 2.718 and 7.389, with sum 11.107, giving approximately $(0.090, 0.245, 0.665)$.

Exercise 11.6. As $T \to \infty$ all the scaled logits approach 0, and the output approaches the uniform distribution $\left(\frac{1}{K}, \dots, \frac{1}{K}\right)$. As $T \to 0$ the differences between the scaled logits blow up, and the output approaches the one-hot vector of the largest logit.

Exercise 11.7. Let $m = \max_j z_j$. The sum $\sum_j e^{z_j}$ contains the term e^m and consists of K terms that are each at most e^m, so $e^m \le \sum_j e^{z_j} \le K e^m$. Taking logarithms gives $m \le \ln \sum_j e^{z_j} \le m + \ln K$.

Exercise 11.8. $4 \cdot \frac{1}{4} \cdot 2 = 2$ bits, and $2 \cdot \frac{1}{2} \cdot 1 = 1$ bit (outcomes of probability 0 contribute nothing).

Exercise 11.9. $-\ln 0.7 \approx 0.357$ and $-\ln 0.2 \approx 1.609$.

Exercise 11.10. $\frac{1}{2}\log_2 \frac{1/2}{1/4} + \frac{1}{2}\log_2 \frac{1/2}{3/4} = \frac{1}{2} \cdot 1 + \frac{1}{2}\log_2 \frac{2}{3} \approx 0.5 - 0.292 = 0.208$ bits.

Exercise 11.11. With $q_k = \frac{1}{K}$, $D_{\mathrm{KL}}(\boldsymbol{p} \,\|\, \boldsymbol{q}) = \sum_k p_k \log(K p_k) = \log K - H(\boldsymbol{p})$, using $\sum_k p_k = 1$. Since this is at least 0, $H(\boldsymbol{p}) \le \log K$, with equality only for the uniform distribution.

Exercise 11.12. $e^{2.3} \approx 10$: on average it is as uncertain as a uniform choice among about 10 tokens.

Exercise 11.13. $2\boldsymbol{x} + 3\boldsymbol{a}$, by parts (3) and (1) of Theorem 11.13.

Exercise 11.14. Apply part (4) of Theorem 11.13 with $\boldsymbol{A} = \boldsymbol{X}$ and $\boldsymbol{b} = \boldsymbol{y}$, and multiply by the constant $\frac{1}{n}$.

Exercise 11.15. $\frac{\partial L}{\partial \boldsymbol{W}} = \boldsymbol{\delta}\boldsymbol{x}^\top = \left(\begin{smallmatrix} 3 & 0 & 1 \\ -6 & 0 & -2 \end{smallmatrix}\right)$; $\frac{\partial L}{\partial \boldsymbol{x}} = \boldsymbol{W}^\top \boldsymbol{\delta} = (1, -2, 2 + 2) = (1, -2, 4)$; $\frac{\partial L}{\partial \boldsymbol{b}} = (1, -2)$.

Exercise 11.16. The gradient is also 128×784 (100 352 numbers); $\boldsymbol{\delta}$ has 128 components, one per neuron.

Exercise 11.17. $\boldsymbol{q} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$, $L = \ln 3 \approx 1.099$ and $\boldsymbol{q} - \boldsymbol{y} = \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right)$.

Exercise 11.18. $\sum_k (q_k - y_k) = 1 - 1 = 0$. Adding the same constant to all logits does not change the loss, so the derivative of the loss in the direction $\mathbf{1} = (1, \dots, 1)$, which is $\mathbf{1}^\top(\boldsymbol{q} - \boldsymbol{y})$, must be 0.

Exercise 11.19. $q = \sigma(2) \approx 0.881$, $L = -\ln(1 - 0.881) \approx 2.127$ and $\frac{\partial L}{\partial z} = q - y \approx 0.881$.

Exercise 11.20. $z = 2 - 1 = 1$, $q = \sigma(1) \approx 0.731$, so $\nabla_{\boldsymbol{w}} L = (q - 1)\boldsymbol{x} \approx -0.269\,(2, 1) = (-0.538, -0.269)$ and $\frac{\partial L}{\partial b} \approx -0.269$.

Exercise 11.21. Then $\hat{y} = y$, the loss is 0, $\delta^{(2)} = 0$, and all gradients are 0: a perfectly predicted example teaches the network nothing.

Exercise 11.22. The first hidden neuron has $z = -1 < 0$, so $\mathrm{ReLU}'(z) = 0$ and $\delta_1^{(1)} = 0$: its weights had no effect on the output for this input. If it were inactive for all examples, its weights would never change: a "dead" ReLU neuron.

Exercise 11.23. Forward: $z_1 = 1$, $h = 1$, $\hat{y} = 3$, $L = \frac{1}{2}$. Backward: $\delta_2 = \hat{y} - y = 1$, so $\frac{\partial L}{\partial w_2} = \delta_2 h = 1$ and $\frac{\partial L}{\partial b_2} = 1$. Then $\delta_1 = w_2 \delta_2 \,\mathrm{ReLU}'(z_1) = 3$, so $\frac{\partial L}{\partial w_1} = \delta_1 x = 3$ and $\frac{\partial L}{\partial b_1} = 3$.

Exercise 11.24. $\boldsymbol{\delta}^{(1)} \in \mathbb{R}^4$ and $\boldsymbol{\delta}^{(2)} \in \mathbb{R}^2$; $\frac{\partial L}{\partial \boldsymbol{W}^{(1)}}$ is 4×3 and $\frac{\partial L}{\partial \boldsymbol{W}^{(2)}}$ is 2×4, the shapes of the weight matrices.

Exercise 11.25. Each layer contributes a factor of at most $\frac{1}{4}$, so the total is at most $\left(\frac{1}{4}\right)^{20} \approx 9.1 \times 10^{-13}$.

Exercise 11.26. Without residual connections the derivative is $(-0.05)^{20} \approx 9.5 \times 10^{-27}$. With them each block has derivative $1 + f'(x) = 0.95$, and the composition has $0.95^{20} \approx 0.36$.

Exercise 11.27. $\|\boldsymbol{g}\| = 10 > 5$, so $\boldsymbol{g} \leftarrow 5 \cdot \frac{(6,-8)}{10} = (3, -4)$.

Exercise 11.28. $\tanh'(0) = 1$, whereas $\sigma'(0) = \frac{1}{4}$: near zero, tanh passes gradients on almost unchanged, while every sigmoid layer divides them by at least 4.

Exercise 11.29. The forward pass creates $u = x \cdot x = 4$ (local derivatives 2 and 2) and $f = u \cdot x = 8$ (local derivatives $x = 2$ with respect to u and $u = 4$ with respect to x). Backward: f gets gradient 1; u gets $1 \cdot 2 = 2$ and x gets $1 \cdot 4 = 4$; then u passes $2 \cdot 2 = 4$ to x twice. In total $4 + 4 + 4 = 12 = 3x^2$.

Exercise 11.30. Compute $t = \tanh(\text{data})$ and return a new `Value` with data t, the single parent `self`, and local derivative $1 - t^2$, since $\tanh'(x) = 1 - \tanh^2(x)$.

Exercise 11.31. Central differences need two evaluations per parameter: about 2×10^9 evaluations. Reverse mode needs one forward and one backward pass, together costing a small multiple of a single evaluation.

Exercise 11.32. Scores $\frac{1}{\sqrt{2}}(0, 1, 1) \approx (0, 0.707, 0.707)$, weights $\approx (0.198, 0.401, 0.401)$, output $\approx 0.198(1, 0) + 0.401(0, 1) + 0.401(1, 1) = (0.599, 0.802)$.

Exercise 11.33. Equal scores give equal weights $\frac{1}{n}$, so the output is the plain average of the values. A dominant score gives a weight close to 1 for that key, so the output is close to its value: attention then acts like a look-up.

Exercise 11.34. The variance is 256, so the standard deviation is 16; after dividing by $\sqrt{256} = 16$ it is 1.

Exercise 11.35. $\boldsymbol{W}_1$ has $16 \times 2 = 32$ entries, $\boldsymbol{b}_1$ has 16, $\boldsymbol{w}_2$ has 16, and b_2 is one number: 65 parameters.

Exercise 11.36. The loss is the *mean* of the n per-example losses, so the derivative of the loss with respect to the logit of example i is $\frac{1}{n}(q_i - y_i)$.

Exercise 11.37. All hidden pre-activations would be 0. The code uses $\text{ReLU}'(0) = 0$, and together with the zero output weights this makes every gradient of $\boldsymbol{W}_1$, $\boldsymbol{b}_1$ and $\boldsymbol{w}_2$ vanish: only b_2 would learn, and the network would predict the same probability everywhere. More generally, identical initial weights give identical neurons, which receive identical gradients and stay identical; random initialisation breaks this symmetry.

Exercise 11.38. Steps would overshoot and the loss would oscillate or diverge instead of decreasing, as in Section 8.4; with ReLU networks, large steps can also push many neurons into the inactive region permanently.

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